

AOS-CX 10.13 Fundamentals Guide

6300, 6400 Switch Series



**Hewlett Packard
Enterprise**

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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- Aruba 6300 Switch Series (JL658A, JL659A, JL660A, JL661A, JL662A, JL663A, JL664A, JL665A, JL666A, JL667A, JL668A, JL762A, R8S89A, R8S90A, R8S91A, R8S92A, S0E91A, S0X44A)
- Aruba 6400 Switch Series (R0X31A, R0X38B, R0X38C, R0X39B, R0X39C, R0X40B, R0X40C, R0X41A, R0X41C, R0X42A, R0X42C, R0X43A, R0X43C, R0X44A, R0X44C, R0X45A, R0X45C, R0X26A, R0X27A, JL741A, S0E48A, S0E48A #0D1, S1T83A, S1T83A #0D1)

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Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ■ <code><example-text></code> ■ <code><example-text></code> ■ <i>example-text</i> ■ <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (<code>< ></code>). Substitute the text—including the enclosing angle brackets—with an actual value. ■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.

Convention	Usage
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

switch(CONTEXT-NAME)#

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

```
member/slot/port
```

On the 6300 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 10. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on member 1.

On the 6400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/1 and 1/2.
 - Line modules are on the front of the switch starting in slot 1/3.
- *port*: Physical number of a port on a line module.

For example, the logical interface **1/3/4** in software is associated with physical port 4 in slot 3 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: *member/power supply*:
 - *member*: 1.
 - *power supply*: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: *member/tray/fan*:
 - *member*: 1.
 - *tray*: 1 to 4.
 - *fan*: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: *member/module*:
 - *member*: 1.
 - *member*: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

AOS-CX is a new, modern, fully programmable operating system built using a database-centric design that ensures higher availability and dynamic software process changes for reduced downtime. In addition to robust hardware reliability, the AOS-CX operating system includes additional software elements not available with traditional systems, including:

- **Automated visibility to help IT organizations scale:** The Aruba Network Analytics Engine allows IT to monitor and troubleshoot network, system, application, and security-related issues easily through simple scripts. This engine comes with a built-in time series database that enables customers and developers to create software modules that allow historical troubleshooting, as well as analysis of historical trends to predict and avoid future problems due to scale, security, and performance bottlenecks.
- **Programmability simplified:** A switch that is running the AOS-CX operating system is fully programmable with a built-in Python interpreter as well as REST-based APIs, allowing easy integration with other devices both on premise and in the cloud. This programmability accelerates IT organization understanding of and response to network issues. The database holds all aspects of the configuration, statistics, and status information in a highly structured and fully defined form.
- **Faster resolution with network insights:** With legacy switches, IT organizations must troubleshoot problems after the fact, using traditional tools like CLI and SNMP, augmented by separate, expensive monitoring, analytics, and troubleshooting solutions. These capabilities are built in to the AOS-CX operating system and are extensible.
- **High availability:** For switches that support active and standby management modules, the AOS-CX database can synchronize data between active and standby modules and maintain current configuration and state information during a failover to the standby management module.
- **Ease of roll-back to previous configurations:** The built-in database acts as a network record, enabling support for multiple configuration checkpoints and the ability to roll back to a previous configuration checkpoint.

AOS-CX system databases

The AOS-CX operating system is a modular, database-centric operating system. Every aspect of the switch configuration and state information is modeled in the AOS-CX switch configuration and state database, including the following:

- Configuration information
- Status of all features
- Statistics

The AOS-CX operating system also includes a time series database, which acts as a built-in network record. The time series database makes the data seamlessly available to Aruba Network Analytics Engine agents that use rules that evaluate network conditions over time. Time-series data about the resources monitored by agents are automatically collected and presented in graphs in the switch Web UI.

Aruba Network Analytics Engine introduction

The Aruba Network Analytics Engine is a first-of-its-kind built-in framework for network assurance and remediation. Combining the full automation and deep visibility capabilities of the AOS-CX operating system, this unique framework enables monitoring, collecting network data, evaluating conditions, and taking corrective actions through simple scripting agents.

This engine is integrated with the AOS-CX system configuration and time series databases, enabling you to examine historical trends and predict future problems due to scale, security, and performance bottlenecks. With that information, you can create software modules that automatically detect such issues and take appropriate actions.

With the faster network insights and automation provided by the Aruba Network Analytics Engine, you can reduce the time spent on manual tasks and address current and future demands driven by Mobility and IoT.

AOS-CX CLI

The AOS-CX CLI is an industry standard text-based command-line interface with hierarchical structure designed to reduce training time and increase productivity in multivendor installations.

The CLI gives you access to the full set of commands for the switch while providing the same password protection that is used in the Web UI. You can use the CLI to configure, manage, and monitor devices running the AOS-CX operating system.

Aruba CX mobile app

The Aruba CX mobile app enables you to use a mobile device to configure or access a supported AOS-CX switch. You can connect to the switch through Bluetooth or Wi-Fi.

You can use this application to do the following:

- Connect to the switch for the first time and configure basic operational settings—all without requiring you to connect a terminal emulator to the console port.
- View and change the configuration of individual switch features or settings.
- Manage the running configuration and startup configuration of the switch, including the following:
 - Transferring files between the switch and your mobile device
 - Sharing configuration files from your mobile device
 - Copying the running configuration to the startup configuration
- Access the switch CLI.

For more information about the Aruba CX mobile app, see:

www.arubanetworks.com/products/networking/switches/cx-mobileapp.

Aruba NetEdit

Aruba NetEdit enables the automation of multidevice configuration change workflows without the overhead of programming.

The key capabilities of NetEdit include the following:

- Intelligent configuration with validation for consistency and compliance
- Time savings by simultaneously viewing and editing multiple configurations

- Customized validation tests for corporate compliance and network design
- Automated large-scale configuration deployment without programming
- Ability to track changes to hardware, software, and configurations (whether made through NetEdit or directly on the switch) with automated versioning

For more information about Aruba NetEdit, search for NetEdit at the following website:

www.hpe.com/support/hpesc

Ansible modules

Ansible is an open-source IT automation platform.

Aruba publishes a set of Ansible configuration management modules designed for switches running AOS-CX software. The modules are available from the following places:

- The `arubanetworks.aoscx_role` role in the Ansible Galaxy at: https://galaxy.ansible.com/arubanetworks/aoscx_role
- The `aoscx-ansible-role` at the following GitHub repository: <https://github.com/aruba/aoscx-ansible-role>

AOS-CX Web UI

The Web UI gives you quick and easy visibility into what is happening on your switch, providing faster problem detection, diagnosis, and resolution. The Web UI provides dashboards and views to monitor the status of the switch, including easy to read indicators for: power supply, temperature, fans, CPU use, memory use, log entries, system information, firmware, interfaces, VLANs, and LAGs. In addition, you use the Web UI to access the Network Analytics Engine, run certain diagnostics, and modify some aspects of the switch configuration.

AOS-CX REST API

Switches running the AOS-CX software are fully programmable with a REST (REpresentational State Transfer) API, allowing easy integration with other devices both on premises and in the cloud. This programmability—combined with the Aruba Network Analytics Engine—accelerates network administrator understanding of and response to network issues.

The AOS-CX REST API enables programmatic access to the AOS-CX configuration and state database at the heart of the switch. By using a structured model, changes to the content and formatting of the CLI output do not affect the programs you write. And because the configuration is stored in a structured database instead of a text file, rolling back changes is easier than ever, thus dramatically reducing a risk of downtime and performance issues.

The AOS-CX REST API is a web service that performs operations on switch resources using HTTPS `POST`, `GET`, `PUT`, and `DELETE` methods.

A switch resource is indicated by its Uniform Resource Identifier (URI). A URI can be made up of several components, including the host name or IP address, port number, the path, and an optional query string. The AOS-CX operating system includes the AOS-CX REST API Reference, which is a web interface based on the Swagger UI. The AOS-CX REST API Reference provides the reference documentation for the REST API, including resources URIs, models, methods, and errors. The AOS-CX REST API Reference shows most of the supported read and write methods for all switch resources.

In-band and out-of-band management

Management communications with a managed switch can be either of the following:

In band

In-band management communications occur through ports on the line modules of the switch, using common communications protocols such as SSH and SNMP.

When you use an in-band management connection, management traffic from that connection uses the same network infrastructure as user data. User data uses the `data plane`, which is responsible for moving data from source to destination. Management traffic that uses the data plane is more likely to be affected by traffic congestion and other issues affecting the user network.

Out of band

OOBM (out-of-band management) communications occur through a dedicated serial or USB console port or through a dedicated networked management port.

OOBM operates on a `management plane` that is separate from the `data plane` used by data traffic on the switch and by in-band management traffic. That separation means that OOBM can continue to function even during periods of traffic congestion, equipment malfunction, or attacks on the network. In addition, it can provide improved switch security: a properly configured switch can limit management access to the management port only, preventing malicious attempts to gain access through the data ports.

Networked OOBM typically occurs on a management network that connects multiple switches. It has the added advantage that it can be done from a central location and does not require an individual physical cable from the management station to the console port of each switch.

SNMP-based management support

The AOS-CX operating system provides SNMP read access to the switch. SNMP support includes support of industry-standard MIB (Management Information Base) plus private extensions, including SNMP events, alarms, history, statistics groups, and a private alarm extension group. SNMP access is disabled by default.

User accounts

To view or change configuration settings on the switch, users must log in with a valid account. Authentication of user accounts can be performed locally on the switch, or by using the services of an external TACACS+ or RADIUS server.

Two types of user accounts are supported:

- **Operators:** Operators can view configuration settings, but cannot change them. No operator accounts are created by default.
- **Administrators:** Administrators can view and change configuration settings. A default locally stored administrator account is created with username set to **admin** and no password. You set the administrator account password as part of the initial configuration procedure for the switch.

Perform the initial configuration of a factory default switch using one of the following methods:

- Load a switch configuration using zero-touch provisioning (ZTP). When ZTP is used, the configuration is loaded from a server automatically when the switch booted from the factory default configuration.
- Connect to the switch wirelessly with a mobile device through Bluetooth, and use the Aruba CX Mobile App to deploy an initial configuration from a provided template. The template you choose during the deployment process determines how the management interface is configured. Optionally, as the final deployment step, you can select to import the switch into NetEdit through a WiFi connection to the NetEdit server.

Alternatively, you can use the Aruba CX Mobile App to manually configure switch settings and features for a subset of the features you can configure using the CLI. You can also access the CLI through the mobile application.

- Connect the management port on the switch to your network, and then use SSH client software to reach the switch from a computer connected to the same network. This requires that a DHCP server is installed on the network. Configure switch settings and features by executing CLI commands.
- Connect a computer running terminal emulation software to the console port on the switch. Configure switch settings and features by executing CLI commands.



Reserved ports or ports used by other applications/services within the system are not recommended to be used for other services. When two services use the same port there is a chance of unexpected behaviors from these services. Best practice is to use a unique port for each service across the system.

Initial configuration using ZTP

Zero Touch Provisioning (ZTP) configures a switch automatically from a remote server.

Prerequisites

- The switch must be in the factory default configuration.

Do not change the configuration of the switch from its factory default configuration in any way, including by setting the administrator password.

- Your network administrator or installation site coordinator must provide a Category 6 (Cat6) cable connected to the network that provides access to the servers used for Zero Touch Provisioning (ZTP) operations.

Procedure

1. Connect the network cable to the out-of-band management port on the switch. If your network administrator or installation site coordinator has instructed you to connect network cable to a data port, connect the cable to that data port instead.

See the *Installation Guide* for switch to determine the location of the switch ports.

2. If the switch is powered on, power off the switch.
3. Power on the switch. During the ZTP operation, the switch might reboot if a new firmware image is being installed. ZTP goes to "Failed" state if the switch receives DHCP IP for vlan1 and does not receive any ZTP options within 60 seconds.

Initial configuration using the Aruba CX mobile app

This procedure describes how to use your mobile device to connect to the Bluetooth interface of the switch to connect to the switch for the first time so that you can configure basic operational settings using the Aruba CX mobile app.

Prerequisites

- You have obtained the USB Bluetooth adapter that was shipped with the switch. Information about the make and model of the supported adapter is included in the information about the Aruba CX mobile app in the Apple Store or Google Play.
- The Aruba CX mobile app must be installed on your mobile device.
- Bluetooth must be enabled on your mobile device.
- Your mobile device must be within the communication range of the Bluetooth adapter.
- If you are planning to import the switch into NetEdit, your mobile device must be able to use a Wi-Fi connection—not Bluetooth—to access the NetEdit server.

If your mobile device does not support simultaneous Bluetooth and Wi-Fi connections, you must use the NetEdit interface to import the switch at a later time. You can use the **Devices** tab to display the IP address of the switches you configured using your mobile device.

- The switch must be installed and powered on, with the network operating system boot sequence complete.

For information about installing and powering on the switch, see the *Installation Guide* for the switch.

Because you are using this mobile application to configure the switch through the Bluetooth interface, it is not necessary to connect a console to the switch.

- Bluetooth and USB must be enabled on the switch. On switches shipped from the factory, Bluetooth and USB are enabled by default.

Procedure

1. Install the USB Bluetooth adapter in the USB port of the switch.

For switches that have multiple management modules, you must install the USB Bluetooth adapter in the USB port of the active management module. Typically, the active management module is the module in slot 5. On the 6400, the active management module is typically installed in slot 1. When configuring a stack, a USB Bluetooth adapter must be installed on each 6300 switch in the stack.

Switches shipped from the factory have both USB and Bluetooth enabled by default.

For information about the location of the USB port on the switch, see the *Installation Guide* for the switch.

2. Use the Bluetooth settings on your mobile device to pair and connect the switch to your mobile device.

If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

Switch_model - Serial_number

For example: 8325-987654X1234567 or 8320-AB12CDE123

A switch supports one active Bluetooth connection at a time.

On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

3. Open the Aruba CX mobile app on your mobile device.



The application attempts to connect to the switch using the switch Bluetooth IP address and the default switch login credentials. The **Home** screen of the application shows the status of the connection to the switch:

- If the login attempt was successful, the Bluetooth icon is displayed and the status message shows the Bluetooth IP address of the switch. In addition, the connection graphic is green. You can continue to the next step.
 - If the login attempt was not successful, but a response was received, the Bluetooth icon is displayed, but the status message is: `Login Required`. You can continue to the next step. When you tap one of the tiles, you will be prompted for login credentials.
 - If the login attempt did not receive a response, the Bluetooth icon is not displayed, and the status message is: `No Connection`.
4. Create the initial switch configuration:
 - You can deploy an initial configuration to the switch. Through this process, you supply the information required by a configuration template that you choose from a list of templates provided by the application. Then you deploy the configuration to the switch and, optionally, import the switch into NetEdit.



When you deploy a switch configuration, it becomes the running configuration, replacing the entire existing configuration of the switch. All changes previously made to the factory default configuration are overwritten.

If you plan to both deploy a switch configuration and customize the configuration of switch features, deploy the initial configuration first.

To deploy an initial switch configuration, tap: **Initial Config** and follow the instructions in the application.

- Alternatively, you can complete the initial configuration of the switch by tapping **Modify Config** and then selecting the features and settings to configure.
- You can also use the **Modify Config** feature to configure some switch features after the initial configuration is complete. For more information about what you can configure using the Aruba CX mobile app, see the online help for the application.

Troubleshooting Bluetooth connections

Bluetooth connection IP addresses

The Bluetooth connection uses IP addresses in the 192.168.99.0/24 subnet.

Switch

192.168.99.1

Mobile device

192.168.99.10

Bluetooth is connected but the switch is not reachable

Symptom

The mobile device settings indicate that the device is connected to the switch through Bluetooth. However, the mobile application indicates that the switch is not reachable.

Solution 1

Cause

The mobile device is paired with a different nearby switch.

Action

1. Verify the model number and serial number of the switch to which you are attempting to connect.
2. Use the Bluetooth settings on your mobile device to pair and connect the switch to your mobile device.

If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

Switch_model-Serial_number

For example: 8325-987654X1234567 or 8320-AB12CDE123

A switch supports one active Bluetooth connection at a time.

On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 2

Cause

The mobile device is connected to a different network—such as through a Wi-Fi connection—that conflicts with the subnet used for the switch Bluetooth connection.

Action

Disconnect the mobile device from the network that is using the conflicting subnet.

For example, use the mobile device settings to turn off or disable Wi-Fi. If you choose to disable Wi-Fi on the mobile device, and you are not able to access cellular service, you will not be able to connect to the NetEdit server to import the switch, but you can still deploy a switch configuration.

Bluetooth is not connected

Symptom

Your mobile device cannot establish a Bluetooth connection to the switch.

Solution 1

Cause

Bluetooth is not enabled on your mobile device.

Action

- Use your mobile device settings application to enable Bluetooth.
- Use the Bluetooth settings on your mobile device to pair and connect the switch to your mobile device.

If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

Switch_model-Serial_number

For example: 8325-987654X1234567 or 8320-AB12CDE123

A switch supports one active Bluetooth connection at a time.

On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 2

Cause

Your mobile device is not within the broadcast range of the Bluetooth adapter.

Action

Move closer to the switch.

Devices can communicate through Bluetooth when they are close, typically within a few feet of each other.

Solution 3

Cause

Your mobile device is not paired with the switch.

Action

1. Use your mobile device settings application to enable Bluetooth.
2. Use the Bluetooth settings on your mobile device to pair and connect the switch to your mobile device.

If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

Switch_model-Serial_number

For example: 8325-987654X1234567 or 8320-AB12CDE123

A switch supports one active Bluetooth connection at a time.

3. On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 4

Cause

Bluetooth is not enabled on the switch.

New switches are shipped from the factory with the USB port and Bluetooth enabled. However, an installed switch might have been configured to disable Bluetooth or disable the USB port, which the USB Bluetooth adapter uses.

Action

Use a different CLI connection to enable Bluetooth on the switch.

- Use the `show bluetooth` CLI command to show the Bluetooth configuration and the status of the Bluetooth adapter.
- To enable the USB port, enter the CLI command: `usb`
- An inserted USB drive must be mounted each time the switch boots or fails over to a different management module. To mount the drive, enter the CLI command: `usb mount`
- To enable Bluetooth, enter the CLI command: `bluetooth enable`

Solution 5

Cause

Another mobile device has already connected to the switch through Bluetooth. This cause is likely if your device is repeatedly disconnected within 1-2 seconds of establishing a connection.

Action

1. Use a different CLI connection to see if there is another device connected:
Use the `show bluetooth` CLI command to show the Bluetooth configuration and the status of the Bluetooth adapter.
2. Either disconnect the other device or use that device to communicate with the switch.
A switch can use Bluetooth to connect to one mobile device at a time.

Solution 6

Cause

The switch has been restarted since the mobile device was last paired with the switch, and the device is having difficulty establishing the Bluetooth connection.

Action

1. Use the Bluetooth mobile device settings to forget the switch device.
2. Use your mobile device settings application to disable Bluetooth.
Use your mobile device settings application to enable Bluetooth.
If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:
Switch_model-Serial_number
For example: 8325-987654X1234567 or 8320-AB12CDE123
A switch supports one active Bluetooth connection at a time.
On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 7

Cause

The USB Bluetooth adapter is not installed in the switch.

If the switch has multiple management modules, the USB Bluetooth adapter might be installed in the management module that is not the active management module.

Action

Install the USB Bluetooth adapter in the USB port of the switch.

For switches that have multiple management modules, you must install the USB Bluetooth adapter in the USB port of the active management module. Typically, for new switches, the active management module is the module in slot 5 (Aruba 8400 switches) or slot 1 (Aruba 6400 switches).

For information about the location of the USB port on the switch, see the *Installation Guide* for the switch.

Solution 8

Cause

A problem occurred with the Bluetooth feature on the switch. For example, the software daemon was stopped and then restarted.

Action

1. Use a different connection to the switch CLI to disable and then enable Bluetooth.

```
switch(config)# bluetooth disable  
switch(config)# bluetooth enable
```

2. Use the Bluetooth mobile device settings to forget the switch device.
3. Use your mobile device settings application to disable Bluetooth.
4. Use your mobile device settings application to enable Bluetooth.
5. Use your mobile device settings application to enable Bluetooth.

If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

Switch_model-Serial_number

For example: 8325-987654X1234567 or 8320-AB12CDE123

A switch supports one active Bluetooth connection at a time.

On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 9

Cause

A switch that is member of a stack (but is not the conductor switch), has a USB Bluetooth adapter installed, but mobile application has lost contact with that switch.

Action

Remove and then reinstall the USB Bluetooth adapter.

Do not remove the USB Bluetooth adapter from the conductor switch.

Initial configuration using the CLI

This procedure describes how to connect to the switch for the first time and configure basic operational settings using the CLI. In this procedure, you use a computer to connect to the switch using the either the console port or management port.

Procedure

1. Connect to the [console port](#) or the [management port](#).
2. [Log into the switch for the first time](#).
3. [Configure switch time using the NTP client](#).

Connecting to the console port

Prerequisites

- A switch installed as described in its hardware installation guide.
- A computer with terminal emulation software.
- A JL448A Aruba X2 C2 RJ45 to DB9 console cable. (6400 only), or a USB-C cable (6300/6400).

Procedure

1. Connect the console port on the switch to the serial port on the computer using a console cable, or connect the USB-C port on the switch to the USB-C port on the computer using a USB-C cable.
2. Start the terminal emulation software on the computer and configure a new serial session with the following settings:
 - Speed: 115200 bps
 - Data bits: 8
 - Stop bits: 1
 - Parity: None
 - Flow control: None
3. Start the terminal emulation session.
4. Press **Enter** once. If the connection is successful, you are prompted to login.

Optional console port speed setting

If desired, the console port speed can be set with the `console baud-rate` command. For example, setting the console port speed to 9600 bps:

```
switch(config)# console baud-rate 9600
This command will configure the baud rate immediately for the active serial
console session. After the command is executed the user will be prompted to
re-login. The serial console will be inaccessible until the terminal client
settings are updated to match the baud rate of the switch.
Continue (y/n)? y
```

Showing the console port current speed:

```
switch# show console
Baud Rate: 9600
```

For details on the `console baud-rate` and `show console` commands, see [Switch system and hardware commands](#).

Connecting to the management port

Prerequisites

- Two Ethernet cables
- SSH client software

Procedure

1. By default, the management interface is set to automatically obtain an IP address from a DHCP server, and SSH support is enabled. If there is no DHCP server on your network, you must configure a static address on the management interface:
 - a. Connect to the [console port](#)
 - b. Configure the [management interface](#).
2. Use an Ethernet cable to connect the management port to your network.
3. Use an Ethernet cable to connect your computer to the same network.
4. Start your SSH client software and configure a new session using the address assigned to the management interface. (If the management interface is set to operate as a DHCP client, retrieve the IP address assigned to the management interface from your DHCP server.)
5. Start the session. If the connection is successful, you are prompted to log in.

Configure using DHCP or static IP



Users can use any data ports for in-band management purposes. IP DHCP is supported on interface VLAN 1 only. All switch ports are part of access VLAN 1 by default. Static IP address and IP DHCP configuration can co-exist on VLAN 1, however static addresses take precedence whenever configured.

DHCP Configuration

```
switch#: config
switch(config)#: vlan 1
Switch(config-vlan-1)#: description Management VLAN
Switch(config-vlan-1)#: end
Switch#
!
Switch(config)#: interface 1/1/1
Switch(config-if)#: description IN-BAND Management Port
Switch(config-if)#: vlan access 1
Switch(config-if)#: no shutdown
Switch(config-if)#: end
Switch#
!
Switch(config)#: interface vlan 1
Switch(config-if-vlan)#: description IN-BAND Management Interface
Switch(config-if-vlan)#: ip dhcp
Switch(config-if-vlan)#: no shutdown
Switch(config-if-vlan)#: end
Switch#
!
```

Without DHCP Configuration

```
switch#: config
switch(config)#: vlan 1
Switch(config-vlan-1)#: description Management VLAN
Switch(config-vlan-1)#: end
Switch#
```

```

!
Switch(config)#: interface 1/1/1
Switch(config-if)#: description IN-BAND Management Port
Switch(config-if)#: vlan access 1
Switch(config-if)#: no shutdown
Switch(config-if)#: end
Switch#
!
Switch(config)#: interface vlan 1
Switch(config-if-vlan)#: description IN-BAND Management Interface
Switch(config-if-vlan)#: no ip dhcp
Switch(config-if-vlan)#: ip address 192.168.10.1/24
Switch(config-if-vlan)#: no shutdown
Switch(config-if-vlan)#: end
Switch#

```

Logging into the switch for the first time

The first time you log in to the switch you must use the default administrator account. This account has no password, so you will be prompted on login to define one to safeguard the switch.

Procedure

1. When prompted to log in, specify **admin**. When prompted for the password, press **ENTER**. (By default, no password is defined.)

For example:

```

switch login: admin
password:

```

2. Define a password for the **admin** account. The password can contain up to 32 alphanumeric characters in the range ASCII 32 to 127, which includes special characters such as asterisk (*), ampersand (&), exclamation point (!), dash (-), underscore (_), and question mark (?).

For example:

```

Please configure the 'admin' user account password.
Enter new password: *****
Confirm new password: *****
switch#

```

3. You are placed into the manager command context, which is identified by the prompt: `switch#`, where `switch` is the model number of the switch. Enter the command `config` to change to the global configuration context `config`.

For example:

```

switch# config
switch(config)#

```

Setting switch time using the NTP client

Prerequisites

- The IP address or domain name of an NTP server.
- If the NTP server uses authentication, obtain the password required to communicate with the NTP server.

Procedure

1. If the NTP server requires authentication, define the authentication key for the NTP client with the command `ntp authentication`.
2. Configure an NTP server with the command `ntp server`.
3. By default, NTP traffic is sent on the default VRF. If you want to send NTP traffic on the management VRF, use the command `ntp vrf`.
4. Review your NTP configuration settings with the commands `show ntp servers` and `show ntp status`.
5. See the current switch time, date, and time zone with the command `show clock`.

Example

This example creates the following configuration:

- Defines the authentication key **1** with the password **myPassword**.
- Defines the NTP server **my-ntp.mydomain.com** and makes it the preferred server.
- Sets the switch to use the management VRF (**mgmt**) for all NTP traffic.

```
switch(config)# ntp authentication-key 1 md5 myPassword
switch(config)# ntp server my-ntp.mydomain.com key 10 prefer
switch(config)# ntp vrf mgmt
```

Configuring banners

1. Configure the banner that is displayed when a user connects to a management interface. Use the command `banner motd`. For example:

```
switch(config)# banner motd ^
Enter a new banner. Terminate the banner with the delimiter you have chosen.
>> This is an example of a banner text which a connecting user
>> will see before they are prompted for their password.
>>
>> As you can see it may span multiple lines and the input
>> will be terminated when the delimiter character is
>> encountered.^
Banner updated successfully!
```

2. Configure the banner that is displayed after a user is authenticated. Use the command `banner exec`. For example:

```
switch(config)# banner exec &
Enter a new banner. Terminate the banner with the delimiter you have chosen.
>> This is an example of a different banner text. This time
>> the banner entered will be displayed after a user has
```

```
>> authenticated.  
>>  
>> & This text will not be included because it comes after the '&'  
Banner updated successfully!
```

Configuring in-band management on a data port

Prerequisites

- A connection to the CLI via either the console port or the management port
- Ethernet cable

Procedure

1. Use an Ethernet cable to connect a data port to your network.
2. [Configure a layer 3 interface](#) on the data port.
3. Enable SSH support on the interface (on the default VRF) with the command `ssh server vrf default`.

For example:

```
switch# config  
switch(config)# ssh server vrf default
```

4. Enable the Web UI on the interface (on the default VRF) with the command `https-server vrf default`.

For example:

```
switch(config)# https-server vrf default
```

Using the Web UI

The Web UI is disabled by default. Follow these steps to enable it on the management port and log in. The Web UI is enabled by default on the default VRF.

Prerequisites

- A connection to the switch CLI.

Procedure

1. Log in to the CLI.
2. Switch to `config` context and enable the Web UI on the management port VRF with the command `https-server vrf mgmt`.

For example:

```
switch# config  
switch(config)# https-server vrf mgmt
```

3. Start your web browser and enter the IP address of the management port in the address bar,
For example: **https://192.168.1.1**
4. The Web UI starts and you are prompted to log in.

Configuring the management interface

Prerequisites

A connection to the console port.

Procedure

1. Switch to the management interface context with the command `interface mgmt`.
2. By default, the management interface on the management port is enabled. If it was disabled, re-enable it with the command `no shutdown`.
3. Use the command `ip dhcp` to configure the management interface to automatically obtain an address from a DHCP server on the network (factory default setting). Or, assign a static IPv4 or IPv6 address, default gateway, and DNS server with the commands `ip address`, `ipv6 address`, `ip static`, `default-gateway`, and `nameserver`.
4. SSH is enabled by default on the management VRF. If disabled, enable SSH with the command `ssh server vrf mgmt`.

Examples

This example enables the management interface with dynamic addressing using DHCP:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# no shutdown
switch(config-if-mgmt)# ip dhcp
```

This example enables the management interface with static addressing creating the following configuration:

- Sets a static IPv4 address of **198.168.100.10** with a mask of **24** bits.
- Sets the default gateway to **198.168.100.200**.
- Sets the DNS server to **198.168.100.201**.

```
switch(config)# interface mgmt
switch(config-if-mgmt)# no shutdown
switch(config-if-mgmt)# ip static 198.168.100.10/24
switch(config-if-mgmt)# default-gateway 198.168.100.200
switch(config-if-mgmt)# nameserver 198.168.100.201
```

IP prefix priority

IP prefix priority configuration is used to optimize route lookup based on the IP address prefix length, sometimes known as the IP subnet mask length. Switch models supporting IP prefix priority configuration have six or seven lookup tables with the first five or six being configurable for specific prefix lengths and the last one being used for all other prefix lengths.

The default configuration of the five or six exact IP prefix tables use common lengths typical to many networks. For sites that make extensive use of atypical IP prefix lengths (lengths that differ from the defaults), the exact prefix tables can be customized by configuring IP prefix priority. Doing so avoids over filling the “other” table and keeps as many routes as possible in the fast-path.

Commands `show ip prefix-priority` and `show ipv6 prefix-priority` show the default, current, and any pending IP prefix priority configuration for a switch.

See also [IP Prefix priority commands](#).

System profiles and configurable IP prefix priority

(Applicable to the 6400 Switch Series.) System profiles set the overall capabilities and capacities of the switch (including IP prefix partitions), based on the selected profile used at boot time. Only certain profiles support configurable IP prefix priority.

To use configurable IP prefix priority, the switch must be booted into a capable profile. If a switch is booted into a profile that does not support configurable IP prefix priority, a warning is displayed before the reboot, and upon reboot, the default IP prefix tables are used and any configured IP prefix priority is discarded.

See also [Selecting the system profile](#).

IP prefix priority defaults



(Applicable to the 6400 Switch Series.) This default IP prefix priority information only applies to system profiles specifically named here. Any profile unnamed here does not support configurable IP prefix priority. Some profiles require six IP prefix values and others require five IP prefix values.

6300 Switch Series (no profiles), (six prefix values required):

- **IPv4:** 24, 23, 22, 21, 16, 8
- **IPv6:** 64, 48, 46, 44, 40, 36

6400 Switch Series, V1 modules, profile: **default**, v2 modules profiles: **default**, **V2-default** (six prefix values required):

- **IPv4:** 24, 23, 22, 21, 16, 8
- **IPv6:** 64, 48, 46, 44, 40, 36

6400 Switch Series, V2 modules, profile: **v2-Core-High-Bandwidth** (five prefix values required):

- **IPv4:** 24, 23, 22, 21, 20
- **IPv6:** 64, 48, 44, 36, 32

IP Prefix priority commands

ip prefix-priority

```
ip prefix-priority <PREFIX-LENGTHS>
no ip prefix-priority [<PREFIX-LENGTHS>]
```

Description

Configures custom IPv4 route prefix lengths for the exact prefix match tables of the switch. The switch must be rebooted to apply the change.



Following the reboot, IPv4 prefix priorities will remain in a pending state until at least one route is learned. A connected route counts so this can be as simple as having an L3 interface with an IP address in the up state.

The no form of this command resets the prefix lengths to their default

Parameter	Description
<PREFIX-LENGTHS>>	Specifies a space-separated list of exactly five or six prefix lengths, in descending order. Range: 8 to 31. On the 6300 Switch Series, six prefix lengths are used. On the 6400 Switch Series, six prefix lengths are used for profiles default and V2-default, and five prefix lengths are used for profile v2-Core-High-Bandwidth.

Examples

Configuring custom IPv4 route prefix lengths:

```
switch(config)# ip prefix-priority 29 28 27 24 23 16
Save this config and reboot the switch for the changes to take effect
...
```

Resetting IPv4 route prefix lengths to their default:

```
switch(config)# no ip prefix-priority
Save this config and reboot the switch for the changes to take effect
...
```

Attempting to configure custom IPv4 route prefix lengths with some lengths not in descending order:

```
switch(config)# ip prefix-priority 28 29 27 23 24 16
Prefix lengths must be specified in descending order
```

Attempting to configure eight prefix lengths:

```
switch(config)# ip prefix-priority 29 28 27 24 23 16 12 8
Invalid input: 12
```

Attempting to configure three prefix lengths:

```
switch(config)# ip prefix-priority 29 28 27
% Command incomplete.
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

ipv6 prefix-priority

```
ipv6 prefix-priority <PREFIX-LENGTHS>  
no ipv6 prefix-priority [<PREFIX-LENGTHS>]
```

Description

Configures custom IPv6 route prefix lengths for the exact prefix match tables of the switch. The switch must be rebooted to apply the change.



Following the reboot, IPv6 prefix priorities will remain in a pending state until at least one route is learned. A connected route counts so this can be as simple as having an L3 interface with an IP address in the up state.

The no form of this command resets the prefix lengths to their default

Parameter	Description
<PREFIX-LENGTHS>	Specifies a space-separated list of exactly five or six prefix lengths, in descending order. Range: 8 to 64. On the 6300 Switch Series, six prefix lengths are used. On the 6400 Switch Series, six prefix lengths are used for profiles default and V2-default, and five prefix lengths are used for profile v2-Core-High-Bandwidth.

Examples

Configuring custom IPv6 route prefix lengths:

```
switch(config)# ipv6 prefix-priority 64 63 62 32 31 28  
Save this config and reboot the switch for the changes to take effect  
...
```

Resetting IPv6 route prefix lengths to their default:

```
switch(config)# no ipv6 prefix-priority  
Save this config and reboot the switch for the changes to take effect  
...
```

Attempting to configure custom IPv6 route prefix lengths with some lengths not in descending order:

```
switch(config)# ipv6 prefix-priority 64 62 63 31 32 28  
Prefix lengths must be specified in descending order
```

Attempting to configure eight prefix lengths:

```
switch(config)# ipv6 prefix-priority 64 63 62 32 31 28 24 23  
Invalid input: 24
```

Attempting to configure three prefix lengths:

```
switch(config)# ipv6 prefix-priority 64 63 31
% Command incomplete.
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

show ip prefix-priority

```
show ip prefix-priority
```

Description

Shows the configuration, status, and defaults of the IPv4 route prefix lengths for the exact prefix match tables of the switch.

Examples

Showing configured IPv4 route prefix lengths that are pending until the switch is rebooted and at least one route is learned:

```
switch# show ip prefix-priority

IP Exact-Prefix Table Information

Configuration Status: Ready to apply on next reboot

  Table    Default   Current   Pending
         Length   Length   Length
-----
  1         24        24        29*
  2         23        23        28*
  3         22        22        27*
  4         21        21        24*
  5         16        16        23*
  6          8         8        16*

* Pending values will be applied on the next reboot
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ipv6 prefix-priority

```
show ipv6 prefix-priority
```

Description

Shows the configuration, status, and defaults of the IPv6 route prefix lengths for the exact prefix match tables of the switch.

Examples

Showing configured IPv6 route prefix lengths that are pending until the switch is rebooted and at least one route is learned:

```
switch# show ipv6 prefix-priority

IPv6 Exact-Prefix Table Information

Configuration Status: Ready to apply on next reboot

  Table    Default   Current   Pending
  Table    Length    Length    Length
  -----
  1         64        64        64
  2         48        48        63*
  3         46        46        62*
  4         44        44        32*
  5         40        40        31*
  6         36        36        28*

* Pending values will be applied on the next reboot
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Selecting the system profile



System profiles are applicable to the 6400 Switch Series.

System profiles set the overall capabilities and capacities of the switch, based on the selected profile used at boot time. System profiles set capacities such as that of the hardware forwarding table.

System profiles provide you with the flexibility to configure switches based on their location in the network (for example, core, spine, leaf). When a switch boots without a profile specifically configured, it boots with the default profile. When a switch is configured with a non-default profile, the switch requires a reboot for the profile to be applied.

Command `show profiles available` shows all profiles available on a particular switch.

See also [System profile commands](#).

Procedure

1. Set the system profile with the command `profile`.
2. Reboot the switch for the profile change to take effect with the command `boot system`.

Examples

Selecting the **v2-Aggregation-High-Bandwidth** profile and then rebooting the system:

```
switch(config)# profile v2-Aggregation-High-Bandwidth
switch(config)# exit
switch# boot system
switch(config)# The config will be cleared, and the switch will be
rebooted with the v2-Aggregation-High-Bandwidth profile
Continue(y/n) ...
```

Selecting the **default** profile and then rebooting the system:

```
switch(config)# profile default
switch(config)# exit
switch# boot system
switch(config)# The config will be cleared, and the switch will be
rebooted with the Default profile
Continue(y/n) ...
```

About 6400 Switch Series module versions and the default system profile

The 6400 Switch Series adds support for V2 modules as of AOS-CX software version 10.10. Prior AOS-CX software versions only support 6400 Series Switches V1 modules.

V1 and V2 modules are identified as follows:

- V1 modules model numbers end with the letter A or B.
- V2 modules model numbers end with the letter C.

The default system profile is decided as follows:

- A 6400 Switch Series V1 chassis uses the **default** system profile as the default. This means that an existing V1 chassis continues to work with existing modules without requiring a configuration change or chassis reboot.

- A 6400 Switch Series V2 chassis uses the **v2-default** system profile as the default. This means that a V2 chassis containing all-V2 modules will work without requiring a configuration change or chassis reboot.
- A V1 or V2 6400 Switch Series chassis can be configured to use any profile provided that the installed modules support the selected profile.
- In the special case where you run pre-10.10 software on a V2 6400 Switch Series chassis (which means that all modules must be V1) and you then upgrade the software to 10.10 or greater, the V2 6400 Switch Series chassis automatically selects the **default** system profile and then triggers a reboot cycle for the configuration to take effect.

System profile commands

profile

```
profile <PROFILE-NAME>
no profile [<PROFILE-NAME>]
```

Description

Selects the system profile. System profiles set the overall capabilities and capacities of the switch based on the selected profile used at boot time. Switch profiles set capacities such as that of the hardware forwarding table. Use command [show profiles available](#) to show the details of each available profile.



When a switch is configured with a non-default profile, the switch requires a reboot for the profile to be applied. You are prompted for the reboot.

The no form of this command resets the specified profile to its defaults.

Profile names	Description
default	Selects the original default. v1 and v2 modules supported.
v2-default	(The default.) Selects the v2 default. Only v2 modules are supported.
v2-Aggregation-High-Bandwidth	Selects aggregation high bandwidth. Only R0X44C (1G/10G/25G) and R0X45C (40G/100G) v2 modules are supported.
v2-Core-High-Bandwidth	Selects core high bandwidth. Only R0X44C (1G/10G/25G) and R0X45C (40G/100G) v2 modules are supported.
v2-Leaf-Extended-High-Bandwidth	Selects leaf extended high bandwidth. Only R0X44C (1G/10G/25G) and R0X45C (40G/100G) v2 modules are supported.

Examples

Selecting the v2-Aggregation-High-Bandwidth profile and then rebooting the system:

```

switch(config)# profile v2-Aggregation-High-Bandwidth
switch(config)# exit
switch# boot system
switch(config)# The config will be cleared, and the switch will be
rebooted with the v2-Aggregation-High-Bandwidth profile
Continue(y/n) ...

```

Selecting the default profile and then rebooting the system:

```

switch(config)# profile default
switch(config)# exit
switch# boot system
switch(config)# The config will be cleared, and the switch will be
rebooted with the Default profile
Continue(y/n) ...

```

Command History

Release	Modification
10.10	Command introduced on the 6400.

Command Information

Platforms	Command context	Authority
6400	config	Administrators or local user group members with execution rights for this command.

show profiles available

show profiles available

Description

Shows all system profile names available and profile details.

Examples

Showing all available profiles:

```

switch# show profiles available

Available Profiles
-----
Default          v1 and v2 modules supported
                  32768 L2 entries, 49152 Host entries, 65536 Route entries
v2-Aggregation-  Only R0X44C(1G/10G/25G) & R0X45C(40G/100G)v2 modules
supported
  High-Bandwidth  114688 L2 entries, 163840 Host entries, 65536 Route entries
                  enhanced feature set
v2-Core-High-Bandwidth Only R0X44C(1G/10G/25G) & R0X45C(40G/100G)v2 modules
supported
                  32768 L2 entries, 65536 Host entries, 630784 Route entries
                  enhanced feature set

```


v2-Default	Only v2 modules supported (Default) 32768 L2 entries, 49152 Host entries, 65536 Route entries enhanced feature set (Default)
v2-Leaf-Extended- supported High-Bandwidth	Only R0X44C(1G/10G/25G) & R0X45C(40G/100G) v2 modules 212992 L2 entries, 16384 Host entries, 65536 Route entries enhanced feature set

Note: Not all profiles are supported by all the modules, in order for the profile to perform as expected, the profile must be supported by the module.

Command History

Release	Modification
10.10	Command introduced on the 6400.

Command Information

Platforms	Command context	Authority
6400	Manager (#)	Administrators or local user group members with execution rights for this command.

show profile current

show profile current

Description

Shows the current system profile.

Examples

Showing the current profile:

```
switch# show profile current

Current profile
-----
v2-Aggregation-High-Bandwidth
```

Command History

Release	Modification
10.10	Command introduced on the 6400.

Command Information

Platforms	Command context	Authority
6400	Manager (#)	Administrators or local user group members with execution rights for this command.

Restoring the switch to factory default settings

Prerequisites

You are connected to the switch through its Console port.



This procedure erases all user information and configuration settings. Consider backing up your running configuration first.

1. Optionally, back up the running configuration with either `copy running-config <REMOTE-URL>` or `copy running-config <STORAGE-URL>`. The `json` storage format is required for later configuration restoration.
2. Switch to the configuration context with the command `config`.
3. Erase all user information and configuration, restoring the switch to its factory default state with the command `erase all zeroize`. Enter `Y` when prompted to continue. The switch automatically restarts.
4. Optionally restore your saved configuration (it must be in `json` format) with either `copy <REMOTE-URL> running-config` or `copy <STORAGE-URL> running-config` followed by `copy running-config startup-config`.

Example

Backing up the running configuration to a file on a remote server (using TFTP), resetting the switch to its factory default state, and then restoring the saved configuration.

```
switch# copy running-config tftp://192.168.1.10/backup_cfg json vrf mgmt

  % Total      % Received % Xferd  Average Speed   Time    Time       Time  Current
   %    100      0       0    100    10340      0   1329k  --:--:--  --:--:--  --:--:--  1329k
  100    100      0       0    100    10340      0   1313k  --:--:--  --:--:--  --:--:--  1313k
switch#
switch#
switch# erase all zeroize
This will securely erase all customer data and reset the switch
to factory defaults. This will initiate a reboot and render the
switch unavailable until the zeroization is complete.
This should take several minutes to one hour to complete.
Continue (y/n)? y
The system is going down for zeroization.
[ OK ] Stopped PSPO Module Daemon.
[ OK ] Stopped AOS-CX Switch Daemon for BCM.
...
[ OK ] Stopped Remount Root and Kernel File Systems.
[ OK ] Reached target Shutdown.
reboot: Restarting system
Press Esc for boot options
ServiceOS Information:
  Version:          GT.01.03.0006
```

```

Build Date:      2018-10-30 14:20:44 PDT
Build ID:        ServiceOS:GT.01.03.0006:8ee0faaa52da:201810301420
SHA:             xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
...

##### Preparing for zeroization #####

##### Storage zeroization #####
##### WARNING: DO NOT POWER OFF UNTIL #####
##### ZEROIZATION IS COMPLETE #####
##### This should take several minutes #####
##### to one hour to complete #####

##### Restoring files #####

Boot Profiles:

0. Service OS Console
1. Primary Software Image [XL.10.02.0010]
2. Secondary Software Image [XL.10.02.0010]

Select profile(primary):

Booting primary software image...
Verifying Image...

Image Info:
  Name: AOS-CX
  Version: XL.10.02.0010
  Build Id: AOS-CX:XL.10.02.0010:feaf5b9b7f09:201901292014
  Build Date: 2019-01-29 12:43:50 PST

Extracting Image...
Loading Image...
Done.
kexec_core: Starting new kernel
System is initializing
fips_post_check[5473]: FIPS_POST: Cryptographic selftest started...SUCCESS
[ OK ] Started Login banner readiness check.
...
8400X login: admin
Password:

switch#
switch#
switch# copy tftp://192.168.1.10/backup_cfg running-config json vrf mgmt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total  Spent    Left  Speed
100 10340  100 10340    0     0  2858k      0  --:--:-- --:--:-- --:--:-- 2858k
100 10340  100 10340    0     0  2804k      0  --:--:-- --:--:-- --:--:-- 2804k
Large configuration changes will take time to process, please be patient.
switch#
switch#
switch# copy running-config startup-config
Large configuration changes will take time to process, please be patient.
switch#

```

Management interface commands

default-gateway

```
default-gateway <IP-ADDR>
```

```
no default-gateway <IP-ADDR>
```

Description

Assigns an IPv4 or IPv6 default gateway to the management interface. An IPv4 default gateway can only be configured if a static IPv4 address was assigned to the management interface. An IPv6 default gateway can only be configured if a static IPv6 address was assigned to the management interface. The default gateway should be on the same network segment.

The **no** form of this command removes the default gateway from the management interface.

Parameter	Description
<IP-ADDR>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

Examples

Setting a default gateway with the IPv4 address of **198.168.5.1**:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# default-gateway 198.168.5.1
```

Setting an IPv6 address of **2001:DB8::1**:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# default-gateway 2001:DB8::1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-if-mgmt	Administrators or local user group members with execution rights for this command.

ip static

```
ip static <IP-ADDR>/<MASK>
no ip static <IP-ADDR>/<MASK>
```

Description

Assigns an IPv4 or IPv6 address to the management interface.

The **no** form of this command removes the IP address from the management interface and sets the interface to operate as a DHCP client.

Parameter	Description
<IP-ADDR>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.
<MASK>	Specifies the number of bits in an IPv4 or IPv6 address mask in CIDR format (x), where x is a decimal number from 0 to 32 for IPv4, and 0 to 128 for IPv6.

Examples

Setting an IPv4 address of **198.51.100.1** with a mask of **24** bits:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# ip static 198.51.100.1/24
```

Setting an IPv6 address of **2001:DB8::1** with a mask of **32** bits:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# ip static 2001:DB8::1/32
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-if-mgmt	Administrators or local user group members with execution rights for this command.

nameserver

```
nameserver <PRIMARY-IP-ADDR> [ <SECONDARY-IP-ADDR> ]
no nameserver <PRIMARY-IP-ADDR> [ <SECONDARY-IP-ADDR> ]
```

Description

Assigns a primary or secondary IPv4 or IPv6 DNS server to the management interface. IPv4 DNS servers can only be configured if a static IPv4 address was assigned to the management interface. IPv6 DNS servers can only be configured if a static IPv6 address was assigned to the management interface. The default gateway should be on the same network segment.

The **no** form of this command removes the DNS servers from the management interface.

Parameter	Description
<PRIMARY-IP-ADDR>	Specifies the IP address of the primary DNS server. Specify the

Parameter	Description
	address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.
<SECONDARY-IP-ADDR>	Specifies the IP address of the secondary DNS server. Specify the address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

Examples

Setting primary and secondary DNS servers with the IPv4 addresses of **198.168.5.1** and **198.168.5.2** :

```
switch(config)# interface mgmt
switch(config-if-mgmt)# nameserver 198.168.5.1 198.168.5.2
```

Setting primary and secondary DNS servers with the IPv6 addresses of **2001:DB8::1** and **2001:DB8::2**:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# nameserver 2001:DB8::1 2001:DB8::2
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-if-mgmt	Administrators or local user group members with execution rights for this command.

show interface mgmt

show interface mgmt [vsx-peer]

Description

Shows status and configuration information for the management interface.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

```
switch# show interface mgmt

Address Mode           : static
Admin State            : up
Mac Address            : 02:42:ac:11:00:02
IPv4 address/subnet-mask : 192.168.1.10/16
Default gateway IPv4    : 192.168.1.1
IPv6 address/prefix     : 2001:db8:0:1::129/64
IPv6 link local address/prefix: fe80::7272:cfff:fe8d:e485/64
Default gateway IPv6    : 2001:db8:0:1::1
Primary Nameserver      : 2001::1
Secondary Nameserver    : 2001::2
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

NTP commands

ntp authentication

```
ntp authentication
no ntp authentication
```

Description

Enables support for authentication when communicating with an NTP server.
The **no** form of this command disables authentication support.

Examples

Enabling authentication support:

```
switch(config)# ntp authentication
```

Disabling authentication support:

```
switch(config)# no ntp authentication
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ntp authentication-key

```
ntp authentication-key <KEY-ID> {md5 | sha1}
    [{ <PLAINTEXT-KEY> [trusted] | ciphertext <ENCRYPTED-KEY> }]
no ntp authentication-key <KEY-ID> {md5 | sha1}
    [{ <PLAINTEXT-KEY> [trusted] | ciphertext <ENCRYPTED-KEY> }]
```

Description

Defines an authentication key that is used to secure the exchange with an NTP time server. This command provides protection against accidentally synchronizing to a time source that is not trusted. The **no** form of this command removes the authentication key.

Parameter	Description
<KEY-ID>	Specifies the authentication key ID. Range: 1 to 65534.
md5	Selects MD5 key encryption.
sha1	Specifies SHA1 key encryption.
<PLAINTEXT-KEY>	Specifies the plaintext authentication key. Range: 8 to 40 characters. The key may contain printable ASCII characters excluding "#" or be entered in hex. Keys longer than 20 characters are assumed to be hex. To use an ASCII key longer than 20 characters, convert it to hex.
trusted	Specifies that this is a trusted key. When NTP authentication is enabled, the switch only synchronizes with time servers that transmit packets containing a trusted key.
ciphertext <ENCRYPTED-KEY>	Specifies the ciphertext authentication key in Base64 format. This is used to restore the NTP authentication key when copying configuration files between switches or when uploading a previously saved configuration. NOTE: When the key is not provided on the command line, plaintext key prompting occurs upon pressing Enter, followed by prompting as to whether the key is to be trusted. The entered key characters are masked with asterisks.

Examples

Defining key 10 with MD5 encryption and a provided plaintext trusted key:


```
switch(config)# ntp authentication-key 10 md5 F82#450b trusted
```

Defining key 5 with SHA1 encryption and a prompted plaintext trusted key:

```
switch(config)# ntp authentication-key 5 sha1
Enter the NTP authentication key: *****
Re-Enter the NTP authentication key: *****

Configure the key as trusted (y/n)? y
```

Removing key 10:

```
switch(config)# no ntp authentication-key 10
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ntp disable

ntp disable

Description

Disables the NTP client on the switch. The NTP client is disabled by default.

Examples

Disabling the NTP client.

```
switch(config)# ntp disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ntp enable

```
ntp enable
no ntp enable
```

Description

Enables the NTP client on the switch to automatically adjust the local time and date on the switch. The NTP client is disabled by default.

The **no** form of this command disables the NTP client.

Examples

Enabling the NTP client.

```
switch(config)# ntp enable
```

Disabling the NTP client.

```
switch(config)# no ntp enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ntp conductor

```
ntp conductor vrf <VRF-NAME> {stratum <NUMBER>}
no ntp conductor vrf <VRF-NAME> {stratum <NUMBER>}
```

Description

Sets the switch as the conductor time source for NTP clients on the specified VRF. By default, the switch operates at stratum level 8. The switch cannot function as both NTP conductor and client on the same VRF.

The **no** form of this command stops the switch from operating as the conductor time source on the specified VRF.

Parameter	Description
<code>vrf <VRF-NAME></code>	Specifies the VRF on which to act as conductor time source.
<code>stratum <NUMBER></code>	Specifies the stratum level at which the switch operates. Range: 1 - 15. Default: 8.

Examples

Setting the switch to act as conductor time source on VRF **primary-vrf** with a stratum level of **9**.

```
switch(config)# ntp conductor vrf primary-vry stratum 9
```

Stops the switch from acting as conductor time source on VRF **primary-vrf**.

```
switch(config)# no ntp conductor vrf primary-vry
```

Command History

Release	Modification
10.08	Inclusive language.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

ntp server

```
ntp server <IP-ADDR> [key <KEY-NUM>] [minpoll <MIN-NUM>] [maxpoll <MAX-NUM>] [burst |
iburst] [prefer] [version <VER-NUM>]
no ntp server <IP-ADDR> <IP-ADDR> [key <KEY-NUM>] [minpoll <MIN-NUM>] [maxpoll <MAX-NUM>]
[burst | iburst] [prefer] [version <VER-NUM>]
```

Description

Defines an NTP server to use for time synchronization, or updates the settings of an existing server with new values. Up to eight servers can be defined.

The **no** form of this command removes a configured NTP server.



The default NTP version is 4; it is backwards compatible with version 3.

Parameter	Description
<code>server <IP-ADDR></code>	Specifies the address of an NTP server as a DNS name, an IPv4 address (x.x.x.x), where x is a decimal number from 0 to 255, or

Parameter	Description
	an IPv6 address (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. When specifying an IPv4 address, you can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100. When specifying an IPv6 address, you can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.
<code>key <KEY-NUM></code>	Specifies the key to use when communicating with the server. A trusted key must be defined with the command ntp authentication-key and authentication must be enabled with the command ntp authentication . Range: 1 to 65534.
<code>minpoll <MIN-NUM></code>	Specifies the minimum polling interval in seconds, as a power of 2. Range: 4 to 17. Default: 6 (64 seconds).
<code>maxpoll <MAX-NUM></code>	Specifies the maximum polling interval in seconds, as a power of 2. Range: 4 to 17. Default: 10 (1024 seconds).
<code>burst</code>	Send a burst of packets instead of just one when connected to the server. Useful for reducing phase noise when the polling interval is long.
<code>iburst</code>	Send a burst of six packets when not connected to the server. Useful for reducing synchronization time at startup.
<code>prefer</code>	Make this the preferred server.
<code>version <VER-NUM></code>	Specifies the version number to use for all outgoing NTP packets. Range: 3 or 4. Default: 4.
	NOTE: NTP is backwards compatible.

Usage

For features such as Activate and ZTP, a switch that has a factory default configuration will automatically be configured with pool.ntp.org. NTP server configurations via DHCP options are supported. The DHCP server can be configured with maximum of two NTP server addresses which will be supported on the switch. Only IPV4 addresses are supported.

NTP uses a stratum to describe the distance between a network device and an authoritative time source:

- A stratum 1 time server is directly attached to an authoritative time source (such as a radio or atomic clock or a GPS time source).
- A stratum 2 NTP server receives its time through NTP from a stratum 1 time server.

When using multiple servers with same stratum setting, the best practice to configure a preferred server, so NTP will attempt to use the preferred server as the primary NTP connection. If a preferred server is not manually set when NTP is enabled, the configured server with the lowest stratum will automatically be set as the preferred server. If there are servers with the same stratum, this auto prefer status will prevent AOS-CX from toggling between different servers as the primary server. Auto prefer

selection of servers with same stratum (if not manually selected) may change after reconfiguring the switch, or after executing the **reboot** command.

Examples

Defining the ntp server pool.ntp.org, using iburst, and NTP version 4.

```
switch(config)# ntp server pool.ntp.org iburst version 4
```

Removing the ntp server pool.ntp.org.

```
switch(config)# no ntp server pool.ntp.org
```

Defining the ntp server my-ntp.mydomain.com and makes it the preferred server.

```
switch(config)# ntp server my-ntp.mydomain.com prefer
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ntp trusted-key

```
ntp trusted-key <KEY-ID>  
no ntp trusted-key <KEY-ID>
```

Description

Sets a key as trusted. When NTP authentication is enabled, the switch only synchronizes with time servers that transmit packets containing a trusted key.

The **no** form of this command removes the trusted designation from a key.

Parameter	Description
<KEY-ID>	Specifies the identification number of the key to set as trusted. Range: 1 to 65534.

Examples

Defining key 10 as a trusted key.

```
switch(config)# ntp trusted-key 10
```

Removing trusted designation from key 10:

```
switch(config)# no ntp trusted-key 10
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ntp vrf

```
ntp vrf <VRF-NAME>  
no ntp vrf <VRF-NAME>
```

Description

Specifies the VRF on which the NTP client communicates with an NTP server. The switch cannot function as both NTP conductor and client on the same VRF.
The **no** form of the command returns to default VRF.

Parameter	Description
<VRF-NAME>	Specifies the name of a VRF.

Example

Setting the switch to use the default VRF for NTP client traffic.

```
switch(config)# ntp vrf default
```

Setting the switch to use the default management VRF for NTP client traffic.

Returning the switch to use the default VRF for NTP client traffic.

```
switch(config)# no ntp vrf
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show ntp associations

```
show ntp associations [vsx-peer]
```

Description

Shows the status of the connection to each NTP server. The following information is displayed for each server:

- Tally code : The first character is the Tally code:
 - (blank): No state information available (e.g. non-responding server)
 - x : Out of tolerance (discarded by intersection algorithm)
 - . : Discarded by table overflow (not used)
 - - : Out of tolerance (discarded by the cluster algorithm)
 - + : Good and a preferred remote peer or server (included by the combine algorithm)
 - # : Good remote peer or server, but not utilized (ready as a backup source)
 - * : Remote peer or server presently used as a primary reference
 - o : PPS peer (when the prefer peer is valid)
- ID: Server number.
- NAME: NTP server FQDN/IP address (Only the first 24 characters of the name are displayed).
- REMOTE: Remote server IP address.
- REF_ID: Reference ID for the remote server (Can be an IP address).
- ST: (Stratum) Number of hops between the NTP client and the reference clock.
- LAST: Time since the last packet was received in seconds unless another unit is indicated.
- POLL: Interval (in seconds) between NTP poll packets. Maximum (1024) reached as server and client sync.
- REACH: 8-bit octal number that displays status of the last eight NTP messages (377 = all messages received).

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

```
switch# show ntp associations
-----
ID           NAME           REMOTE           REF-ID ST LAST POLL REACH
-----
```

```

1          192.0.1.1          192.0.1.1          .INIT. 16    -    64    0
* 2  time.apple.com    17.253.2.253          .GPSs.  2    70   128   377
-----

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ntp authentication-keys

show ntp authentication-keys [vsx-peer]

Description

Shows the currently defined authentication keys.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

```

switch# show ntp authentication-keys
-----
Auth key  Trusted  MD5 password
-----
10        No      *****
20        Yes      *****

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show ntp servers

`show ntp servers [vsx-peer]`

Description

Shows all configured NTP servers, including any DHCP servers, default pool servers or any server with the status **auto prefer**.

Parameter	Description
<code>vsx-peer</code>	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

```
switch# show ntp servers
-----
      NTP SERVER KEYID MINPOLL MAXPOLL OPTION VER
-----
      192.0.1.18      -        5      10 iburst   3
      192.0.1.19      -        6      10  none    4
      192.0.1.20      -        6        8  burst   3 prefer
-----
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ntp statistics

`show ntp statistics [vsx-peer]`

Description

Shows global NTP statistics. The following information is displayed:

- Rx-pkts: Total NTP packets received.
- Current Version Rx-pkts: Number of NTP packets that match the current NTP version.
- Old Version Rx-pkts: Number of NTP packets that match the previous NTP version.
- Error pkts: Packets dropped due to all other error reasons.
- Auth-failed pkts: Packets dropped due to authentication failure.
- Declined pkts: Packets denied access for any reason.
- Restricted pkts: Packets dropped due to NTP access control.
- Rate-limited pkts: Number of packets discarded due to rate limitation.
- KOD pkts: Number of Kiss of Death packets sent.

Parameter	Description
<code>vsx-peer</code>	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

```
switch(config)# show ntp statistics
Rx-pkts 100
Current Version Rx-pkts 80
Old Version Rx-pkts 20
Err-pkts 2
Auth-failed-pkts 1
Declined-pkts 0
Restricted-pkts 0
Rate-limited-pkts 0
KoD-pkts 0
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ntp status

```
show ntp status [vsx-peer]
```

Description

Shows the status of NTP on the switch.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Displaying the status information when the switch is not synced to an NTP server:

```
switch# show ntp status
NTP is enabled.
NTP authentication is enabled.
NTP is using the default VRF for NTP server connections.

Wed Nov 23 23:29:10 PDT 2016
NTP uptime: 187 days, 1 hours, 37 minutes, 48 seconds

Not synchronized with an NTP server.
```

Displaying the status information when the switch is synced to an NTP server:

```
switch# show ntp status
NTP is enabled.
NTP authentication is enabled.
NTP is using the default VRF for NTP server connections.

Wed Nov 23 23:29:10 PDT 2016
NTP uptime: 187 days, 1 hours, 37 minutes, 48 seconds

Synchronized to NTP Server 17.253.2.253 at stratum 2.
Poll interval = 1024 seconds.
Time accuracy is within 0.994 seconds
Reference time: Thu Jan 28 2016 0:57:06.647 (UTC)
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Telnet server enables switches to accept Telnet connections from clients to manage the switch. The user authentication is password based authentication (RADIUS, TACACS+ or locally stored password). The Telnet server can be enabled on a desired VRF using the `telnet server` command. The maximum number of Telnet sessions per VRF is five.

In the default configuration, Telnet access is disabled.

Telnet commands

show telnet server

```
show telnet server
```

Description

Shows the Telnet server configuration.

Examples

Showing the Telnet server configuration:

```
switch(config)# show telnet server
TELNET Server Configuration:

  IP Version      : IPv4
  TCP Port        : 23
  Enabled VRFs    : default, vrf1, vrf2,
                  red, green
```

Command History

Release	Modification
10.08.1021	Command introduced on the 6200, 6300, 6400 Switch Series.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show telnet server sessions

show telnet server sessions [vrf <VRF-NAME> | all-vrfs]

Description

Shows all active Telnet sessions for the specified VRF or all VRFs. If no VRF is provided, the Telnet sessions on the **default** VRF are shown.

Parameter	Description
vrf <VRF-NAME>	Specifies the Telnet sessions for a specific VRF.
all-vrfs	Specifies the Telnet sessions for all VRFs

Examples

Showing the Telnet session on the **default** VRF:

```
switch(config)# show telnet server sessions
TELNET sessions on VRF default:

IPv4 TELNET Sessions:
  Server IP      : 10.1.1.1
  Client IP      : 10.1.1.2
  Client Port    : 58835
```

Showing the Telnet sessions on all VRFs:

```
switch(config)# show telnet server sessions all-vrfs
TELNET sessions on VRF mgmt:

IPv4 TELNET Sessions:
  Server IP      : 10.1.1.1
  Client IP      : 10.1.1.2
  Client Port    : 58835

TELNET sessions on VRF default:

IPv4 TELNET Sessions:
  Server IP      : 20.1.1.1
  Client IP      : 20.1.1.2
  Client Port    : 58837
```

Command History

Release	Modification
10.08.1021	Command introduced on the 6200, 6300, 6400 Switch Series.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

telnet server

```
telnet server vrf <VRF-NAME>  
no telnet server vrf <VRF-NAME>
```

Description

Enables the Telnet server on the desired VRF. Telnet server is disabled by default.
The **no** form of this command disables the Telnet server.

Parameter	Description
vrf <VRF-NAME>	Specifies the VRF on which the Telnet server will be enabled or disabled.

Examples

Configuring the Telnet server on the **mgmt** VRF:

```
switch(config)# telnet server vrf mgmt
```

Command History

Release	Modification
10.08.1021	Command introduced on the 6200, 6300, 6400 Switch Series.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

Configuring a layer 2 interface

Procedure

1. Change to the interface configuration context for the interface with the command `interface`.
2. Set the interface MTU (maximum transmission unit) with the command `mtu`.
3. Review interface configuration settings with the command `show interface`.

Example

On the 6300 switch series:

```
switch(config)# interface 1/1/1
switch(config-if)# mtu 1900
```

On the 6400 switch series:

```
switch(config)# interface 1/3/1
switch(config-if)# mtu 1900
```

Configuring a layer 3 interface

Procedure

1. Change to the interface configuration context for the interface with the command `interface`.
2. Enable routing support with the command `routing`.
3. Assign an IPv4 address with the command `ip address`, or an IPv6 address with the command `ipv6 address`.
4. If required, enable support for layer 3 counters with the command `l3-counters`.
5. If required, set the IP MTU with the command `ip mtu`.
6. Review interface configuration settings with the command `show interface`.

Examples

This example creates the following configuration on the 6300 Switch Series:

- Configures interface **1/1/1** as a layer 3 interface.
- Defines an IPv4 address of **10.10.20.209** with a 24-bit mask.

```
switch# config
switch(config)# interface 1/1/1
```

```
switch(config-if) # routing  
switch(config-if) # ip address 10.10.20.209/24
```

This example creates the following configuration on the 6400 Switch Series:

- Configures interface **1/3/1** as a layer 3 interface.
- Defines an IPv6 address of **2001:0db8:85a3::8a2e:0370:7334** with a 24-bit mask.
- Enables layer 3 transmit and receive counters.

```
switch# config  
switch(config) # interface 1/3/1  
switch(config-if) # routing  
switch(config-if) # ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24  
switch(config-if) # l3-counters tx  
switch(config-if) # l3-counters rx
```

Single source IP address

Certain IP-based protocols used by the switch (such as RADIUS, sFlow, TACACS, and TFTP), use a client-server model in which the client's source IP address uniquely identifies the client in packets sent to the server. By default, the source IP address is defined as the IP address of the outgoing switch interface on which the client is communicating with the server. Since the switch can have multiple routing interfaces, outgoing packets can potentially be sent on different paths at different times. This can result in different source IP addresses being used for a client, which can create a client identification problem on the server. For example, it can be difficult to interpret system logs and accounting data on the server when the same client is associated with multiple IP addresses.

To resolve this issue, you can use the commands `ip source-interface` and `ipv6 source-interface` to define a single source IP address that applies to all supported protocols (RADIUS, sFlow, TACACS, and TFTP), or an individual address for each protocol. This ensures that all traffic sent by a client to a server uses the same IP address.

Forward error correction

Forward error correction (FEC) is used to control errors in transmissions where the source sends redundant data and the destination only recognizes the data portion that contains no apparent errors. FEC does not require a handshake between the source and destination at the cost of a higher forward channel bandwidth. It is therefore best used in scenarios where re-transmissions are costly or impossible, such as using multicast one-way communication.

Unsupported transceiver support

Transceiver products (optical, DAC, AOCs) that are listed as supported by a switch model are detailed in the *Transceiver Guide*. Transceiver products that are not listed, are considered unsupported; this would include transceivers that are:

- Non-Aruba branded products
- HPE branded products that were designed for non-AOS-CX switch models (e.g. Comware)
- HPE branded products designated for use in HPE Compute Servers or Storage

- Transceivers originally designated for use in Aruba WLAN controllers or former Mobility Access Switch (MAS) products
- End-of-life Aruba Transceivers

The unsupported transceiver mode (UT-mode) is designed to allow the possible use of these unsupported products. Not all unsupported products can be recognized and enabled; they may be unable to be identified (do not follow the proper MSA standards for identification). These unsupported transceiver products are enabled only on a best-effort basis and there are no guarantees implied for their continued operation.

This feature is enabled by default. A periodic system log will be generated by default at an interval of 24 hours listing the ports on which unsupported transceivers are present. The log interval is configurable and can be disabled by setting the log-interval to `none`.

Configuring an interface persona

A persona is a template interface. It is a mechanism intended to ease the process of configuring one or multiple interfaces that behave in the same manner. For example, a persona could be an uplink interface with a well-known collection of VLANs to which it belongs. Instead of manually configuring each interface one-by-one, the persona collects the common configuration settings and allows them to be applied on the interface or a collection of interfaces.

Persona configuration is similar to configuring other interfaces. Most of the commands in the interface context are available. Because of the nature of the commands, several of them do not apply to the persona (for example, applying the same IP address configuration to several ports) and have been removed from the available list.

The following commands are not supported within the interface context when configuring a persona:

```
aaa authentication port-access dot1x authenticator macsec
arp ipv4 <IPV4_ADDR> mac <MAC_ADDR>
downshift-enable
energy-efficient-ethernet
error-control { auto | none | base-r-fec | rs-fec }
ip bootp-gateway <IPV4-ADDR>
ip forward-protocol udp <IPV4-ADDR> {<PORT-NUM> | <PROTOCOL>}
ip helper-address <IPV4-ADDR> [vrf <VRF-NAME>]
ip igmp router-alert-check [enable | disable]
ip igmp snooping [auto vlan <VLAN-LIST>]
ip igmp snooping [blocked vlan <VLAN-LIST>]
ip igmp snooping [fastleave vlan <VLAN-LIST>]
ip igmp snooping [forced-fastleave vlan <VLAN-LIST>]
ip igmp snooping [forward vlan <VLAN-LIST>]
ip ospf <PROCESS-ID> area <AREA-ID>
ip ospf passive
ip rip <PROCESS-ID> {all-ip | ip-address}
ip rip all-ip disable
ip rip all-ip enable
ip rip all-ip receive disable
ip rip all-ip send disable
ipv6 helper-address multicast {all-dhcp-servers | <MULTICAST-IPV6-ADDR>} egress <PORT-NUM>
ipv6 mld snooping [auto vlan <VLAN-LIST>]
ipv6 mld snooping [blocked vlan <VLAN-LIST>]
ipv6 mld snooping [fastleave vlan <VLAN-LIST>]
ipv6 mld snooping [forced-fastleave vlan <VLAN-LIST>]
ipv6 mld snooping [forward vlan <VLAN-LIST>]
ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
ipv6 ospfv3 <PROCESS-ID> area <AREA-ID>
```

```

ipv6 ospfv3 passive
ipv6 ripng <PROCESS-ID>
lacp port-id <PORT-ID>
lacp port-priority <PORT-PRIORITY>
lag <ID>
link-poe
lldp dot3 eee
lldp dot3 poe
lldp med poe
lldp med poe [priority-override]
persona {access | uplink | custom <PERSONA-NAME>} [copy | attach]
port-access device-profile <DEVICE-PROFILE-NAME>
power-over-ethernet
power-over-ethernet allocate-by {usage | class}
power-over-ethernet assigned-class {3 | 4 | 6 | 8}
power-over-ethernet pd-class-override
power-over-ethernet power-pairs {alt-a|alt-a-and-alt-b}
power-over-ethernet pre-std-detect
power-over-ethernet priority {critical|high|low}
ptp lag-role {primary | secondary}
spanning-tree cost <PORT-COST>
spanning-tree instance <INSTANCE-ID> cost <PORT-COST>
spanning-tree instance <INSTANCE-ID> port-priority <PRIORITY-MULTIPLIER>
spanning-tree port-priority <PRIORITY-MULTIPLIER>
spanning-tree vlan <A:1-4094> cost <0-200000000>
spanning-tree vlan <A:1-4094> port-priority <0-15>
split [confirm]
track by <OBJECT-ID>
vsx-sync {access-lists | qos | rate-limits | vlans | policies | irdp}

```

Modes

There are two supported modes:

1. Copy—The `copy` mode is a one-step configuration that copies the persona configuration to an interface. Further changes to the persona will not be applied to the interfaces using that mode.
2. Attach—Unlike the `copy` mode, besides applying the configuration to the interface immediately, the `attach` mode also follows the persona configuration. It means that the subsequent changes to the persona will be applied to the interfaces attached to it.

Predefined and custom persona names

There are two predefined interface persona names:

- uplink
- access

These names have no predefined configuration. To use them, they must be manually configured as needed. You can also create personas with a custom name. These custom personas can be created and configured in the same manner as the predefined ones. The only difference is the command used to apply them to the interface. The procedure below provides the details.

Creating and configuring an interface persona

Follow these steps to create and configure an interface persona:

1. Create the interface **persona** with the command `interface persona <PERSONA-NAME>`.
2. Set the interface configuration as any other physical interface.
3. Review the configuration with the command `show running-configuration current-context`.
4. Switch to an interface context with the command `interface <PORT>`.
5. Apply the persona configuration to the interface and set the mode with the command `persona custom <PERSONA-NAME> <mode>`. Note that `<custom>` is an optional argument, required only if the persona is not one of the predefined names (neither **uplink** nor **access**).

For information on this feature, see the related video on the [Aruba AirHeads Broadcasting Channel](#).

Examples

To copy a predefined persona name configuration to an interface:

1. Configure the interface persona:

```
switch(config)# interface persona uplink
switch(config-if)# no shutdown
switch(config-if)# no routing
switch(config-if)# vlan access 100
switch(config-if)# exit
```

2. Apply the configuration with **copy** mode:

```
switch(config)# interface 1/1/1
switch(config-if)# persona custom mypersona copy
switch(config-if)# exit
```

To attach a custom persona name to several interfaces simultaneously:

1. Configure the interface persona:

```
switch(config)# interface persona mytemplate
switch(config-if)# no shutdown
switch(config-if)# vrf attach upstream
switch(config-if)# exit
```

2. Apply the configuration with **attach** mode:

```
switch(config)# interface 1/1/1-1/1/24
switch(config-if)# persona custom mypersona attach
switch(config-if)# exit
```

Monitor mode

Monitor mode displays interface statistics in real time, updating frequently. The update interval varies by product, number of visible ports, and display options in use. View the current update interval in the help menu.

Formatting options can be supplied either at command launch or while monitor mode is active. Press `?` to see the available formatting options available in the current command context. Exit the help menu with `q`.

Monitor mode detects terminal size to adjust the number of interfaces and statistics viewable on screen. Additional interfaces or statistics columns can be viewed with the navigation keys: Arrow keys, PageUp, PageDown, Home, and End. Serial console does not automatically detect terminal setting changes, resulting in unexpected output. Recommended recovery steps are to exit and restart. Monitor mode refreshes data automatically until exited with `q`.

Interface commands

allow-unsupported-transceiver

```
allow-unsupported-transceiver [confirm | log-interval {none | <INTERVAL>}]  
no allow-unsupported-transceiver
```

Description

Allows unsupported transceivers to be enabled or establish connections. Transceivers with speeds up to 100G are enabled by this command.



The following models will enable unsupported transceivers for speeds up to 100G when running AOS-CX 10.10 or later:

- 6300 Series Switches—Up to 50G
- 6400 Series Switches—Up to 100G



This command is enabled by default, allowing the use of third party transceiver products without adding the command in the configuration. Disabling this command with the **no** form will now disable the command in the running and stored configurations.

The **no** form of this command disallows using unsupported transceivers.

Parameter	Description
<code>confirm</code>	Specifies that unsupported transceiver warnings are to be automatically confirmed.
<code>log-interval none</code>	Disables unsupported transceiver logging.
<code>log-interval <INTERVAL></code>	Sets the unsupported transceiver logging interval in minutes. Default: 1440 minutes. Range: 1440 to 10080 minutes.

Usage

When none of the parameters are specified it will display a warning message to accept the warranty terms. With **confirm** option the warning message is displayed but the user is not prompted to **(y/n)** answering. Warranty terms must be agreed to as part of enablement and the support is on best effort basis.

Examples

Allowing unsupported transceivers with follow-up confirmation:

```
switch(config)# allow-unsupported-transceiver
Warning: The use of unsupported transceivers, DACs, and AOCs is at your
own risk and may void support and warranty. Please see HPE Warranty terms
and conditions.

Do you agree and do you want to continue (y/n)? y
```

Allowing unsupported transceivers with confirmation in command syntax:

```
switch(config)# allow-unsupported-transceiver confirm
Warning: The use of unsupported transceivers, DACs, and AOCs is at your
own risk and may void support and warranty. Please see HPE Warranty terms
and conditions.
```

Configuring unsupported transceiver logging with an interval of every 48 hours:

```
switch(config)# allow-unsupported-transceiver log-interval 2880
```

Disabling unsupported transceiver logging:

```
switch(config)# allow-unsupported-transceiver log-interval none
```

Disallowing unsupported transceivers with follow-up confirmation:

```
switch(config)# no allow-unsupported-transceivers
Warning: Unsupported transceivers, DACs, and AOCs will be disabled,
which could impact network connectivity. Use 'show allow-unsupported-transceiver'
to identify unsupported transceivers, DACs, and AOCs.

Continue (y/n)? y
```

Disallowing unsupported transceivers with confirmation in command syntax:

```
switch(config)# no allow-unsupported-transceiver confirm
Warning: Unsupported transceivers, DACs, and AOCs will be disabled,
which could impact network connectivity. Use 'show allow unsupported-transceiver'
to identify unsupported transceivers, DACs, and AOCs.

switch(config)#
```

Command History

Release	Modification
10.10	Up to 100G support enabled for unsupported transceivers on 6300 (up to 50G) and 6400 (up to 100G) series switches in UT mode.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

default interface

default interface <INTERFACE-ID>

Description

Sets an interface (or a range of interfaces) to factory default values.

Parameter	Description
<INTERFACE-ID>	Specifies the ID of a single interface or range of interfaces. Format: member/slot/port or member/slot/port-member/slot/port to specify a range.

Examples

On the 6400 Switch Series, interface identification differs.

Resetting an interface:

```
switch(config)# default default interface 1/1/1
```

Resetting an range of interfaces:

```
switch(config)# default default interface 1/1/1-1/1/10
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

description

description <DESCRIPTION>
no description

Description

Associates descriptive information with an interface to help administrators and operators identify the purpose or role of an interface.

The **no** form of this command removes a description from an interface.

Parameter	Description
<DESCRIPTION>	Specify a description for the interface. Range: 1 to 64 ASCII characters (including space, excluding question mark).

Examples

Setting the description for an interface to **DataLink 01**:

```
switch(config-if) # description DataLink 01
```

Removing the description for an interface.

```
switch(config-if) # no description
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

energy-efficient-ethernet

energy-efficient-ethernet

Description

Enables auto-negotiation of Energy-Efficient Ethernet (EEE) on an interface. EEE Negotiation is established only on auto-link negotiation with supported link partners.

Examples

Configuring an interface:

```
switch(config) # interface 1/1/1
switch(config-if) # energy-efficient-ethernet
```

Disabling Energy Efficient Ethernet on an interface:

```
switch(config) # interface 1/1/1
switch(config-if) # no energy-efficient-ethernet
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

error-control

```
error-control {auto | none | base-r-fec | rs-fec}
no error-control {auto | none | base-r-fec | rs-fec}
```

Description

Configures the forward error correction (FEC) mode to use for an interface. When not configured, the system will automatically select the FEC mode based on the installed transceiver. In most cases, the standard FEC mode will work best, but certain link partners may require a non-standard mode.

The **no** and **auto** forms of this command configure the interface to automatically use the standard FEC mode of the currently installed transceiver.



FEC configuration only applies to transceivers, DACs, or AOCs running at 25G or 100G. 100G DACs are a special case. They can only set FEC to none when auto-negotiation is disabled through the **speed override** command. The default for the installed transceiver is used in all other cases.



Transceivers for which FEC is auto-negotiated will request the mode configured by this command, but may resolve to a different mode. The applied FEC mode is displayed as a commented line in the configuration shown with the **show run** command. It is also displayed with **show interface** command.

Parameter	Description
auto	Use the transceiver default.
none	Do not use any FEC.
base-r-fec	Use IEEE BASE-R (Firecode) FEC.
rs-fec	Use IEEE RS (Reed-Solomon) FEC.

Command History

Release	Modification
10.11	Command enabled on 6400 and 8400 Switch Series.
10.08.1021	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

flow-control

```
flow-control rtxx  
no flow-control rtxx
```

Description

Command **flow-control** enables negotiation of IEEE 802.3x link-level flow control on the current interface. The switch advertises link-level flow control support to the link partner. The final configuration is determined based on the capabilities of both partners.

Each invocation of this command replaces the previous configuration.

The **no** form of these commands disables any configured flow control on the selected interface.

Parameter	Description
rtxx	Enables the ability to honor received and to transmit IEEE 802.3x LLFC pause frames to the remote device.

Usage (flow control)

- For interfaces that auto-negotiate, link-level flow control is subject to negotiation, plus speed and other parameters. Both ends of the link must negotiate the same flow control mode for it to be applied.
- For interfaces that do not auto-negotiate, the configured link-level flow control mode is always applied and the user is responsible for ensuring that both ends of the link are configured for the same mode.
- All members of a LAG must have the same flow control configuration.
- Lossless flow control is only supported for single destination unicast traffic. Replicated traffic (for example, broadcast, multicast, mirroring) cannot be guaranteed to be lossless.
- Lossless behavior is not supported when operating in a VSF stack configuration.
- Lossless flow control will only operate correctly when both the ingress and egress interfaces have flow control enabled.

Examples

Enabling support for RXTX flow control:

```
switch(config)# interface 1/1/1  
switch(config-if)# flow-control txrx
```

Disabling support for RXTX flow control:

```
switch(config)# interface 1/1/1  
switch(config-if)# no flow-control txrx
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

interface

interface <PORT-NUM>

Description

Switches to the **config-if** context for a physical port. This is where you define the configuration settings for the logical interface associated with the physical port.

Parameter	Description
<PORT-NUM>	Specifies a physical port number. Format: member/slot/port.

Examples

On the 6400 Switch Series, interface identification differs.

Configuring an interface:

```
switch(config)# interface 1/1/1
switch(config-if)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

interface loopback

```
interface loopback <ID>
no interface loopback <ID>
```

Description

Creates a loopback interface and changes to the **config-loopback-if** context. Loopback interfaces are layer 3.

The **no** form of this command deletes a loopback interface.

Parameter	Description
<code><INSTANCE></code>	Specifies the loopback interface ID. Range: 1 to 256

Examples

```
switch# config
switch(config)# interface loopback 1
switch(config-loopback-if)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

interface vlan

```
interface vlan <VLAN-ID>
no interface vlan <VLAN-ID>
```

Description

Creates an interface VLAN also know as an SVI (switched virtual interface) and changes to the **config-if-vlan** context. The specified VLAN must already be defined on the switch.

The **no** form of this command deletes an interface VLAN.

Parameter	Description
<code><VLAN-ID></code>	Specifies the loopback interface ID.
none	Do not reserve any internal VLANs.

Examples

```
switch# config
switch(config)# vlan 10
switch(config-vlan-10)# exit
switch(config)# interface vlan 10
switch(config-if-vlan)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

interpacket-gap

```
interpacket-gap [high average]
no interpacket-gap
```

Description

Configures the interpacket gap (IPG) between transmitted packets for the interface. The **no** form of this command sets IPG as default.

Parameter	Description
<i>high average</i>	Use a higher average IPG at certain speeds.

Examples

Allow the interface to use a higher IPG.

```
switch(config)# interface 1/1/1
switch(config-if)# interpacket-gap high-average
```

Use default IPG.

```
switch(config)# interface 1/1/1
switch(config-if)# no interpacket-gap
```

Output for **show interface** when **high-average** mode is configured and the port is linked at 1 Gbps.

```
switch(# show interface 1/1/1
Interpacket gap: high-average
```

Command History

Release	Modification
10.13.1060	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

ip address

```
ip address <IPV4-ADDR>/<MASK> [secondary]
no ip address <IPV4-ADDR>/<MASK> [secondary]
```

Description

Sets an IPv4 address for the current layer 3 interface.

The **no** form of this command removes the IPv4 address from the interface.

Parameter	Description
<IPV4-ADDR>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100 .
<MASK>	Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.
secondary	Specifies a secondary IP address.

Examples

On the 6400 Switch Series, interface identification differs.

Creating a layer 3 interface setting its IP address to **192.168.100.1** with a mask of **24** bits.

```
switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if)# ip address 192.168.100.1/24
```

Assigning the IP address **192.168.20.1** with a mask of **24** bits to loopback interface **1**:

```
switch(config)# interface loopback 1
switch(config-loopback-if)# routing
switch(config-loopback-if)# ip address 192.168.20.1/24
```

Assigning the IP address **192.168.199.1** with a mask of **24** bits to interface VLAN **10**:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# ip address 192.168.199.1/24
```

Removing the IP address **192.168.199.1** with a mask of **24** bits from interface VLAN **10**:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no ip address 192.168.199.1/24
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if config-loopback-if config-if-vlan	Administrators or local user group members with execution rights for this command.

ip mtu

ip mtu <VALUE>

no ip mtu

Description

Sets the IP MTU (maximum transmission unit) for an interface. This defines the largest IP packet that can be sent or received by the interface. This value should be less than or equal to the overall MTU for the interface.

The **no** form of this command sets the IP MTU to the default value 1500. This command is only allowed when routing is enabled on the interface.

Parameter	Description
<VALUE>	Specifies the IP MTU in bytes. Range: 68 to 9198. Default: 1500.

Usage

The IP MTU value for subinterface must be less than or equal to the parent MTU for the subinterface. The subinterface uses its IP MTU value and not the parent IP MTU value.

Examples

Setting the IP MTU to 576 bytes:

```
switch(config-if)# ip mtu 576
```

Setting the IP MTU to the default value:

```
switch(config-if)# no ip mtu
```

Setting the IP MTU value on a subinterface:

```
switch(config)# interface 1/1/1.10
switch(config-subif)# ip mtu 6000
```

Command History

Release	Modification
10.08	Subinterface support added.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if config-if-vlan config-subif	Administrators or local user group members with execution rights for this command.

ip source-interface

```
ip source-interface {sflow | tftp | radius | tacacs | ntp | syslog | ubt | dhcp-relay |
simplivity | dns | all} {interface <IFNAME> | <IPv4-ADDR>} [vrf <VRF-NAME>]
no ip source-interface {sflow | tftp | radius | tacacs | ntp | syslog | ubt | dhcp-
relay | simplivity | dns | all} [interface <IFNAME> | <IPv4-ADDR>] [vrf <VRF-NAME>]
```

Description

Sets a single source IP address for a feature on the switch. This ensures that all traffic sent the feature has the same source IP address regardless of how it egresses the switch. You can define a single global address that applies to all supported features, or an individual address for each feature.

This command provides two ways to set the source IP addresses: either by specifying a static IP address, or by using the address assigned to a switch interface. If you define both options, then the static IP address takes precedence.

The `no` form of this command deletes the single source IP address for all supported services, or a specific service.

Parameter	Description
sflow tftp radius tacacs ntp syslog ubt dhcp-relay simplivity dns all	Sets a single source IP address for a specific service. The <code>all</code> option sets a global address that applies to all protocols that do not have an address set. For DHCP relay, the address is used as both the source IP and GIADDR.
interface <IFNAME>	Specifies the name of the interface from which the specified service obtains its source IP address. The interface must have a valid IP address assigned to it. If the interface has both a primary and secondary IP address, the primary IP address is used.
<IPv4-ADDR>	Specifies the source IP address to use for the specified service. The IP address must be defined on the switch, and it must exist on the specified VRF (which is the <code>default</code> VRF, if the <code>vrf</code> option is not used). Specify the address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
vrf <VRF-NAME>	Specifies the name of a VRF.

Examples

Setting the IPv4 address 10.10.10.5 as the global single source address:

```
switch# config
switch(config)# ip source-interface all 10.10.10.5
```

Setting the secondary IPv4 address 10.10.10.5 on interface 1/1/1 as the global single source address. (On the 6400 Switch Series, interface identification differs.)

```
switch# config
switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if)# ip address 10.10.10.1/24
switch(config-if)# ip address 10.10.10.5/24 secondary
switch(config)# exit
switch(config)# ip source-interface all 10.10.10.5
```

Clearing the global single source IP address **10.10.10.5**:

```
switch(config)# no ip source-interface all 10.10.10.5
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ipv6 address

```
ipv6 address <IPv6-ADDR>/<MASK>{eui64 | [tag <ID>]}
no ipv6 address <IPv6-ADDR>/<MASK>
```

Description

Sets an IPv6 address on the interface.

The **no** form of this command removes the IPv6 address on the interface.



This command automatically creates an IPv6 link-local address on the interface. However, it does not add the **ipv6 address link-local** command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the **ipv6 address link-local** command.

Parameter	Description
<IPv6-ADDR>	Specifies the IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a

Parameter	Description
	hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55 .
<MASK>	Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.
eui64	Configure the IPv6 address in the EUI-64 bit format.
tag <ID>	Configure route tag for connected routes. Range: 0 to 4294967295. Default: 0.

Examples

Setting the IPv6 address **2001:0db8:85a3::8a2e:0370:7334** with a mask of 24 bits:

```
switch(config-if)# ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Removing the IP address **2001:0db8:85a3::8a2e:0370:7334** with mask of 24 bits:

```
switch(config-if)# no ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

ipv6 source-interface

```
ipv6 source-interface {sflow | tftp | radius | tacacs | ntp | syslog | ubt | dhcp-relay | simplivity | dns | all} [interface <IFNAME> | <IPv6-ADDR>] [vrf <VRF-NAME>]
no ipv6 source-interface {sflow | tftp | radius | tacacs | ntp | syslog | ubt | dhcp-relay | simplivity | dns | all} [interface <IFNAME> | <IPv6-ADDR>] [vrf <VRF-NAME>]
```

Description

Sets a single source IP address for a feature on the switch. This ensures that all traffic sent the feature has the same source IP address regardless of how it egresses the switch. You can define a single global address that applies to all supported features, or an individual address for each feature.

This command provides two ways to set the source IP addresses: either by specifying a static IP address, or by using the address assigned to a switch interface. If you define both options, then the static IP address takes precedence.

The `no` form of this command deletes the single source IP address for all supported protocols, or a specific protocol.

Parameter	Description
<code>sflow tftp radius tacacs ntp syslog ubt dhcp-relay simplivity dns all</code>	Sets a single source IP address for a specific protocol. The <code>all</code> option sets a global address that applies to all protocols that do not have an address set.
<code>interface <IFNAME></code>	Specifies the name of the interface from which the specified protocol obtains its source IP address.
<code><IPv6-ADDR></code>	Specifies the source IP address to use for the specified protocol. The IP address must be defined on the switch, and it must exist on the specified VRF (which is the default VRF, if the <code>vrf</code> option is not used). Specify the IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.
<code>vrf <VRF-NAME></code>	Specifies the name of the VRF from which the specified protocol sets its source IP address.

Examples

Configuring the IPv6 address `2001:DB8::1` as the global single source address:

```
switch# config
switch(config)# ip source-interface all 2001:DB8::1/32
```

Configuring the IPv6 address `2001:DB8::1` on VRF `sflow-vrf` on interface `1/1/2` as the single source address for sFlow:

```
switch(config)# vrf sflow-vrf
switch(config-vrf)# exit
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# vrf attach sflow-vrf
switch(config-if)# ipv6 address 2001:DB8::1/32
switch(config-if)# exit
switch(config)# ip source-interface sflow interface 1/1/2 vrf sflow-vrf
```

Stop the source IP address from using the IP address on interface `1/1/1` on VRF `one`.

```
switch(config)# no ip source-interface all interface 1/1/1 vrf one
```

Clear the source IP address `2001:DB8::1`.

```
switch(config)# no ip source-interface all 2001:DB8::1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

l3-counters

```
l3-counters [rx | tx]
no l3-counters [rx | tx]
```

Description

Enables counters on a layer 3 interface. By default, all interfaces are layer 3. To change a layer 2 interface to layer 3, use the **routing** command.

The **no** form of this command, with no specification, disables both transmit and receive counters on a layer 3 interface. To disable transmit (**tx**) or receive (**rx**) counters only, specify the counter type you want to disable.

Parameter	Description
rx	Specifies receive counters.
tx	Specifies transmit counters.

Examples

Enabling layer 3 transmit counters

On the 6300 Switch Series:

```
switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if)# l3-counters
```

On the 6400 Switch Series:

```
switch(config)# interface 1/3/1
switch(config-if)# routing
switch(config-if)# l3-counters
```

Enabling layer 3 transmit counters on subinterfaces

On the 6300 Switch Series:

```
switch(config)# interface 1/1/1.10
switch(config-if)# routing
switch(config-if)# l3-counters tx
```

On the 6400 Switch Series:

```
switch(config)# interface 1/3/1.10
switch(config-if)# routing
switch(config-if)# 13-counters tx
```

Command History

Release	Modification
10.08	Added support for 13 counters on subinterfaces
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-if config-subif	Administrators or local user group members with execution rights for this command.

leader-follower-mode

```
leader-follower-mode
forced-follower
forced-leader
no ...
preferred-follower
preferred-leader
```

Description

Configure the leader-follower mode for 1000BASE-T, 2.5GBASE-T, 5GBASE-T, and 10GBASE-T on Aruba CX 6300 Switch Series JL659A & JL660A. The leader uses an external clock for generating its clock signals to determine the timing of the transmitter and receiver operations. The follower recovers its clock from the received signal from the leader and uses it to determine the timing of the transmitter operations. As a multiport device, the switch operates as the **preferred-leader** by default.



Configuring both the ends of the link to the same forced leader-follower role will result in the failure to link-up.

Parameter	Description
forced-follower	Force the interface to be the follower.
forced-leader	Force the interface to be the leader.
no ...	Negates any configured parameter.
preferred-follower	Advertise preference to be the follower.
preferred-leader	Advertise preference to be the leader (default).

Example

The following example configures the **preferred-follower** mode:

```
switch(config)# interface 1/1/1
switch(config-if)# leader-follower-mode preferred-follower
```

Applying this configuration on interfaces with different technologies like 100BASE-TX and 10BASE-T will have no effect.

Command History

Release	Modification
10.10.1100	Command introduced.

Command Information

Platforms	Command context	Authority
6300 JL659A & JL660A	config-if	Administrators or local user group members with execution rights for this command.

link-clock-wider-tolerance

[no]link-clock-wider-tolerance

Description

Configure a lower tolerance level for link partner clock variance for a link operating at 100Mbps on Smart Rate ports. Wider tolerance allows greater variation in link partner clock accuracy while still maintaining the link.

Wider tolerance is enabled by default. Use the **no** prefix to configure narrower tolerance, which may improve link reliability with certain 100Mbps devices. Note that this command only applies to Smart Rate ports and only when linked at 100Mbps. It has no effect at other speeds or on other ports.

Example

The following example allows less partner link clock variance.

```
switch(config)# interface 1/1/1
switch(config-if)# no link-clock-wider-tolerance
```

Issuing this command without the **no** parameter returns the tolerance level to its default value.

```
switch(config)# interface 1/1/1
switch(config-if)# link-clock-wider-tolerance
```

Command History

Release	Modification
10.10.1100	Command introduced.

Command Information

Platforms	Command context	Authority
6300 JL659A & JL660A	config-if	Administrators or local user group members with execution rights for this command.

mtu

mtu <VALUE>
no mtu

Description

Sets the MTU (maximum transmission unit) for an interface. This defines the maximum size of a layer 2 (Ethernet) frame. Frames larger than the MTU (1500 bytes by default) are dropped and cause an ICMP fragmentation-needed message to be sent back to the originator.

To support jumbo frames (frames larger than 1522 bytes), increase the MTU as required by your network. A frame size of up to 9198 bytes is supported.

The largest possible layer 1 frame will be 18 bytes larger than the MTU value to allow for link layer headers and trailers.

The **no** form of this command sets the MTU to the default value 1500.

Parameter	Description
<VALUE>	Specifies the MTU in bytes. Range: 46 to 9198. Default: 1500.

Examples

On the 6400 Switch Series, interface identification differs.

Setting the MTU on interface **1/1/1** to 1000 bytes:

```
switch(config)# interface 1/1/1
switch(config-if)# no routing
switch(config-if)# mtu 1000
```

Setting the MTU on interface **1/1/1** to the default value:

```
switch(config)# interface 1/1/1
switch(config-if)# no routing
switch(config-if)# no mtu
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

persona

```
persona {access | uplink | custom <PERSONA-NAME>} [copy | attach]
no persona {access | uplink | custom <PERSONA-NAME>} [copy | attach]
```

Description

Associates one of three persona types with an interface to classify the purpose or role of an interface. On the 10000 Switch Series, “access” persona ports are typically connected to workloads / VMs, and the “uplink” (fabric) persona ports are connected to the core / spine.

The **no** form of this command removes the interface persona.

Parameter	Description
access	Selects the access persona type.
uplink	Selects the uplink persona type.
custom <PERSONA-NAME>	Selects the custom persona type with a user-provided name. Range: 1 to 64 printable ASCII characters including space.
copy	Specifies the mode: copies settings from the persona interface of the same name.
attach	Specifies the mode: attaches the specified interface to the persona interface of the same name.

Usage

- If the mode is specified, either copy or attach, the interface configuration is dependent on the interface template whose name is "access", "uplink", or "<PERSONA-NAME>". On the other hand, if the mode is not specified, then the persona is just a label in the interface, and its configuration is not modified even if the interface persona exists. When configuring the mode, one of the following options is possible:
 - The **copy** option performs a one-time copy of the template interface. Subsequent changes to the template are not copied and the 'persona' setting is just a label. If the mode is set to **copy** and the interface persona does not exist, then the CLI command fails with the message "Interface persona not found".
 - The **attach** option performs a copy of the template interface, and subsequent changes to the template interface configuration are immediately applied to all attached interfaces. The template interface does not need to exist before attaching other interfaces to it. After attaching a template, the copied settings can be modified for an individual interface. However, any change in the attached template will overwrite the modified values with the new template values.
- When a mode is specified, it should match an interface created with the command interface persona <PERSONA-NAME>. The only exception to this rule is when the mode is set to **attach** and the persona does not already exist.

- The mode is only available to be configured for an interface that meets the following conditions:
 - IS a physical interface
 - IS NOT a LAG member
 - IS NOT a persona interface

Examples

Configuring an access persona:

```
switch(config)# interface 1/1/1  
switch(config-if)# persona access
```

Configuring an uplink persona:

```
switch(config)# interface 1/1/1  
switch(config-if)# persona uplink
```

Configuring a custom persona named "mypersona":

```
switch(config)# interface 1/1/1  
switch(config-if)# persona custom mypersona
```

Removing the persona setting.

```
switch(config-if)# no persona
```

Copying a predefined persona name configuration to an interface:

1. Configuring the interface persona:

```
switch(config)# interface persona uplink  
switch(config-if)# no shutdown  
switch(config-if)# no routing  
switch(config-if)# vlan access 100  
switch(config-if)# exit
```

2. Applying the configuration from the persona named "mypersona" with **copy** mode:

```
switch(config)# interface 1/1/1  
switch(config-if)# persona custom mypersona copy  
switch(config-if)# exit
```

Attaching a custom persona name named "mypersona" to several interfaces simultaneously:

1. Configuring an interface persona named "mypersona":

```
switch(config)# interface persona mypersona  
switch(config-if)# no shutdown  
switch(config-if)# vrf attach upstream  
switch(config-if)# exit
```


2. Applying the "mypersona" configuration with **attach** mode:

```
switch(config)# interface 1/1/1-1/1/24
switch(config-if)# persona custom mypersona attach
switch(config-if)# exit
```

Command History

Release	Modification
10.10	Added optional parameters: attach, copy.
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

rate-interval

```
rate-interval <VALUE>
```

```
no rate-interval
```

Description

This command sets the time interval to calculate interface rates. Lower intervals are more useful for detecting traffic bursts, but may increase computation load to the overall system. Intervals must be a multiple of five seconds. The command-line interface will not accept a rate interval value that is not a multiple of five.

The **no** form of this command sets the rate collection interval to the default value of 300 seconds.

Parameter	Description
<VALUE>	The statistics rate collection interval in seconds. The supported range is 5-300 seconds, where the number of seconds is a multiple of five. NOTE: The supported range for 6400 and 8400 switch series is 30 - 300 seconds.

Examples

Setting the rate collection interval to 50 seconds

```
switch(config)# interface 1/1/1
switch(config-if)# rate-interval 50
```

Setting the rate collection interval to the default value:

```
switch(config-if)# no rate-interval
```

The following example shows the command-line interface warning that appears while configuring an invalid rate-interval.

```
switch(config)# interface 1/1/1
switch(config-if)# rate interval 6
The interval must be a multiple of 5.
```

Usage

The rate collection interval must be configured in the multiples of 5. Any other value will be rejected and the CLI will display the error message, **The interval must be a multiple of 5.**

Command History

Release	Modification
10.12.1000	Command supported on all platforms.
10.12	Command Introduced on 6300 and 8360 Switch series.

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

routing

routing
no routing

Description

Enables routing support on an interface, creating a L3 (layer 3) interface on which the switch can route IPv4/IPv6 traffic to other devices.

By default, routing is disabled on all interfaces.

The **no** form of this command disables routing support on an interface, creating a L2 (layer 2) interface.

Examples

Enabling routing support on an interface:

```
switch(config-if)# routing
```

Disabling routing support on an interface:

```
switch(config-if)# no routing
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

show allow-unsupported-transceiver

show allow-unsupported-transceiver

Description

Displays configuration and status of unsupported transceivers.

Examples

Showing unallowed unsupported transceivers:

```
switch(config)# show allow-unsupported-transceiver
Allow unsupported transceivers : no
Logging interval           : 1440 minutes
-----
Port      Type      Status
-----
1/1/31    SFP-SX      unsupported
1/1/32    SFP-1G-BXD  unsupported
1/1/2     SFP28DAC3   unsupported
```

Showing allowed unsupported transceivers:

```
switch# show allow-unsupported-transceiver
Allow unsupported transceivers : yes
Logging interval           : 1440 minutes
-----
Port      Type      Status
-----
1/1/31    SFP-SX      unsupported-allowed
1/1/32    SFP-1G-BXD  unsupported-allowed
1/1/2     SFP28DAC3   unsupported
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

show interface

```
show interface [<IFNNAME>|<IFRANGE>] [brief | physical]
show interface [<IFNNAME>|<IFRANGE>] [extended [non-zero] | [human-readable]]
show interface [<IFNNAME>] monitor [human-readable]
show interface [lag | loopback | tunnel | vlan ] [<ID>] [brief]
show interface lag [<LAG-ID>] [extended [non-zero] | [human-readable]]
show interface lag [<LAG-ID>] monitor [human-readable]
```

Description

Shows active configurations and operational status information for interfaces.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
brief	Shows brief info in tabular format.
physical	Shows the physical connection info in tabular format.
extended	Shows additional statistics, including the tx filtered and rx filtered counters. - Rx filter packets are protocol packets received when the protocol is disabled on the switch and there is only one port in the VLAN. Protocols include OSPF, PIM, RIP, LACP, and LLDP. - An example of a Tx filtered packet would be a multicast packet being filtered from going out of the ingress port.
human-readable	Shows statistics rounded to the nearest power of 1000, for example, 1K, 345M, 2G. This is available only in the CLI interface output.
non-zero	Shows only non zero statistics.
LAG	Shows LAG interface information.
monitor	Continuously monitor interface statistics.
LOOPBACK	Shows loopback interface information.
TUNNEL	Shows tunnel interface information.
VLAN	Shows VLAN interface information.
<LAG-ID>	Specifies the LAG number. Range: 1-256
<LOOPBACK-ID>	Specifies the LOOPBACK number. Range: 0-255
<TUNNEL-ID>	Specifies the tunnel ID. Range: 1-255

Parameter	Description
<VLAN-ID>	Specifies the VLAN ID. Range: 1-4094
VXLAN	Shows the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1

Examples

Showing interface information when it is configured as a route-only port:

```
switch# show interface 1/1/1
Interface 1/1/1 is up
Admin state is up
Link state: up for 2 days (since Sun Jun 21 05:30:22 UTC 2020)
Link transitions: 1
Description: backup data center link
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 1500
Type 1GbT
Full-duplex
qos trust none
Speed 1000 Mb/s
Auto-negotiation is on
Flow-control: off
Error-control: off
Energy-Efficient Ethernet is enabledMDI mode: MDIX
L3 Counters: Rx Enabled, Tx Enabled
Rate collection interval: 300 seconds
```

Rates	RX	TX	Total (RX+TX)
Mbits / sec	0.00	0.00	0.00
KPkts / sec	0.00	0.00	0.00
Unicast	0.00	0.00	0.00
Multicast	0.00	0.00	0.00
Broadcast	0.00	0.00	0.00
Utilization %	0.00	0.00	0.00

Statistics	RX	TX	Total
Packets	0	0	0
Unicast	0	0	0
Multicast	0	0	0
Broadcast	0	0	0
Bytes	0	0	0
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0
Pause Frames	0	0	0
L3 Packets	0	0	0
L3 Bytes	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0
Other	0	0	0

Showing information when the interface is currently linked at a downshifted speed:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active
```

Showing information when the interface is currently linked with energy-efficient-ethernet negotiated:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Energy-Efficient Ethernet is enabled and active
```

Showing information when the interface is shut down during a VSX split:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is down
Admin state is up
State information: Disabled by VSX
Link state: down for 3 days (since Tue Mar 16 05:20:47 UTC 2021)
Link transitions: 0
Description:
Hardware: Ethernet, MAC Address: 04:09:73:62:90:e7
MTU 1500
Type SFP+DAC3
Full-duplex
qos trust none
Speed 0 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: native-untagged
Native VLAN: 1
Allowed VLAN List: 1502-1505
Rate collection interval: 300 seconds
```

Rate	RX	TX	Total (RX+TX)
Mbits / sec	0.00	0.00	0.00
KPkts / sec	0.00	0.00	0.00
Unicast	0.00	0.00	0.00
Multicast	0.00	0.00	0.00
Broadcast	0.00	0.00	0.00
Utilization	0.00	0.00	0.00

Statistic	RX	TX	Total
Packets	0	0	0
Unicast	0	0	0
Multicast	0	0	0
Broadcast	0	0	0
Bytes	0	0	0
Jumbos	0	0	0
Dropped	0	0	0
Pause Frames	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0

Giants	0	n/a	0
--------	---	-----	---

Showing information when the interface is configured with EEE and the EEE has auto-negotiated:

```
switch(config-if)# show interface 1/1/1 physical
```

EEE			Link	Admin	Speed		Flow-Control	
Port	Type	PoE Power	Status	Config	Port	Status Config	Status Config	
Status Config		(Watts)	State	Information	Description			
1/1/1	1GbT		up	up	1G		off	off
on	on	--	10M/100M/1G		--	auto		

Showing the monitor information:



In monitor mode, the CLI refreshes data automatically until it is exited by entering **q**. Pressing **?** opens the help menu to display which options are available in this context.

Interface 1/1/1 is up			
Rate	RX	TX	Total (RX+TX)
-----	-----	-----	-----
MBits / sec	30196.43	30196.43	60392.85
MPkts / sec	58977.39	58977.40	117954.79
Unicast	0.00	0.00	0.00
Multicast	58977.39	58977.40	117954.79
Broadcast	0.00	0.00	0.00
Utilization %	75.49	75.49	150.98
Statistic	RX	TX	Total (RX+TX)
-----	-----	-----	-----
Packets	4756527649	4756527865	9513055514
Unicast	0	0	0
Multicast	4756527649	4756527865	9513055514
Broadcast	2	0	2
Bytes	304417778668	304417795428	608835574096
Jumbos	0	0	0
Dropped	0	19028847730	19028847730
Pause Frames	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
help: ?, quit: q			

```

Help for Interface Monitor
h  Toggle human-readable mode
c  Clear interface statistics
Does not apply to rates
Arrows, PgUp, PgDn, Home, End
Navigate interface statistics
Delay: 2
help: ?, quit: q

```

Showing the output for interface 1/1/1 in human-readable format:



In human-readable format, the **< 1** symbol for **Utilization** indicates that the amount of packets is between zero and one. This is true in cases where the number of bytes increases but the number of packets and the **Utilization** value is not displayed even in the normal output, where the human-readable parameter is not included in the command.

```
switch(config-if)# show interface 1/1/1 human-readable
Interface 1/1/1 is up

```

Rate	RX	TX	Total (RX+TX)
Bits / sec	3M	3M	6M
Pkts / sec	316	316	633
Unicast	319	319	638
Multicast	0	0	0
Broadcast	0	0	0
Utilization %	< 1	< 1	< 1

```
Statistic
```

	RX	TX	Total
Packets	577K	577K	1M
Unicast	577K	577K	1M
Multicast	0	51	51
Broadcast	0	15	15
Bytes	744M	745M	1G
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0
Pause Frames	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0

Showing information about extended counters:



The output of the `show interface extended` command varies depending on the switch model and configuration.

```
switch(config-if)# show interface 1/1/17 extended
-----
Interface 1/1/17
-----

```

Statistics	Value
Dot1d Tp Port In Frames	547
Dot1d Tp Port Out Frames	608
Dot3 In Pause Frames	0
Dot3 Out Pause Frames	0
Ethernet Stats Broadcast Packets	19
Ethernet Stats Bytes	40162
Ethernet Stats Packets	342
...	

```
-----
Error-Statistics
```

	Value
Dot1d Base Port MTU Exceeded Discards	0
Dot3 Control In Unknown Opcodes	0


```

Dot3 Stats Alignment Errors          0
Dot3 Stats FCS Errors                0
Dot3 Stats Frame Too Longs          0
Dot3 Stats Internal Mac Transmit Errors 0
Ethernet RX Oversize Packets        0
...

```

Command History

Release	Modification
10.11	Added <code>monitor</code> parameter.
10.10	Added <code>human-readable</code> parameter.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface dom

```
show interface [<INTERFACE-ID>] dom [detail] [vsx-peer]
```

Description

Shows diagnostics information and alarm/warning flags for the optical transceivers (SFP, SFP+, QSFP+). This information is known as DOM (Digital Optical Monitoring). DOM information also consists of vendor determined thresholds which trigger high/low alarms and warning flags.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: member/slot/port .
detail	Show detailed information.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

On the 6400 Switch Series, interface identification differs.

```
switch# show interface dom
```

Port	Type	Channel	Temperature (Celsius)	Voltage (Volts)	Tx Bias (mA)	Rx Power (mW/dBm)	Tx Power (mW/dBm)
1/1/1	SFP+SR		47.65	3.31	8.40	0.08, -10.96	0.63, -2.49
1/1/2	SFP+SR		n/a	n/a	n/a	n/a	n/a
1/1/3	SFP+DA3		42.10	3.24	n/a	n/a	n/a
1/1/4	QSFP+SR4	1	44.46	3.30	6.12	0.08, -10.96	0.63, -1.95
2			44.46	3.30	6.04	0.08, -10.96	0.63, -2.00
3			44.46	3.30	6.51	0.08, -10.96	0.60, -2.16
4			44.46	3.30	6.19	0.08, -10.96	0.63, -1.94

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface energy-efficient ethernet

```
show interface [<IFNAME>|<IFRANGE>] energy-efficient-ethernet
```

Description

Displays Energy-Efficient Ethernet information for the interface.

Parameter	Description
<IFNAME>	Specifies the name of an interface on the switch. Use the format member/slot/port (for example, 1/1/1).
<IFRANGE>	Specifies the port identifier range of an interface on the switch. Use the format member/slot/port (for example, 1/1/1).

Example

The following example shows when the interfaces are Energy-Efficient Ethernet capable.

```
switch# show interface energy-efficient-ethernet
```

Port	Enabled	Negotiated	Speed (MB/s)	TX Wake Time (us)	RX Wake Time (us)
1/1/1	no	no	--	--	--
1/1/2	yes	yes	100	36	36
1/1/3	yes	yes	1000	17	17

1/1/4	no	no	--	--	--
1/1/5	yes	no	1000	--	--

The following example shows when the interface is not Energy-Efficient Ethernet capable :

```
switch# show interface 1/1/1 energy-efficient-ethernet
Port 1/1/1 does not support Energy-Efficient-Ethernet
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface flow-control

show interface [<IFNNAME>|<IFRANGE>] flow-control [detail]

Description

Shows the flow control configuration, status, and statistics of the specified interface for interfaces on which flow control is enabled.



If **detail** is not specified, this command shows a summary of all flow controlled interfaces with one interface per line. If **detail** is specified, this command shows flow control detailed statistics.



As of AOS-CX 10.10, the separate **show flow-control** command has been removed, with it being effectively replaced by this command.

Parameter	Description
<IFNNAME> <IFRANGE>	Specifies the interface (port) name or range. When no interface range is specified, only interfaces with flow control enabled in the configuration or status are shown.
detail	Shows detailed information.

Examples

Showing summary flow control information:

```
switch# show interface flow-control
```

```
-----  
Port      Flow  
          Control  
-----
```

```
1/1/1      config: llfc rx  
           status: llfc rx  
1/1/2      config: llfc rx  
           status: none
```

Showing summary flow control information with PFC:

```
switch# show interface flow-control
```

```
-----  
Port      Flow  
          Control  
-----
```

```
1/1/1      config: pfc rxtx-1,2  
           status: pfc rxtx-1,2  
1/1/2      config: pfc rxtx-5  
           status: none
```

Showing summary flow control information with PFC:

```
switch# show interface flow-control
```

```
Flow Control Watchdog Settings
```

```
Trigger Timeout: 100 milliseconds
```

```
Resume Time: 100 milliseconds
```

```
-----  
Port      Flow      Watchdog      Watchdog  
          Control    Status        Timeouts  
-----  
1/1/1      config: llfc rx  
           status: llfc rx  
1/1/2      config: llfc rx      incompatible    0  
           status: llfc rx  
1/1/10     config: pfc rxtx-1,2  enabled         1234  
           status: pfc rxtx-1,2  
1/1/12     config: pfc rxtx-1,2  error           0  
           status: pfc rxtx-1,2  
1/1/32:4   config: pfc rxtx-5  
           status: pfc rxtx-5
```

Showing summary flow control information where the configuration does not match status due to a reboot required to apply PFC configuration in hardware:

```
switch# show interface flow-control
```

```
Flow Control Watchdog Settings
```

```
Trigger Timeout: 100 milliseconds (actual: not applied)
```

```
Resume Time: 100 milliseconds (actual: not applied)
```

```
-----  
Port      Flow      Watchdog      Watchdog  
          Control    Status        Timeouts  
-----
```

1/1/1	config: llfc rx status: llfc rx		
1/1/2	config: llfc rx status: llfc rx	incompatible	0
1/1/10	config: pfc rxtx-1,2 status: none	pending	1234
1/1/12	config: pfc rxtx-1,2 status: none	pending	0
1/1/32:4	config: pfc rxtx-5 status: none		

Showing detailed flow control information with RX flow control enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: llfc rx

Statistics                               RX
-----
Dot3 Pause Frames                        0
```

Showing detailed flow control information with RX flow control enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: llfc rx
Flow-control watchdog: disabled

Statistics                               RX
-----
Dot3 Pause Frames                        0
```

Showing detailed flow control information with RXTX flow control enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: llfc rxtx

Statistics                               RX                               TX
-----
Dot3 Pause Frames                        0                               0
```

Showing detailed flow control information with PFC enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: pfc rxtx-4,5
```

Statistics	RX	TX
-----	-----	-----
Priority 0 Pauses	0	0
Priority 1 Pauses	0	0
Priority 2 Pauses	0	0
Priority 3 Pauses	0	0
Priority 4 Pauses	0	0
Priority 5 Pauses	0	0
Priority 6 Pauses	0	0
Priority 7 Pauses	0	0
Total Pause Frames	0	0

Showing detailed flow control information with PFC enabled and flow control watchdog disabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: pfc rxtx-4,5
Flow-control watchdog: disabled
```

Statistics	RX	TX
-----	-----	-----
Priority 0 Pauses	0	0
Priority 1 Pauses	0	0
Priority 2 Pauses	0	0
Priority 3 Pauses	0	0
Priority 4 Pauses	0	0
Priority 5 Pauses	0	0
Priority 6 Pauses	0	0
Priority 7 Pauses	0	0
Total Pause Frames	0	0

```
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: pfc rxtx-4,5
Flow-control watchdog: disabled
```

Statistics	RX	TX
-----	-----	-----
Priority 0 Pauses	0	0
Priority 1 Pauses	0	0
Priority 2 Pauses	0	0
Priority 3 Pauses	0	0
Priority 4 Pauses	0	0
Priority 5 Pauses	0	0
Priority 6 Pauses	0	0
Priority 7 Pauses	0	0
Total Pause Frames	0	0

Showing detailed flow control information with both PFC and flow control watchdog enabled:

```

switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: pfc rxtx-4,5
Flow-control watchdog: enabled

```

Statistics	RX	TX
Priority 0 Pauses	0	0
Priority 1 Pauses	0	0
Priority 2 Pauses	0	0
Priority 3 Pauses	0	0
Priority 4 Pauses	0	0
Priority 5 Pauses	0	0
Priority 6 Pauses	0	0
Priority 7 Pauses	0	0
Total Pause Frames	0	0

Queue	Watchdog Timeouts
Queue 0	0
Queue 1	0
Queue 2	0
Queue 3	0
Queue 4	0
Queue 5	0
Queue 6	0
Queue 7	0

Showing detailed flow control information when flow control watchdog is enabled in the configuration but it could not be applied because the configured flow control mode is not compatible with watchdog:

```

switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: llfc rx
Flow-control watchdog: incompatible

```

Showing detailed flow control information when flow control watchdog is enabled in the configuration but could not be applied because a compatible flow control mode first requires a reboot:

```

switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: off
Flow-control watchdog: pending

```

Command History

Release	Modification
10.10	Examples updated with new and changed output elements.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface statistics

```
show interface [<IFNAME>|<IFRANGE>] statistics [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] statistics monitor [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] error-statistics [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] error-statistics monitor [non-zero] [human-readable]
show interface lag [<LAG-ID>] statistics [non-zero] [human-readable]
show interface lag [<LAG-ID>] statistics monitor [non-zero] [human-readable]
show interface lag [<LAG-ID>] error-statistics [non-zero] [human-readable]
show interface lag [<LAG-ID>] error-statistics monitor [non-zero] [human-readable]
show interface vxlan <VXLAN-ID> statistics [non-zero] [human-readable]
```

Description

Shows statistics for switch interfaces such as packets transmitted and received, bytes transmitted and received, broadcast and multicast packets.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
LAG	Shows LAG interface information.
<LAG-ID>	Specifies the LAG number. Range: 1-256
VXLAN	Shows the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1
monitor	Continuously monitor interface statistics.
human-readable	Shows statistics rounded to the nearest power of 1000, for example, 1K, 345M, 2G.
non-zero	Shows only non zero statistics.

Examples

Showing statistics of all interfaces:


```
show interface statistics
```

Interface	RX Bytes	RX Packets	RX Drops	TX Bytes	TX Packets	TX Drops	RX Broadcast	RX Multicast	TX Broadcast	TX Multicast	RX Pause	TX Pause
1/1/1	2727136	1975	0	17796	195	0	82	1788	96	54	0	0
1/1/10	0	0	0	0	0	0	0	0	0	0	0	0
1/1/11	0	0	0	0	0	0	0	0	0	0	0	0
1/1/12	0	0	0	0	0	0	0	0	0	0	0	0
...												
1/1/30 - lag1	0	0	0	11271	92	0	0	0	0	51	0	0
1/1/31 - lag2	2360	25	50	2732119	2040	0	0	0	178	1839	0	0
1/1/32 - lag2	0	0	0	11373	93	0	0	0	0	51	0	0
vlan1	0	0	0	0	0	0	0	0	0	0	0	0

Showing statistics of all interfaces with only non-zero statistics:

```
show interface statistics non-zero
```

Interface	RX Bytes	RX Packets	RX Drops	TX Bytes	TX Packets	TX Drops	RX Broadcast	RX Multicast	TX Broadcast	TX Multicast	RX Pause	TX Pause
1/1/1	2727136	1975	0	17796	195	0	82	1788	96	54	0	0
1/1/30 - lag1	0	0	0	11271	92	0	0	0	0	51	0	0
1/1/31 - lag2	2360	25	50	2732119	2040	0	0	0	178	1839	0	0
1/1/32 - lag2	0	0	0	11373	93	0	0	0	0	51	0	0

Showing statistics of all interfaces in the human-readable format:

```
show interface statistics human-readable
```

Interface	RX Bytes	RX Pkts	RX Drops	TX Bytes	TX Pkts	TX Drops	RX Bcast	RX Mcast	TX Bcast	TX Mcast	RX Pause	TX Pause
1/1/1	744M	577K	0	745M	578K	0	0	0	73	287	0	0
1/1/2	474M	367K	0	475M	369K	0	0	0	73	288	0	0
1/1/3	0	0	0	0	0	0	0	0	0	0	0	0

Showing statistics of a single interfaces:

```
show interface 1/1/2 statistics
```

Interface	RX Bytes	RX Packets	RX Drops	TX Bytes	TX Packets	TX Drops	RX Broadcast	RX Multicast	TX Broadcast	TX Multicast	RX Pause	TX Pause
1/1/2	2725080	1931	0	25877	253	0	21	1788	65	55	0	0

Showing statistics of all members of a LAG interface:

```
show interface lag1 statistics
```

Interface	RX Bytes	RX Packets	RX Drops	TX Bytes	TX Packets	TX Drops	RX Broadcast	RX Multicast	TX Broadcast	TX Multicast	RX Pause	TX Pause
1/1/3 - lag1	2424	26	0	2734082	2062	0	0	0	191	1848	0	0
1/1/30 - lag1	0	0	0	12383	100	0	0	0	0	59	0	0
lag1	2424	26	0	2746465	2162	0	0	0	191	1907	0	0

Showing error statistics of all interfaces:

```
show interface error-statistics
```

Interface	RX Errors	TX Errors	Giants	Runts	CRC/FCS	Collisions
1/1/1	190	20	100647	0	0	0
1/1/10	0	0	100	290	7165	949
1/1/11	0	0	0	0	0	0
1/1/12	0	0	0	0	0	0
...						
1/1/30 - lag1	1500	500	45800	0	0	0
1/1/31 - lag2	0	0	11	27	0	0
1/1/32 - lag2	0	0	0	0	6	18

Showing monitor statistics:



The rows and columns of show interface monitor statistics depends on the length of width of the client terminal. The CLI can be navigated using the arrow keys as well as the PageUp, PageDown, Home, and End keys.

```
show interface statistics monitor
```

Interface	RX Bytes	RX Packets >>
1/1/1	3440525421984	53758209526
1/1/2	3440526607008	53758228042
1/1/3	3440527785312	53758246453
1/1/30	3440559671264	53758744653
1/1/31	3440560851680	53758763098
1/1/32	3440562028704	53758781489

```
help: ?, quit: q
```

Help for Interface Monitor

f Toggle full statistics
h Toggle human-readable mode
n Toggle non-zero mode
r Toggle rate display

c Clear interface statistics
Does not apply to rates

Arrows, PgUp, PgDn, Home, End
Navigate interface statistics

Delay:2

```
help: ?, quit: q
```

Showing monitor error statistics in human-readable format:

```
show interface 1/1/1-1/1/3,1/1/30-1/1/32 error-statistics monitor human-readable
```

Interface	RX Errors	TX Errors	RX Giants	RX Runts	CRC/FCS	Collisions
1/1/1	0	0	0	0	0	0
1/1/2	0	0	0	0	0	0
1/1/3	0	0	0	0	0	0
1/1/30	0	0	0	0	0	0
1/1/31	0	0	0	0	0	0
1/1/32	0	0	0	0	0	0

Human-readable

```
help: ?, quit: q
```

Help for Interface Monitor

h Toggle human-readable mode
n Toggle non-zero mode

c Clear interface statistics
Does not apply to rates

Arrows, PgUp, PgDn, Home, End
Navigate interface statistics

Delay:2

```
help: ?, quit: q
```

Command History

Release	Modification
10.11	Added <code>moitor</code> parameter.
10.10	Added <code>human-readable</code> parameter.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface transceiver

```
show interface [<INTERFACE-ID>] transceiver [detail | threshold-violations] [vsx-peer]
```

Description

Displays information about transceivers present in the switch. The information shown varies for different transceiver types and manufacturers. Only basic information is shown for unsupported HPE and third-party transceivers installed in the switch and they are also identified with an asterisk in the output.

Parameter	Description
<INTERFACE-ID>	Specifies the name or range of an interface on the switch. Use the format member/slot/port (for example, 1/3/1).
detail	Show detailed information for the interfaces.
threshold-violations	Show threshold violations for transceivers.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

On the 6400 Switch Series, interface identification differs.

Showing summary transceiver information with identification of unsupported transceivers:

```
switch(config)# show interface transceiver
```

Port Number	Type Number	Product Number	Serial	Part
1/1/1	SFP+SR	J9150A	MYxxxxxxxxx	1990-3657
1/1/2	SFP+ER*	--	--	--
1/2/1	QSFP+SR4	JH233A	MYxxxxxxxxx	2005-1234
1/2/2	QSFP+ER4*	--	--	--

```
1/3/1      SFP28DAC3      844477-B21      MYxxxxxxxxx      77fc-7ce7
* unsupported transceiver
```

Showing detailed transceiver information:

```
switch(conf#) show interface transceiver detailing
Transceiver in 1/1/1
  Interface Name      : 1/1/1
  Type                : SFP+SR
  Connector Type      : LC
  Wavelength          : 850nm
  Transfer Distance   : 0m (SMF), 30m (OM1), 80m (OM2), 300m (OM3)
  Diagnostic Support   : DOM
  Product Number      : J9150A
  Serial Number       : MYxxxxxxxxx
  Part Number         : 1990-3657
```

```
Status
  Temperature : 47.65C
  Voltage     : 3.31V
  Tx Bias     : 8.40mA
  Rx Power    : 0.08mW, -10.96dBm
  Tx Power    : 0.56mW, -2.49dBm
```

```
Recent Alarms :
  Rx power low alarm
  Rx power low warning
Recent Errors :
  Rx loss of signal
```

```
Transceiver in 1/1/2
  Interface Name      : 1/1/2
  Type                : unknown
  Connector Type      : ??
  Wavelength          : ??
  Transfer Distance   : ??
  Diagnostic Support   : ??
  Product Number      : ??
  Serial Number       : ??
  Part Number         : ??
```

```
Transceiver in 1/2/1
  Interface Name      : 1/2/1
  Type                : QSFP+SR4
  Connector Type      : MPO
  Wavelength          : 850nm
  Transfer Distance   : 0m (SMF), 0m (OM1), 0m (OM2), 100m (OM3)
  Diagnostic Support   : DOM
  Product Number      : JH233A
  Serial Number       : MYxxxxxxxxx
  Part Number         : 2005-1234
```

```
Status
  Temperature : 44.46C
  Voltage     : 3.30V
```

```
-----
      Tx Bias  Rx Power      Tx Power
Channel#  (mA)   (mW/dBm)   (mW/dBm)
-----
```

```

1          6.12      0.00, -inf    0.63, -1.95
2          6.04      0.00, -inf    0.63, -2.00
3          6.51      0.00, -inf    0.60, -2.16
4          6.19      0.00, -inf    0.63, -1.94

```

Recent Alarms :

```

Channel 1 :
  Rx power low alarm
  Rx power low warning
Channel 2 :
  Rx power low alarm
  Rx power low warning
Channel 3 :
  Rx power low alarm
  Rx power low warning
Channel 4 :
  Rx power low alarm
  Rx power low warning

```

Recent Errors :

```

Channel 1 :
  Rx Loss of Signal
Channel 2 :
  Rx Loss of Signal
Channel 3 :
  Rx Loss of Signal
Channel 4 :
  Rx Loss of Signal

```

Transceiver in 1/2/2

```

Interface Name      : 1/2/2
Type                : unknown
Connector Type      : ??
Wavelength          : ??
Transfer Distance    : ??
Diagnostic Support   : ??
Product Number      : ??
Serial Number       : ??
Part Number         : ??

```

Transceiver in 1/3/1

```

Interface Name      : 1/3/1
Type                : SFP28DAC3
Connector Type      : Copper Pigtail
Transfer Distance    : 0.00km (SMF), 0m (OM1), 0m (OM2), 0m (OM3)
Diagnostic Support   : None
Product Number      : 844477-B21
Serial Number       : MYxxxxxxx
Part Number         : 77fc-7ce7

```

Showing detailed transceiver information with identification of unsupported transceivers:

Transceiver in 1/1/2

```

Interface Name      : 1/1/2
Type                : SFP+ER (unsupported)
Connector Type      : LC
Wavelength          : 3590nm
Transfer Distance    : 80m (SMF), 0m (OM1), 0m (OM2), 0m (OM3)
Diagnostic Support   : DOM
Vendor Name         : INNOLIGHT
Vendor Part Number   : TR-PX15Z-NHP

```

```
Vendor Part Revision: 1A
Vendor Serial number: MYxxxxxxx
```

```
Status
Temperature : 28.88C
Voltage      : 3.30V
Tx Bias      : 65.53mA
Rx Power     : 0.00mW, -inf
Tx Power     : 1.47mW, 1.67dBm
```

```
Recent Alarms:
Rx Power low alarm
Rx Power low warning
Recent Errors:
```

Showing transceiver threshold-violations:

```
switch(config)# show interface transceiver threshold-violations
-----
Port      Type      Channel  Type(s) of Recent
              Threshold Violation(s)
-----
1/1/1     SFP+SR                      Tx bias high warning
                    50.52 mA > 40.00 mA
1/1/2     SFP+ER*                      ??
1/2/1     QSFP+SR4    1      Tx power low alarm
                    -17.00 dBm < -0.50 dBm
                    2      Tx bias low warning
                    3.12 mA < 4.00 mA
1/2/2     QSFP+ER4*                      ??
1/3/1     SFP28DAC3                      n/a
* unsupported transceiver
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface utilization

```
show interface [<IFNAME>|<IFRANGE>] utilization [non-zero]
```

Description

Displays physical port throughput and utilization.

Parameter	Description
<IFNAME>	Specifies an interface name.
<IFRANGE>	Specifies the port identifier range.
utilization	Displays utilization statistics.
non-zero	Displays non-zero statistics

Examples

The following example shows port utilization of all interfaces:

```
switch# show interface utilization
-----|-----|-----|-----
Interval |-----|
Interface |          RX          |          TX          | Total (RX+TX) |
          seconds | Mbps  KPkt/s  Util % | Mbps  KPkt/s  Util % | Mbps
KPkt/s   Util % | Description
-----|-----|-----|-----
1/1/1      300  9578.02  788.70  95.78   25.70   45.89   0.26  9603.72
834.59    96.04  Aruba-AP
1/1/2      300   25.71   45.90   0.26  9581.09  788.96   95.81  9606.80
834.86    96.07  Aruba2530-AP-conce...
1/1/3 - lag123  300    0.00    0.00   0.00    0.00    0.00    0.00    0.00
0.00     0.00  ISL: SWRTS-0064-1
1/1/4      300  9261.79  804.52  92.62  9496.70  823.97   94.97  18758.50
1628.48   187.58  Backup data center...
1/1/5      300  9496.70  823.97  94.97  9261.79  804.52   92.62  18758.50
1628.48   187.58  --
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ip interface

```
show ip interface <INTERFACE-ID> [vsx-peer]
```

Description

Shows status and configuration information for an IPv4 interface.

Parameter	Description
<INTERFACE-ID>	Specifies the name of an interface. Format: member/slot/port .
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

On the 6400 Switch Series, interface identification differs.

```
switch# show ip interface 1/1/1
Interface 1/1/1 is up
Admin state is up
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
IPv4 address 192.168.1.1/24
MTU 1500
RX
    0 packets, 0 bytes
TX
    0 packets, 0 bytes
```

```
switch# show interface <intfid>.id
Interface 1/1/14.1 is up
Admin state is up
IP MTU 1500
Description:
Hardware: Ethernet, MAC Address: b8:6a:97:22:2f:42
Encapsulation dot1q ID: 20
IPv4 address 30.0.0.1/24
L3 Counters: Rx Disabled, Tx Disable
```

```
switch# show interface lag2.1
Interface lag2.1 is up
Admin state is up
IP MTU 1500
Description:
Hardware: Ethernet, MAC Address: b8:6a:97:22:2f:42
Encapsulation dot1q ID: 30
L3 Counters: Rx Disabled, Tx Disabled
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ip source-interface

```
show ip source-interface {sflow | tftp | radius | tacacs | all} [vrf <VRF-NAME>]
[vsx-peer]
```

Description

Shows single source IP address configuration settings.

Parameter	Description
sflow tftp radius tacacs all	Shows single source IP address configuration settings for a specific protocol. The all option shows the global setting that applies to all protocols that do not have an address set.
vrf <VRF-NAME>	Specifies the name of a VRF.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

On the 6400 Switch Series, interface identification differs.

```
Showing single source IP address configuration settings for sFlow:
switch# show ip source-interface sflow
Source-interface Configuration Information
-----
Protocol      Source Interface
-----
sflow         10.10.10.1
```

Showing single source IP address configuration settings for all protocols:

```
switch# show ip source-interface all
Source-interface Configuration Information
-----
Protocol      Source Interface
-----
all           1/1/1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show ipv6 interface

show ipv6 interface <INTERFACE-ID> [vsx-peer]

Description

Shows status and configuration information for an IPv6 interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface ID. Format: member/slot/port .
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

On the 6400 Switch Series, interface identification differs.

```
switch#
switch# show ipv6 interface 1/1/1
Interface 1/1/1 is up
  Admin state is up
  IPv6 address:
    2001:0db8:85a3:0000:0000:8a2e:0370:7334/24 [VALID]
  IPv6 link-local address: fe80::1e98:ecff:fee3:e800/64 (default) [VALID]
  IPv6 virtual address configured: none
  IPv6 multicast routing: disable
  IPv6 Forwarding feature: enabled
  IPv6 multicast groups locally joined:
    ff02::ff70:7334 ff02::ffe3:e800 ff02::1 ff02::1:ff00:0
    ff02::2
  IPv6 multicast (S,G) entries joined: none
  IPv6 MTU: 1524 (using link MTU)
  IPv6 unicast reverse path forwarding: none
  IPv6 load sharing: none
  RX
    0 packets, 0 bytes
  TX
    0 packets, 0 bytes
```

```
switch# show ipv6 interface <intfid>.id
Interface 1/1/14.1 is up
  Admin state is up
  IPv6 address:
    30::1/64 [VALID]
  IPv6 link-local address: fe80::b86a:97c0:122:2f42/64 [VALID]
```

```

IPv6 virtual address configured: none
IPv6 multicast routing: disable
IPv6 Forwarding feature: enabled
IPv6 multicast groups locally joined:
    ff02::1 ff02::1:ff22:2f42 ff02::1:ff00:1 ff02::1:ff00:0
    ff02::2
IPv6 multicast (S,G) entries joined: none
IPv6 MTU 1500
IPv6 unicast reverse path forwarding: none
IPv6 load sharing: none
Encapsulation dot1q ID: 20

```

```

switch# show ipv6 interface lag2.1
Interface lag2.1 is up
Admin state is up
IPv6 address:
    40::1/64 [VALID]
IPv6 link-local address: fe80::b86a:97c0:122:2f42/64 [VALID]
IPv6 virtual address configured: none
IPv6 multicast routing: disable
IPv6 Forwarding feature: enabled
IPv6 multicast groups locally joined:
    ff02::1 ff02::1:ff22:2f42 ff02::1:ff00:1 ff02::1:ff00:0
    ff02::2
IPv6 multicast (S,G) entries joined: none
IPv6 MTU 1500
IPv6 unicast reverse path forwarding: none
IPv6 load sharing: none
Encapsulation dot1q ID: 30

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ipv6 source-interface

```

show ipv6 source-interface {sflow | tftp | radius | tacacs | all} [vrf <VRF-NAME>]
[vsx-peer]

```

Description

Shows single source IP address configuration settings.

Parameter	Description
sflow tftp radius tacacs all	Shows single source IP address configuration settings for a specific protocol. The all option shows the global setting that applies to all protocols that do not have an address set.
vrf <VRF-NAME>	Specifies the name of a VRF.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

On the 6400 Switch Series, interface identification differs.

Showing single source IP address configuration settings for sFlow:

```
switch# show ipv6 source-interface sflow

Source-interface Configuration Information
-----
Protocol      Source Interface
-----
sflow         2001:DB8::1
```

Showing single source IP address configuration settings for all protocols:

```
switch# show ipv6 source-interface all

Source-interface Configuration Information
-----
Protocol      Source Interface
-----
all           1/1/1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

shutdown

shutdown
no shutdown

Description

Disables an interface. Interfaces are disabled by default when created.
The **no** form of this command enables an interface.

Examples

Disabling an interface:

```
switch(config-if) # shutdown
```

Enabling an interface:

```
switch(config-if) # no shutdown
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

speed

```
speed {<SPEED> | <SPEED-DUPLEX> | auto [<SPEED>] }  
no speed
```

Description

Configures the link speed, duplex, and auto-negotiation settings for an interface.
The no form of this command removes the configurations and returns to the defaults.

Parameter	Description
Speed	Configures interface speed, duplex, and auto-negotiation.
10-full	10 Mbps, full duplex, no auto-negotiation
10-half	10 Mbps, half duplex, no auto-negotiation
100-full	100 Mbps, full duplex, no auto-negotiation
100-half	100 Mbps, half duplex, no auto-negotiation
1000-full	1000 Mbps, full duplex, no auto-negotiation
10g	10 Gbps, full duplex, no auto-negotiation

Parameter	Description
25g	25 Gbps, full duplex, no auto-negotiation
40g	40 Gbps, full duplex, no auto-negotiation
50g	50 Gbps, full duplex, no auto-negotiation
100g	100 Gbps, full duplex, no auto-negotiation
200g	200 Gbps, full duplex, no auto-negotiation
	NOTE: Not applicable for override.
400g	400 Gbps, full duplex, no auto-negotiation
	NOTE: Not applicable for override.
auto	Auto-negotiate speed and duplex. More than one speed can be set at a time.
10m	Allow interface to link at 10 Mbps.
100m	Allow interface to link at 100 Mbps.
1g	Allow interface to link at 1 Gbps.
2.5g	Allow interface to link at 2.5 Gbps.
5g	Allow interface to link at 5 Gbps.
10g	Allow interface to link at 10 Gbps.
25g	Allow interface to link at 25 Gbps.
40g	Allow interface to link at 40 Gbps.
50g	Allow interface to link at 50 Gbps.
100g	Allow interface to link at 100 Gbps.
200g	Allow interface to link at 200 Gbps.
400g	Allow interface to link at 400 Gbps.

Usage

The following options can be configured for an interface. The option available is based on the interface type.

`speed <SPEED-DUPLEX>`

Uses a fixed speed and duplex mode with no auto-negotiation. Half-duplex is only supported for 10 Mbps and 100 Mbps link speeds.

`speed <SPEED>`

Uses a fixed speed with no auto-negotiation. If the currently installed transceiver does not support the speed, the setting is ignored and the port will use the highest speed that is supported.

`speed auto`

Uses auto-negotiation and offers all speeds supported by the port and transceiver. This is the default. If the link technology does not support auto-negotiation this setting is ignored, and the port uses the highest possible fixed speed.

`speed auto <SPEED>`

Uses auto-negotiation and offers the specified speeds only. For ports that support pluggable transceivers, only speeds supported by the transceiver are offered and other speeds are ignored. If the link technology does not support auto-negotiation, this setting is ignored and the port uses the highest possible fixed speed.

Examples

Configuring an interface to operate at a fixed speed of 1000 Mbps with full duplex and no auto-negotiation:

```
switch(config)# interface 1/1/1
switch(config-if)# speed 1000-full
```

Configuring an interface to operate at a fixed speed of 10 Gbps with no auto-negotiation:

```
switch(config)# interface 1/1
switch(config-if)# speed 10g
```

Configuring an interface to auto-negotiate and advertise only 1 Gbps and 2.5 Gbps speeds:

```
switch(config)# interface 1/1/1
switch(config-if)# speed auto 1g 2.5g
```

Configuring an interface to override the detected transceiver speed and use the configured speed if the installed transceiver does not support auto-negotiation:

```
switch(config)# interface 1/1/1
switch(config-if)# speed auto 50g override
```

Configuring an interface to use default settings for speed, duplex, and auto-negotiation:

```
switch(config)# interface 1/1/1
switch(config-if)# no speed
```

Command History

Release	Modification
10.09.0001	Speeds not supported by hardware hidden by CLI.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

system interface-group

This command only applies to the following 6400 Switch Series modules:

R0X44A Aruba 6400 48p 10G/25G SFP28 Module

R0X44C Aruba 6400 48p 1G/10G/25G SFP28 v2 Module

```
system interface-group <GROUP> line-module <SLOT-ID> speed <SPEED>
no system interface-group <GROUP> line-module <SLOT-ID> speed <SPEED>
system interface group member <MEMBER-ID>
no system interface group member <MEMBER-ID>
```

Description

Configures the speed for an interface group. After changing group speed, only transceivers compatible with the new speed will be enabled.

- (Applies to the 6400 Switch Series): R0X44C (version 2) is the only module that can apply 50G speed as an option. If the command is attempted to any other type of module, the command is ignored.
- All speed-mismatched interfaces in the group will be disabled.
- This command can interrupt active network links, user confirmation is required to proceed.

The **no** form of this command resets the specified interface group to its default.

Parameter	Description
<GROUP>	Specifies the interface group to configure.
<SPEED>	Configures transceiver speed (10g, 25g or 50g) for a group. Default is 25g (see the Transceiver Guide for further detail). On 6400 Switch Series: 25g allows transceivers up to 25Gbps. 50g allows 50Gbps transceivers and DACs on the R0X44C version 2 module. This command is ignored on any other type of module (including R0X44A version 1).
<SLOT-ID>	Specifies the slot ID of the line module.
member <MEMBER-ID>	Specifies the VSF member ID of the VSF member of the group. (For 6300 Switch Series only.)

Examples

Configuring interface group 1 on line-module 1/1 to allow 10Gbps and slower transceivers:


```
switch(config)# system interface-group 1 line-module 1/1 speed 10g  
Changing the group speed will disable all member interfaces that do not match the  
new speed.  
  
Continue (y/n)? y
```

Command History

Release	Modification
10.11	Command introduced on 6300 Switch series.
10.09.0002	Command introduced on 6400 and 8400 Switch series.

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

A subinterface is a virtual interface created by dividing one physical interface into multiple logical interfaces. Subinterfaces use the parent physical interface for sending and receiving data.

Supported features

The following features are supported on L3 subinterfaces:

- RoP, L3 LAG and Hydra interface (split cable) support
- IPv4/IPv6 addressing
- ARP/ND
- Static unicast routing (IPv4/IPv6)
- Unicast routing (IPv4/IPv6) - OSPF and VRRP
- Unicast routing (IPv4/IPv6) - BGP and IVRL
- IPv4/IPv6 multicast routing (IGMP and PIM)
- Ingress ACLs and Ingress Policies
- MTU (maximum transmission unit)
- L3 counters
- VSX keep alive links
- SNMP read
- REST support

Configuring subinterfaces



- Subinterfaces can be configured for physical ports, split children of physical ports and L3 LAG interfaces.
- Subinterfaces on multiple ports can be assigned the same VLAN ID (there is no bridging between subinterfaces or between subinterfaces and SVIs) using `encapsulation vlan-id` `encapsulation vlan-id`. Each subinterface is considered to be in a separate bridge domain.
- The IP-MTU of a subinterface can be configured independent of the parent port's IP-MTU configuration.

Procedure

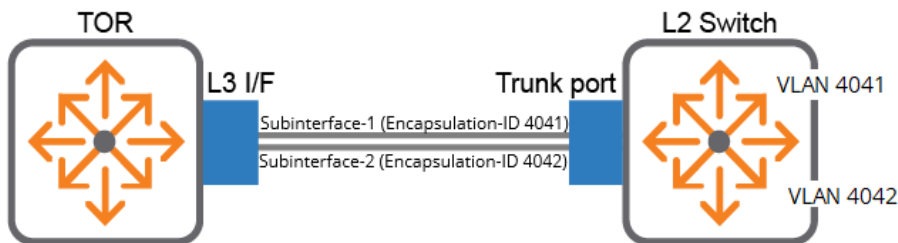
One router with one physical interface needs to be connected to two IP networks:

1. Create two subinterfaces within the physical interface.
2. Assign each subinterface an IP address within each subnet.
3. Route packets between the two subnets.

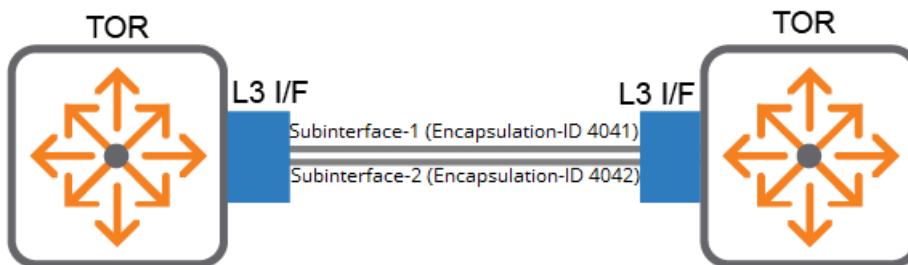
Limitations

- Subinterfaces can only be configured on L3 ports with routing enabled.
- A subinterface cannot be a member of a LAG.
- An L3 parent interface with subinterfaces cannot be a member of a LAG.
- An L3 parent interface with subinterfaces cannot be used for L3 services (for example IP address configuration is not supported on an L3 interface if the interface is configured with subinterfaces).
- On physical interfaces, each subinterface must have a unique encapsulation ID.
- BFD and IPDB are not supported on subinterfaces.

Subinterface in a router-on-a-stick deployment



- Top-of-rack switch/router with an L3 interface connected to the trunk port of an L2 switch.
- Routing tables configured to forward outgoing traffic through a subinterface while applying a VLAN ID tag.
- All outgoing traffic from the L3 interface is tagged with a VLAN ID which enables the switch to forward traffic through different VLANs.



- Top-of-rack switch/router with an L3 interface connected to the trunk port of an L3 interface.
- Routing tables configured to forward outgoing traffic through a subinterface while applying a VLAN ID tag.
- All outgoing traffic from the L3 interface is tagged with a VLAN ID which enables the switch to forward traffic through different VLANs.

Subinterface commands

encapsulation dot1q

```
encapsulation dot1q <VLAN-ID>
no encapsulation dot1q <VLAN-ID>
```

Description

Configures 802.1Q encapsulation on a subinterface.

The **no** form of this command removes 802.1Q encapsulation on a subinterface.

Parameter	Description
<VLAN-ID>	Specifies encapsulation VLAN ID. Range 1 to 4094. NOTE: The encapsulation VLAN ID should be unique within an L3 LAG subinterface. The same encapsulation VLAN ID can be configured among different parent interfaces, but the encapsulation VLAN ID should not be configured in the internal VLAN range. (Encapsulation VLAN IDs and static VLANs are entirely different and do not coincide.)

Usage

Associates an 802.1Q VLAN ID with a subinterface.

Examples

Configuring 802.1Q encapsulation on a subinterface:

```
switch(config)# interface 1/1/1.201
switch(config-subif)# encapsulation dot1q 10
```

Removing 802.1Q encapsulation on a subinterface:

```
switch(config-subif)# no encapsulation dot1q 10
```

Command History

Release	Modification
10.08	Command introduced for the 6300, 6400switch series.

Command Information

Platforms	Command context	Authority
6300 6400	config-subif	Administrators or local user group members with execution rights for this command.

interface

```
interface <IFNAME>.<ID>
no interface <IFNAME>.<ID>
```

```
interface lag <LAGNUM>.<ID>
no interface lag <LAGNUM>.<ID>
```

Description

Creates a subinterface on an L3 interface and enters subinterface configuration mode. The subinterface name consists of the parent interface name (for example, 1/1/1) followed by a period and a unique ID number.

The **no** form of these commands deletes a subinterface from an L3 interface.

Parameter	Description
<IFNAME>	Specifies L3 interface name.
<ID>	Specifies subinterface ID. Range 1 to 4094.
<LAGNUM>	Specifies L3 LAG interface number.

Usage

To create a LAG subinterface, the parent LAG must exist before creating the subinterface.

Examples

Creating a subinterface on L3 interface 1/1/1.201 and entering subinterface configuration mode:

```
switch(config)# interface 1/1/1.201
switch(config-subif)#
```

Deleting subinterface on L3 interface 1/1/1.201:

```
switch(config)# no interface 1/1/1.201
```

Creating a subinterface on an L3 LAG port and entering subinterface configuration mode:

```
switch(config)# interface lag 1
switch(config-if)# interface lag 1.201
switch(config-subif)#
```

Deleting subinterface on an L3 LAG port :

```
switch(config)# no interface lag 1.201
```

Command History

Release	Modification
10.08	Command introduced for the 6300, 6400 switch series.

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

show capacities subinterface

show capacities subinterface

Description

Displays maximum subinterface capacity.

Examples

Showing maximum subinterface capacity:

```
switch# show capacities subinterface

System Capacities: Filter Subinterface
Capacities Name
Value
-----
-
Maximum number of LAG subinterfaces for the entire system
256
Maximum number of LAG members when the LAG has subinterfaces
4
Maximum number of normal subinterfaces for the entire system
1024
Maximum number of subinterface resources for the entire system (normal+(4*LAG)
1024
```

Command History

Release	Modification
10.08	Command introduced for the 6300, 6400 switch series.

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

show interface

```
show interface <IFNAME>.<ID>
show interface lag <LAGNUM>.<ID>
```

Description

Displays a subinterface configuration.

Parameter	Description
<IFNAME>	Specifies L3 interface name.
<ID>	Specifies subinterface ID.
<LAGNUM>	Specifies L3 LAG interface number.

Examples

Showing subinterface configuration:

```
switch# show interface 1/1/1.201
Interface 1/1/1.201 is down
Admin state is up
State information: Waiting for link
Description:
Hardware: Ethernet, MAC Address: 38:21:c7:5a:80:80
Encapsulation dot1Q ID: 10
Statistic                RX                TX                Total
-----
L3 Packets                0                0                0
L3 Bytes                  0                0                0
```

Showing subinterface LAG configuration:

```
switch# show interface lag1.1
Interface lag1.1 is down
Admin state is up
Description:
Hardware: Ethernet, MAC Address: 38:21:c7:5a:80:80
Encapsulation dot1Q ID: 2
Statistic                RX                TX                Total
-----
L3 Packets                0                0                0
L3 Bytes                  0                0                0
```

Command History

Release	Modification
10.08	Command introduced for the 6300, 6400 switch series.

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

The source IP address is determined by the system and is typically the IP address of the outgoing interface in the routing table. However, routing switches may have multiple routing interfaces and outgoing packets can potentially be sent by different paths at different times. This results in different source IP addresses.

AOS-CX provides a configuration model that allows the selection of an IP address to use as the source address for all outgoing traffic. This allows unique identification of the software application on the server site regardless of which local interface has been used to reach the destination server. The source interface selection supports selecting an IP address or interface name.



If the source interface and source IP are configured, Source IP will have priority.

Source-interface selection commands

ip source-interface (protocol <ip-addr>)

```
ip source-interface <PROTOCOL> <IP-ADDR> [vrf <VRF-NAME>]
no ip source-interface <PROTOCOL> <IP-ADDR> [vrf <VRF-NAME>]
```

Description

Configures the source-interface IPv4 address to use for the specified protocol. If a VRF is not given, the default VRF applies. If no interface option is given, the device floods through interfaces and VRFs to reach Aruba Central. Whichever reaches Aruba Central will be picked automatically.

The `no` form of this command removes all configurations.

Parameter	Description
<PROTOCOL>	Specifies the protocol to configure. <code>all</code> Selects all protocols that can be configured by this command. <code>central</code> Selects Aruba Central. <code>dhcp_relay</code> Selects DHCP relay. <code>dns</code> Selects DNS. <code>ntp</code> Selects NTP. <code>radius</code> Selects radius. <code>sflow</code>

Parameter	Description
	Selects sFlow. sFlow
	Selects simplivity. sFlow
	Selects syslog. syslog
	Selects TACACS. tacacs
	Selects TFTP. tftp
<IP-ADDR>	Specifies the IPv4 address.
vrf <VRF-NAME>	Specifies the VRF name.

Examples

Configuring source-interface IPv4 10.1.1.1 to use for the TFTP protocol:

```
switch(config)# ip source-interface tftp 10.1.1.1
```

Configuring source-interface IPv4 10.1.1.2 to use for the TFTP protocol on VRF green :

```
switch(config)# ip source-interface tftp 10.1.1.2 vrf green
```

Removing source-interface IPv4 10.1.1.1 configuration for the TFTP protocol:

```
switch(config)# no ip source-interface tftp 10.1.1.1
```

Removing source-interface IPv4 10.1.1.2 configuration for TFTP protocol on VRF green:

```
switch(config)# no ip source-interface tftp 10.1.1.2 vrf green
```

Configuring source-interface IPv4 10.1.1.1 to use for the DNS protocol:

```
switch(config)# ip source-interface dns 10.1.1.1
```

Configuring source-interface IPv4 10.1.1.2 to use for the DNS protocol on VRF green :

```
switch(config)# ip source-interface dns 10.1.1.2 vrf green
```

Removing source-interface IPv4 10.1.1.1 configuration for the DNS protocol:

```
switch(config)# no ip source-interface dns 10.1.1.1
```

Removing source-interface IPv4 10.1.1.2 configuration for the DNS protocol on VRF green:

```
switch(config)# no ip source-interface dns 10.1.1.2 vrf green
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip source-interface

```
ip source-interface <PROTOCOL> {<IP-ADDR>|interface <IFNAME>} [vrf <VRF-NAME>]  
no ip source-interface <PROTOCOL> interface <IFNAME> [vrf <VRF-NAME>]
```

Description

Configures the IPv4 source-interface interface to use for the specified protocol. If a VRF is not given, the default VRF applies.

The **no** form of this command removes the specified configuration.

Parameter	Description
<PROTOCOL>	Specifies the protocol to configure. all Selects the source for all protocols covered by this command. central Selects Aruba Central. dhcp_relay Selects DHCP relay. When you configure a dhcp_relay source interface, you must also enable DHCP relay Option 82 using the dhcp-relay option 82 source-interface command. dns Selects DNS. http Selects HTTP. ntp Selects NTP. ptp Selects PTP. radius Selects RADIUS. sflow

Parameter	Description
	Selects sFlow. sftp-scp
	Selects SFTP and SCP. ssh-client
	Selects SSH Client. syslog
	Selects the source for syslog packets. tacacs
	Selects the source for TACACS packets. tftp
	Selects TFTP. ubt
	Selects UBT.
	Specifies the VRF name.
<IFNAME>	Specifies the interface name.
<IP-ADDR>	Specifies the IPv4 address.
vrf <VRF-NAME>	Specifies the VRF name.

Examples

Configuring IPv4 source-interface interface 1/1/1 to use for the TFTP protocol:

```
switch(config)# ip source-interface tftp interface 1/1/1
```

Configuring IPv4 source-interface interface 1/1/2 to use for the TFTP protocol on VRF green :

```
switch(config)# ip source-interface tftp interface 1/1/2 vrf green
```

Removing IPv4 source-interface 1/1/1 configuration for the TFTP protocol:

```
switch(config)# no ip source-interface tftp interface 1/1/1
```

Removing source-interface interface 1/1/2 configuration for TFTP protocol on VRF green:

```
switch(config)# no ip source-interface tftp interface 1/1/2 vrf green
```

Configuring source-interface IPv4 10.1.1.1 to use for the TFTP protocol:

```
switch(config)# ip source-interface tftp 10.1.1.1
```

Configuring source-interface IPv4 10.1.1.2 to use for the TFTP protocol on VRF green :

```
switch(config)# ip source-interface tftp 10.1.1.2 vrf green
```

Removing source-interface IPv4 10.1.1.1 configuration for the TFTP protocol:

```
switch(config)# no ip source-interface tftp 10.1.1.1
```

Removing source-interface IPv4 10.1.1.2 configuration for TFTP protocol on VRF green:

```
switch(config)# no ip source-interface tftp 10.1.1.2 vrf green
```

Configuring source-interface IPv4 10.1.1.1 to use for the DNS protocol:

```
switch(config)# ip source-interface dns 10.1.1.1
```

Configuring source-interface IPv4 10.1.1.2 to use for the DNS protocol on VRF green :

```
switch(config)# ip source-interface dns 10.1.1.2 vrf green
```

Removing source-interface IPv4 10.1.1.1 configuration for the DNS protocol:

```
switch(config)# no ip source-interface tftp 10.1.1.1
```

Removing source-interface IPv4 10.1.1.2 configuration for the DNS protocol on VRF green:

```
switch(config)# no ip source-interface dns 10.1.1.2 vrf green
```

Command History

Release	Modification
10.12.1000	Added <code>cental</code> , <code>sftp-scp</code> , and <code>ssh-client</code> parameters.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

ipv6 source-interface

```
ipv6 source-interface <PROTOCOL> <IPV6-ADDR> [vrf <VRF-NAME>]  
no source-interface <PROTOCOL> <IPV6-ADDR> [vrf <VRF-NAME>]
```

Description

Configures the source-interface IPv6 address to use for the specified protocol. If a VRF is not given, the default VRF applies.

The `no` form of this command removes the specified protocol configuration.

Parameter	Description
<code><PROTOCOL></code>	Specifies the protocol to configure. ▪ <code>all</code>:Selects all protocols supported by this command. ▪ <code>central</code>:Selects Aruba Central. ▪ <code>ntp</code>:Selects NTP. ▪ <code>radius</code>:Selects radius. ▪ <code>sflow</code>:Selects sFlow. ▪ <code>syslog</code>:Selects syslog. ▪ <code>tacacs</code>:Selects TACACS. ▪ <code>tftp</code>:Selects TFTP.
<code><IPv6-ADDR></code>	Specifies the IPv6 address.
<code>vrf <VRF-NAME></code>	Specifies the VRF name.

Examples

Configuring source-interface IPv6 1111:2222 to use for the TFTP protocol:

```
switch(config)# ipv6 source-interface tftp 1111:2222
```

Configuring source-interface IPv6 1111:3333 to use for TFTP protocol on VRF green :

```
switch(config)# ipv6 source-interface tftp 1111:3333 vrf green
```

Removing source-interface IPv6 1111:2222 configuration for TFTP protocol:

```
switch(config)# no ipv6 source-interface tftp 1111:2222
```

Removing source-interface IPv6 1111:3333 configuration for TFTP protocol on VRF green:

```
switch(config)# no ipv6 source-interface tftp 1111:3333 vrf green
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ipv6 source-interface dns

```
ipv6 source-interface {dns | all} {interface | X:X::X:X} [vrf <VRF-NAME>]
[no] ipv6 source-interface {dns | all} {interface | X:X::X:X} [vrf <VRF-NAME>]
```

Description

Configures the IPv6 source-interface or source IP for IPv6 DNS clients.

The **no** form of this command removes all configurations.

Parameter	Description
<PROTOCOL>	Specifies the protocol to configure. all Selects all protocols supported by this command. central Selects Aruba Central. dhcp_relay Selects DHCP relay. dns Selects DNS packet source. http Selects HTTP. ntp Selects NTP. radius Selects radius. sftp-scp Selects SFTP and SCP. sflow Selects sFlow. ssh-client Selects SSH Client. syslog Selects syslog. tacacs Selects TACACS. tftp SelectsTFTP. ubt SelectsUBT.
<IPv6-ADDR>	Specifies the IPv6 address.
vrf <VRF-NAME>	Specifies the VRF name.

Examples

Configuring IPv6 source-interface dns :

```
switch(config)# ipv6 source-interface
all           All protocols
central       Aruba Central protocol
dhcp_relay    DHCP_RELAY protocol
dns           DNS protocol
http          HTTP protocol
ntp           NTP protocol
radius        RADIUS protocol
sflow         sFlow protocol
sftp-scp      SFTP and SCP protocols
ssh-client    SSH Client protocol
syslog        syslog protocol
tacacs        TACACS protocol
tftp          TFTP protocol
```

Configuring IPv6 source -interface dns:

```
switch(config)# ipv6 source-interface dns
X:X::X:X      Specify an IPv6 address
interface     Interface information
```

Configuring IPv6 source-interface dns on 1::1:

```
switch(config)# ipv6 source-interface dns 1::1
vrf           VRF Configuration
<cr>
```

Configuring IPv6 source-interface dns on 1::1: vrf:

```
switch(config)# ipv6 source-interface dns 1::1 vrf
VRF_NAME      VRF name
```

Configuring IPv6 source-interface dns on 1::1 vrf BLUE

```
switch(config)# ipv6 source-interface dns 1::1 vrf BLUE
switch(config)# ipv6 source-interface dns interface vlan10
vrf           VRF Configuration
<cr>
```

Configuring IPv6 source-interface dns on vlan10 vrf BLUE:

```
switch(config)# ipv6 source-interface dns interface vlan10 vrf BLUE
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ipv6 source-interface

```
ipv6 source-interface <PROTOCOL> {<IPV6-ADDR>|interface <IFNAME>} [vrf <VRF-NAME>]
no ipv6 source-interface <PROTOCOL> {<IPV6-ADDR>|interface <IFNAME>} [vrf <VRF-NAME>]
```

Description

Configures the IPv6 source-interface interface to use for the specified protocol. If a VRF is not given, the default VRF applies.

The **no** form of this command removes all configurations.

Parameter	Description
<PROTOCOL>	Specifies the protocol to configure. all Selects all protocols supported by this command. central Selects Aruba Central. dhcp_relay Selects DHCP relay. dns Selects DNS packets http Selects HTTP. ntp Selects NTP. radius Selects radius. sftp-scp Selects SFTP and SCP. sflow Selects sFlow. ssh-client Selects SSH Client. syslog Selects syslog. tacacs Selects TACACS. tftp SelectsTFTP.

Parameter	Description
<IPv6-ADDR>	Specifies the IPv6 address.
<IFNAME>	Specifies the interface name.
vrf <VRF-NAME>	Specifies the VRF name.

Examples

Configuring IPv6 source-interface interface 1/1/1 to use for the TFTP protocol :

```
switch(config)# ipv6 source-interface tftp interface 1/1/1
```

Configuring IPv6 source-interface interface 1/1/2 to use for the TFTP protocol on VRF green :

```
switch(config)# ipv6 source-interface tftp interface 1/1/2 vrf green
```

Removing IPv6 source-interface interface 1/1/1 configuration for the TFTP protocol:

```
switch(config)# no ipv6 source-interface tftp interface 1/1/1
```

Removing IPv6 source-interface interface 1/1/2 configuration for the TFTP protocol on VRF green:

```
switch(config)# no ipv6 source-interface tftp interface 1/1/2 vrf green
```

Configuring source-interface IPv6 1111:2222 to use for the TFTP protocol:

```
switch(config)# ipv6 source-interface tftp 1111:2222
```

Configuring source-interface IPv6 1111:3333 to use for TFTP protocol on VRF green :

```
switch(config)# ipv6 source-interface tftp 1111:3333 vrf green
```

Removing source-interface IPv6 1111:2222 configuration for TFTP protocol:

```
switch(config)# no ipv6 source-interface tftp 1111:2222
```

Removing source-interface IPv6 1111:3333 configuration for TFTP protocol on VRF green:

```
switch(config)# no ipv6 source-interface tftp 1111:3333 vrf green
```

Command History

Release	Modification
10.13.0001	Added the dns protocol parameter.
10.12.1000	Added central , sftp-scp , dhcp_relay and ssh-client parameters.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show ip source-interface

```
show ip source-interface <PROTOCOL> [vrf <VRF-NAME> | all-vrfs]
```

Description

Displays the source interface information for all VRFs or a specific VRF.

If a VRF is not specified, the default is displayed.

Parameter	Description
<PROTOCOL>	Specifies the protocol to show.
all	Shows the source interface configuration for all other protocols.
central	Shows the source interface configuration for Aruba Central.
dhcp relay	Shows the source interface configuration for DHCP relay.
dns	Shows the source interface configuration for DNS.
http	Shows the source interface configuration for HTTP.
ntp	Shows the source interface configuration for NTP.
ptp	Shows the source interface configuration for PTP.
radius	Shows the source interface configuration for radius.
sflow	Shows the source interface configuration for sFlow.
sftp-scp	Shows source interface configuration for SFTP and SCP.
ssh-client	

Parameter	Description
	Shows source interface configuration for SSH Client. syslog
	Shows the source interface configuration for syslog. tacacs
	Shows the source interface configuration for TACACS. tftp
	Shows the source interface configuration for TFTP. ubt
	Shows the source interface configuration for PTP.
<code>vrf <VRF-NAME></code>	Specifies the VRF name.
<code>all-vrfs</code>	Shows the source interface configuration for all VRFs.

Examples

Displaying all source-interface protocol configurations for VRF red:

```
switch# show ip source-interface all vrf red
Source-interface Configuration Information
-----
Protocol      Src-Interface      Src-IP      VRF
-----
all           1/1/1              red
switch#
```

Displaying all source-interface protocol configurations for default VRF:

```
switch# show ip source-interface all
Source-interface Configuration Information
-----
Protocol      Src-Interface      Src-IP      VRF
-----
all           1.1.1.1            default
switch#
```

Displaying all source-interface protocol configurations for all VRFs:

```
switch# show ip source-interface all all-vrfs
Source-interface Configuration Information
-----
Protocol      Src-Interface      Src-IP      VRF
-----
all           2.2.2.2            all-vrfs
all           1.1.1.1            default
all           1/1/1/1            red
switch#
```

Command History

Release	Modification
10.12.1000	Added <code>central</code> , <code>sftp-scp</code> , and <code>ssh-client</code> parameters.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show ipv6 source-interface

```
show ipv6 source-interface <PROTOCOL> [detail] [vrf <VRF-NAME> | all-vrfs]
```

Description

Displays the IPV6 source interface information configured in the router for all VRFs or a specific VRF. If a VRF is not specified, the default is displayed.

Parameter	Description
<code><PROTOCOL></code>	Specifies the protocol to show. <code>all</code> Shows the source interface configuration for all other protocols. <code>central</code> Shows the source interface configuration for Aruba Central. <code>dhcp_relay</code> Shows the source interface configuration for DHCP relay. <code>dns</code> Shows the source interface configuration for DNS. <code>http</code> Shows the source interface configuration for HTTP. <code>ntp</code> Shows the source interface configuration for NTP. <code>radius</code> Shows the source interface configuration for radius. <code>sflow</code> Shows the source interface configuration for sFlow. <code>sftp-scp</code> Shows source interface configuration for SFTP and SCP. <code>ssh-client</code> Shows source interface configuration for SSH Client. <code>syslog</code> Shows the source interface configuration for syslog. <code>tacacs</code>

Parameter	Description
	Shows the source interface configuration for TACACS. tftp
	Shows the source interface configuration for TFTP.
vrf <VRF-NAME>	Specifies the VRF name.
all-vrfs	Shows the source interface configuration for all VRF.

Examples

Displaying all IPv6 source-interface protocol configurations for default VRF:

```
switch# show ipv6 source-interface all
Source-interface Configuration Information
-----
Protocol      Src-Interface      Src-IP              VRF
-----
all           1111::2222         default
switch#
```

Displaying all IPv6 source-interface protocol configuration for VRF red:

```
switch# show ipv6 source-interface all vrf red
Source-interface Configuration Information
-----
Protocol      Src-Interface      Src-IP              VRF
-----
all           1/1/1             2005::2             red
switch#
```

Displaying all IPv6 source-interface protocol configurations for all VRFs:

```
switch# show ipv6 source-interface all all-vrfs
Source-interface Configuration Information
-----
Protocol      Src-Interface      Src-IP              VRF
-----
all           2222::3333        all-vrfs
all           1111::2222        default
all           1/1/1             2::2               red
switch#
```

Displaying all IPv6 source-interface protocol configurations for dns all VRFs:

```
switch# show ipv6 source-interface dns all-vrfs
Source-interface Configuration Information
-----
--
Protocol      Src-Interface      Src-IP              VRF
-----
--
dns           1::3              blue
dns           1::4              default
dns           1::2              red
switch#
```

Command History

Release	Modification
10.13	Added <code>dns</code> parameters.
10.12.1000	Added <code>central</code> , <code>sftp-scp</code> , and <code>ssh-client</code> parameters.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config

`show running-config`

Description

Displays the current running configuration.

Examples

Displaying the running configuration (only items of interest to source interface selection are shown in this example output command):



Aruba Central is the priority agent. If no command is specified for `ip source-interface`, Central will choose the command automatically if it is reachable on any of the known ports.

```
switch# show running-config
vrf green
ip source-interface tftp interface 1/1/2 vrf green
ip source-interface radius interface 1/1/2 vrf green
ip source-interface ntp interface 1/1/2 vrf green
ip source-interface tacacs interface 1/1/2 vrf green
ip source-interface dns interface 1/1/2 vrf green
ip source-interface central interface 1/1/2 vrf green
ip source-interface all interface 1/1/2 vrf green
ipv6 source-interface tftp 2222::3333 vrf green
ipv6 source-interface radius 2222::3333 vrf green
ipv6 source-interface ntp 2222::3333 vrf green
ipv6 source-interface tacacs 2222::3333 vrf green
ipv6 source-interface central 2222::3333 vrf green
ipv6 source-interface all 2222::3333 vrf green
ip source-interface tftp 10.20.3.1
ip source-interface radius 10.20.3.1
ip source-interface ntp 10.20.3.1
ip source-interface tacacs 10.20.3.1
ip source-interface dns 10.20.3.1
ip source-interface central 10.20.3.1
ip source-interface all 10.20.3.1
interface 1/1/1
```

```
no shutdown
ip address 10.20.3.1/24
interface 1/1/2
vrf attach green
ip address 20.1.1.1/24
ipv6 address 2222::3333/64
interface 1/1/45
no shutdown
ip address 100.1.0.1/24
ipv6 address 1111::2222/64
ip route 100.2.0.0/24 10.20.3.2
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

VLANs are primarily used to provide network segmentation at layer 2. VLANs enable the grouping of users by logical function instead of physical location. They make managing bandwidth usage within networks possible by:

- Allowing grouping of high-bandwidth users on low-traffic segments
- Organizing users from different LAN segments according to their need for common resources and individual protocols
- Improving traffic control at the edge of networks by separating traffic of different protocol types.
- Enhancing network security by creating subnets to control in-band access to specific network resources

VLANs are generally assigned on an organizational basis rather than on a physical basis. For example, a network administrator could assign all workstations and servers used by a particular workgroup to the same VLAN, regardless of their physical locations.

Hosts in the same VLAN can directly communicate with one another. A router or a Layer 3 switch is required for hosts in different VLANs to communicate with one another.

VLANs help reduce bandwidth waste, improve LAN security, and enable network administrators to address issues such as scalability and network management.



Refer to the Layer 2 Bridging Guide for VLAN configuration and commands.

Precision Time Protocol (PTP)



PTP is supported only on the Aruba 6300 Switch Series.



If PTP is configured on MACsec-supported ports, the switch can experience higher jitter or lower PTP accuracy.

Precision Time Protocol (PTP) is defined in the IEEE 1588 standard (Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems). PTP synchronizes clocks in packet-based networks that include distributed device clocks of varying precision and stability. On a local area network, it achieves clock accuracy in the sub-microsecond range, making it suitable for measurement and control systems. PTP is currently employed to synchronize financial transactions, mobile phone tower transmissions, and networks that require precise timing but lack access to satellite navigation signals.

PTP is not supported on EVPN/VXLAN tunnels.

PTP clocks

A PTP network consists of PTP-enabled devices and devices that are not using PTP. The PTP-enabled devices typically consist of clock-aware devices such as ordinary clocks (which are usually single-port end-stations) and one or more grandsource clocks, transparent clocks, and boundary clocks (multi-port L2/L3 time-aware devices).

The basic clock node types include:

Grandsource clock—A grandsource clock is the primary source of time for the downstream devices. This is a device with greater clock quality which may have direct access to a reference clock.

Ordinary clock—An ordinary clock is a single port end-station (which can include a GM as the originating PTP time-aware device).

Transparent clock—A transparent clock can have multiple network port connections but it does not act as either a clock-source or a clock-sink. Rather, it updates the correction field within the PTP event messages (`SYNC/FOLLOW_UP`, `DELAY_REQUEST`) to compensate for the transit time delay. Transparent clocks compensate for switch latency and jitter, making network devices appear transparent to other PTP time-aware devices. They help in reducing the end-station time errors and improving synchronization quality. But a transparent clock itself does not synchronize its time.

There are two types of transparent clocks:

- **End-to-End (E2E) transparent clock**—Updates the `correctionField` field in the PTP messages with the total time the PTP packet was resident in the network device. This is called resident time correction.
- **Peer-to-Peer (P2P) transparent clock**—Updates the `correctionField` field in the PTP messages with the sum of upstream link delay and the resident time. The upstream link delay is the estimated packet propagation delay between the upstream neighbor P2P clock and the local transparent clock.

Boundary clock—A boundary clock implements a local PTP clock where one port acts as clock-sink which synchronizes itself with the clock-source while other ports act as clock-source ports to its

downstream clock-aware devices. The clock-source port is used to redistribute the clock to another set of clock-sinks. Boundary clocks can also use E2E or P2P delay-mechanism, and can be configured based on the selected PTP profile. The best clock source algorithm is used by the boundary clock to select the best or most precise configured acceptable clock-source clock.

Based on hardware capability, the switch supports either boundary clock, transparent clock, or both modes.

Best clock-source algorithm

The best clock-source algorithm helps in choosing the source of timing on your network. It runs independently on each clock in a domain. This algorithm specifies the way that a local clock can determine which of all the clocks (including itself) is the best. Since it runs continuously, it continually re-adapts to changes in the network or the clocks.

Each clock sends a message to the network describing its own properties. The best clock-source algorithm running in the clock compares these properties to determine the best clock.

The comparisons of attributes happens with the following precedence :

Priority1: user configurable absolute priority

ClockClass: Attribute defining a clock's TAI traceability

Time Source: Attribute defining the accuracy of a clock

Variance: An attribute defining the precision of a clock

Priority2: This is a user configurable designation that provides finer grained ordering among otherwise equivalent clocks

Clock Identity : A tiebreaker consisting of the MAC address of the clock

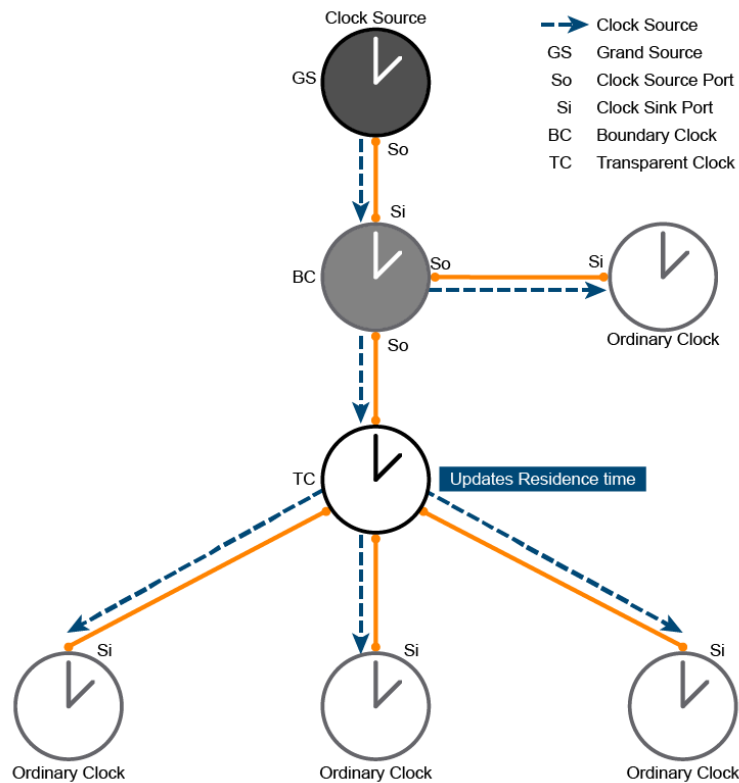
In addition to this precedence order, the distance measured by the number of boundary clocks between the local clock and the best clock is used when two Announce messages reflect the same best clock.

The distance is indicated in the stepsRemoved field of announce messages.

PTP network diagram

The following diagram illustrates how the various PTP clock nodes are connected and how the timing information flows from the origin to the end-stations to achieve the time synchronization. This diagram depicts the PTP clocks in a source-sink hierarchy.

Figure 1 Example PTP network with clocks in a source-sink hierarchy



Configuration examples

Configuring an end-to-end transparent clock (TC E2E)

Follow these steps to configure the E2E transparent clock:

1. Configure the mandatory commands:
 - a. Configure the PTP mode and the delay mechanism.
 - b. Configure the PTP clock-step mode.
 - c. Configure the transport protocol.
 - d. Configure PTP globally.

```
Switch config:
switch(config)# ptp profile 1588v2
switch(config-ptp)# mode transparent end-to-end
switch(config-ptp)# clock-step one-step
switch(config-ptp)# transport-protocol ethernet
switch(config-ptp)# enable
```

2. Enable PTP on the connected interfaces :

```
switch(config)# int 1/1/1
switch(config-if)# ptp enable
switch(config-if)# int 1/1/2
switch(config-if)# ptp enable
```

Configuring an end-to-end boundary clock (BC E2E)

Follow these steps to configure the end-to-end boundary clock:

1. Create the PTP context using the specified profile.

```
switch(config)# ptp profile 1588v2  
switch(config-ptp)#
```

2. Configure the mandatory commands:
 - a. Configure the PTP mode.
 - b. Configure the PTP clock-step mode.
 - c. Configure the transport protocol.
 - d. Configure PTP globally.

```
switch(config-ptp)# mode boundary end-to-end  
switch(config-ptp)# clock-step one-step  
switch(config-ptp)# transport-protocol ethernet  
switch(config-ptp)# enable
```

3. Configure the optional commands which participate in the best clock source algorithm:
 - a. Configure priority1 value.
 - b. Configure priority2 value.

```
switch(config-ptp)# priority1 1  
switch(config-ptp)# priority2 10
```

4. Enable PTP on the connected interfaces:

```
switch(config)# int 1/1/1  
switch(config-if)# ptp enable  
switch(config-if)# int 1/1/2  
switch(config-if)# ptp enable
```

5. Optional: changing packet interval rate for various PTP parameters:

```
switch(config-if)# ptp sync-interval 1588v2 1  
switch(config-if)# ptp announce-interval 1588v2 -5  
switch(config-if)# ptp announce-timeout 1588v2 4  
switch(config-if)# ptp delay-req-interval 1588v2 -3
```

PTP over VSF

- When PTP is configured on a Virtual Switch Framework (VSF) stack, each member of the stack calculates the residence-time of the PTP packets.
- All VSF ports are PTP-enabled (by default) when transparent clock is configured for a VSF stack. If the VSF link is up, the interface status is displayed as **Running** in the `show ptp interface brief` command output.

PTPv1 passthrough

As part of the PTPv1 passthrough feature, PTPv1 packets are allowed to pass through both PTP and non-PTP aware devices. PTPv1 traffic is forwarded based on the destination MAC address and the switches do not process or route the PTPv1 traffic.



As per the IEEE standard, PTPv1 packets are expected to have a TTL value of 0, which makes them operational only in LAN environment and non-routable.

Hardware considerations

The Aruba 6300 Switch Series downlink ports with speed 1G (Smart Rate, non-Smart Rate and XCVR 1G BT) plus 2.5G Smart Rate and 5G Smart Rate have a slightly higher jitter compared to Aruba 6300 Switch Series uplink ports, so the PTP offset of Clients connected on 1G, 2.5G, and 5G downlink ports of the Aruba 6300 Switch Series is slightly higher compared to non-MACsec-supported uplink ports.



Use of third party transceivers may introduce unknown latency and have an effect on the accuracy of PTP deployments.

Configuration recommendations



On the Aruba 6300 Switch Series, boundary clock (one-step and two-step modes) are available only on models R8S89A and R8S90A. All other Aruba 6300 Switch Series models support only transparent clock (E2E and P2P). A VSF stack only supports E2E mode.

PTP CoPP class configuration recommendations

Configuration recommendations for a boundary clock

The PTP CoPP class must be adjusted based on the number of clients associated with the boundary clock and the configured packet rate. For example, if there are 1000 clients with a configured packet rate of 2pps, and a default CoPP limit of 1000, packet drops will be observed. In such instances the CoPP limit should be increased to more than 2000.

The `show copp statistics class ptp` command can be used to monitor whether the CoPP policy must be adjusted. For example:

```
Statistics for CoPP policy 'default':
Class: ptp
Description: Precision Time Protocol (PTP) .
  priority           : 5
  rate (pps)         : 1000
  burst size (pkts)  : 250

  packets passed    : 611153           packets dropped : 0
```



Increasing the PTP CoPP values may help in scaling the clients but needs to be planned and configured properly so as to not to impact other protocols.

QoS prioritization configuration recommendations for a transparent clock

```
class ip PTP
  10 match udp any any eq 319 count
policy PTP-POL
  10 class ip PTP action local-priority 6
policy test
interface lag 240
  apply policy PTP-POL in
interface 1/1/26
  apply policy PTP-POL in
interface 2/1/26
  apply policy PTP-POL in
```



PTP Event messages carrying a critical timestamp use UDP port 319.

General guidelines for PTP IPv4 multicast

- For IP multicast-based PTP time distribution, it is recommended to use PIM Sparse-Mode.
- When connecting transparent clock (TC) and boundary clock (BC), ensure that the TC becomes the DR by setting the DR priority.
- Ensure the mroutes are programmed on TCs so that there is reachability for PTP streams from the upstream.
- Configure static-igmp groups on TCs if the clients themselves cannot send IGMP joins for the PTP multicast group.
- The `QoS trust dscp` command needs to be explicitly configured on all non-BC switches in the network to ensure that the incoming DSCP value of PTP traffic is honored.
- If Peer-to-Peer (P2P) does not happen with the immediate peer (for example some configurations with Unicast mode that skip over intermediate-nodes, which could be one or more Transparent Clocks) the offsets may go high.
- Transparent clock peer-to-peer using IPv4 transport requires source IP configuration to send IPv4 PTP messages over L2 MCLAG.

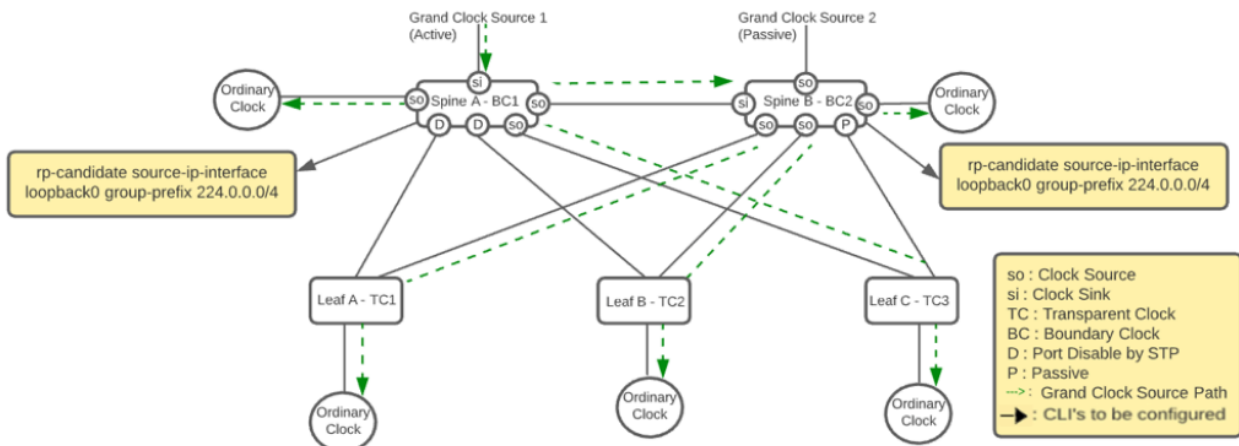
Use cases



On the Aruba 6300 Switch Series, boundary clock (one-step and two-step modes) are available only on models R8S89A and R8S90A. All other Aruba 6300 Switch Series models support only transparent clock (E2E and P2P). A VSF stack only supports E2E mode.

Use case: PTP – IPv4 over L2 spine leaf topology

Figure 1 Grand clock source connected to boundary clock (spines), followed by transparent clock leaves



The following considerations and best practices apply to this use case illustrated in the above topology:

- Configure candidate-RP on switches that are connected to a grand clock source. The candidate-RP needs to be configured on the BC1 Spine A and BC2 Spine B.
- When the PTP clients are not capable of sending IGMP joins, be sure to configure `ip igmp static-group 224.0.1.129` on the VLAN interface of a TC, where PTP is configured.

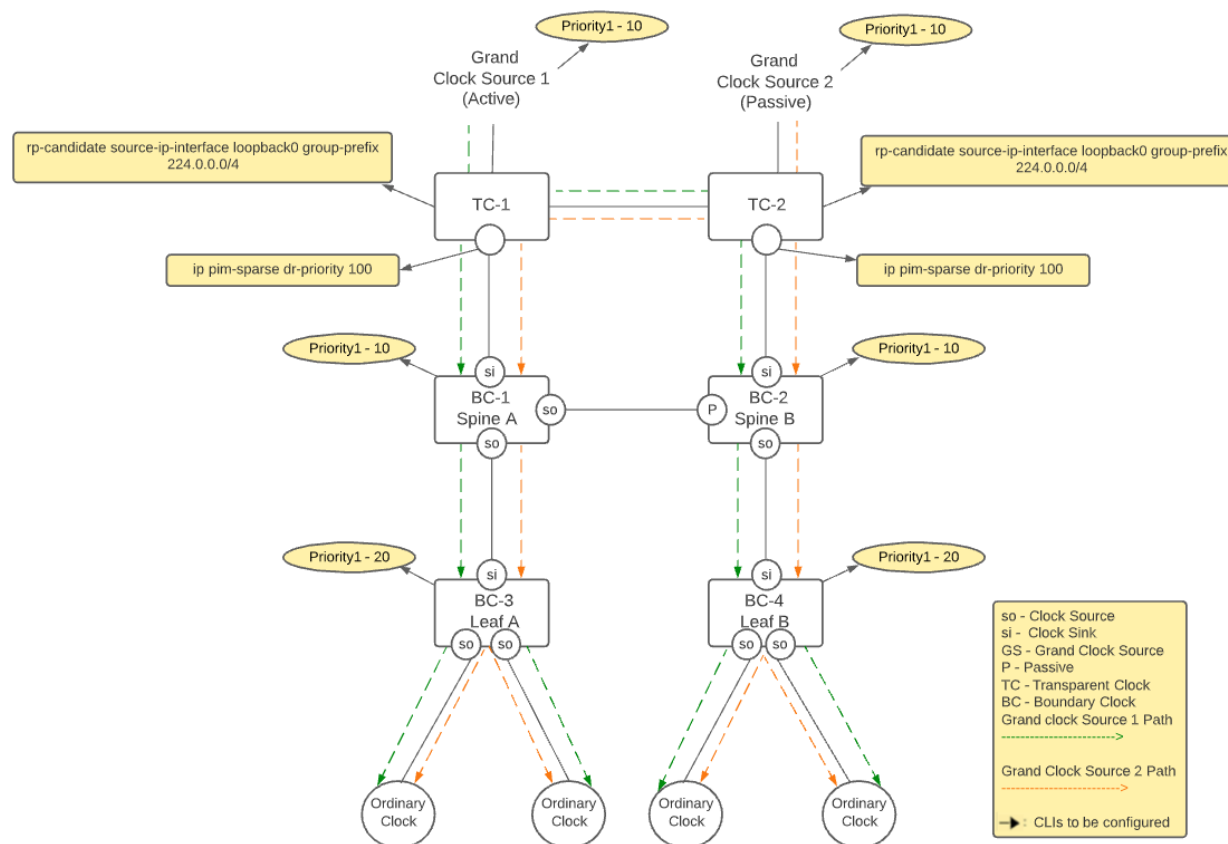


In this case, it is an L2 switch in leaf so `ip igmp snooping enable` also needs to be configured on the transparent clock VLAN interface.

- Ensure that `ip source-interface ptp` is configured on both BC switches on the VLAN interface where PTP is configured.
- Ensure that `spanning tree` is enabled (the default) to avoid unwanted loops.
- In this use case topology, It is recommended to have VRRP on boundary clock switches to have L3 redundancy for PTP end clients.

Use case: PTP – L3 spine leaf topology

Figure 1 Grand clock source connected to transparent clock, followed by boundary clock spines and boundary clock leaves

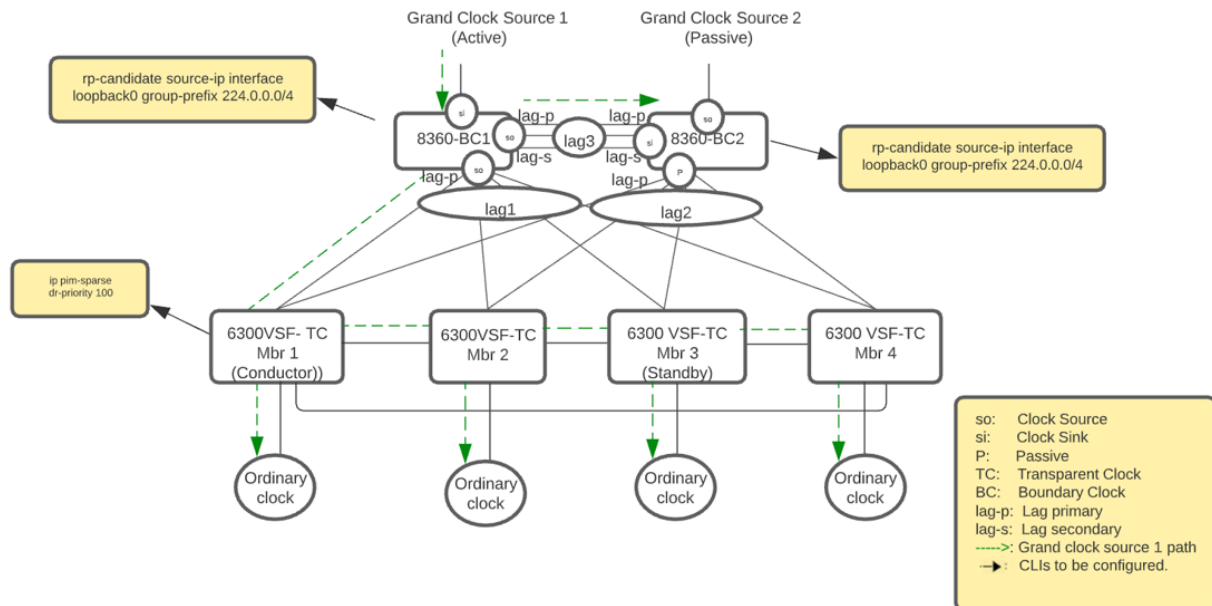


The following considerations and best practices apply to this use case illustrated in the above topology:

- Configure candidate-RP on switches which are connected to a grand clock source. Candidate-RP needs to be configured on TC1 and TC2. Each candidate-RP controls multicast traffic forwarding for one or more multicast groups.
- Configure higher DR priority on switches in which candidate-RP is configured. The DR priority needs to be configured on the TC1 and TC2 on the BC facing ports.
- When the PTP clients are not capable of sending IGMP joins, ensure that `ip igmp static-group 224.0.1.129` is configured on all PTP-enabled interfaces of a TC.
- The `igmp static-group` configuration is not needed on BC switches. In this case, it is an L3 network so `ip igmp enable` also needs to be configured on all PTP-enabled L3 interfaces of the TC.

Use case: PTP – BC and TC (VSF) topology connected via LAG

Figure 1 Grand clock source connected to a boundary clock serving time to other boundary clocks and VSF transparent clocks in the network



The following considerations and best practices apply to this use case illustrated in the above topology:

- Configure candidate-RP on switches that are connected to a grand clock source. The Candidate-RP needs to be configured on the BC1 and BC2 switches. Each candidate-RP controls multicast traffic forwarding for one or more multicast groups.
- When the PTP clients are not capable of sending IGMP joins, ensure that `ip igmp static-group 224.0.1.129` is configured on all PTP-enabled interfaces of a TC.



`igmp static-group` config is not needed on boundary clock switches. In this case, it is an L3 network so `ip igmp enable` also needs to be configured on all PTP enabled L3 interfaces of the transparent clock.

- For the primary or secondary LAG roles, ensure that the same link ports are configured on both ends of the LAG across BC (this command is applicable only on BC switches).



When both the primary and secondary LAG role links are down, PTP packets will not be forwarded even if other LAG links are up.

- Configure `ip pim-sparse dr-priority <PRIORITY-VALUE>` to configure higher dr-priority on the links originating from the VSF (which are facing the BC switches).

PTP commands

clear ptp statistics

```
clear ptp statistics [<IFNAME>]
```

Description

Clears PTP counters for the given interface.

Parameter	Description
<IFNAME>	Optional: Specifies the interface name.

Examples

Clearing PTP counters for the given interface:

```
switch# clear ptp statistics 1/1/8
switch# clear ptp statistics lag1
switch# clear ptp statistics
```

Command History

Release	Modification
10.10.1000	Added boundary clock support on the 6300 Switch Series.
10.10	Command introduced on the 6300 Switch Series for transparent clock.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Administrators or local user group members with execution rights for this command.

clock-domain

```
clock-domain <DOMAIN-NUMBER>
no clock-domain
```

Description

Configures the PTP clock domain to a specified value.

The **no** form of this command removes the PTP domain configuration of the PTP clock.

Parameter	Description
<DOMAIN-NUMBER>	Sets the PTP clock domain. Range: 0 to 254. Value configurable subject to limits established by the PTP profile.

Usage

- The one-step end-to-end transparent clock works across domains.
- For boundary clocks, the clock-domain has to be identical with the domain used in the network.
- All PTP devices must be within same domain to be able to sync with each other.
- This command is only enabled in the PTP profile context.
- For PTP transparent clock, you must configure the same clock-domain as on clients and GM to synchronize.

Examples

Entering the PTP profile context and setting the PTP clock domain value:

```
switch(config)# ptp profile aes-r16
switch(config-ptp)#
switch(config-ptp)#clock-domain 4
switch(config-ptp)#
```

Removing the PTP clock domain value:

```
switch(config-ptp)# no clock-domain
switch(config-ptp)#
```

Command History

Release	Modification
10.10.1000	Command introduced on the 6300 Switch Series.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	config-ptp	Administrators or local user group members with execution rights for this command.

clock-step

```
clock-step {one-step|two-step}
no clock-step
```

Description

Configures the clock step mode that determines when the egress-time information is sent.

The 6300 Switch Series (models R8S89A and R8S90A) support both one-step and two-step modes for boundary clocks. Transparent clocks only support one-step mode. All other 6300 Switch Series models support only transparent clock one-step mode.

The **no** form of this command removes the PTP clock-step configuration of the PTP clock.

Parameter	Description
one-step	Sets the PTP clock-step mode to one-step messaging in which egress-time information is sent along with the SYNC message.
two-step	Sets the PTP clock-step mode to two-step messaging in which egress-time information is sent a subsequent follow-up message with the egress timestamp of the previously sent SYNC message.

Usage

- Mandatory command to start the PTP clock.
- Boundary clocks can inter-operate with different step modes upstream or downstream.

Example

Setting the clock-step mode to one-step messaging:

```
switch(config-ptp)# clock-step one-step
```

Removing the clock-step mode configuration:

```
switch(config-ptp)# no clock-step
```

Setting the clock-step mode to two-step messaging:

```
switch(config-ptp)# clock-step two-step
```

Command History

Release	Modification
10.11	For boundary clock, added support for the two-step parameter on the 6300 Switch Series (models R8S89A and R8S90A).
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config-ptp	Administrators or local user group members with execution rights for this command.

enable

enable
no enable

Description

Enables the PTP profile globally. However, the PTP clock is started only when all the mandatory commands are set.

The **no** form of this command disables the PTP profile globally.

Usage

Mandatory command to start the PTP clock.

Examples

Enabling the PTP profile:

```
switch(config)# ptp profile 1588v2  
switch(config-ptp)# enable
```

Disabling the PTP profile:

```
switch(config)# ptp profile 1588v2  
switch(config-ptp)# no enable
```

Command History

Release	Modification
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config-ptp	Administrators or local user group members with execution rights for this command.

ip source-interface

```
ip source-interface {ptp | all} interface <IFNAME> [vrf <VRF-NAME>  
ip source-interface {ptp | all} <IPV4-ADDR> [vrf <VRF-NAME>]  
no ip source-interface {ptp | all} interface <IFNAME> [vrf <VRF-NAME>  
no ip source-interface {ptp | all} <IPV4-ADDR> [vrf <VRF-NAME>]
```

Description

Configures the source IP address to be used when sending PTP messages. Use the **ptp** keyword to set source IP address specific to the PTP feature. If the feature-specific configuration is not available, the source IP address corresponding to the **all** option will be used.

The **no** form of this command removes the configuration of the source IP address used by the PTP feature.

Parameter	Description
ptp	Selects the PTP protocol.
all	Selects all protocols that can be configured by this command.
interface <IF-NAME>	Specifies the name of the interface from which the source IP

Parameter	Description
	address is obtained. The interface must have a valid IP address assigned to it. If the interface has both a primary and secondary IP address, the primary IP address is used.
<code>vrf <VRF-NAME></code>	Specifies the VRF name.
<code><IPv4-ADDR></code>	Specifies the source IPv4 address to be used. The IP address must be defined on the switch, and it must exist on the specified VRF (which is the default VRF, if the vrf option is not used). Specify the address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

Usage

- This command must be configured on the switch when PTP is enabled on VLAN trunk or access ports, and the transport protocol is IPv4.
- In the current version of PTP, only the default VRF is supported.
- This command is not applicable to the end-to-end transparent clock.

Examples

Configuring the source IP address for sending PTP messages:

```
switch(config)# ip source-interface ptp interface 1/1/1
switch(config)# ip source-interface ptp 10.10.10.1
switch(config)# ip source-interface ptp interface vlan10
```

Command History

Release	Modification
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.

mode

```
mode boundary {end-to-end | peer-to-peer}
no mode boundary {end-to-end | peer-to-peer}
mode transparent {end-to-end | peer-to-peer}
no mode transparent {end-to-end | peer-to-peer}
no mode
```

Description

Configures the switch PTP clock mode, either boundary or transparent, with a delay-request mechanism of either end-to-end or peer-to-peer. A device in transparent clock mode does not synchronize (syntonize) itself to a grandsource clock.



On the Aruba 6300 Switch Series, boundary clock (one-step and two-step modes) are available only on models R8S89A and R8S90A. All other Aruba 6300 Switch Series models support only transparent clock (E2E and P2P). A VSF stack only supports E2E mode.

The **no** form of this command unconfigures the PTP clock mode and delay-request mechanism.

Parameter	Description
boundary	Selects boundary clock mode.
transparent	Selects transparent clock mode.
end-to-end	Selects the end-to-end delay-request mechanism.
peer-to-peer	Selects the peer-to-peer delay-request mechanism. Not supported with VSF.

Examples

Configuring PTP boundary clock mode with the end-to-end delay-request mechanism:

```
switch(config-ptp) # mode boundary end-to-end
```

Unconfiguring PTP boundary clock mode with the end-to-end delay-request mechanism:

```
switch(config-ptp) # no mode boundary end-to-end
```

Configuring PTP boundary clock mode with the peer-to-peer delay-request mechanism:

```
switch(config-ptp) # mode boundary peer-to-peer
```

Unconfiguring PTP boundary clock mode with the peer-to-peer delay-request mechanism:

```
switch(config-ptp) # no mode boundary peer-to-peer
```

Configuring PTP transparent with the end-to-end delay-request mechanism:

```
switch(config-ptp) # mode transparent end-to-end
```

Unconfiguring PTP transparent with the end-to-end delay-request mechanism:

```
switch(config-ptp) # no mode transparent end-to-end
```

Configuring PTP transparent with the peer-to-peer delay-request mechanism:

```
switch(config-ptp) # mode transparent peer-to-peer
```

Unconfiguring PTP transparent with the peer-to-peer delay-request mechanism:

```
switch(config-ptp) # no mode transparent peer-to-peer
```

Command History

Release	Modification
10.11	Added support for the peer-to-peer delay-request mechanism.
10.10.1000	Added boundary clock to the 6300 Switch Series models R8S89A and, R8S90A.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config-ptp	Administrators or local user group members with execution rights for this command.

priority1

```
priority1 <PRIORITY>  
no priority1
```

Description

Configures the PTP clock **priority1** value of the device. This value is operational when the device is in boundary clock mode and participating in the Best Clock Source Algorithm (BMCA). This value is used to indicate priority to its downstream clock-aware devices.

The **no** form of this command removes the PTP **priority1** configuration of the PTP clock and sets it to the default value of 128.

Parameter	Description
<PRIORITY>	Sets the priority value. Default 128.

Usage

This value can be configured only for the boundary clock.

Examples

Configuring PTP priority1 value:

```
switch(config-ptp) # priority1 129
```

Removing PTP priority1 configuration:


```
switch(config-ptp) # no priority1
```

Command History

Release	Modification
10.10.1000	Command introduced on the 6300 Switch Series.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	config-ptp	Administrators or local user group members with execution rights for this command.

priority2

```
priority2 <PRIORITY>  
no priority2 <PRIORITY>
```

Description

Configures the PTP clock **priority2** value of the device. This value is operational when the device is in boundary clock mode and participating in the Best Clock Source Algorithm (BMCA). This value is used to indicate priority to its downstream clock-aware devices.

The **no** form of this command removes the PTP **priority2** configuration of the PTP clock and sets it to the default value of 128.

Parameter	Description
<PRIORITY>	Sets the priority value. Default 128.

Usage

This value can be configured only for the boundary clock.

Examples

Configuring PTP **priority1** value:

```
switch(config-ptp) # priority2 129
```

Removing PTP **priority2** configuration:

```
switch(config-ptp) # no priority2
```

Command History

Release	Modification
10.10.1000	Command introduced on the 6300 Switch Series.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	config-ptp	Administrators or local user group members with execution rights for this command.

ptp announce-interval

```
ptp announce-interval {1588v2 | aes67 | aes-r16 | dot1as | smpte} <LOG-SECONDS>
no ptp announce-interval {1588v2 | aes67 | aes-r16 | dot1as | smpte}
```

Description

Sets the announce message transmit interval on a PTP-enabled interface for a specific PTP profile.

The **no** form of this command removes the announce message transmit interval configuration on a PTP-enabled interface and sets a profile specific default value.

Parameter	Description
1588v2	Specifies the PTP 1588v2 profile timers. Default: 1.
aes67	Specifies the PTP AES67 profile timers. Default: 1.
aes-r16	Specifies the PTP AES-R16 profile timers. Default: 1.
dot1as	Specifies the PTP 802.1 AS profile timers. Default: 0.
smpte	Specifies the PTP SMTPE profile timers. Default: -2.
<LOG-SECONDS>	Sets the announce message interval in log seconds.

Usage

This value can be configured only for the boundary clock.

Examples

Setting the PTP 802.1 AS profile timers:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp announce-interval dot1as 1
```

Setting the PTP AES67 profile timers:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp announce-interval aes67 2
```

Removing the PTP AES67 profile timer configuration:

```
switch(config)# interface 1/1/1
switch(config-if)# no ptp announce-interval aes67
```

Command History

Release	Modification
10.13	Support extended for 802.1AS profile.
10.10.1000	Command introduced on the 6300 Switch Series.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	config-if	Administrators or local user group members with execution rights for this command.

ptp announce-timeout

```
ptp announce-timeout {1588v2| aes67 | aes-r16 | dot1as | smpte} <COUNT>
no ptp announce-timeout {1588v2| aes67 | aes-r16 | dot1as | smpte}
```

Description

Sets the announce message receipt timeout on a PTP-enabled interface for a specific PTP profile.

The **no** form of this command resets the announce message receipt timeout configuration on a PTP-enabled interface and sets a profile-specific default value.

Parameter	Description
1588v2	Specifies the PTP 1588v2 profile timers. Default: 3.
aes67	Specifies the PTP AES67 profile timers. Default: 3.
aes-r16	Specifies the PTP AES-R16 profile timers. Default: 3.
dot1as	Specifies the PTP 802.1AS profile timers. Default: 3.
smpte	Specifies the PTP SMTPE profile timers. Default: 3.
<LOG-SECONDS>	Specifies the number of announcement intervals.

Usage

This value can be configured only for the boundary clock.

Examples

Setting the PTP 802.1 AS profile timer:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp announce-timeout dot1as 4
```

Setting the PTP AES67 profile timer:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp announce-timeout aes67 4
```

Resetting the PTP AES67 profile timer:

```
switch(config)# interface 1/1/1
switch(config-if)#no ptp announce-timeout aes67
```

Command History

Release	Modification
10.13	Support extended for 802.1AS profile.
10.10.1000	Command introduced on the 6300 Switch Series.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	config-if	Administrators or local user group members with execution rights for this command.

ptp clock-source-only

```
ptp clock-source-only
no ptp clock-source-only
```

Description

Configures the PTP port state to clock_source state. This prohibits the port from entering into a clock_sink or passive state.

The **no** form of this command removes the clock_source state configuration on the port and returns it to normal BMCA operation.

Usage

This can only be configured for the boundary clock.

Examples

Configuring the clock_source only role for the port:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp clock-source-only
```

Removing the configuration of clock_source only role for the port:

```
switch(config)# interface 1/1/1
switch(config-if)# no ptp clock-source-only
```

Command History

Release	Modification
10.10.1000	Command introduced on the 6300 Switch Series.
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	config-if	Administrators or local user group members with execution rights for this command.

ptp delay-req-interval

```
ptp delay-req-interval {1588v2 | aes67 | aes-r16 | smpte} <LOG-SECONDS>
no ptp delay-req-interval {1588v2 | aes67 | aes-r16 | smpte}
```

Description

Sets the **delay_req** message transmit interval on a PTP-enabled interface for a specific PTP profile. The **no** form of this command removes the **delay_req** message transmit interval configuration on a PTP-enabled interface and sets a profile specific default value.

Parameter	Description
1588v2	Specifies the PTP 1588v2 profile timers. Default 0.
aes67	Specifies the PTP AES67 profile timers. Default 0.
aes-r16	Specifies the PTP AES-R16 profile timers. Default 0.
smpte	Specifies the PTP SMTPE profile timers. Default -3.
<LOG-SECONDS>	Sets the delay_req message interval in log seconds.

Usage

- Use this command for end-to-end (E2E) mode and use command **ptp pdelay-interval** for peer-to-peer mode.
- This command is only for boundary clock.

Examples

Setting the PTP AES67 profile timers:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp delay-req-interval aes67 2
```

Removing the PTP AES67 profile timer configuration:

```
switch(config)# interface 1/1/1
switch(config-if)# no ptp delay-req-interval aes67
```

Command History

Release	Modification
10.10.1000	Command introduced on the 6300 Switch Series.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	config-if	Administrators or local user group members with execution rights for this command.

ptp enable

```
ptp enable
no ptp enable
```

Description

Enables PTP on the interface. The **no** form of this command disables PTP on the interface. PTP can be enabled only on physical L2 or L3 interfaces and LAG L2 or L3 interfaces.

Examples

Enabling PTP on a physical interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp enable
```

Disabling PTP on the interface context:

```
switch(config)# interface 1/1/1
switch(config-if)# no ptp enable
```

Command History

Release	Modification
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config-if	Administrators or local user group members with execution rights for this command.

ptp lag-role

```
ptp lag-role {primary | secondary}
no ptp lag-role
```

Description

Configures the PTP role for the member interfaces of a Link Aggregation (LAG) . When there are two or more member interfaces for a LAG, only one link can be configured as primary and only one other link can be configured as secondary. The primary member interface is used for transmitting the PTP packets generated by the boundary clock. When the primary member goes down, the secondary member is used for PTP packet transmission. If both primary and secondary members go down, PTP does not flip over to the other links of the LAG.

The **no** form of this command removes the PTP role configuration for the LAG member interface.



This command is not supported when configured as a transparent clock.

Parameter	Description
primary	Sets the primary PTP lag-role for the LAG member interface.
secondary	Sets the secondary PTP lag-role for the LAG member interface.

Usage

- LAG roles must be configured for boundary clock.
- For the primary or secondary LAG roles, ensure that the same link ports are configured on both ends of the LAG.

Examples

Setting the primary PTP lag-role for the LAG member interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp lag-role primary
```

Setting the secondary PTP lag-role for the LAG member interface:

```
switch(config)# interface 1/1/2
switch(config-if)# ptp lag-role secondary
```

Removing the PTP lag-role configuration for the LAG member interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ptp lag-role
```

Command History

Release	Modification
10.10.1000	Command introduced on the 6300 Switch Series.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	config-if	Administrators or local user group members with execution rights for this command.

ptp neighbor-propagation-delay-threshold

`ptp neighbor-propagation-delay-threshold <threshold value>`

Description

Configures PTP neighbor propagation delay threshold in nanoseconds.

The **no** form of this command removes the PTP neighbor propagation delay threshold configuration.

Parameter	Description
<code><threshold value></code>	Sets the PTP neighbor propagation delay threshold in nanoseconds. The supported range is 0-2147483648. Default threshold value: 800 nanoseconds.

Examples

Setting the PTP neighbor-propagation-delay-threshold:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp neighbor-propagation-delay-threshold 200
```

Removing the PTP neighbor-propagation-delay-threshold:

```
switch(config)# interface 1/1/1
switch(config-if)# no ptp neighbor-propagation-delay-threshold
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	config-if	Administrators or local user group members with execution rights for this command.

ptp pdelay-req-interval

```
ptp pdelay-req-interval {1588v2| aes67 | aes-r16 | dot1as | smpte} <LOG-SECONDS>
no ptp pdelay-req-interval {1588v2| aes67 | aes-r16 | dot1as | smpte}
```

Description

Sets the **pdelay_req** message transmit interval on a PTP-enabled interface for a specific PTP profile. The **no** form of this command removes the **pdelay_req** message transmit interval configuration on a PTP-enabled interface and sets a profile specific default value.

Parameter	Description
1588v2	Specifies the PTP 1588v2 profile timers. Default 0.
aes67	Specifies the PTP AES67 profile timers. Default 0.
aes-r16	Specifies the PTP AES-R16 profile timers. Default 0.
dot1as	Specifies the PTP 802.1AS profile timers. Default: 0.
smpte	Specifies the PTP SMTPE profile timers. Default -3.
<LOG-SECONDS>	Sets the delay_req message interval in log seconds.

Usage

- Use this command for peer-to-peer (P2P) mode and use command **ptp delay-interval** for end-to-end (E2E) mode.

Examples

Setting the PTP 802.1 AS profile timers:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp pdelay-req-interval dot1as 2
```

Removing the PTP 802.1AS profile timer configuration:

```
switch(config)# interface 1/1/1
switch(config-if)# no ptp pdelay-req-interval dot1as
```

Setting the PTP AES67 profile timers:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp pdelay-req-interval aes67 1
```

Removing the PTP AES67 profile timer configuration:

```
switch(config)# interface 1/1/1
switch(config-if)# no ptp pdelay-req-interval aes67
```

Setting the PTP smpte profile timers:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp pdelay-req-interval smpte 1
```

Removing the PTP smpte profile timer configuration:

```
switch(config)# interface 1/1/1
switch(config-if)# no ptp pdelay-req-interval smpte
```

Command History

Release	Modification
10.13	Support extended for 802.1AS profile.
10.11	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	config-if	Administrators or local user group members with execution rights for this command.

ptp peer ip

```
ptp peer ip <IP-ADDRESS>
no ptp peer ip <IP-ADDRESS>
```

Description

Configures destination IP addresses for the interfaces in unicast transmission. The **no** form of this command removes the PTP destination IP address configuration for the interfaces in unicast transmission.

Parameter	Description
ip <IP-ADDRESS>	Specifies the peer IPv4 address. Syntax: A.B.C.D

Usage

- This command has no effect when configured as a transparent clock.

Example

Configuring `ptp peer ip` on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp peer ip 10.0.0.1
```

Removing `ptp peer ip` on the interface:

```
switch(config-if)# no ptp peer ip 10.0.0.1
```

Command History

Release	Modification
10.10.1000	Command introduced on the 6300 Switch Series.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	config-if	Administrators or local user group members with execution rights for this command.

ptp profile

```
ptp profile {<PROFILE NAME>}
no ptp profile
```

Description

Enters the PTP context to configure the PTP profile in which the device will operate.

Configure PTP profile before configuring mode or other profile-specific parameters. The device can be operating in any one profile at a given point of time. The **no** form of this command removes the PTP profile configuration in which the device will operate. This command clears the PTP profile and all parameters related to that profile.

Parameter	Description
<PROFILE NAME>	Specifies the profile to be used. Profiles include: ▪ 1588v2: Specifies the IEEE 1588-2008 profile to be used. ▪ aes-r16: Specifies the IEEE AES-R16-2016 profile to be used. ▪ aes67: Specifies the IEEE AES67 profile to be used. ▪ dot1as: Specifies the IEEE 802.1AS profile to be used. ▪ smpte: Specifies the IEEE SMPTE-ST-2059-2 profile to be used.

Parameter	Description
	NOTE: The 802.1AS (2011) PTP profile specification supports only two-step clock, Ethernet transport, and peer-to-peer delay mechanism. The 802.1AS PTP profile is supported only on following SKUs: JL717C, JL718C, JL719C, JL721C, JL722C, R8S89A, and R8S90A.

Usage

Configure PTP profile before configuring mode or other profile-specific parameters.

Example

Configuring PTP profiles:

```
switch(config)# ptp profile 1588v2
```

Configuring PTP profiles:

```
switch(config)# ptp profile dot1as
```

Configuring more than one PTP profile:

```
switch(config)# ptp profile 1588v2
switch(config-ptp)# exit
switch(config)# ptp profile smpte
switch(config-ptp)#
The existing profile must be removed using the 'no ptp profile' command before
configuring a different profile.
```

Configuring Ethernet transport for the 802.1AS PTP profile:

```
switch(config)# ptp profile dot1as
switch(config-ptp)# transport-protocol ethernet
```

Configuring peer-to-peer delay mechanism for the 802.1AS PTP profile:

```
switch(config)# ptp profile dot1as
switch(config-ptp)# mode boundary peer-to-peer
```

Configuring two-step clock for the 802.1AS PTP profile:

```
switch(config)# ptp profile dot1as
switch(config-ptp)# clock-step two-step
```

Removing the PTP profile:

```
switch(config-ptp)# no ptp profile
```

Command History

Release	Modification
10.13	Support extended for 802.1AS profile.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.

ptp sync-interval

```
ptp sync-interval {1588v2 | dot1as | aes67 | smpte} <LOG-SECONDS>  
no ptp sync-interval {1588v2 | dot1as | aes67 | smpte}
```

Description

Sets the sync message transmit interval on a PTP-enabled interface for a specific PTP profile.

The **no** form of this command removes the sync message transmit interval configuration on a PTP enabled interface and sets it to a profile-specific default value.

Parameter	Description
1588v2	Specifies the PTP 1588v2 profile timers. Default 0.
dot1as	Specifies the PTP 802.1 AS profile timers. Default -3.
aes67	Specifies the PTP AES67 profile timers. Default -3.
smpte	Specifies the PTP SMTPE profile timers. Default -3
<LOG-SECONDS>	Sets the sync message interval in log seconds.

Examples

Setting the PTP 802.1 AS sync interval :

```
switch(config)# interface 1/1/1  
switch(config-if)# ptp sync-interval dot1as -2
```

Setting the PTP 1588v2 sync interval :

```
switch(config)# interface 1/1/1  
switch(config-if)# ptp sync-interval 1588v2 2
```

Setting the PTP AES67 sync interval :

```
switch(config)# interface 1/1/1
switch(config-if)# ptp sync-interval aes67 -2
```

Removing the PTP AES67 sync interval:

```
switch(config)# interface 1/1/1
switch(config-if)# no ptp sync-interval aes67
```

Command History

Release	Modification
10.13	Support extended for 802.1AS profile.
10.10.1000	Command introduced on the 6300 Switch Series.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	config-if	Administrators or local user group members with execution rights for this command.

ptp sync-timeout

```
ptp sync-timeout dotlas <COUNT>
no ptp sync-timeout dotlas
```

Description

Sets the synchronization message receipt timeout on a PTP enabled interface.

The **no** form of this command resets the synchronization message receipt timeout configuration on a PTP-enabled interface and sets a profile-specific default value.

Parameter	Description
dotlas	Specifies the PTP 802.1AS profile timers. Default: 3.
<LOG-SECONDS>	Specifies the number of announcement intervals.

Examples

Setting the PTP 802.1 AS profile timer:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp sync-timeout dotlas 4
```

Resetting the PTP 802.1 AS profile timer:

```
switch(config)# interface 1/1/1
switch(config-if)# no ptp sync-timeout dot1as
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	config-if	Administrators or local user group members with execution rights for this command.

ptp vlan

```
ptp vlan <VLAN-ID>
no ptp vlan
```

Description

Configures a VLAN for PTP messages. It is necessary when the boundary clock port is a VLAN trunk L2 interface (**no routing**). The **no** form of this command removes the VLAN configuration for PTP messages.

Parameter	Description
<VLAN-ID>	Specifies a VLAN. Range: 1-4094.

Usage

- This configuration has no bearing on the one-step transparent clock.
- In boundary clock mode, only PTP packets in PTP VLAN are processed; PTP packets from other VLANs are dropped.
- **ptp vlan** should be configured on interfaces only when the specific VLAN is a trunk/tagged member of that port. This configuration should not be performed on an access port.

Examples

Configuring a specific VLAN for PTP messages:

```
switch(config)# interface 1/1/1
switch(config-if)# ptp vlan 4
```

Removing the VLAN configuration for PTP messages:

```
switch(config)# interface 1/1/1
switch(config-if)# no ptp vlan
```

Command History

Release	Modification
10.10.1000	Command introduced on the 6300 Switch Series.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	config-if	Administrators or local user group members with execution rights for this command.

show ptp clock

show ptp clock

Description

Shows PTP clock-related information.

Example

Showing PTP transparent clock information:

```
switch# show ptp clock
PTP Profile           : smpte
PTP Mode              : transparent
Delay Mechanism       : end-to-end
Clock Identity        : NA
Network Transport Protocol : ipv4
Clock Step            : One
Clock Domain          : NA
Number of PTP Ports   : 1
Priority1              : NA
Priority2              : NA
Clock Quality :
  Class               : NA
  Accuracy             : NA
  Offset (log variance) : NA
Offset From Clock-Source : NA
Mean Delay            : NA
Steps Removed         : NA
```

Showing PTP boundary clock information (boundary clock is available only on the 6300 Switch Series models R8S89A and R8S90A that first released with AOS-CX 10.10.1000):

```
switch# show ptp clock
PTP Profile           : aes67
PTP Mode              : boundary
Delay Mechanism       : end-to-end
Clock Identity        : 00:fd:45:ff:fe:68:f3:00
Network Transport Protocol : ipv4
Clock Step            : Two
```



```

Clock Domain                : 0
Number of PTP Ports         : 3
Priority1                    : 128
Priority2                    : 128

Clock Quality :
  Class          : 248
  Accuracy       : 49
  Offset (log variance) : 52592

Offset From Clock-Source    : - 0.000000006 (s)
Mean Delay                  : + 0.000000277 (s)
Steps Removed               : 1

```

Command History

Release	Modification
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

show ptp foreign-clock-sources

show ptp foreign-clock-sources

Description

Shows the priority1, priority2, class, accuracy, offset-scaled-log-variance (OSLV), and steps removed information for foreign clock-source nodes.

Example

Showing PTP foreign clock-source information:

```

switch(config-if)# show ptp foreign-clock-sources
Pl=Priority1, P2=Priority2, C=Class, A=Accuracy,
OSLV=Offset-scaled-log-variance, SR=Steps-removed

```

```

-----
Interface  Foreign Port ID          Clock Source ID          P1  P2  C  A  OSLV  SR
-----
1/1/4      00:00:00:00:00:00:01(0x0001)  00:00:00:00:00:00:01  0   0   6  35  0    1
1/1/5      b4:99:ba:ff:fe:54:2b:00(0x0002)  00:00:00:00:00:00:01  0   0   6  35  0    2

```

Command History

Release	Modification
10.10.1000	Command introduced on the 6300 Switch Series.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

show ptp interface

show ptp interface [<IFNAME> profile-data dotlas | [brief]

Description

Shows PTP port-related information.

Parameter	Description
<IFNAME>	Specifies the interface name.
profile-data	Shows additional information of the profile.
dotlas	Shows port data specific to the 802.1 AS profile.
brief	Shows information in a brief format.

Examples

Showing PTP port information when the switch is acting as a transparent clock:

```
switch# show ptp interface 1/1/1
Port Identity           : 00:00:00:00:00:00:00:00 (0x0000)
Port Number             : 0
PTP Version             : 2
PTP Enable              : Enabled
PTP Transport           : ipv4
Port State              : Disabled
Delay Mechanism          : peer-to-peer
Announce Interval (log mean) : 0
Announce Receipt Timeout : 0
Sync Interval (log mean)  : 0
Sync Timeout            : NA
Delay Request Interval (log mean) : NA
Peer Delay Request Interval (log mean) : 0
Mean Path Delay         : 0 (ns)
switch#

switch# show ptp interface lag20
Port Identity           : NA
Port Number             : NA
PTP Version             : 2
```

```

PTP Enable                : Enabled
Transport of PTP          : ipv4
Port State                : NA
Delay Mechanism           : end-to-end
Announce Interval (log mean) : NA
Announce Receipt Timeout  : NA
Sync Interval (log mean)  : NA
Sync Timeout             : NA
Delay Request Interval (log mean) : N
Peer Delay Request Interval (log mean) : 0
Mean Path Delay          : 0 (ns)
switch#

```

Showing PTP port information when the switch is acting as a boundary clock (boundary clock is available only on the 6300 Switch Series models R8S89A and R8S90A that first released with AOS-CX 10.10.1000):

```

switch# show ptp interface 1/1/1
Interface : 1/1/1
Port Identity      : 88:3a:30:ff:fe:05:c9:80 (port: 0x0002)
Port Number       : 2
PTP Version       : 2
PTP Enable        : Enabled
Transport of PTP  : ethernet
Port State        : Clock Source
Delay Mechanism   : end-to-end
Announce Interval (log mean) : 0
Announce Receipt Timeout  : 3
Sync Interval (log mean)  : -3
Sync Timeout        : NA
Delay Request Interval (log mean) : 0
Peer Delay Request Interval (log mean) : 0
Mean Path Delay     : 0 (ns)

```

```

switch# show ptp interface lag1
Port Identity      : 00:fd:45:ff:fe:68:f3:00 (port: 0x0002)
Port Number       : 2
PTP Version       : 2
PTP Enable        : Enabled
Transport of PTP  : ipv4
Port State        : Clock Source
Delay Mechanism   : end-to-end
Announce Interval (log mean) : 0
Announce Receipt Timeout  : 3
Sync Interval (log mean)  : -3
Sync Timeout        : NA
Delay Request Interval (log mean) : -3
Peer Delay Request Interval (log mean) : 0
Mean Path Delay     : 0 (ns)
Primary Interface  : 1/1/5
Secondary Interface : 1/1/6

```

```

switch# show ptp interface
Interface lag20:
Port Identity      : 00:fd:45:ff:fe:68:f3:00 (port: 0x0002)
Port Number       : 2
PTP Version       : 2
PTP Enable        : Enabled
Transport of PTP  : ipv4
Port State        : Clock Source
Delay Mechanism   : end-to-end
Announce Interval (log mean) : 0

```

```

Announce Receipt Timeout      : 3
Sync Interval (log mean)     : -3
Sync Timeout                  : NA
Delay Request Interval (log mean) : -3
Peer Delay Request Interval (log mean) : 0
Mean Path Delay               : 0 (ns)
Primary Interface             : 1/1/5
Secondary Interface           : 1/1/6

Member Interface 1/1/5:
Port Identity                  : 00:fd:45:ff:fe:68:f3:00 (port: 0x0002)
Port Number                    : 2
PTP Version                    : 2
PTP Enable                     : Enabled
Transport of PTP               : ipv4
Port State                     : Running
Delay Mechanism                 : end-to-end
Announce Interval (log mean)   : 0
Announce Receipt Timeout      : 3
Sync Interval (log mean)      : -3
Sync Timeout                   : NA
Delay Request Interval (log mean) : -3
Peer Delay Request Interval (log mean) : 0
Mean Path Delay               : 0 (ns)

Member Interface 1/1/6:
Port Identity                  : 00:fd:45:ff:fe:68:f3:00 (port: 0x0003)
Port Number                    : 3
PTP Version                    : 2
PTP Enable                     : Enabled
Transport of PTP               : ipv4
Port State                     : Not Running
Delay Mechanism                 : end-to-end
Announce Interval (log mean)   : 0
Announce Receipt Timeout      : 3
Sync Interval (log mean)      : -3
Sync Timeout                   : NA
Delay Request Interval (log mean) : -3
Peer Delay Request Interval (log mean) : 0
Mean Path Delay               : 0 (ns)

Interface 1/1/15:
Port Identity                  : 00:fd:45:ff:fe:68:f3:00 (port: 0x0001)
Port Number                    : 1
PTP Version                    : 2
PTP Enable                     : Enabled
Transport of PTP               : ipv4
Port State                     : Clock Sink
Delay Mechanism                 : end-to-end
Announce Interval (log mean)   : 0
Announce Receipt Timeout      : 3
Sync Interval (log mean)      : -3
Sync Timeout                   : NA
Delay Request Interval (log mean) : -3
Peer Delay Request Interval (log mean) : 0
Mean Path Delay               : 0 (ns)

```

Showing PTP port information (in brief form) when the switch is acting as a boundary clock (boundary clock is available only on the 6300 Switch Series models R8S89A and R8S90A that first released with AOS-CX 10.10.1000):

```
switch# show ptp interface brief
Interface      PTP State
-----
1/1/11        Clock Sink
1/1/12        Clock Source
1/1/15        Clock Source
1/1/16        Clock Source
```

Showing PTP port information of 802.1AS profile:

```
switch# show ptp int 1/1/1 profile-data dot1as
asCapable      : TRUE
Compute Neighbor RateRatio : TRUE
Neighbor RateRatio : 1.2
Compute Neighbor PropDelay : TRUE
Neighbor Propagation Delay : 35 ns
PDelay Lost Response Threshold : 3
Lost Response Threshold Exceeded : FALSE
```

Showing PTP port information in brief format:

Showing PTP port information in brief format:

```
switch# show ptp interface brief
Interface      PTP State
-----
1/1/1          Running
1/1/2          Running
```

Command History

Release	Modification
10.13	Support extended for 802.1AS profile.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

show ptp parent

show ptp parent

Description

Shows parent node information for the PTP device.

Example

Showing PTP parent node information:

```
switch# show ptp parent
PTP Parent Properties

Parent Clock
-----
Parent Clock Identity          : 00:00:00:00:00:00:00:01
Parent Port Number             : 0x0001
Observed Parent Offset (log variance) : 65535
Observed Parent Clock Phase Change Rate: 2147483647

Grandsource Clock
-----
Grandsource Clock Identity      : 00:00:00:00:00:00:00:01
Grandsource Clock Quality
  Class                         : 6
  Accuracy                      : 35
  Offset (log variance)         : 0
Priority1                       : 0
Priority2                       : 0
```

Command History

Release	Modification
10.10.1000	Command introduced on the 6300 Switch Series.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

show ptp statistics

show ptp statistics [<IFNAME>]

Description

Shows PTP port statistics.

Parameter	Description
<IFNAME>	Optional. Specifies the interface name.

Examples

Showing PTP port statistics:

```
switch# show ptp statistics
```

PTP Interface Statistics

	Received Packets	Sent Packets	Discarded Packets	Lost Packets
Interface: 1/1/15				
Announce	0	1019	0	0
Sync	0	2038	0	0
Signaling	0	0	0	0
DelayReq	0	0	0	0
DelayResp	0	0	0	0
FollowUp	0	0	0	0
PdelayReq	81957	655750	0	0
PdelayResp	655750	81957	0	0
PdelayRespFollowUp	655749	81957	0	0
Management	0	0	0	0

	Received Packets	Sent Packets	Discarded Packets	Lost Packets
Interface: 1/1/16				
Announce	0	1019	0	0
Sync	0	2038	0	0
Signaling	0	0	0	0
DelayReq	0	0	0	0
DelayResp	0	0	0	0
FollowUp	0	0	0	0
PdelayReq	81957	655750	0	0
PdelayResp	655750	81957	0	0
PdelayRespFollowUp	655749	81957	0	0
Management	0	0	0	0

Showing PTP port statistics for the specified interface:

```
switch# show ptp statistics 1/1/15
```

PTP Interface Statistics

	Received Packets	Sent Packets	Discarded Packets	Lost Packets
Interface: 1/1/15				
Announce	0	1024	0	0
Sync	0	2048	0	0
Signaling	0	0	0	0
DelayReq	0	0	0	0
DelayResp	0	0	0	0
FollowUp	0	0	0	0
PdelayReq	81957	655750	0	0
PdelayResp	655750	81957	0	0
PdelayRespFollowUp	655749	81957	0	0
Management	0	0	0	0

Command History

Release

Modification

10.08

Command introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Administrators or local user group members with execution rights for this command.

show ptp time-property

show ptp time-property

Description

Shows PTP clock-time properties for the PTP device.

Parameter	Description
-----------	-------------

Example

Showing PTP clock time properties:

```
switch # show ptp time-property
PTP Clock Time Property
-----
Current UTC Offset Valid : FALSE
Current UTC Offset      : 37
Leap59                  : FALSE
Leap61                  : FALSE
Time Traceable          : FALSE
Frequency Traceable     : FALSE
PTP Timescale           : FALSE
Synchronization Uncertain : FALSE
Time Source             : 160
```

Command History

Release	Modification
10.10.1000	Command introduced on the 6300 Switch Series.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300 (only R8S89A, R8S90A)	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

show running-config ptp

show running-config ptp

Description

Shows PTP running configuration related information.

Example

Showing PTP running configuration information (boundary clock is available only on the 6300 Switch Series models R8S89A and R8S90A that first released with AOS-CX 10.10.1000):

```
switch# show running-config ptp
ntp profile smpte
enable
clock-step two-step
transport-protocol ipv4
mode boundary peer-to-peer
interface 1/1/15
no shutdown
ip address 30.1.1.1/16
ntp enable
```

Command History

Release	Modification
10.08 or later	Command introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

transport-protocol

```
transport-protocol {ethernet | ipv4}
no transport-protocol
```

Description

Sets the transport protocol for PTP packets. In the case of IPv4, the UDP check-sum is reset. There is no default transport-protocol. The **no** form of this command disconnects the clock from its source.

Parameter	Description
ethernet	Specifies the Ethernet (Layer 2) transport protocol.
ipv4	Specifies the IPv4 transport protocol.

Usage

Mandatory command to start the PTP clock.

Example

Setting the Ethernet transport protocol for PTP packets:

```
switch(config-ptp)# transport-protocol ethernet
```

Removing the transport protocol for PTP packets:

```
switch(config-ptp) # no transport-protocol
```

Command History

Release	Modification
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config-ptp	Administrators or local user group members with execution rights for this command.

Audio-Video Bridging (AVB)

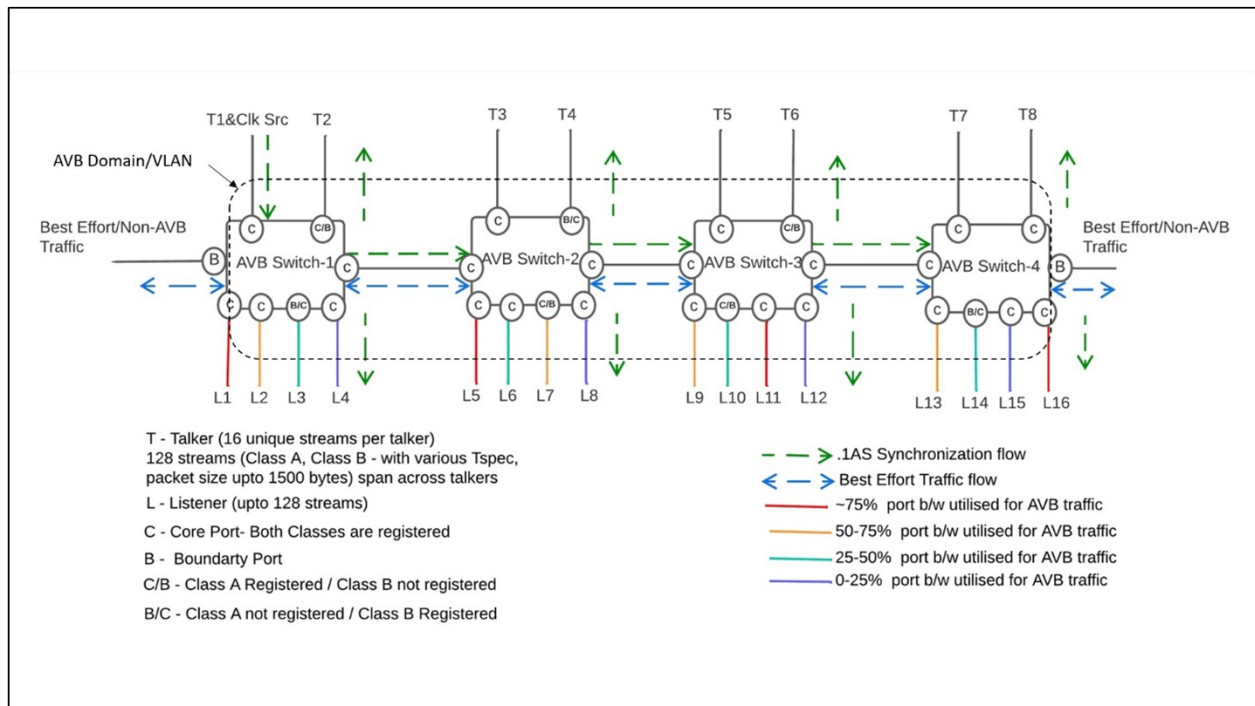


AVB is supported only on the Aruba 6300 and 8360 Switch Series (JL717C, JL718C, JL721C, JL722C, R8S89A, and R8S90A).

Audio-Video Bridging (AVB) defines features, options, configurations, defaults, protocols, and procedures of bridges, stations, and Local Area Networks (LANs) that are necessary to build networks that are capable of transporting time-sensitive audio and/or video data streams. AVB operates on the concept of stream reservation. Stream Reservation Protocol (SRP) is an amendment to the IEEE 802.1Q standard (standardized separately as 802.1Qat) to provide end-to-end management of resource reservations for data streams requiring guaranteed Quality of Service (QoS) in Local Area Networks (LANs).

AOS-CX switch series support only as AV bridge. AVB is supported only over default STP instance(CIST). AVB is activated through feature-pack.

Figure 1 *Illustrative AVB topology with the validated profile*



Multiple Stream Registration Protocol

Multiple Stream Registration Protocol (MSRP) is a signaling protocol that provides end-stations with the ability to reserve network resources that guarantee the transmission and reception of data streams across a bridged network with the requested QoS. The end stations are referred to as Talkers (devices that produce data streams) and Listeners (devices that consume data streams). MSRP defines payloads and rules for propagating registrations for both Talkers and Listeners. MSRP relies on Multiple Reservation Protocol (MRP) for its communication. MRP (IEEE 802.1Q-2018 section 10) is a robust, efficient protocol for declaring attributes to be registered in a database in each port of each bridge (optionally, station) in a bridged network.

Stream-ID should always be generated from Source MAC of talker and multicast AVB streams are supported. MSRP rank-based prioritization and pre-emption is not supported. MSRP Listener registration is considered when the corresponding talker stream is registered. Streams with non-AVB VLAN are sent as Talker failed with error code 16 (Changes in FirstValue for a registered StreamID). As per protocol functionality, standard failure codes 1, 5, 8, 14, 15, 16, and 19 are generated.

MSRP timer configuration

Whenever MSRP timer config is changed, for the protocol to operate more efficiently, and with less likelihood of unwanted deregistrations, the following relationships should be maintained between timer values used by peer Participants:

LeaveTime should be at least twice the maximum JoinTime, plus six times the timer resolution, to allow reregistration after a Leave or LeaveAll even if a message is lost.

To minimize the volume of rejoining traffic generated following a LeaveAll, the value chosen for LeaveAllTime should be large relative to LeaveTime.

Forwarding and queueing for time sensitive streams

Forwarding and Queueing for Time-Sensitive Streams (FQTSS) is an IEEE 802.1Qav standard to describe a set of procedures and tools that can be used to support the forwarding and queueing requirements for AVB.

QoS Configuration

AVB requires the Class-A and Class-B traffic to be given the highest priority and thus manages these class-queues and a control-queue (reserved/future-use). While AVB manages the (3) AVB queues, flexibility is available to customize the mapping of the local-priority to any (5) non-AVB queues. The following is one of the suggested templates for Queue and Schedule profile:

Among the (5) non-AVB queues, queue 4 has highest priority and all protocol control packet (BPDUs) exchange happens in queue 4. So transmitting data traffic at line rate in queue 4 can cause control packet (BPDUs) to get dropped and result in loss of protocol convergence. It is recommended to keep the overall control and data traffic within 25% port speed that are egressing through AVB egress port.

Ensure that the QoS configuration is applied before AVB enablement for proper queueing and bandwidth allocation. It is suggested not to change the Queue-mapping or add conflicting ACLs when AVB is enabled as it can lead to undefined traffic-handling, drops, or affect the latency.

If the user-configured Queue-profile is incorrect when AVB is operational, AVB must be disabled, then the valid 5-Queue profile must be correctly applied, and AVB re-enabled. Do not change the Queue/Schedule profile when AVB is operational. PCP values for class-A and class-B are fixed to 3 and 2, respectively.

The example queue-profile and schedule-profile configuration is given below:

```
switch(config)# qos queue-profile newQprofile
switch(config-queue) # map queue 4 local-priority 7
switch(config-queue) # map queue 4 local-priority 6
switch(config-queue) # map queue 3 local-priority 5
switch(config-queue) # map queue 3 local-priority 4
switch(config-queue) # map queue 2 local-priority 3
switch(config-queue) # map queue 2 local-priority 2
switch(config-queue) # map queue 1 local-priority 1
switch(config-queue) # map queue 0 local-priority 0
switch(config-queue)# exit
switch(config-queue)# qos schedule-profile tmp_msrp-profile
switch(config-schedule)# strict queue 4
switch(config-schedule)# dwrr queue 3 weight 1
switch(config-schedule)# dwrr queue 2 weight 1
switch(config-schedule)# dwrr queue 1 weight 1
switch(config-schedule)# dwrr queue 0 weight 1
switch(config-schedule)#exit
switch(config)# apply qos queue-profile newQprofile schedule-profile tmp_msrp-profile
```



It is suggested to keep the top user-manageable queue (i.e. Queue-4) as Strict-Priority (SP) to avoid drops in the highest priority non-AVB and control traffic.

AVB caps the maximum reservation to 75% of port bandwidth so that non-AVB traffic is not starved. In case of congestion or over-subscription, the non-AVB traffic >25% is affected in accordance to the user-defined queue-mapping with Queue-0 getting the lowest precedence.

Traffic gets remapped on entering an SRP domain carrying the priority value associated with domain is forward by the bridge at the domain boundary using a lower priority. Any non-AVB traffic having PCP values 2 and 3 with the same as SRVLAN ID, Cos value is remarked to 0 at the egress.

In case of any replication traffic congestion, packet drops can be seen across the queues/priorities.

Generalized precision time protocol configuration

gPTP (generalized Precision Time Protocol) incorporates the functionality of PTP as defined by IEEE 802.1as. It operates over a LAN network and takes care of timing, synchronization and PTP functionality requirements over the AVB network.

AVB requires dot1as profile to be configured for its timing and synchronization requirements. If no PTP profile is enabled, enabling AVB enables the dot1as profile internally. If any other PTP profile is configured, AVB enablement fails, the PTP profile needs to be removed or fixed and AVB needs to be re-enabled. Removing or disabling the dot1as profile results in AVB going to non-operational.

VLAN configuration

AVB enable internally enables the AVB VLAN if it is not configured. AVB VLAN is a mandatory requirement for the functionality. Removal of AVB VLAN results in AVB going to non-operational. If MVRP is enabled globally, then **AVB enable** does not internally enable the static VLAN and waits for AVB VLAN to be created dynamically. If AVB is enabled first followed by MVRP, then the static VLAN is already created by **AVB enable**. In this scenario, AVB continues to operate with the static VLAN. Manually remove the AVB static VLAN using **no vlan** if dynamic creation of AVB VLAN (through MVRP registration) is expected.

Feature exclusions

AVB supports static or dynamic (based on MVRP registration) AVB VLAN configuration and only one AVB VLAN is supported.

User should not configure the following for AVB:

- LAG
- VSF
- VSX (ISL and MCLAGs)
- Split cables
- MTU, EEE, Flow control

AVB commands

avb

```
avb
no avb
```

Description

Enables the AVB feature and creates the AVB context. Enables user to exercise various configuration options available under AVB.

The **no** form of this command removed the AVB configuration in the global context.

Examples

Enable AVB:

```
switch(config)# avb
switch(config-avb)#
```

Disable AVB:

```
switch(config)# no avb
AVB configuration will be deleted.
Continue (y/n)? y
```

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.

clear msrp statistics

clear msrp statistics [<IF-NAME>]

Description

Clears the MSRP counters for the given interfaces.

Parameter	Description
<IFNAME>	Specifies the interface name.

Examples

Clear the MSRP counter for interface 1/1/1:

```
switch# clear msrp statistics 1/1/1
```

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Administrators or local user group members with execution rights for this command.

enable

enable
no enable

Description

Enables AVB globally.
The **no** form of this command disables AVB globally.

Examples

Enable audio video bridging:

```
switch(config)# avb
switch(config-avb)# enable
```

Disable audio video bridging:

```
switch(config)# avb
switch(config-avb)# no enable
```

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.

max fan in ports

max-fan-in-ports <NUMPORTS>
no max-fan-in-ports <NUMPORTS>

Description

Configures the AVB maximum number of fan in ports globally. The default maximum is 10. Whenever the max-fan-in-ports is changed, AVB disable and enable are required to make it operational.
The **no** form of this command sets the AVB maximum number of fan in ports to default.

Parameter	Description
<NUMPORTS>	Specifies the maximum number of fan in ports. Range: 1-10. Default: 10.

Examples

Enable AVB max-fan-in-ports 3:

```
switch(config)# avb
switch(config-avb)# max-fan-in-ports 3
Disable and Enable AVB to reflect max-fan-in-ports configuration as operational.
```

Disable AVB VLAN 3 max-fan-in-ports and setting to default:

```
switch(config)# avb
switch(config-avb)# no max-fan-in-ports 3
Disable and Enable AVB to reflect max-fan-in-ports configuration as operational.
```

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.

msrp

```
msrp
no msrp
```

Description

Configures the MSRP protocol on the interface
The **no** form of this command disables the MSRP protocol on the interface.

Examples

Enable MSRP protocol on the interface:

```
switch(config-if)# msrp
```

Disable MSRP protocol on the interface:

```
switch(config-if)# no msrp
```


Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	config-if	Administrators or local user group members with execution rights for this command.

msrp timer join

```
msrp timer join <TIME-INTERVAL>  
no msrp timer join <TIME-INTERVAL>
```

Description

Configures the MSRP join timer on a MSRP enabled interface. The timer configures the time to wait for the MSRP Protocol Data Units (PDUs) to be sent out of the interface. The timer units are in centiseconds. The default is 20 centiseconds.

The **no** form of this command sets the configuration back to the default value.

Parameter	Description
<TIME-INTERVAL>	Specifies the time interval in centiseconds. Range: 20-100. Default: 20.

Examples

Configure MSRP timer join to 50 centiseconds:

```
switch(config-if)# msrp timer join 50
```

Remove MSRP timer join configuration and set to default timers:

```
switch(config-if)# no msrp timer join 50
```

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	config-if	Administrators or local user group members with execution rights for this command.

msrp timer leave

```
msrp timer leave <TIME-INTERVAL>
no msrp timer leave <TIME-INTERVAL>
```

Description

Configures the MSRP leave timer on a MSRP enabled interface. The timer configures the time to wait for the MSRP registrar state to move from LEAVE state to EMPTY state on the interface. The timer units are in centiseconds. The default is 300 centiseconds.

The **no** form of this command sets the configuration back to the default value.

Parameter	Description
<TIME-INTERVAL>	Specifies the time interval in centiseconds. Range: 40-1000000. Default: 300.

Examples

Configure MSRP timer leave to 500 centiseconds:

```
switch(config-if)# msrp timer leave 500
```

Remove MSRP timer leave configuration and set to the default configuration:

```
switch(config-if)# no msrp timer leave 500
```

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	config-if	Administrators or local user group members with execution rights for this command.

msrp timer leaveall

```
msrp timer leaveall <TIME-INTERVAL>
no msrp timer leaveall <TIME-INTERVAL>
```

Description

Configures the MSRP leaveall timer on a MSRP enabled interface. The timer configures the time to wait for the leaveall messages to be sent on the interface. The timer units are in centiseconds. The default is 1000 centiseconds.

The **no** form of this command sets the configuration back to the default value.

Parameter	Description
<TIME-INTERVAL>	Specifies the time interval in centiseconds. Range: 500-1000000. Default: 1000.

Examples

Configure MSRP leaveall timer to 500 centiseconds:

```
switch(config-if)# msrp timer leaveall 500
```

Remove MSRP leaveall timer configuration and set to default configuration:

```
switch(config-if)# no msrp timer leaveall 500
```

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	config-if	Administrators or local user group members with execution rights for this command.

show avb domain

```
show avb domain
```

Description

Displays the global AVB domain status.

Examples

Display the global AVB domain status:

```
switch# show avb domain
AVB state                : operational
AVB VLAN                 : 2
Max Fan in Ports         : 10
AVB Class-A
  Priority Code Point     : 3
  Number of Core Ports    : 1
  Number of Boundary Ports : 1
AVB Class-B
  Priority Code Point     : 2
  Number of Core Ports    : 1
  Number of Boundary Ports : 1
```

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show avb interface

show avb interface [<IFNAME> | brief]

Description

Displays the AVB interface information.

Parameter	Description
<IFNAME>	Specifies the interface name.
brief	Shows information in brief format.

Examples

Display the AVB interface status:

```
switch# show avb interface 1/1/2
AVB state : Enabled
MSRP State : Enabled
PTP State : Enabled, asCapable
  Neighbor Propagation Delay : 98
  Port State : clock_source
  Peer Mean Path Delay : 99
AVB readiness state : operational
Allocated BW (Kbit/s) : 7500000
Used BW (Kbits/s) : 27456
Available BW (Kbits/s) : 7472544
Per-class value
-----
Tx srClassVID      2          2
Rx srClassVID      2          2
Tx PCP             3          2
Rx PCP             3          -
Domain State       Core      Boundary
```

Display the AVB interface status in brief:

```
switch# show avb interface brief
Ethernet  Peer PCP  Peer  AVB Core  Allocated  Available
Used
```

Interface (Kbit/s)	A B	VLAN	A B	(Kbit/s)	(Kbit/s)

1/1/1 0	3 2	2	YES YES	0	0
1/1/2 27456	3 2	2	YES NO	7500000	7472544

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show msrp interface

show msrp interface [<IFNAME> | brief]

Description

Displays MSRP interface information.

Parameter	Description
<IFNAME>	Specifies the interface name.
brief	Shows information in brief format.

Examples

Display the MSRP interface 1/1/1 information:

```
switch# show msrp interface 1/1/1
Stream Id           : 00:11:01:00:00:01:00:01
Stream Age          : N/A
Peer Participant    : Talker
Registration Attribute
Type                : Talker-advertise
Registrar State     : Registered(IN)
Last registered Event : JoinIn
Declaration Attribute
Type                : Listener-ready
Applicant State     : Quiet Active(QA)
Last Declared Event : JoinIn
```

Declared Failure Info : N/A

Display the MSRP interface 1/1/2 information in brief:

```
switch# show msrp interface 1/1/2 brief
Stream-Id          Peer      Applicant  Registrar  Reservation
Failure
-----
-----
00:11:01:00:00:01:00:01 Listener  qa         in         Reserved
N/A
```

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show msrp statistics

show msrp statistics [<IF-NAME>]

Description

Displays the MSRP statistics for MSRP enabled interfaces.

Parameter	Description
<IFNAME>	Specifies the interface name.

Examples

Display the MSRP interface 1/1/1 statistics:

```
switch# show msrp statistics 1/1/1
Total PDU Transmitted : 7029
Total PDU Received    : 7036
Leaveall Tx count      : 2242
Leaveall Rx count      : 2260
Domain Tx count       : 2242
Domain Rx count       : 4908
```

	TALKER-ADV	TALKER-FAILED	LISTENER-ASKING-FAILED	LISTENER-READY-FAILED
LISTENER-READY				
Rx-New 0	0	3	0	0
Rx-In 0	0	0	0	0
Rx-Empty 0	0	0	0	0
Rx-JoinEmpty 0	6545	0	0	0
Rx-JoinIn 0	7170	0	0	0
Rx-Leave 0	3	0	0	0
Tx-New 6	0	0	2	0
Tx-In 0	0	0	0	0
Tx-Empty 0	0	0	0	0
Tx-JoinEmpty 6367	0	0	79	0
Tx-joinIn 7170	0	0	105	0
Tx-Leave 3	0	0	0	0

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show msrp state

show msrp state <STREAM-ID>

Description

Displays all information about a given stream.

Parameter	Description
<STREAM-ID>	Specifies the Stream ID.

Examples

Display the MSRP stream information:

```

switch# show msrp state 00:11:01:00:00:01:00:01
Stream-ID      : 00:11:01:00:00:01:00:01
Stream Talker Port : 1/1/32
Stream Creation time: 1 minute
Destination MAC  : 91:e0:f0:00:fe:00
VLAN            : 2
Priority         : 3 (class-A)
Rank            : low
Accumulated Latency : 20 ns
Max frame size   : 100
Max frame interval : 1 (frames/125 us)
Bandwidth        : 9152 (Kbit/s)
Status          : active
Failure Information : N/A
Failure Bridge    : N/A

```

```

-----
Port   Reg   Appl  Peer      Reg      Decl      Rsvn
  State State Participant Attribute Attribute
-----
status
-----
1/1/31  in    qa    listener  listener_ready  talker_advertise
Reserved
1/1/32  in    qa    talker    talker_advertise  listener_ready    N/A

```

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show msrp streams

```
show msrp streams
```

Description

Displays all information about MSRP streams.

Examples

Display the MSRP stream information:

```

switch# show msrp streams
Stream-Id      DMAC      VLAN  Priority Rank Accumulated
              (Kbit/s)   Status  Talker Port      (class)      Latency

```



```

-----
00:11:01:00:00:01:00:01  91:e0:f0:00:fe:00  2      3 (A)   low  150020
                        9152      active  1/1/1
00:11:01:00:00:01:00:05  91:e0:f0:00:fe:02  2      3 (A)   low  150020
                        9152      active  1/1/1
00:11:01:00:00:01:00:03  91:e0:f0:00:fe:01  2      3 (A)   low  150020
                        9152      active  1/1/1

```

Note: One extra byte (per packet) is considered (to offset the clock diff from the neighbor device) for MSRP stream bandwidth.

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show msrp streams statistics

show msrp streams statistics

Description

Displays the traffic statistics for all streams.

Examples

Display the MSRP traffic statistics:

```

switch# show msrp streams statistics
Stream-Id          DMAC          Class      In Packets  Drop Packets
    In Bytes      Drop bytes
-----
00:11:01:00:00:01:00:01  91:e0:f0:00:fe:00  3 (A)           0           0
                        0           0
00:11:01:00:00:01:00:05  91:e0:f0:00:fe:02  3 (A)           0           0
                        0           0
00:11:01:00:00:01:00:03  91:e0:f0:00:fe:01  3 (A)           0           0
                        0           0
Note: The drop counters are pointing to the ACL drops associated to a given
stream.

```

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show running-config avb

show running-config avb

Description

Displays all configured commands under the AVB context.

Examples

Display configured commands under the AVB context:

```
switch# show running-config avb
avb
    enable
    vlan 3
    max-fan-in-ports 4
```

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

vlan

vlan <VLAN ID>
no vlan <VLAN ID>

Description

Configures the AVB VLAN globally. The default AVB VLAN is 2. Whenever the VLAN is changed, AVB disable and enable are required to make it operational. The **no** form of this command sets the AVB VLAN to default.

Parameter	Description
<VLAN ID>	Specifies the VLAN. Range: 1-4094. Default: 2.

Examples

Enable AVB VLAN 3:

```
switch(config)# avb
switch(config-avb)# vlan 3
Disable and Enable AVB to reflect VLAN configuration as operational.
```

Disable AVB VLAN 3 and setting to default:

```
switch(config)# avb
switch(config-avb)# no vlan 3
Disable and Enable AVB to reflect VLAN configuration as operational.
```

Command History

Release	Modification
10.13.1000	Featured introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.

Upgrade and downgrade scenarios

Upgrade and downgrade scenarios are provided by the Configuration Migration Framework (CMF).

Upgrades

All upgrade scenarios are supported and configuration is migrated.

Downgrades

The following scenario is supported:

Create a checkpoint before upgrading.

Upgrade.

Perform config changes.

Set the startup configuration to the checkpoint created in step 1.

Downgrade



Configuration changes that occur after the checkpoint in step 1 will be lost during the downgrade.

Limitations

The following are the limitations of upgrade and downgrade scenarios provided by CMF.

Table 1: Configuration from and to the same build

From/to	Checkpoint	Running	Startup
Checkpoint	N/A	Supported	Supported
Running	Supported	N/A	Supported
Startup	Supported	Supported	N/A
Config from URL in CLI Format	Supported	Supported	Supported
Config from URL in JSON Format	Supported	Supported	Supported

Table 2: Configuration from an older build to a newer build (upgrade)

From/to	Checkpoint	Running	Startup
Config from URL in CLI Format	Not supported	Not supported	Not supported

From/to	Checkpoint	Running	Startup
Config from URL in JSON Format	Not supported	Not supported	Not supported
Startup	Not supported	Supported	Supported

Table 3: Configuration from a newer build to an older build (downgrade)

From/to	Checkpoint	Running	Startup
Config from URL in CLI Format	Not supported	Not supported	Not supported
Config from URL in JSON Format	Not supported	Not supported	Not supported
Startup	Not supported	Not supported	Not supported

In service software upgrade

In-Service Software Upgrade (ISSU) is the name given to a mechanism used to upgrade a device without affecting the service it is providing. In the case of switches, this means that the traffic won't be interrupted during the upgrade and if an error happens and traffic is interrupted to keep the interruption as short as possible.

Aruba's switch architecture decouples the control plane and the data plane, so it is possible to implement an ISSU mechanism by keeping the data plane up while the control plane manages the upgrade.



ISSU is supported on the Aruba 6400 Switch Series.



ISSU is supported for upgrades between minor releases.



ISSU is supported for VSF. Please refer to the [Limitations](#) section of this document for additional information.

Problem description

The main problem ISSU solves is the need to reboot the switch when doing an upgrade. Before ISSU is implemented, the only supported way of upgrading is through a reboot. ISSU allows this upgrade operation to happen without interrupting traffic, even when there is a reboot of the management modules and an upgrade of line card modules.

Use cases

ISSU is designed to upgrade the current device's firmware without interrupting traffic. This attribute is critical in some scenarios where a network outage would mean a very disruptive event. The following are examples of such scenarios:

- **Health care networks:** In this scenario, the medical devices connected to the switch must not stop sending information to monitoring stations. These devices could be connected to the switch directly or through a wireless access point. In this case, the access points must not be interrupted.

- **IoT networks:** Many small devices depend on network traffic to remain connected to a central command center. If traffic is interrupted, these IoT devices could lose the connection to the central place and stop sending critical data.

Upgrade scope

The current ISSU feature is aimed to do upgrades between major and minor releases. This is suitable for stable networks that need critical bug fixes or security fixes instead of full version upgrades to get new functionality.

Operation

ISSU in the 6400 consists in a set of steps designed to keep the system in a stable state while the firmware is upgraded and to recover this state after such upgrade.

Step 1: Image download and validation

The first step of the process is to download the image with the new version of the firmware and verify it complies with the ISSU rules and the security checks (e.g., valid signed image). These rules are internal checks which verify that the ISSU operation can be performed between the current image running in the switch and the new one.

Upon successful verification the new image will be installed in the bank that doesn't store the image currently running on the switch. The old image will be kept in the alternative bank in case a fail back process needs to be triggered due to an error.

Step 2: Install the image in the standby management module

Once the new image is verified, the Standby Management Module (SMM) will be upgraded with this image. Note that during this process there is no redundancy between MM's and any fatal error in the Active Management Module (AMM) will mean a switch reboot (more on this in the Errors and Recovery section).

Step 3: Freeze the system

Once the SMM is successfully upgraded, the system's state will be frozen. This means that no further configuration will be allowed until ISSU is complete. In fact, no configuration via CLI, REST, or SNMP is allowed once the ISSU process starts (Step 1) to make sure the system doesn't become unstable once ISSU has started. Informational queries are still allowed. Any change in the network will not be reflected until ISSU is complete. E.g., MAC learning. Please refer to [Information not updated during ISSU](#) for additional information.

Step 4: Perform the upgrade for the line card modules

The Line Card (LC) modules will be upgraded at this time, however they will not be rebooted.

Step 5: Recover the previous state

The previous state will be recovered and synchronized with the new firmware running in the SMM. Some features can't recover the previous state at this point and therefore do not support uninterrupted operation during ISSU. Please refer to [Features that don't support uninterrupted operation during ISSU](#) for more information.

Step 6: Failover occurs between the AMM and the SMM

The SMM will take the Active Management Module role using the new firmware version. All sessions will be closed and the user will be required to login again.

Step 7: The previous AMM will upgrade its image

The previous AMM will upgrade its image and take the role of Standby Management Module. The ISSU is complete at this point.

Considerations

The first requirement for ISSU is to ensure traffic continues to pass through the switch uninterrupted during the upgrade operation. If something goes wrong, traffic must be restored as soon as possible. This could mean falling back to the previous image, or in some cases to fall forward to the new one.

The upgrade process can take minutes to complete. During that time as many control plane protocols as possible will remain operational, however some of them need to stop processing packets or avoid reacting to events. Please refer to [Information not updated during ISSU](#) for additional information. The amount of time where they are not operational will be kept as small as possible. The only full control plane disruption is similar to the disruption caused by an HA failover event. This means that ISSU will be as hitless as an HA failover. If a feature is not hitless during an HA failover then it won't be hitless during ISSU. Please refer to [Features that don't support uninterrupted operation during ISSU](#) for additional information.

In order to start ISSU, all modules (i.e., LCs, fabric cards, management modules) must be operational and running. If any of them is in another state (e.g., failover, line cards not ready, etc.) the ISSU process will not start.

Limitations

The ISSU process currently has the following limitations:

Management modules

In order for ISSU to work the switch chassis needs to have two management modules installed.

Line card support

The ISSU version released from 10.11 supports line cards with model numbers ending in A, B, or C i.e. LCs with model numbers following the format "ROX_A", "ROX_B", "ROX_C". Mixed mode (i.e. a chassis with A, B, or C line cards and newer line cards) is supported in ISSU.

Hardware faults

It is recommended that hardware faults are resolved prior to performing ISSU. Fan faults and abnormal temperature statuses should be addressed. Fans may increase in speed during the ISSU process.

Prohibited operations

During ISSU the following operations are not permitted:

- Any hardware hot swap
- Pressing any button

Performing any of these operations during an ISSU could put the system in an error state that could adversely affect the ISSU.

The system may not be able to address failures during the ISSU. E.g., if network protocols require blocking a port, flush MAC, or routing tables, etc. the switch would not be able to react to those tasks until the ISSU is complete.

Scheduled jobs

Since no configuration is allowed during the ISSU, it is possible that some scheduled jobs will fail if they are attempted while the ISSU is in progress. Users can check for the "ISSU is in-progress, configuration is not allowed" message to find out if any scheduled job has failed. It is recommended to not execute any scheduled job during ISSU.

Information not updated during ISSU

During ISSU the following information may not be current until after ISSU completes:

- Link state. Transitions may not be reflected until ISSU is done.
- Temperatures.
- Power consumption values.
- System state as indicated by LEDs.
- MAC Learning.
- Change in MAC for a neighbor via Gratuitous ARP.
- Some counters will stop increasing their value during ISSU.

Show tech

The show tech command is enabled during ISSU using the CLI, however it isn't possible to use it with the REST API.

Features that don't support uninterrupted operation during ISSU

During ISSU the following features may be interrupted until after ISSU completes:

PIM with MSDP

Currently, PIM is not supported in ISSU on RP routers, therefore traffic loss is expected during ISSU on RP routers where PIM and MSDP are enabled.

Dynauthz ISSU

All Dynamic Authorization requests will be rejected with NAK during ISSU operation. Captive portal workflow will be affected due to NAK received for Dynauthz requests during ISSU.

BFD ISSU

It's important to note that this mechanism is only recommended for sessions that have a failure detection time higher than 6 seconds (e.g. Tx=Rx=3000 and Detect Multiplier=2, or Tx=Rx=2000 and Detect Multiplier=3). The setup should have less than 64 sessions, otherwise the ISSU failover might not be hitless for all sessions.

NAE

NAE will have a momentary disruption during ISSU events due to the failover event. See the Network Analytics Engine Guide for more information.

MACsec

ISSU is not allowed when MACsec is enabled on any port of the system.

Feature readiness check

ISSU will fail if the feature ready check does not pass. The feature ready check validates that none of the following features are in a failed state:

- ACLs, object groups, and their applications
- Classifier policies, classes, and their applications
- Port-access policies, classes, and their applications
- Port-access group-based policies, classes, and their applications
- Active PBR actions and their applications

In general, if any of the above is in a failed state Aruba recommends fixing it and then upgrading to the newer operating system image using the system reboot method instead of ISSU. Some of the failed state will remain even after it is fixed on a running system and a reboot is good best practice for clearing it.

There are system validation checks to ensure that all line card part numbers end in an A or B.

VSX environments

In order to minimize a potential downtime, it is strongly recommended that in a VSX topology ISSU is performed at different times for each switch. E.g., run ISSU on the primary switch, verify that it was successful, and then perform ISSU for the secondary switch.

Errors and recovery

During ISSU there are several different error recovery actions, depending on the ISSU step where the error occurs. The system may be able to recover from a single failure, but it is not possible to recover from a double failure. A double failure would require a full switch reboot.

There are two types of errors during ISSU: recoverable errors and unrecoverable errors.

Recoverable errors are handled by aborting the ISSU process and getting back to a stable system without impacting the data plane. Recoverable errors fall back to the current version.

Unrecoverable errors require either a LC reboot or an MM reboot depending on where the error occurs. If some conditions are met (e.g., double failure, MM-LC failure) it is possible that a full switch reboot is needed.

The following are scenarios that would mean an error during ISSU:

Firmware errors

- New version is not a valid target
- SMM fatal errors before the failover

SMM errors

- SMM doesn't boot correctly
- Critical process crash

LC upgrade errors

- An LC failing while executing any of the ISSU steps

Error handling definitions

Possible ways to handle an ISSU error:

Abort Hitless

- ISSU is aborted with no traffic interruption
- No traffic impact
- Recoverable
- Abort hitless before the SMM software upgrade means that ISSU is not started
- Abort hitless after the SMM software upgrade means that SMM reboots to current version
- No upgrade occurs

Abort with non-recoverable errors

- ISSU is aborted and traffic was interrupted
- Traffic Impact
- All affected LCs are moved to an error state
- Affected LCs and SMM rebooted to current version
- No upgrade occurs

Complete with non-recoverable errors

- Upgrade was done, but with errors
- Traffic impact
- All affected LCs are moved to an error state
- Affected LCs rebooted to new version after ISSU is completed
- Upgrade complete

Fail forward with non-recoverable errors

- ISSU operation fails and new image is used for recovery
- Traffic impact
- Chassis reboots to new version (AMM, SMM, LCs)
- Upgrade complete

Error Handling

Each step of the ISSU process has different ways to handle errors. Errors will be managed based on which stage of the ISSU process the system is currently on.

Image download and validation

If during .swi file validation the system finds out that the target image is not suitable for ISSU or that it is not compatible with the current version, the ISSU process will not be started. The error will be handled as abort hitless.

Install image in the SMM

If during the SMM software upgrade the new image fails to install or to boot to a valid state, then the ISSU process will be aborted. The error will be handled as abort hitless and the SMM reboots to current version.

System freeze

If the error happens while the system state is frozen, then the system needs to be thawed. This means taking it out of the frozen state and restoring the previous state without interrupting traffic. Any change done by the ISSU must be reverted. The ISSU process is aborted. The error will be handled as abort hitless. There are some cases where the system cannot be thawed. In this case the LCs will reboot to the new image since the system cannot recover the previous state. The error will be handled as fail forward.

Perform the upgrade for the line card modules

If any of the modules running in the LCs fail to upgrade, the line card will be marked as in an error state and it will wait until the ISSU process is done. The error will be handled as complete with errors.

Recover the previous state

Once the upgrade is complete, the system will start to recover the previous state. If there is an error recovering the line card's previous state, the line card will be marked as in an error state and it will wait until ISSU is over before coming back. The error will be handled as complete with errors.

If the previous state cannot be recovered for all LCs or for the whole system, a system reboot is required to reach a known state. The system will boot with the new image. The error will be handled as fail forward.

Failover

The last step in the ISSU process is to do a failover where the SMM becomes the AMM. If the failover fails, the whole chassis will be rebooted. The system will boot with the new image. The error will be handled as fail forward and the previous AMM will upgrade its image. After the failover, the ISSU process is almost done and the LCs are already running with the new firmware, so a failure when the previous AMM is upgrading would mean a reboot of the MM which restarts with the new image.

Global failures

ISSU is a very complex feature that includes many modules. If a fatal failure (e.g., an OS crash) happens in any MM during any stage of ISSU it would mean an ISSU abort and a system reboot. This is a fail forward scenario, where ISSU fails and the switch reboots to the new version. If the global error happens in the AMM, the error will be handled as fail forward. If the global error happens in the SMM before it is upgraded, the error will be handled as abort hitless. If the global error happens in the SMM after it was upgraded or if all line cards fail at any stage then the error will be handled as fail forward.

Configuration task list

The ISSU feature doesn't have many configuration tasks and it consists mainly of setting the system to a stable state and then triggering the ISSU process.

Pre-ISSU

1. Download the new image and store in the secondary bank.
2. Check management and line modules. The `show modules` command can be used for verification.

```
switch# show module
```

```
Management Modules
```

```
=====
```

Name	Product Number	Description	Serial Number	Status
1/1	R0X31A	6400 Management Module	SG9ZKMG327	Active (local)
1/2	R0X31A	6400 Management Module	SG9ZKMG405	Standby

```
Line Modules
```

```
=====
```

Name	Product Number	Description	Serial Number	Status
1/3	R0X42A	6400 24p 10GT 4SFP56 Mod	SG0ZKMS156	Ready
1/4	R0X41A	6400 48p SR5 CL6 PoE 4SFP56 Mod	SG0ZKMR193	Ready

Start ISSU

The command to start the ISSU process is `issu update-software`:

```
switch# issu update-software

This command will perform an in-service software upgrade
using pre-staged secondary operating system image
FL.10.10.1000
This will save the current running configuration

Continue (y/n)? y

Starting in-service software upgrade.
```

Starting an ISSU will cause the running configuration to be saved in case an error is encountered that requires a reboot to recover the switch. Additionally, a special configuration checkpoint will be stored to disk, which can be used to go back to the pre-ISSU firmware. During the ISSU process all management methods (CLI, REST, WebUI) will be blocked from making configuration changes to the switch. Only non-config commands are allowed. The configuration block is active from the time ISSU starts until the time the ISSU failover is complete.

Troubleshooting

Once ISSU is started, it cannot be stopped. The current status of the ISSU process can be checked with the `show issu` and `show issu brief` commands.

In progress ISSU:

```
switch# show issu
ISSU Summary
=====
ISSU Status      : In Progress
Current Version  : FL.10.10.0001 Upgrade Version : FL.10.10.1000
Upgrade Image    : secondary          Start Date      : 2022-05-26 20:27:01
Last ISSU Result: -- (--)

ISSU Progress
=====
Upgrade Operation      Status      Start Date
-----
Initiate ISSU          Complete    2022-05-26 20:27:01
Validate System Readiness Complete    2022-05-26 20:27:01
Upgrade Standby Management Module In Progress 2022-05-26 20:27:09
Upgrade Line Modules    Pending     --
Prepare for Switchover  Pending     --
Finalize Upgrade        Pending     --
ISSU Complete          Pending     --
```

```
switch# show issu brief
ISSU Summary
=====
```

```
ISSU Status      : In Progress
Current Version  : FL.10.10.0001 Upgrade Version : FL.10.10.1000
Upgrade Image    : secondary          Start Date      : 2022-05-26 20:27:01
Last ISSU Result: -- (--)
```

Complete ISSU

```
switch# show issu
ISSU Summary
=====
ISSU Status      : Ready
Current Version  : FL.10.10.1000 Upgrade Version : --
Upgrade Image    : --          Start Date      : --
Last ISSU Result: Completed (Without errors)

ISSU Progress
=====
Upgrade Operation          Status          Start Date
-----
Initiate ISSU              Complete      2022-05-26 20:27:01
Validate System Readiness  Complete      2022-05-26 20:27:01
Upgrade Standby Management Complete      2022-05-26 20:27:09
Upgrade Line Modules       Complete      2022-05-26 20:28:48
Prepare for Switchover     Complete      2022-05-26 20:30:21
Finalize Upgrade           Complete      2022-05-26 20:30:23
ISSU Complete              Complete      2022-05-26 20:32:07
```

```
switch# show issu brief
ISSU Summary
=====
ISSU Status      : Ready
Current Version  : FL.10.11.0000-3140- Upgrade Version : --
Upgrade Image    : --          Start Date      : --
Last ISSU Result: Completed (Without errors)
```

The output of these commands will show the current step in the ISSU process and the current state. E.g., if an error occurs and the error action taken.

There is also a debug command: `debug issu all severity debug`. This command will show information useful for troubleshooting and for technical support.

Frequently asked questions

Can ISSU be stopped?

Not currently.

What happens if a network event occurs during ISSU (e.g., a route reconvergence, a BFD update)?

Depending on the current ISSU step, these events could not be attended. Once ISSU is complete the system will start attending such events, but it is possible that some of them are missing. A network reconvergence may occur and possibly even some disruption (e.g., BFD blocking a port) is possible. It is important to note that all the limitations that apply to a regular failover apply to ISSU as well.

Can ISSU be performed between major versions?

Currently, ISSU is only supported to upgrade between minor versions. 10.10.0001 is the first release enabling ISSU support. ISSU is meant to be used to upgrade from this version to future minor releases.

Hot-patch software

Overview

Hot-patching provides users of AOS-CX with a way to update running software without rebooting their system. Hot-patches are distributed as signed .patch files and are applied on top of an official AOS-CX image.

A hot-patch can be downloaded from a remote server onto an AOS-CX switch then applied without rebooting the switch. By default, the software patch remains on the switch after it is applied, and is automatically reapplied when the switch reboots. In order to revert an applied hot-patch update, the patch must be disabled. When the hot-patch is disabled, rebooting the box is not necessary, though the hot-patch will still remain on the system. The hot-patch can be removed from the system after disabling the hot-patch.

Once a downloaded hot-patch is removed, it must be downloaded again before it can be reapplied.



Hot-patches are cumulative, meaning that each subsequent patch is a superset of the previous patch(es) for a given software image. When multiple hot-patches are required, the first should be removed after the second has been applied.

Prerequisites

- AOS-CX hot-patch software can be obtained from Aruba customer support, and is identified with a **.patch** extension. The name of the patch indicates the version of AOS-CX to which it can be applied. For example, the patch **FL.10.12.XXXX-YYYY.patch** indicates that the patch can be applied on top of switch image **FL.10.12.XXXX.swi** and brings software up to date with the **FL.10.12.YYYY** release.
- Hot-patch software can be applied only after it has been downloaded to the switch.

Caveats and limitations

- Hot-patch files cannot be managed when the switch is loaded with a .swi image that does not support the hot-patch feature.
- Only restartable daemons can be updated with a hot-patch file.
- Each hot-patch applies to only one release image and each subsequent hot-patch for that release image is cumulative. This means that applying the most recent hot-patch provides all the same fixes included in previous hot-patches. As such, only one hot-patch can be applied at a time.
- incompatible and un-applied hot-patches are allowed to remain on the system and in the config to support rollback to the previous version of the software, and to automatically apply any hot-patches that have been previously applied.
- A maximum of ten hot-patches are allowed to be present or preconfigured on the system at a time.
- A daemon/service may restart if it has a hard dependency on another service that gets restarted by being hot-patched. Upon rebooting, hot-patches are reapplied after daemons have started.
- The **dpsed** and **hpe-cardd** daemon cannot have a hot-patch applied while a switchover or failover is in progress.

To apply a hot-patch update

1. Request a hot-patch from Aruba customer support, then place the hot-patch on an FTP-SFTP server.
2. Use the **copy** command to copy the hot-patch from the remote server to the standalone switch or VSF.
3. Use the **hot-patch apply** command to apply the patch.

To unapply hot-patch update

This option only unapplies the patch, but does not remove the patch from the device. The patch can be reapplied again since it is still on the device. If an unapplied patch needs to be removed/deleted from the switch, it can be removed with the command **hot-patch remove <patch>.patch**.

```
switch(config)# show hot-patch
Name                               Status
-----
FL_10_12_0001-0002.patch          Not applied
switch(config)# hot-patch apply FL_10_12_0001-0002.patch
switch(config)#
switch(config)# show hot-patch
Name                               Status
-----
FL_10_12_0001-0002.patch          Applied

switch(config)# no hot-patch apply FL_10_12_0001-0002.patch
switch(config)# show hot-patch
Name                               Status
-----
FL_10_12_0001-0002.patch          Not applied

switch(config)#
switch(config)# hot-patch remove FL_10_12_0001-0002.patch
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.
switch(config)#
switch(config)# show hot-patch
No hot-patch found or configured
switch(config)#
```

To remove/delete a patch:

An applied hot-patch can be both unapplied and removed from the device with the command **hot-patch remove <patch-name>.patch**. The patch cannot be reapplied unless it is downloaded again.

Checkpoints

A checkpoint is a snapshot of the running configuration of a switch and its relevant metadata during the time of creation. Checkpoints can be used to apply the switch configuration stored within a checkpoint whenever needed, such as to revert to a previous, clean configuration. Checkpoints can be applied to other switches of the same platform. A switch is able to store multiple checkpoints.

Checkpoint types

The switch supports two types of checkpoints:

- **System generated checkpoints:** The switch automatically generates a system checkpoint whenever a configuration change occurs.

- **User generated checkpoints:** The administrator can manually generate a checkpoint whenever required.

Maximum number of checkpoints

- Maximum checkpoints: 64 (including the startup configuration)
- Maximum user checkpoints: 32
- Maximum system checkpoints: 32

User generated checkpoints

User checkpoints can be created at any time, as long as one configuration difference exists since the last checkpoint was created. Checkpoints can be applied to either the running or startup configurations on the switch.

All user generated checkpoints include a time stamp to identify when a checkpoint was created.

A maximum of 32 user generated checkpoints can be created.

System generated checkpoints

System generated checkpoints are automatically created by default. Whenever a configuration change occurs, the switch starts a timeout counter (300 seconds by default). For each additional configuration change, the timeout counter is restarted. If the timeout expires with no additional configuration changes being made, the switch generates a new checkpoint.

System generated checkpoints are named with the prefix `CPC` followed by a time stamp in the format `<YYYYMMDDHHMMSS>`. For example: `CPC20170630073127`.

System checkpoints can be applied using the checkpoint rollback feature or copy command.

A maximum of 32 system checkpoints can be created. Beyond this limit, the newest system checkpoint replaces the oldest system checkpoint.

Supported remote file formats

You can restore a switch configuration by copying a switch configuration stored on a USB drive or a remote network device through SFTP/TFTP. The remote file formats that the switch supports depends on where you plan to restore the checkpoint.

Restoring a checkpoint to a...	File type supported
Running configuration	▪ CLI ▪ JSON ▪ Checkpoint
Startup configuration	▪ JSON ▪ Checkpoint
Specified checkpoint	Specified checkpoint

Rollback

The term rollback is used to refer to when a switch configuration is reverted to a pre-existing checkpoint.

For example, the following command applies the configuration from checkpoint `ckpt1`. All previous configurations are lost after the execution of this command: `checkpoint rollback ckpt1`

You can also specify the rollback of the running configuration or of the startup configuration with a specified checkpoint, as shown with the following command: `copy checkpoint <checkpoint-name> {running-config | startup-config}`

Checkpoint auto mode

Checkpoint auto mode configures the switch with failover support, causing it to automatically revert to a previous configuration if it becomes inoperable or inaccessible due to configuration changes that are being made.

After entering checkpoint auto mode, you have a set amount of time to add, remove, or modify the existing switch configuration. To save your changes, you must execute the `checkpoint auto confirm` command before the auto mode timer expires. If you do not execute the `checkpoint auto confirm` command within the specified time, all configuration changes you made are discarded and the running configuration reverts to the state it was before entering checkpoint auto mode.

Testing a switch configuration in checkpoint auto mode

Process overview:

1. Enable the checkpoint auto mode.
2. To save the configuration, enter the `checkpoint auto confirm` command before the specified time set in step 1.

Checkpoint commands

checkpoint auto

`checkpoint auto <TIME-LAPSE-INTERVAL>`

Description

Starts auto checkpoint mode. In auto checkpoint mode, the switch temporarily saves the runtime configuration as a checkpoint only for the specified time lapse interval. Configuration changes must be saved before the interval expires, otherwise the runtime configuration is restored from the temporary checkpoint.

Parameter	Description
<code><TIME-LAPSE-INTERVAL></code>	Specifies the time lapse interval in minutes. Range: 1 to 60.

Usage

To save the runtime checkpoint permanently, run the **checkpoint auto confirm** command during the time lapse interval. The filename for the saved checkpoint is named **AUTO <YYYYMMDDHHSS>**. If the **checkpoint auto confirm** command is not entered during the specified time lapse interval, the previous runtime configuration is restored.

Examples

Confirming the auto checkpoint:

```
switch# checkpoint auto 20
Auto checkpoint mode expires in 20 minute(s)
switch# WARNING Please "checkpoint auto confirm" within 2 minutes
switch# checkpoint auto confirm
checkpoint AUTO20170801011154 created
```

In this example, the runtime checkpoint was saved because the **checkpoint auto confirm** command was entered within the value set by the **time-lapse-interval** parameter, which was 20 minutes.

Not confirming the auto checkpoint:

```
switch# checkpoint auto 20
Auto checkpoint mode expires in 20 minute(s)
switch# WARNING Please "checkpoint auto confirm" within 2 minutes
WARNING: Restoring configuration. Do NOT add any new configuration.
Restoration successful
```

In this example, the runtime checkpoint was reverted because the **checkpoint auto confirm** command was not entered within the value set by the **time-lapse-interval** parameter, which was 20 minutes.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint auto confirm

checkpoint auto confirm

Description

Signals to the switch to save the running configuration used during the auto checkpoint mode. This command also ends the auto checkpoint mode.

Usage

To save the runtime checkpoint permanently, run the **checkpoint auto confirm** command during the time lapse value set by the **checkpoint auto TIME-LAPSE-INTERVAL** command. The generated checkpoint name will be in the format **AUTO <YYYYMMDDHHSS>**. If the **checkpoint auto confirm** command is not entered during the specified time lapse interval, the previous runtime configuration is restored.

Examples

Confirming the auto checkpoint:

```
switch# checkpoint auto confirm
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint diff

```
checkpoint diff {<CHECKPOINT-NAME1> | running-config | startup-config}
               {<CHECKPOINT-NAME2> | running-config | startup-config}
```

Description

Shows the difference in configuration between two configurations. Compare checkpoints, the running configuration, or the startup configuration.

Parameter	Description
{<CHECKPOINT-NAME1> running-config startup-config}	Selects either a checkpoint, the running configuration, or the startup configuration as the baseline.
{<CHECKPOINT-NAME2> running-config startup-config}	Selects either a checkpoint, the running configuration, or the startup configuration to compare.

Usability

The output of the **checkpoint diff** command has several symbols:

- The plus sign (+) at the beginning of a line indicates that the line exists in the comparison but not in the baseline.
- The minus sign (-) at the beginning of a line indicates that the line exists in the baseline but not in the comparison.

Examples

In the following example, the configurations of checkpoints **cp1** and **cp2** are displayed before the **checkpoint diff** command, so that you can see the context of the **checkpoint diff** command.

```
switch# show checkpoint cp1
Checkpoint configuration:
!
!Version AOS-CX XL.10.00.0002
!Schema version 0.1.8
```

```

module 1/1 product-number j1363a
!
!
!
!
!
!
vlan 1,200
interface 1/1/1
    no shutdown
    ip address 1.0.0.1/24
interface 1/1/2
    no shutdown
    ip address 2.0.0.1/24

```

```

switch# show checkpoint cp2
Checkpoint configuration:
!
!Version AOS-CX XL.10.00.0002
!Schema version 0.1.8
module 1/1 product-number j1363a
!
!
!
!
!
!
vlan 1,200,300
interface 1/1/1
    no shutdown
    ip address 1.0.0.1/24
interface 1/1/2
    no shutdown
    ip address 2.0.0.1/24

```

```

switch# checkpoint diff cp1 cp2
--- /tmp/chkpt11501550258421    2017-08-01 01:17:38.420514016 +0000
+++ /tmp/chkpt21501550258421    2017-08-01 01:17:38.420514016 +0000
@@ -9,7 +9,7 @@
!
!
!
-vlan 1,200
+vlan 1,200,300
 interface 1/1/1
     no shutdown
     ip address 1.0.0.1/24

```

```

switch# checkpoint diff chkpt01 chkpt02
--- /tmp/chkpt011607564301327
+++ /tmp/chkpt021607564301353
@@ -1,7 +1,7 @@
!
!Version AOS-CX PL.10.06.0100V
!export-password: default
-hostname Switch
+hostname Switch1
 user admin group administrators password ciphertext

```

```
AQBapTyg9tpaiAaTfSVV5eNdFzOORRvZ6CMpglh1P+LQUHQLYgAAAGAhmRqFbkNvrgy2SBVk7H8C5hvg/I
ib8rWYFZLEaSCrobNP9EwMu+hLNM0xmsh45yG8dncP7WkxjwrW4p4Qra6dVfr0EW8xh/lpQf8F/2Wki20L
c9JLXiYge7ti0H6cVn+G
radius-server tracking interval 60
no usb

switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint post-configuration

checkpoint post-configuration

no checkpoint post-configuration

Description

Enables creation of system generated checkpoints when configuration changes occur. This feature is enabled by default.

The **no** form of this command disables system generated checkpoints.

Usage

System generated checkpoints are automatically created by default. Whenever a configuration change occurs, the switch starts a timeout counter (300 seconds by default). For each additional configuration change, the timeout counter is restarted. If the timeout expires with no additional configuration changes being made, the switch generates a new checkpoint.

System generated checkpoints are named with the prefix **CPC** followed by a time stamp in the format **<YYYYMMDDHHMMSS>**. For example: **CPC20170630073127**.

System checkpoints can be applied using the checkpoint rollback feature or copy command.

A maximum of 32 system checkpoints can be created. Beyond this limit, the newest system checkpoint replaces the oldest system checkpoint.

Examples

Enabling system checkpoints:

```
switch(config)# checkpoint post-configuration
```

Disabling system checkpoints:

```
switch(config)# no checkpoint post-configuration
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint post-configuration timeout

```
checkpoint post-configuration timeout <TIMEOUT>
```

```
no checkpoint post-configuration timeout <TIMEOUT>
```

Description

Sets the timeout for the creation of system checkpoints. The timeout specifies the amount of time since the latest configuration for the switch to create a system checkpoint.

The **no** form of this command resets the timeout to 300 seconds, regardless of the value of the **<TIMEOUT>** parameter.

Parameter	Description
timeout <TIMEOUT>	Specifies the timeout in seconds. Range: 5 to 600. Default: 300.

Examples

Setting the timeout for system checkpoints to 60 seconds:

```
switch(config)# checkpoint post-configuration timeout 60
```

Resetting the timeout for system checkpoints to 300 seconds:

```
switch(config)# no checkpoint post-configuration timeout 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint rename

```
checkpoint rename <OLD-CHECKPOINT-NAME> <NEW-CHECKPOINT-NAME>
```

Description

Renames an existing checkpoint.

Parameter	Description
<OLD-CHECKPOINT-NAME>	Specifies the name of an existing checkpoint to be renamed.
<NEW-CHECKPOINT-NAME>	Specifies the new name for the checkpoint. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-). NOTE: Do not start the checkpoint name with CPC because it is used for system-generated checkpoints.

Examples

Renaming checkpoint **ckpt1** to **cfg001**:

```
switch# checkpoint rename ckpt1 cfg001
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint rollback

```
checkpoint rollback {<CHECKPOINT-NAME> | startup-config}
```

Description

Applies the configuration from a pre-existing checkpoint or the startup configuration to the running configuration.

Parameter	Description
<CHECKPOINT-NAME>	Specifies a checkpoint name.
startup-config	Specifies the startup configuration.

Examples

Applying a checkpoint named **ckpt1** to the running configuration:

```
switch# checkpoint rollback ckpt1
Success
```

Applying a startup checkpoint to the running configuration:

```
switch# checkpoint rollback startup-config
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy checkpoint <CHECKPOINT-NAME> <REMOTE-URL>

```
copy checkpoint <CHECKPOINT-NAME> <REMOTE-URL> [vrf <VRF-NAME>]
```

Description

Copies a checkpoint configuration to a remote location as a file. The configuration is exported in checkpoint format, which includes switch configuration and relevant metadata.

Parameter	Description
<CHECKPOINT-NAME>	Specifies the name of a checkpoint.
<REMOTE-URL>	Specifies the remote destination and filename using the syntax: TFTP format: tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME> SFTP format: sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME> SCP format: scp://USER@{IP HOST}[:PORT]/FILE

Parameter	Description
<code>vrf <VRF-NAME></code>	Specifies a VRF name.

Examples

Copying checkpoint configuration to remote file through TFTP:

```
switch# copy checkpoint ckpt1 tftp://192.168.1.10/ckptmeta vrf default
##### 100.0%
Success
```

Copying checkpoint configuration to remote file through SFTP:

```
switch# copy checkpoint ckpt1 sftp://root@192.168.1.10/ckptmeta vrf default
The authenticity of host '192.168.1.10 (192.168.1.10)' can't be established.
ECDSA key fingerprint is SHA256:FtOm6Uxuxumil7VCwLnhz92H9LkjY+eURbdddOETy50.
Are you sure you want to continue connecting (yes/no)? yes
root@192.168.1.10's password:
sftp> put /tmp/ckptmeta ckptmeta
Uploading /tmp/ckptmeta to /root/ckptmeta
Warning: Permanently added '192.168.1.10' (ECDSA) to the list of known hosts.
Connected to 192.168.1.10.
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy checkpoint <CHECKPOINT-NAME> {running-config | startup-config}

```
copy checkpoint <CHECKPOINT-NAME> {running-config | startup-config}
```

Description

Copies an existing checkpoint configuration to the running configuration or to the startup configuration.

Parameter	Description
<code><CHECKPOINT-NAME></code>	Specifies the name of an existing checkpoint.

Parameter	Description
	Selects whether the running configuration or the startup configuration receives the copied checkpoint configuration. If the startup configuration is already present, the command overwrites the startup configuration.

Examples

Copying **ckpt1** checkpoint to the running configuration:

```
switch# copy checkpoint ckpt1 running-config
Success
```

Copying **ckpt1** checkpoint to the startup configuration:

```
switch# copy checkpoint ckpt1 startup-config
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy checkpoint <CHECKPOINT-NAME> <STORAGE-URL>

```
copy checkpoint <CHECKPOINT-NAME> <STORAGE-URL>
```

Description

Copies an existing checkpoint configuration to a USB drive. The file format is defined when the checkpoint was created.

Parameter	Description
<CHECKPOINT-NAME>	Specifies the name of the checkpoint to copy. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-).
<STORAGE-URL>>	Specifies the name of the target file on the USB drive using the following syntax: <code>usb: / <FILE></code> The USB drive must be formatted with the FAT file system.

Examples

Copying the **test** checkpoint to the **testCheck** file on the USB drive:

```
switch# copy checkpoint test usb:/testCheck
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy <REMOTE-URL> checkpoint <CHECKPOINT-NAME>

```
copy <REMOTE-URL> checkpoint <CHECKPOINT-NAME> [vrf <VRF-NAME>]
```

Description

Copies a remote configuration file to a checkpoint. The remote configuration file must be in checkpoint format.

Parameter	Description
<REMOTE-URL>	Specifies a remote file using the following syntax: TFTP format: <pre>tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME></pre> SFTP format: <pre>sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME></pre> SCP format: <pre>scp://USER@{IP HOST}[:PORT]/FILE</pre>
<CHECKPOINT-NAME>	Specifies the name of the target checkpoint. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-). Required. NOTE: Do not start the checkpoint name with <code>CPC</code> because it is used for system-generated checkpoints.
vrf <VRF-NAME>	Specifies a VRF name. Default: default.

Examples

Copying a checkpoint format file to checkpoint **ckpt5** on the default VRF:

```
switch# copy tftp://192.168.1.10/ckptmeta checkpoint ckpt5
##### 100.0%
100.0%
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy <REMOTE-URL> {running-config | startup-config}

```
copy <REMOTE-URL> {running-config | startup-config } [vrf <VRF-NAME>]
```

Description

Copies a remote file containing a switch configuration to the running configuration or to the startup configuration.

Parameter	Description
<REMOTE-URL>	Specifies a remote file with the following syntax: TFTP format: tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME> SFTP format: sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME> SCP format: scp://USER@{IP HOST}[:PORT]/FILE
{running-config startup-config}	Selects whether the running configuration or the startup configuration receives the copied checkpoint configuration. If the startup configuration is already present, the command overwrites the startup configuration.
vrf <VRF-NAME>	Specifies the name of a VRF. Default: default.

Usage

The switch copies only certain file types. The format of the file is automatically detected from contents of the file. The **startup-config** option only supports the JSON file format and checkpoints, but the **running-config** option supports the JSON and CLI file formats and checkpoints.

When a file of the CLI format is copied, it overwrites the running configuration. The CLI command does not clear the running configuration before applying the CLI commands. All of the CLI commands in the file are applied line-by-line. If a particular CLI command fails, the switch logs the failure and it continues to the next line in the CLI configuration. The event log (**show events -d hpe-config**) provides information as to which command failed.

Examples

Copying a JSON format file to the running configuration:

```
switch# copy tftp://192.168.1.10/runjson running-config
##### 100.0%
Configuration may take several minutes to complete according to configuration file
size
--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%--
Success
```

Copying a CLI format file to the running configuration with an error in the file:

```
switch# copy tftp://192.168.1.10/runcli running-config
##### 100.0%
Configuration may take several minutes to complete according to configuration file
size
--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%--
Some of the configuration lines from the file were NOT applied. Use 'show
events -d hpe-config' for more info.
##### 100.0%
Configuration may take several minutes to complete according to configuration file
size
--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%--
Some of the configuration lines from the file were NOT applied. Use 'show
events -d hpe-config' for more info.
```

Copying a CLI format file to the startup configuration:

```
switch# copy tftp://192.168.1.10/startjson startup-config
##### 100.0%
100.0%
Success
```

Copying an unsupported file format to the startup configuration:

```
switch# copy tftp://192.168.1.10/startfile startup-config
##### 100.0%
100.0%
unsupported file format
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy running-config {startup-config | checkpoint <CHECKPOINT-NAME>}

copy running-config {startup-config | checkpoint <CHECKPOINT-NAME>}

Description

Copies the running configuration to the startup configuration or to a new checkpoint. If the startup configuration is already present, the command overwrites the existing startup configuration.

Parameter	Description
startup-config	Specifies that the startup configuration receives a copy of the running configuration.
checkpoint <CHECKPOINT-NAME>	Specifies the name of a new checkpoint to receive a copy of the running configuration. The checkpoint name can be comprised of alphanumeric character, underscores (_) and dashes (-), and must be 32 characters or fewer. NOTE: Do not start the checkpoint name with CPC because it is used for system-generated checkpoints.

Examples

Copying the running configuration to the startup configuration:

```
switch# copy running-config startup-config
Success
```

Copying the running configuration to a new checkpoint named **ckpt1**:

```
switch# copy running-config checkpoint ckpt1
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy {running-config | startup-config} <REMOTE-URL>

copy {running-config | startup-config} <REMOTE-URL> {cli | json} [vrf <VRF-NAME>]

Description

Copies the running configuration or the startup configuration to a remote file in either CLI or JSON format.

Parameter	Description
{running-config startup-config}	Selects whether the running configuration or the startup configuration is copied to a remote file.
<REMOTE-URL>	Specifies the remote file using the syntax: TFTP format: tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME> SFTP format: sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME> SCP format: scp://USER@{IP HOST}[:PORT]/FILE
{cli json}	Selects the remote file format: P: CLI or JSON.
vrf <VRF-NAME>	Specifies the name of a VRF. Default: default.

Examples

Copying a running configuration to a remote file in CLI format:

```
switch# copy running-config tftp://192.168.1.10/runcli cli
##### 100.0%
Success
```

Copying a running configuration to a remote file in JSON format:

```
switch# copy running-config tftp://192.168.1.10/runjson json
##### 100.0%
Success
```

Copying a startup configuration to a remote file in CLI format:

```
switch# copy startup-config sftp://root@192.168.1.10/startcli cli
root@192.168.1.10's password:
sftp> put /tmp/startcli startcli
Uploading /tmp/startcli to /root/startcli
Connected to 192.168.1.10.
Success
```

Copying a startup configuration to a remote file in JSON format:

```
switch# copy startup-config sftp://root@192.168.1.10/startjson json
root@192.168.1.10's password:
sftp>
root@192.168.1.10's password:
sftp> put /tmp/startjson startjson
Uploading /tmp/startjson to /root/startjson
Connected to 192.168.1.10.
```

Success

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy {running-config | startup-config} <STORAGE-URL>

copy {running-config | startup-config} <STORAGE-URL> {cli | json}

Description

Copies the running configuration or a startup configuration to a USB drive.

Parameter	Description
{running-config startup-config}	Selects the running configuration or the startup configuration to be copied to the switch USB drive.
<STORAGE-URL>	Specifies a remote file with the following syntax: usb:/<file>
{cli json}	Selects the format of the remote file: CLI or JSON.

Usage

The switch supports JSON and CLI file formats when copying the running or starting configuration to the USB drive. The USB drive must be formatted with the FAT file system.

The USB drive must be enabled and mounted with the following commands:

```
switch(config)# usb
switch(config)# end
switch# usb mount
```

Examples

Copying a running configuration to a file named **runCLI** on the USB drive:

```
switch# copy running-config usb:/runCLI cli
Success
```

Copying a startup configuration to a file named **startCLI** on the USB drive:


```
switch# copy startup-config usb:/startCLI cli
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy startup-config running-config

copy startup-config running-config

Description

Copies the startup configuration to the running configuration.

Examples

```
switch# copy startup-config running-config
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy <STORAGE-URL> running-config

copy <STORAGE-URL> {running-config | startup-config | checkpoint <CHECKPOINT-NAME>}

Description

This command copies a specified configuration from the USB drive to the running configuration, to a startup configuration, or to a checkpoint.

Parameter	Description
<code><STORAGE-URL></code>	Specifies the name of a configuration file on the USB drive with the syntax: usb:/<FILE>
<code>running-config</code>	Specifies that the configuration file is copied to the running configuration. The file must be in CLI, JSON, or checkpoint format or the copy will fail. the copy will not work.
<code>startup-config</code>	Specifies that the configuration file is copied to the startup configuration. The switch stores this configuration between reboots. The startup configuration is used as the operating configuration following a reboot of the switch. The file must be in JSON or checkpoint format or the copy will fail.
<code>checkpoint <CHECKPOINT-NAME></code>	Specifies the name of a new checkpoint file to receive a copy of the configuration. The configuration file on the USB drive must be in checkpoint format. NOTE: Do not start the checkpoint name with CPC because it is used for system-generated checkpoints.

Usage

This command requires that the USB drive is formatted with the FAT file system and that the file be in the appropriate format as follows:

- **running-config:** This option requires the file on the USB drive be in CLI, JSON, or checkpoint format.
- **startup-config:** This option requires the file on the USB drive be in JSON or checkpoint format.
- **checkpoint <checkpoint-name>:** This option requires the file on the USB drive be in checkpoint format.

Examples

Copying the file **runCli** from the USB drive to the running configuration:

```
switch# copy usb:/runCli running-config
Configuration may take several minutes to complete according to configuration
file size
--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%--
Success
```

Copying the file **startUp** from the USB drive to the startup configuration:

```
switch# copy usb:/startUp startup-config
Success
```

Copying the file **testCheck** from the USB drive to the **abc** checkpoint:

```
switch# copy usb:/testCheck checkpoint abc
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

erase

```
erase
  checkpoint <checkpoint-name>
  core-dump all|daemon|dsm|kernel|vsf
  startup-config
  all
```

Description

Deletes an existing checkpoint, startup configuration, or core-dump.

Parameter	Description
<code>checkpoint <CHECKPOINT-NAME></code>	Specifies the name of a checkpoint.
<code>core-dump</code> <code>all daemon <daemon-name></code> <code> kernel vsf</code>	Erase one of the following sets of core-dump files: - all: Erase all core-dump files. - daemon <daemon-name>: Erase daemon core-dump files. - kernel: Erase the kernel core-dump. - vsf: Erase daemon core-dump files for VSF. (For 6300 Switches only.)
<code>startup-config</code>	Specifies the startup configuration.
<code>all</code>	Specifies all checkpoints.

Examples

Erasing checkpoint **ckpt1**:

```
switch# erase checkpoint ckpt1
```

Erasing the startup configuration:

```
switch# erase startup-config
```

Erasing all checkpoints:

```
switch# erase checkpoint all
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint <CHECKPOINT-NAME>

show checkpoint <CHECKPOINT-NAME> [json]

Description

Shows the configuration of a checkpoint.

Parameter	Description
<CHECKPOINT-NAME>	Specifies the name of a checkpoint.
[json]	Specifies that the output is displayed in JSON format.

Examples

Showing the configuration of the **ckpt1** checkpoint in CLI format:

```
switch# show checkpoint ckpt1
Checkpoint configuration:
!
!Version AOS-CX PL.10.07.0000K-75-g55e5193
!export-password: default
lacp system-priority 65535
user admin group administrators password ciphertext
AQBapQjwipebv36io0jFfde7ZzrHckncal1D+3n8XFTZKQdmYgAAADetYOeHSme93xzdD0uz6Vr9K1+XBz
B+2GB0UBxSF7rvgn2x8KSgkqv7iqXVQ0Te6LkSMnH4BdNaT3Bf25qyvOqmr4YakO1V3rg8zAOADkPktQD8
joTHXflzwomoIzcmv/uX
cli-session
    timeout 0
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
interface lag 1
    no shutdown
    vlan access 1
interface lag 128
    no shutdown
    vlan access 1
interface lag 129
    shutdown
    vlan access 1
    lacp mode active
```

```

interface 1/1/1
    no shutdown
    lag 128
    lacp port-id 65535
interface 1/1/2
    no shutdown
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
interface 1/1/14
    no shutdown
    vlan access 1
interface 1/1/15
    no shutdown
    vlan access 1
interface 1/1/16
    no shutdown
    vlan access 1
interface vlan 1
    ip dhcp
snmp-server vrf default
!
!
!
!
!
https-server vrf default

```

Showing the configuration of the **ckpt1** checkpoint in JSON format:

```
switch# show checkpoint ckpt1 json
Checkpoint configuration:
{
  "AAA_Server_Group": {
    "local": {
      "group_name": "local"
    },
    "none": {
      "group_name": "none"
    }
  },
  ...
  ...
  ...
  ...
}
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint <CHECKPOINT-NAME> hash

show checkpoint <CHECKPOINT-NAME> hash [cli | json]

Description

Shows a configuration checkpoint hash calculated with the SHA-256 algorithm. When the output format is not specified, the CLI format is used. This enables you to determine whether there has been a configuration change since a previous hash was calculated.

Parameter	Description
<CHECKPOINT-NAME>	Specifies an existing checkpoint name.
[cli json]	Selects either the CLI or JSON format.

Examples

Showing a checkpoint SHA-256 hash in JSON format:

```
switch# show checkpoint ckpt1 hash json
Calculating the hash: [Success]

The SHA-256 hash of the checkpoint in JSON format, created in image XX.10.08.xxxx:
cc7a57a9bbb4e6600d3b4180296a35f6af9e797ce9c439955dfe5de58b06da9e
```

This hash is only valid for comparison to a baseline hash if the configuration has not been explicitly changed (such as with a CLI command, REST operation, etc.) or implicitly changed (such as by changing a hardware module, upgrading the SW version, etc.).

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint post-configuration

show checkpoint post-configuration

Description

Shows the configuration settings for creating system checkpoints.

Examples

```
switch# show checkpoint post-configuration

Checkpoint Post-Configuration feature
-----

Status           : enabled
Timeout (sec)    : 300
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint

show checkpoint

Description

Shows a detailed list of all saved checkpoints.

Examples

Showing a detailed list of all saved checkpoints:

```
switch# show checkpoint
```

NAME	TYPE	WRITER	DATE (YYYY/MM/DD)	IMAGE VERSION
ckpt1	checkpoint	User	2017-02-23T00:10:02Z	XX.01.01.000X
ckpt2	checkpoint	User	2017-03-08T18:10:01Z	XX.01.01.000X
ckpt3	checkpoint	User	2017-03-09T23:11:02Z	XX.01.01.000X
ckpt4	checkpoint	User	2017-03-11T00:00:03Z	XX.01.01.000X
ckpt5	latest	User	2017-03-14T01:12:27Z	XX.01.01.000X

Command History

Release	Modification
10.08	Command syntax <code>show checkpoint list all</code> is replaced with <code>show checkpoint</code> .
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint date

```
show checkpoint date <START-DATE> <END-DATE>
```

Description

Shows detailed list of all saved checkpoints created within the specified date range.

Parameter	Description
<START-DATE>	Specifies the starting date for the range of saved checkpoints to show. Format: YYYY-MM-DD .
<END-DATE>	Specifies the ending date for the range of saved checkpoints to show. Format: YYYY-MM-DD .

Examples

Showing a detailed list of saved checkpoints for a specific date range:

```
switch# show checkpoint date 2017-03-08 2017-03-12
```


NAME	TYPE	WRITER	DATE (YYYY/MM/DD)	IMAGE VERSION
ckpt2	checkpoint	User	2017-03-08T18:10:01Z	XX.01.01.000X
ckpt3	checkpoint	User	2017-03-09T23:11:02Z	XX.01.01.000X
ckpt4	checkpoint	User	2017-03-11T00:00:03Z	XX.01.01.000X

Command History

Release	Modification
10.08	Command syntax <code>show checkpoint list date <START-DATE> <END-DATE></code> is replaced with <code>show checkpoint date <START-DATE> <END-DATE></code>
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config hash

`show running-config hash [cli | json]`

Description

Shows the running-config checkpoint hash, calculated with the SHA-256 algorithm. When the output format is not specified, the CLI format is used. This enables you to determine whether there has been a configuration change since a previous hash was calculated.

Parameter	Description
[cli json]	Selects either the CLI or JSON format.

Examples

Showing the running-config checkpoint SHA-256 hash in CLI format:

```
switch# show running-config hash cli
Calculating the hash: [Success]

SHA-256 hash of the config in CLI format:

8db4e7e10f4b7f1a6ab17ad2b4efe0e72f1849103eaf43da62aa1d715075b89e

This hash is only valid for comparison to a baseline hash if the configuration has
not been explicitly changed (such as with a CLI command, REST operation, etc.)
or implicitly changed (such as by changing a hardware module, upgrading the
SW version, etc.).
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show startup-config hash

`show startup-config hash [cli | json]`

Description

Shows the startup-config checkpoint hash, calculated with the SHA-256 algorithm. When the output format is not specified, the CLI format is used. This enables you to determine whether there has been a configuration change since a previous hash was calculated.

Parameter	Description
[cli json]	Selects either the CLI or JSON format.

Examples

Showing the startup-config checkpoint SHA-256 hash in CLI format:

```
switch# show startup-config hash cli
Calculating the hash: [Success]

SHA-256 hash of the config in CLI format:

8db4e7e10f4b7f1a6ab17ad2b4efe0e72f1849103eaf43da62aa1d715075b89e

This hash is only valid for comparison to a baseline hash if the configuration has
not been explicitly changed (such as with a CLI command, REST operation, etc.)
or implicitly changed (such as by changing a hardware module, upgrading the
SW version, etc.).
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

write memory

write memory

Description

Saves the running configuration to the startup configuration. It is an alias of the command **copy running-config startup-config**. If the startup configuration is already present, this command overwrites the startup configuration.

Examples

```
switch# write memory
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Boot commands

boot fabric-module

boot fabric-module <SLOT-ID>

Description

Reboots the specified fabric module.

Parameter	Description
<SLOT-ID>	Specifies the member and slot of the module in the format member/slot . For example, to specify the module in member 1 slot 3, enter 1/3 .

Usage

The **boot fabric-module** command reboots the specified fabric module. Traffic performance is affected while the module is down.

If the specified module is the only fabric module in an up state, rebooting that module stops traffic switching between line modules and the line modules power down. The line modules power up when one fabric module returns to an up state.

This command is valid for fabric modules only.

Examples

Rebooting the fabric module in slot **1/3** when auto-confirm is not enabled:

```
switch# boot fabric-module 1/3
This command will reboot the specified fabric module. Traffic performance may
be affected while the module is down. Rebooting the last fabric module will
stop traffic switching between line modules.
Do you want to continue (y/n)? y

switch#
```

Rebooting the fabric module in slot **1/1** when auto-confirm is enabled:

```
switch# boot fabric-module 1/3
This command will reboot the specified fabric module. Traffic performance may
be affected while the module is down. Rebooting the last fabric module will
stop traffic switching between line modules.
Do you want to continue (y/n) y (auto-confirm)

switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot line-module

boot line-module <SLOT-ID>

Description

Reboots the specified line module.

Parameter	Description
<SLOT-ID>	Specifies the member and slot of the module in the format member/slot . For example, to specify the module in member 1 slot 3, enter 1/3 .

Usage

This command is supported on switches that have multiple line modules.

Reboots the specified line module. Any traffic for the switch passing through the affected module (SSH, TELNET, and SNMP) is interrupted. It can take up to 2 minutes to reboot the module. During that time, you can monitor progress by viewing the event log.

This command is valid for line modules only.

Examples

Reloading the module in slot 1/1:

```
switch# boot line-module 1/1
This command will reboot the specified line module and interfaces on this
module will not send or receive packets while the module is down. Any
traffic passing through the line module will be interrupted. Management
sessions connected through the line module will be affected. It might take
up to 2 minutes to complete rebooting the module. During that time, you can
monitor progress by viewing the event log.
Do you want to continue (y/n)? y
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot management-module

boot management-module {active | standby | <SLOT-ID>}

Description

Reboots the specified management module. Choose the management module to reboot by role (active or standby) or by slot number.

Parameter	Description
<i>active</i>	Selects the active management module.
<i>standby</i>	Selects the standby management module.
<SLOT-ID>	Specifies the member and slot of the management module in the format member/slot . For example, to specify the module in member 1 slot 5, enter 1/5 .

Usage

This command is supported on switches that have multiple management modules.

This command reboots a single management module in a chassis. Choose the management module to reboot by role (active or standby) or by slot number.

You can use the **show images** command to show information about the primary and secondary system images.

If you reboot the active management module and the standby management module is available, the active management module reboots and the standby management module becomes the active management module.

If you reboot the active management module and the standby management module is not available, you are warned, you are prompted to save the configuration, and you are prompted to confirm the operation.

If you reboot the standby management module, the standby management module reboots and remains the standby management module.

If you attempt to reboot a management module that is not available, the **boot** command is aborted.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot** command is aborted.



Hewlett Packard Enterprise recommends that you use the **boot management-module** command instead of pressing the module reset button to reboot a management module because if you are rebooting the only available management module, the **boot management-module** command enables you to save the configuration, cancel the reboot, or both.

Examples

Rebooting the active management module when the standby management module is available:

```
switch# boot management-module active
The management-module in slot 1/5 is going down for reboot now.
```

Rebooting the active management module when the standby management module is not available:

```
switch# boot management-module 1/5
The management module in slot 1/5 is currently active and no
standby management module was found.
This will reboot the entire switch.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot management-module (recovery console)

boot management-module {local|remote}

Description

Reboots the specified management module by specified location (local or remote).

Parameter	Description
<code><local></code>	Reboots the local management module.
<code><remote></code>	Reboots the remote management module.

Usage

This command is supported on switches that have multiple management modules.

This command reboots a single management module in a chassis. Choose the management module to reboot by role (active or standby) or by slot number.

You can use the **show images** command to show information about the primary and secondary system images.

If you reboot the active management module and the standby management module is available, the active management module reboots and the standby management module becomes the active management module.

If you reboot the active management module and the standby management module is not available, you are warned, you are prompted to save the configuration, and you are prompted to confirm the operation.

If you reboot the standby management module, the standby management module reboots and remains the standby management module.

If you attempt to reboot a management module that is not available, the **boot** command is aborted.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot** command is aborted.



Hewlett Packard Enterprise recommends that you use the **boot management-module** command instead of pressing the module reset button to reboot a management module because if you are rebooting the only available management module, the **boot management-module** command enables you to save the configuration, cancel the reboot, or both.

Examples

Booting a remote management module:

```
switch# boot management-module remote
There is no other management module installed.
Aborting.
switch#
```

Command History

Release	Modification
10.12	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot set-default

```
boot set-default {primary | secondary}
```

Description

Sets the default operating system image to use when the system is booted.

Parameter	Description
primary	Selects the primary network operating system image.
secondary	Selects the secondary network operating system image.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

boot system

```
boot system [primary | secondary | serviceos]
```

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Parameter	Description
<code>primary</code>	Selects the primary operating system image for this reboot and sets the configured default operating system image to primary for future reboots.
<code>secondary</code>	Selects the secondary operating system image for this reboot and sets the configured default operating system image to secondary for future reboots.
<code>serviceos</code>	Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovery OS for switches running the AOS-CX operating system and is used in rare cases when troubleshooting a switch.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the **show images** command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the **primary** or **secondary** optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.

You can use the **boot set-default** command to change the configured default operating system image.

- If you select **serviceos** as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the **boot system** command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot system** command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show boot-history

```
show boot-history [all|{vsf member <1-10>}]
```

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.

For switches that support line modules (such as 6400 switch series) including the **all** parameter displays information for the active management module and all available line modules.



To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
all	Optional. Shows boot information for the active management module. For switches that support line modules, including this parameter displays information for and all available line modules.
vsf member <1-10>	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
Index	The position of the boot in the history file. Range: 0 to 3.
Boot ID	A unique ID for the boot . A system-generated 128-bit string.
Current Boot, up for <time>	For the current boot, the show boot-history command shows the number of seconds the module has been running on the current software.
<Timestamp>: boot reason	For previous boot operations, the show boot-history command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values: ▪ <DAEMON-NAME> crash: The daemon identified by <DAEMON-NAME> caused the module to boot. ▪ Kernel crash: The operating system software associated with the module caused the module to boot. ▪ Uncontrolled reboot: The reason for the reboot is not known. ▪ Reboot requested through database: The reboot occurred because of a request made through the CLI or other API. For details, see show boot-history

Table 1: Description of reboots handled through the database

Boot History String	Description
Reboot requested by user	A user requested a switch reboot through the CLI or web UI.
Reset button pressed	The switch detected a short-press of the reset button
Backplane fault	A backplane fault occurred.
Configuration change	A configuration change resulted in a reboot.

Boot History String	Description
Configuration version migration	A configuration version migration occurred which required a reboot.
Console error	The console failed to start.
Fabric fault	A fabric fault occurred.
All line modules faulted	A zero line card condition occurred.
Redundancy switchover requested	A user requested a redundancy switchover.
Redundant Management communication timeout	The standby management module has taken over from an unresponsive active management module.
Redundant Management election timeout	A failure to elect a standby management module in the allotted time.
Critical service fault (error)	A daemon critical to switch operation has stopped functioning. An extra error string may be present to describe the error in detail.
VSF autojoin renumber	Reset triggered by VSF autojoin.
VSF member renumbered	A user requested a renumber of a VSF member.
VSF switchover requested	A user requested a VSF switchover.
VSX software update	Reset triggered by a VSX software update.
Chassis critical temperature	Chassis operating temperature exceeded.
Chassis low critical temperature	Chassis temperature below the minimum operating threshold.
Chassis insufficient fans	Insufficient fans to cool the chassis.
Chassis unsupported PSUs/fans	Unsupported or misconfigured PSUs or system fans.
Management module critical temperature	Management module operating temperature exceeded.
ISSU SMM update	Standby management module reboot triggered by an In-Service Software Upgrade (ISSU).
ISSU switchover	Redundancy switchover triggered by an In-Service Software Upgrade.
ISSU aborted	Standby management module reset triggered by failure during an In-Service Software Upgrade.
Rollback timer expired	Reset triggered by the ISSU rollback timer expiring.

Examples

Showing the boot history of the active management module:

```

switch# show boot-history
Management module
=====

Index : 2
Boot ID : c34a2c2499004a02bbeeff4992e1fdbd
Current Boot, up for 1 days 13 hrs 13 mins 27 secs

Index : 1
Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2

Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811dddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1

Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 : Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 : Switch boot count is 3
25 Aug 23 10:29:02 : Primary overtemperature fault detected with 0x2 in PSU 1/1

```

(For 6400 Switch series) Showing the boot history of the active management module and all line modules:

```

switch#
Management module
=====

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database

Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...

Management module
=====

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612

```

```

Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database

Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...

```

In the event of a reset triggered by a power supply unit (PSU), or a PSU input fault, the output of this command also displays information about why the PSU initiated a reboot. The following example displays the boot history of a switch with a reboot initiated by a PSU.

```

switch# show boot-history

Management module
=====
Index : 2
Boot ID : a61ad00d10864c748bc7893a5d4af2e4
15 Dec 23 19:02:02 : Power on reset with 0x1, Version: FL.10.13.1000AF

15 Dec 23 19:02:02 : Switch boot count is 0
15 Dec 23 19:02:17 : PSU 1/1: Fault detected

Index : 1
Boot ID : 30d831bbfdfa425baf50a629ee01b185
15 Dec 23 19:01:58 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:01:58 : Switch boot count is 0

```

The following example displays the boot history for the VSF member 2.

```

switch# show boot-history vsf member 2

Member-2
=====

Index : 0
Boot ID : df99026c194a44f1944a3e7685fb4d90
Current Boot, up for 3 hrs 31 mins 39 secs

Index : 3
Boot ID : 7bf4104903fe4ad1ba4bce40e8099c76
10 Aug 17 10:02:24 : Reboot requested through database
10 Aug 17 10:02:13 : Switch boot count is 2

```

Command History

Release	Modification
10.13.1000	The output of this command is enhanced to display additional information about the reason for the reboot, if available.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Firmware management commands

copy {primary | secondary} <REMOTE-URL>

copy {primary | secondary} <REMOTE-URL> [vrf <VRF-NAME>]

Description

Uploads a firmware image to a TFTP or SFTP server.

Parameter	Description
{primary secondary}	Selects the primary or secondary image profile to upload. Required
<REMOTE-URL>	Specifies the URL to receive the uploaded firmware using SFTP , TFTP or SCP. TFTP format: tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME> SFTP format: sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME> SCP format: scp://USER@{IP HOST}[:PORT]/FILE
vrf <VRF-NAME>	Specifies a VRF name. Default: default.

Examples

TFTP upload:

```
switch# copy primary tftp://192.0.2.0/00_10_00_0002.swi
##### 100.0%
Verifying and writing system firmware...
```

SFTP upload:

```
switch# copy primary sftp://swuser@192.0.2.0/00_10_00_0002.swi
swuser@192.0.2.0's password:
Connected to 192.0.2.0.
sftp> put primary.swi XL_10_00_0002.swi
Uploading primary.swi to /users/swuser/00_10_00_0002.swi
primary.swi                               100% 179MB 35.8MB/s 00:05
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy {primary | secondary} <FIRMWARE-FILENAME>

copy {primary | secondary} <FIRMWARE-FILENAME>

Description

Copies a firmware image to USB storage.

Parameter	Description
{primary secondary}	Selects the primary or secondary image from which to copy the firmware. Required
<FIRMWARE-FILENAME>	Specifies the name of the firmware file to create on the USB storage device. Prefix the filename with usb:/ . For example: usb:/firmware_v1.2.3.swi For information on how to format the path to a firmware file on a USB drive, see USB URL.

Examples

```
switch# copy primary usb:/11.10.00.0002.swi
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy primary secondary

copy primary secondary

Description

Copies the firmware image from the primary to the secondary location.

Examples

```
switch# copy primary secondary
The secondary image will be deleted.

Continue (y/n)? y
Verifying and writing system firmware...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy <REMOTE-URL>

copy <REMOTE-URL> {hot-patch|primary|secondary} [vrf <VRF-NAME>]

Description

Downloads a hot-patch or firmware image from a TFTP or SFTP server.

Parameter	Description
<REMOTE-URL>	Specifies the URL from which to download the firmware using SFTP or TFTP. TFTP format: <pre>tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME></pre> SFTP format: <pre>sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME></pre> SCP format: <pre>scp://USER@{IP HOST}[:PORT]/FILE</pre>

Parameter	Description
<code>{hot-patch primary secondary}</code>	Select a hot-patch or a primary or secondary image profile for receiving the downloaded firmware. Required. NOTE: For more information about hot-patch, see hot-patch .
<code>vrf <VRF-NAME></code>	Specifies the name of a VRF. Default: default.

TFTP usage

To specify a URL with:

- an IPv4 address: **`tftp://192.0.2.1/a.txt`**
- an IPv6 address: **`tftp://[2000::2]/a.txt`**
- a hostname: **`tftp://hpe.com/a.txt`**

To specify TFTP with:

- the port number of the server in the URL: **`tftp://192.0.2.1:12/a.txt`**
- the blocksize in the URL: **`tftp://192.0.2.1;blocksize=1462/a.txt`**

The valid blocksize range is 8 to 65464.

- the port number of the server and blocksize in the URL: **`tftp://192.0.2.1:12;blocksize=1462/a.txt`**

To specify a file in a directory of URL: **`tftp://192.0.2.1/dir/a.txt`**

SFTP usage

To specify:

- A URL with an IPv4 address: **`sftp://user@192.0.2.1/a.txt`**
- A URL with an IPv6 address: **`sftp://user@[2000::2]/a.txt`**
- A URL with a hostname: **`sftp://user@hpe.com/a.txt`**
- SFTP port number of a server in the URL: **`sftp://user@192.0.2.1:12/a.txt`**
- A file in a directory of URL: **`sftp://user@192.0.2.1/dir/a.txt`**
- To specify a file with absolute path in the URL: **`sftp://user@192.0.2.1/home/user/a.txt`**

SCP Usage

To specify:

- A username with an IP address: **`scp://user@192.0.2.1:12/a.txt`**
- A username with a remote host: **`scp://user@hpe.com/a.txt`**

Examples

TFTP download for a hot-patch:

```
switch# copy tftp://192.168.1.1/FL.10.12.0001-0002.patch hot-patch vrf vrf1

Fetching /users/swuser/FL.10.10.0001-0002.patch to hotpatch.dnld.uE2YT1
FL.10.12.0001-0002.patch                               100%   62KB   12.4MB/s   00:00
```

```
Verifying and writing hot-patch...
```

TFTP download for primary software image:

```
switch# copy tftp://192.10.12.0/FL_10_12_0001.swi primary
The primary image will be deleted.

Continue (y/n)? y
##### 100.0%
Verifying and writing system firmware...
```

SFTP download:

```
switch# copy sftp://swuser@192.10.12.0/FL_10_12_0001.swi primary
The primary image will be deleted.

Continue (y/n)? y
The authenticity of host '192.10.12.0 (192.10.12.0)' can't be established.
ECDSA key fingerprint is SHA256:L64khLwlyLgXlARKRMiwcAAK8oRaQ8C0oWP+PkGBXHY.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.10.12.0' (ECDSA) to the list of known hosts.
swuser@192.10.12.0's password:
Connected to 192.10.12.0.
Fetching /users/swuser/ss.10.00.0002.swi to ss.10.00.0002.swi.dnld
/users/swuser/ss.10.00.0002.swi          100% 179MB 25.6MB/s 00:07
Verifying and writing system firmware...
```

Command History

Release	Modification
10.12	The hot-patch parameter is supported on all platforms.
10.10	The hot-patch parameter is introduced on the 6300 series switches.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy secondary primary

```
copy secondary primary
```

Description

Copies the firmware image from the secondary to the primary location.

Examples

```
switch# copy secondary primary
The primary image will be deleted.

Continue (y/n)? y
Verifying and writing system firmware...
```

```
switch# copy sftp://stor@192.22.1.0/im-switch.swi primary vrf mgmt
The primary image will be deleted.

Continue (y/n)? y
The authenticity of host '192.22.1.0 (192.22.1.0)' can't be established.
ECDSA key fingerprint is SHA256:MyIlxbdKnehYut0NLfL69gDpNzCmZqBVvBaRR46m7o8.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.22.1.0' (ECDSA) to the list of known hosts.
stor@192.22.1.0's password:
Connected to 192.22.1.0.
sftp> get c8d5b9f-topflite.swi c8d5b9f-topflite.swi.dnld
Fetching /home/dr/im-switch.swi to c8d5b9f-topflite.swi.dnld
/home/dr/im-switch.swi          100% 226MB 56.6MB/s 00:04

Verifying and writing system firmware...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy <STORAGE-URL>

copy <STORAGE-URL> {hot-patch|primary|secondary}

Description

Copies, verifies, and installs a hot-patch or firmware image from a USB storage device connected to the active management module.

Parameter	Description
<STORAGE-URL>	Specifies the name of the firmware file to copy from the storage device. Required. USB format: usb: / <FILENAME>
{hot-patch primary secondary}	Select a hot-patch image or a primary or secondary profile for receiving the copied firmware. NOTE: For more information about hot-patch, see hot-patch .

USB usage

To specify a file:

- In a USB storage device: **usb:/a.txt**
- In a directory of a USB storage device: **usb:/dir/a.txt**

Examples

```
switch# copy usb:/FL.10.12.0001-0002.patch
```

```
switch# copy usb:/FL.10.12.0001.swi primary
The primary image will be deleted.
```

```
Continue (y/n)? y
```

```
Verifying and writing system firmware...
```

Command History

Release	Modification
10.12	The hot-patch parameter is supported on all platforms.
10.10	The hot-patch parameter is introduced on the 6300 series switches.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy hot-patch

```
copy hot-patch <Word> {<REMOTE-URL>|<Storage-URL>} [vrf <VRF-NAME>]
```

Description

Copies a hot-patch from a switch to the specified remote URL or storage URL.

Parameter	Description
<Word>	Name of the hot-patch software to upload.
<REMOTE-URL>	Specifies the URL to receive the uploaded patch using SFTP or TFTP. For information on how to format the remote URL, see URL formatting for copy commands.

Parameter	Description
<code>vrf <VRF-NAME></code>	[Optional] specify the VRF instance to use for upload.
<code><STORAGE-URL></code>	Specifies the name of the patch file to create on the USB storage device. Prefix the filename with usb:/ , for example, usb:/firmware_FL_10_12_0001-0002.patch .

Examples

```
switch# copy hot-patch FL_10_12_0001-0002.patch tftp:172.21.18.170/FL_10_12_0001-0002.patch vrf vrf1
```

Related Commands

Command	Description
copy <REMOTE-URL>	Downloads a hot-patch image from a TFTP or SFTP server.
hot-patch	Apply a hot-patch image or remove it from the switch.

Command History

Release	Modification
10.12	Hot-patch is now supported on all platforms.
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

hot-patch

```
hot-patch apply|remove <name.patch>
no hot-patch apply <name.patch>
```

Description

Apply hot-patch software or remove it from the switch. The **no** form of the **hot-patch apply** command disables the hot-patch image, but does not remove it from the switch. Rebooting the system after disabling or removing the patch is not required.

Profile names	Description
<code>apply <name.patch></code>	Apply the specified hot-patch image to a standalone switch or VSF stack. AOS-CX hot-patch software images can be obtained from

Profile names	Description
	Aruba customer support, and are identified with a .patch extension.
<code>remove <name.patch></code>	Disables the hot-patch image and removes the patch from the switch. This removal will also disable the patch. Once removed, a hot-patch must be downloaded again in order to be applied.

Usage

A hot-patch can be downloaded from a remote server onto a switch then applied without rebooting the switch. When the hot-patch is disabled, the hot-patch will still remain on the system. The disabled hot-patch can be removed from the system without the need for a reboot of the system.

If a checkpoint configuration that does not contain a hot-patch is restored to a running configuration that does have a hot-patch, the patch is not deleted, it remains as not applied but is present in the device memory.

Examples

```
switch(config)# hot-patch apply FL_10_12_0001-0002.patch
```

Related Commands

Command	Description
<code>copy <REMOTE-URL></code>	Downloads and installs a hot-patch image from a TFTP or SFTP server.
<code>copy hot-patch</code>	Copies a hot-patch software image from a switch to a specified remote URL or storage URL.

Command History

Release	Modification
10.12	Hot-patch is now supported on all platforms.
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

issu rollback-timer

```
issu rollback-timer [wait-time <TIME>]
```

Description

Enables the ISSU rollback timer on the system. The rollback timer automatically rolls the system back to the configuration and OS image used before starting the ISSU, unless the upgrade is confirmed with `issu update-software confirm`. Changing the rollback-timer will not affect an active timer and will apply on the next ISSU. To cancel the active timer, confirm the previous ISSU with `issu update-software confirm`. Disabled by default.

The **no** form of this command disables the rollback timer on the system.

Parameter	Description
<code><TIME></code>	Specifies how many minutes the system will wait for confirmation that the last ISSU is accepted before triggering a system reboot and roll back to the previous configuration and OS version. This change will not affect an active timer and will apply on the next ISSU. Range: 30-1440.

Examples

Enabling the ISSU rollback timer:

```
switch(config)# issu rollback-timer
```

Disabling the ISSU rollback timer:

```
switch(config)# no issu rollback-timer
```

Disabling the ISSU rollback timer on the system while a previous ISSU's timer is active:

```
switch(config)# no issu rollback-timer
The ISSU rollback timer is active. This change will apply on the next ISSU
operation
To cancel the active timer, confirm the previous ISSU with 'issu update-software
confirm'
```

Setting the ISSU rollback timer wait time to 80 minutes:

```
switch# issu rollback-timer wait-time 80
```

Setting the ISSU rollback timer wait time to 81 minutes while a previous ISSU's timer is active:

```
switch# issu rollback-timer wait-time 81
The ISSU rollback timer is active. This change will apply on the next ISSU
operation
```

Resetting the ISSU rollback timer wait time to default:

```
switch(config)# no issu rollback-timer wait-time
```

Related Commands

For information on the VSF Enhanced Software Upgrade (ESU) feature that includes **issu** commands, refer to the AOS-CX VSF User Guide.

Command History

Release	Modification
10.11	Command introduced

Command Information

Platforms	Command context	Authority
6400	config	Administrators or local user group members with execution rights for this command.

issu update-software

`issu update-software [validate|confirm]`

Description

Initiates ISSU to the alternate boot location. The newer operating system image must be downloaded to the alternate boot location prior to running this command. Additionally, the current running operating system version must match the version that is stored in the current boot location or ISSU will not be allowed.



This operation is disruptive and will result in the management interface being disconnected during the process.

Starting an ISSU will cause the running configuration to be saved in case an error is encountered that requires a reboot to recover the switch. In addition, a special configuration checkpoint will be stored to disk and is used to roll back to the pre-ISSU firmware. During the ISSU process all management methods (CLI, REST, WebUI) will be blocked from making configuration changes to the switch. The configuration block is active from the time ISSU starts until the time ISSU switchover is complete. After the ISSU switchover is completed, switch configuration can resume.

Parameter	Description
<code>validate</code>	Runs all pre-ISSU validations without executing the actual upgrade. The validation runs in the background, however its results will be displayed in real time for approximately the next three minutes. If the validation is not finished within that time frame or if the display is aborted with Control+C or Control+Z, the results can be queried using the <code>show issu validation</code> command.
<code>confirm</code>	Confirms the software update and cancels the configured rollback timer. If the rollback timer is configured then this command has to be executed after an ISSU before the timer expires. Else, the pre-ISSU checkpoint is copied to the startup configuration and the system is rebooted to the image used before ISSU.

Parameter	Description
	Note: To perform an intentional system rollback before the timer expires, a manual downgrade must be executed through the following steps: 1. Copy the pre-ISSU checkpoint to the startup configuration using the <code>copy checkpoint pre-issu-startup-config startup-config</code> command. 2. Boot to the previous image using the <code>boot system</code> command.

Usage

Note the following points before running this command:

- The newer operating system image must be downloaded to the alternate boot location prior to running this command.
- The current running operating system version must match the version that is stored in the current boot location or ISSU will not be allowed.
- This operation is disruptive and will result in the management interface being disconnected during the process.
- The running configuration will be stored in case an error is encountered that requires a reboot to recover the switch.
- During the ISSU process all management methods (CLI, REST, WebUI) are blocked from making configuration changes to the switch. The configuration changes are not allowed from the time ISSU starts until ISSU switchover is complete. After the ISSU switchover is completed, switch configuration can resume.
- In case of ISSU, conductor gets transitioned to standby role without reboot.
- The stack topology must be a ring before initiating ISSU. ISSU is not supported in chain topologies and the process is aborted if the ISSU is initiated.
- During the ISSU process, the **show core-dump all**, **show tech all**, and **copy support-files** commands all may fail to run or display correct output.

Examples

Initiating an ISSU:

```
switch# issu update-software

This command will perform an in-service software upgrade
using pre-staged secondary operating system image
FL.10.13.1000M

This will save the current running configuration
WARNING:

The rollback timer is enabled and configured to 30 minutes.

After the upgrade is done, execute "issu update-software confirm"
```

to confirm the new image works as expected. If the command is not entered, the system will be rebooted to the previous version.

Continue (y/n)? y

Starting in-service software upgrade.

Use "show issu" to monitor status and progress.

Use "show events -c issu" to view event notifications.

Initiating an ISSU, but stopping it without confirming the upgrade:

```
switch(config)# issu update-software
```

This command will perform an in-service software upgrade using pre-staged secondary operating system image FL.10.13.1000M

This will save the current running configuration

WARNING:

The rollback timer is enabled and configured to 30 minutes. After the upgrade is done, execute "issu update-software confirm" to confirm the new image works as expected. If the command is not entered, the system will be rebooted to the previous version.

Continue (y/n)? n

In-service software upgrade aborted. No changes were made.

Confirming the ISSU configuration when the rollback timer has been configured and started:

```
switch# issu update-software confirm
```

The ISSU has been confirmed and the rollback timer has been cancelled.

Confirming the ISSU configuration when the rollback timer has not started:

```
switch# issu update-software confirm
```

No rollback timer has been started, no action was done.

Executing an ISSU "dry run" where all pre-ISSU validations are run without executing the actual upgrade:

```
switch# issu update-software validate
```

ISSU Validation

=====

Condition	Status
-----------	--------

```
Current Image Valid      ---
Target Image Valid      ---
Target Version Compatible ---
Management Modules Ready ---
Line Modules Ready      ---
Features Ready          ---

In Progress[/]
```

Executing an ISSU "dry run" where all pre-ISSU validations are run and the user aborts the ISSU validation on screen display without user confirmation:

```
switch# issu update-software validate

ISSU Validation

=====
Condition                      Status
-----
Current Image Valid           Pass
Target Image Valid            Pass
Target Version Compatible     Failed
Management Modules Ready      ---
Line Modules Ready            ---
Features Ready                ---

In Progress[\]
To view the validation progress and results, execute "show issu validation"
```

Executing an ISSU "dry run" where all pre-ISSU validations are run without executing the actual upgrade and the validation progress has finished successfully:

```
switch# issu update-software validate

ISSU Validation

=====
Condition                      Status
-----
Current Image Valid           Pass
Target Image Valid            Pass
Target Version Compatible     Pass
Management Modules Ready      Pass
Line Modules Ready            Pass
Features Ready                Pass

ISSU Validation has completed
```

Executing an ISSU validation while a previous ISSU is unconfirmed, i.e. the rollback timer is still running:

```
switch# issu update-software validate
The previous ISSU has not been confirmed. Please confirm it with
'issu update-software confirm' before starting a new ISSU or running a validation.
```

Executing an ISSU "dry run" when the validations are taking more than three minutes to complete, then checking the result of the validation afterwards:

```
switch# issu update-software validate
ISSU validation is taking longer than expected. Check the final result with 'show
issu validation'

switch# show issu validation

ISSU Validation
=====

Condition                               Status
-----
Current Image Valid                     Pass
Target Image Valid                     Pass
Target Version Compatible               Pass
Management Modules Ready               Pass
Line Modules Ready                     Pass
Features Ready                         Pass
```

Following example shows ISSU performed with chain topology:

```
switch# issu update-software
Stack topology is not a ring. ISSU upgrade aborted.
```

Related Commands

For information on the VSF Enhanced Software Upgrade (ESU) feature that includes **issu** commands, refer to the AOS-CX VSF User Guide.

Command History

Release	Modification
10.11	Validate and confirm parameters added.
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
6400	config	Administrators or local user group members with execution rights for this command.

show hot-patch

show hot-patch [detail]

Parameter	Description
detail	Displays the detailed status of all hot-patches present on the system.

Description

the **show hot-patch** command displays the status of all hot-patches present on the system. The **show hot-patch detail** command displays detailed information for all hot patches present on the system.

Examples

```
switch# show hot-patch

Name
-----
FL_10_12_0001-0002.patch      Status
-----
Applied

switch# show hot-patch detail

Name           : FL_10_12_0001-0002.patch
Status          : Applied
Version         : FL_10_12_0001-0002.patch
Compatible Version : FL.10.12.0001
Issues Fixed    : CR1234, CR2345
Patch Date      : 2022-03-29 20:46:15 UTC
Patch ID        : ArubaOS-CX:FL.10.12.0001-sp1-256-gd457e868d39:202204142009
Patch SHA       : a40438d06a82e5fe7e30d457e868d39e8526185b
```

Related Commands

Command	Description
copy <REMOTE-URL>	Downloads a hot-patch image from a TFTP or SFTP server.
hot-patch	Apply a hot-patch image or remove it from the switch.

Command History

Release	Modification
10.12	Command supported on all platforms.
10.10	Command introduced on 6300 Switch series.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show issu

```
show issu [brief|history|validation]
```

Description

Shows information about the current state of ISSU. If no ISSU is currently in progress, the command displays the progress details of the last ISSU.

The command with the **brief** parameter displays a short summary of the ISSU state and indicates if the system is ready to accept an ISSU command and whether or not an ISSU is in progress. If the **brief** parameter is not included, then more details about an in progress ISSU or the last ISSU are displayed.

Parameter	Description
brief	Shows a short summary of the ISSU state.
history	Shows details of ISSU software update history.
validation	Shows information about the current state of an ISSU validation.

Examples

Showing detailed ISSU status with an ISSU in progress for the first time:

```
switch# show issu
ISSU Summary
=====
ISSU Status      : In Progress
Current Version  : FL.10.13.0005K      Upgrade Version : FL.10.13.1000M
Upgrade Image    : secondary           Start Date      : 2023-11-08 07:01:47
Last ISSU Result: --
Rollback timer   : Not started

ISSU Progress
=====
Upgrade Operation      Status      Start Date
-----
Initiate ISSU          Complete    2023-11-08 07:01:47
Validate System Readiness In Progress 2023-11-08 07:01:47
Upgrade Standby and Member Modules Pending     --
Upgrade Line Module Services Pending         --
Prepare for Switchover Pending            --
Finalize Upgrade       Pending            --
ISSU Complete          Pending            --
```

Showing detailed status for VSF ISSU:

```
switch# show issu
ISSU Summary
=====
ISSU Status      : In Progress
Current Version  : FL.10.11.0001      Upgrade Version : FL.10.11.1000BD
Upgrade Image    : secondary           Start Date      : 2023-02-02 14:22:31
Last ISSU Result: --
Rollback timer   : Not started

ISSU Progress
=====
Upgrade Operation      Status      Start Date
-----
Initiate ISSU          Complete    2023-02-02 14:22:31
Validate System Readiness Complete    2023-02-02 14:22:31
Upgrade Standby and Member Modules In Progress 2023-02-02 14:22:54
Upgrade Line Module Services Pending         --
Prepare for Switchover Pending            --
```

Finalize Upgrade	Pending	--
ISSU Complete	Pending	--

Showing detailed ISSU status with ISSU in progress after successfully completing a previous ISSU:

```
switch# show issu
ISSU Summary
=====
ISSU Status      : In progress
Current Version  : FL.10.10.0001      Upgrade Version : FL.10.10.0002
Upgrade Image    : secondary          Start Date      : 2021-10-15 08:37:49
Last ISSU Result: Completed (Without errors)

ISSU Progress
=====
Upgrade Operation      Status      Start Date
-----
Initiate ISSU          Complete    2021-10-13 23:05:41
Validate System Readiness Complete    2021-10-13 23:05:41
Upgrade Standby Management Module Complete    2021-10-13 23:05:41
Upgrade Line Modules    In Progress 2021-10-13 23:07:07
Prepare for Switchover  Pending     --
Finalize Upgrade        Pending     --
ISSU Complete           Pending     --
```

Showing detailed ISSU status with ISSU in progress after aborting the previous ISSU:

```
switch# show issu
ISSU Summary
=====
ISSU Status      : In progress
Current Version  : FL.10.10.0001      Upgrade Version : FL.10.10.0002
Upgrade Image    : secondary          Start Date      : 2021-10-15 08:37:49
Last ISSU Result: Aborted (One or more line modules are not ready to start ISSU)

ISSU Progress
=====
Upgrade Operation      Status      Start Date
-----
Initiate ISSU          Complete    2021-10-13 23:05:41
Validate System Readiness Complete    2021-10-13 23:05:41
Upgrade Standby Management Module Complete    2021-10-13 23:05:41
Upgrade Line Modules    In Progress 2021-10-13 23:07:07
Prepare for Switchover  Pending     --
Finalize Upgrade        Pending     --
ISSU Complete           Pending     --
```

Showing detailed ISSU status with no ISSU in progress and no previous ISSU performed:

```
switch# show issu
ISSU Summary
=====
ISSU Status      : Ready
Current Version  : FL.10.10.0001      Upgrade Version : --
Upgrade Image    : --                Start Date      : --
Last ISSU Result: -- (--)
```



```

ISSU Progress
=====
Upgrade Operation          Status          Start Date
-----
Initiate ISSU              --              --
Validate System Readiness  --              --
Upgrade Standby Management Module --            --
Upgrade Line Modules       --              --
Prepare for Switchover     --              --
Perform Switchover         --              --
Finalize Upgrade           --              --
ISSU Complete              --              --

```

Showing detailed ISSU status after completion and before system is ready to start a new ISSU:

```

switch# show issu
ISSU Summary
=====
ISSU Status      : Not ready
Current Version  : FL.10.10.0001      Upgrade Version : --
Upgrade Image    : --                  Start Date      : --
Last ISSU Result: Completed (Without errors)

ISSU Progress
=====
Upgrade Operation          Status          Start Date
-----
Initiate ISSU              Complete        2021-10-13 23:05:41
Validate System Readiness  Complete        2021-10-13 23:05:41
Upgrade Standby Management Module Complete        2021-10-13 23:05:41
Upgrade Line Modules       Complete        2021-10-13 23:07:07
Prepare for Switchover     Complete        2021-10-13 23:07:50
Finalize Upgrade           Complete        2021-10-13 23:07:53
ISSU Complete              Complete        2021-10-13 23:08:10

```

Showing detailed ISSU status after an error occurred and the process is aborted:

```

switch# show issu
ISSU Summary
=====
ISSU Status      : Aborted
Current Version  : FL.10.10.0001      Upgrade Version : FL.10.10.0002
Upgrade Image    : secondary           Start Date      : 2021-12-09 19:17:15
Last ISSU Result: Aborted (System failed to prepare for ISSU)

ISSU Progress
=====
Upgrade Operation          Status          Start Date
-----
Initiate ISSU              Complete        2021-12-09 19:17:15
Validate System Readiness  Complete        2021-12-09 19:17:15
Upgrade Standby Management Module Complete        2021-12-09 19:17:15
Upgrade Line Modules       Error            2021-12-09 19:19:22
Prepare for Switchover     Aborted        --
Finalize Upgrade           Aborted        --
ISSU Complete              Aborted        --

```

Showing summary of ISSU status with no ISSU in progress where system is ready to start a new ISSU and with no previous ISSU performed:

```
switch# show issu brief
ISSU Summary
=====
ISSU Status      : Ready
Current Version  : FL.10.10.0001      Upgrade Version : --
Upgrade Image    : --                Start Date       : --
Last ISSU Result: -- (--)
```

Showing summary of ISSU status with no ISSU in progress where system is ready to start a new ISSU and after successfully completing a previous ISSU:

```
switch# show issu brief
ISSU Summary
=====
ISSU Status      : Ready
Current Version  : FL.10.10.0001      Upgrade Version : --
Upgrade Image    : --                Start Date       : --
Last ISSU Result: Completed (Without errors)
```

Showing a summary of ISSU status with no ISSU in progress where system is ready to start a new ISSU and after aborting the previous ISSU:

```
switch# show issu brief
ISSU Summary
=====
ISSU Status      : Ready
Current Version  : FL.10.10.0001      Upgrade Version : --
Upgrade Image    : --                Start Date       : --
Last ISSU Result: Aborted (One or more line modules are not ready to start ISSU)
```

Showing a summary of ISSU status with ISSU in progress:

```
switch# show issu brief
ISSU Summary
=====
ISSU Status      : In progress
Current Version  : FL.10.10.0001      Upgrade Version : FL.10.10.0002
Upgrade Image    : secondary          Start Date       : 2021-10-15 08:37:49
Last ISSU Result: Completed (Without errors)
```

Showing a summary of ISSU status with no ISSU in progress where the system is not ready to start a new ISSU:

```
switch# show issu brief
ISSU Summary
=====
ISSU Status      : Not ready
Current Version  : FL.10.10.0001      Upgrade Version : FL.10.10.0002
Upgrade Image    : secondary          Start Date       : 2021-10-15 08:37:49
Last ISSU Result: Completed (Without errors)
```

Showing summary of ISSU status after an error occurred and the process is aborted:

```
switch# show issu brief
ISSU Summary
=====
ISSU Status      : Aborted
Current Version  : FL.10.10.0001      Upgrade Version : FL.10.10.0002
Upgrade Image    : secondary          Start Date      : 2021-12-09 19:17:15
Last ISSU Result: Aborted (System failed to prepare for ISSU)
```

Showing ISSU validation status per condition/validation:

```
switch# show issu validation

ISSU Validation

=====
Condition                               Status
-----
Current Image Valid                    ---
Target Image Valid                     ---
Target Version Compatible               ---
Management Modules Ready               ---
Line Modules Ready                     ---
Features Ready                         ---
```

Showing ISSU history:

```
switch# show issu history
Upgrade: 1
  From Version : FL.10.11.0001
  To Version   : FL.10.11.1000
  Start Time   : 2022-09-14 15:37:33
  End Time     : 2022-09-14 15:40:45
  Status       : Completed
```

Related Commands

For information on the VSF Enhanced Software Upgrade (ESU) feature that includes **issu** commands, refer to the AOS-CX VSF User Guide.

Command History

Release	Modification
10.11.1000	History parameter introduced.
10.11	Support for 6300 Switch Series added.
10.10	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A	Manager (#)	Administrators or local user group members with execution rights for this command.

Chapter 11

Dynamic Segmentation

For information on dynamic segmentation, view the *Security Guide*.

Simple Network Management Protocol (SNMP) is an Internet-standard protocol used for managing and monitoring the devices connected to a network by collecting, organizing and modifying information about managed devices on IP networks.

Configuring SNMP

(The SNMP agent provides read-only access.)

Procedure

1. Enable SNMP on a VRF using the command `snmp-server vrf`.
2. Set the system contact, location, and description for the switch with the following commands:
 - `snmp-server system-contact`
 - `snmp-server system-location`
 - `snmp-server system-description`
3. If required, change the default SNMP port on which the agent listens for requests with the command `snmp-server agent-port`.
4. By default, the agent uses the community string **public** to protect access through SNMPv1/v2c. Set a new community string with the command `snmp-server community`.
5. Configure the trap receivers to which the SNMP agent will send trap notifications with the command `snmp-server host`.
6. Create an SNMPv3 context and associate it with any available SNMPv3 user to perform context specific v3 MIB polling using the command `snmpv3 user`.
7. Create an SNMPv3 context and associate it with an available SNMPv1/v2c community string to perform context specific v1/v2c MIB polling using the command `snmpv3 context`.
8. Review your SNMP configuration settings with the following commands:
 - `show snmp agent-port`
 - `show snmp community`
 - `show snmp system`
 - `show snmpv3 context`
 - `show snmp trap`
 - `show snmp vrf`
 - `show snmpv3 users`
 - `show tech snmp`

Example 1

This example creates the following configuration:

- Enables SNMP on the out-of-band management interface (VRF **mgmt**).
- Sets the contact, location, and description for the switch to: **JaniceM**, **Building2**, **LabSwitch**.
- Sets the community string to **Lab8899X**.

```
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server system-contact JaniceM
switch(config)# snmp-server system-location Building2
switch(config)# snmp-server system-description LabSwitch
switch(config)# snmp-server community Lab8899X
```

Example 2

This example creates the following configuration:

- Creates an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**.
- Associates the SNMPv3 user `Admin` with a context named `newContext`.

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des
priv-pass plaintext myprivpass
switch(config)# snmpv3 user Admin context newContext
```



Refer to the SNMP Guide for SNMP Commands.

The Aruba Central network management solution, a software-as-a-service subscription in the cloud, provides streamlined management of multiple network devices. AOS-CX switches are able to talk to Aruba Central and utilize cloud-based management functionality. Cloud-based management functionality allows for the deployment of network devices at sites with no or few dedicated IT personnel (branch offices, retail stores, and so forth). AOS-CX switches utilize secure communication protocols to connect to the Aruba Central cloud portal, and can coexist with corporate security standards, such as those mandating the use of firewalls.



When Aruba Central manages AOS-CX switches, it functions as the single source of truth and the Web UI operates in read-only mode.

This feature provides:

- Zero-touch provisioning
- Network Management/Remote monitoring
- Events/alerts notification
- Switch Configuration using templates
- Firmware management

Connecting to Aruba Central

AOS-CX switch downloads the location of Aruba Central server using:

- Command-line interface (CLI).
- Aruba Activate server.
- DHCP options provided during ZTP.

DHCP servers are used to connect to Central on-premise management.

If switch is unable to connect to Activate server, it retries to establish connection in exponential back off of 1s, 2s, 8s, 16s, 32s, 64s, 128s, and 256s. After the maximum back off of 256s, switch retries happen for every 5 minutes.

If the Network Time Protocol (NTP) is not enabled on the switch, it will synchronize the system time with the Activate server.



When a switch is able to connect to Aruba Central, but is not registered in the Aruba Central inventory or does not have a proper license, the switch will get disconnected. If the Aruba Central feature is enabled using this command, the switch will then reconnect back to Aruba Central and will get disconnected again. This connect/disconnect process will continue until the switch is properly registered in Aruba Central. To avoid this unnecessary reconnection cycle, best practices is to [disable Aruba Central](#) until the switch is registered in Aruba Central, or a license is obtained for that device.

Custom CA certificate

To use a custom CA certificate to connect to Aruba Central, AOS-CX switch downloads the certificate from the Aruba Activate server.



- If there is no custom CA provided by Aruba Activate, the CA certificate present in the device is used.
- Duplicate CA certificates from Aruba Activate server will be ignored.
- If CA certificate is absent in consecutive responses from Aruba Activate server, the installed custom CA certificate in device will be removed.
- Switch will have only one custom CA certificate installed from Aruba Activate Server.
- The certificate installed from Aruba Activate server will not be displayed in the show commands.

Support mode in Aruba Central

When the AOS-CX switch is managed by Aruba Central, the switch configuration cannot be modified using other interfaces such as CLI or Web UI. The following command categories are blocked:

- auto-confirm
- boot
- checkpoint
- copy-in commands
- erase
- erps
- https-server
- mfgread
- mfgwrite
- port-access
- vsf
- All configuration commands except the aruba-central command

In cases where a maintenance or troubleshooting activity requires configuration updates, aruba-central support-mode can be enabled to allow these operations.



The aruba-central support-mode enable or disable operation is effective only in the CLI session where it is executed and does not impact the other CLI sessions.

If the user tries to execute any command that is not allowed, an **Invalid input:** error message is displayed.

One Touch Provisioning

One Touch Provisioning (OTP) allows a switch to be provisioned on Aruba Central with the minimum user configuration.

Configuring One Touch Provisioning

To provision a switch to Aruba Central by using an OTP scenario, you can configure:

The Aruba Central Location

Following configurations can be done optionally:

The Domain Name System (DNS) server address.

The HTTP proxy location.

Configuring Aruba Central location

Create the Aruba-Central context.

```
switch(config)# aruba-central
```

Configure the Aruba Central server location

```
switch(config-aruba-central)# location-override <LOCATION> vrf <VRF>
```



IPv4 or IPv6 address can be used.

Validate the connectivity status to Aruba Central

```
switch# show aruba-central
```

Configuring DNS server address

Configure the DNS server address.

```
switch(config)# ip dns server-address <ADDRESS> vrf <VRF>
```



IPv4 or IPv6 address can be used.

Validate the DNS server status

```
switch# show ip dns
```

Configuring HTTP proxy location

Configure the HTTP Proxy location

```
switch(config)# http-proxy <LOCATION> vrf <VRF>
```



LOCATION can be either a FQDN, IPv4 address or IPv6 address

Limitations for Zero Touch Provisioning

The following are the limitations for OTP when provisioning a switch by using IPv6:

- Support for obtaining the Aruba Central location via Activate.
- Support for obtaining the Aruba Central location via DHCP.
- Support for connectivity to an Aruba Central on-premise instance.

Supportability

Debug logging can be enabled with the following command:



The central debug logs are helpful to determine connectivity issues with Aruba Central.

```
switch# debug central all
```

Aruba Central commands

aruba-central

```
aruba-central
no aruba-central
```

Description

Creates or enters the Aruba Central configuration context (`config-aruba-central`).

Example

Creating the Aruba Central configuration context:

```
switch(config)# aruba-central
switch(config-aruba-central)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

aruba-central support-mode

```
aruba-central support-mode
no aruba-central support-mode
```

Description

Allows the device to be writable for all operations in Aruba Central lockout mode for troubleshooting. The no form of this command disables this activity.



Support-mode is disabled by default when the switch is managed by Aruba Central. This command is only effective in the CLI session where it is executed.

Examples

Configuring the device to be writable for all operations in Aruba Central lockout mode:

```
switch# aruba-central support-mode
switch#
```

Removing the configuration that allows the device to be writable for all operations in Aruba Central lockout mode:

```
switch# no aruba-central support-mode
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

configuration-lockout central managed

```
configuration-lockout central managed
no configuration-lockout central managed
```

Description

Configures the device to only be writable from Aruba Central. Aruba Central will be the only agent that can add, modify, or delete configurations on the device. The no form of this command disables this feature.



The no form of this command is only available when the device is disconnected from Aruba Central.

Usage

The AOS-CX switch connects to Aruba Central in either of two modes: monitor or managed. When the device is connected in monitor mode, Aruba Central monitors the configurations on the switch. When the device is connected in managed mode, the **configuration-lockout central managed** command does not allow configuration changes from other interfaces such as CLI or Web UI.

Examples

Configuring the device to only be writable from Aruba Central :

```
switch(config)# configuration-lockout central managed
switch# show configuration-lockout
configuration lockout
-----
central: managed
switch# sh aruba-central
Central admin state           :enable
Central location              :20.0.0.2:8083
VRF for connection            :default
Central connection status     :connected

Central source                 :cli
Central source connection status :connected
Central source last connected on :Tue Feb 9 17:53:13 UTC 2021

Activate Server URL           :devices-v2.arubanetworks.com
CLI location                   :20.0.2:8083
CLI VRF                        :default
switch(config)# end
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

disable

disable

Description

Disables connection to Aruba Central server.

When the connection is disabled, the switch does not attempt to connect to the Aruba Central server or fetch central location from any of the three sources (CLI/Aruba Activate/DHCP). It also disconnects any active connection to the Aruba Central server.

Example

```
switch(config-aruba-central)# disable
switch(config-aruba-central)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-aruba-central	Administrators or local user group members with execution rights for this command.

enable

enable

Description

Enables connection to Aruba Central server. When the connection is enabled, the switch attempts to download the location of the Aruba Central server in one of the following ways at startup and after the connection is lost:

- Using command-line interface (CLI).
- Connecting to Aruba Activate server.
- Using DHCP options provided during ZTP.

DHCP servers provide the options requested by the device to connect to Central, Central On-premise management, or the TFTP server.



When a switch is able to connect to Aruba Central, but is not registered in the Aruba Central inventory or does not have a proper license, the switch will get disconnected. If the Aruba Central feature is enabled using this command, the switch will then reconnect back to Aruba Central and will get disconnected again. This connect/disconnect process will continue until the switch is properly registered in Aruba Central. To avoid this unnecessary reconnection cycle, best practices is to [disable Aruba Central](#) until the switch is registered in Aruba Central, or a license is obtained for that device.

Examples

```
switch(config-aruba-central)# enable
switch(config-aruba-central)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-aruba-central	Administrators or local user group members with execution rights for this command.

location-override

```
location-override <location> [vrf <VRF-NAME>]  
no location-override
```

Description

When **location** and **vrf** are configured, the switch overrides existing connections to Aruba Central. The switch attempts to establish connection to Aruba Central with the specified location and VRF with highest priority.

Location can take one of the following values:

- A fully qualified domain name (FQDN) along with an optional port number.
- An IPv4 address with an optional port number.
- An IPv6 address with an optional port number.

If the port number is not specified, then port 443 is used by default. If the command is executed without the VRF parameter, the switch uses the 'default' VRF.

The **no** form of this command removes location override values from the Aruba Central configuration context.



When you configure an IPv6 address with a port number, specify the address part inside square brackets, optionally followed by the port number, e.g. [2001:0db8:85a3:0000:0000:8a2e:0370:7334]:443.

Parameter	Description
<code><location></code>	Specifies one of these values: ■ <code><FQDN></code>: a fully qualified domain name. ■ <code><IPv4></code>: an IPv4 address. ■ <code><IPv6></code>: an IPv6 address.
<code>vrf <VRF-NAME></code>	Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named <code>default</code> is used.

Examples

Configuring location override with location and VRF:

```
switch(config-aruba-central)# location-override aruba-central.com vrf default  
switch(config-aruba-central)#  
switch(config-aruba-central)# location-override aruba-central.com vrf red  
switch(config-aruba-central)# location-override 10.0.0.1 vrf red  
switch(config-aruba-central)# location-override 10.0.0.1:443 vrf red  
switch(config-aruba-central)# location-override  
2001:0db8:85a3:0000:0000:8a2e:0370:7334 vrf red  
switch(config-aruba-central)# location-override  
[2001:0db8:85a3:0000:0000:8a2e:0370:7334]:443 vrf red
```

Configuring location override with location only:

```
switch(config-aruba-central)# location-override aruba-central.com
switch(config-aruba-central)#
```

Removing location override values from the Aruba Central configuration context:

```
switch(config-aruba-central)# no location-override
switch(config-aruba-central)#
```

Command History

Release	Modification
10.13.1000	Command updated to reflect OTP scenario..
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-aruba-central	Administrators or local user group members with execution rights for this command.

location-override-alternative

```
location-override-alternative <LOCATION> [vrf <VRF>]
no location-override-alternative <LOCATION> [vrf <VRF>]
```

Description

Configures information about Aruba Central connection when the alternative location is used.

The **no** form of this command removes the **location-override-alternative** configuration.

Parameter	Description
<LOCATION>	Specifies the Aruba-Central location.
vrf <VRF>	Specifies the VRF used to connect to Aruba-Central.

Usage

When the main and alternative Aruba Central server locations are specified, the switch attempts to connect to the main Aruba Central server. If there is connectivity failure with the main Aruba Central server location, it attempts to establish a connection with the alternative server location.

If the alternative location is configured without a main location, the user is prompted for confirmation. In this case, there is no redundancy and the switch attempts to connect to the alternative location.

Location can take one of the following values:

- A fully qualified domain name (FQDN) along with an optional port number.
- An IPv4 address with an optional port number.
- An IPv6 address with an optional port number.

If the port number is not specified, then port 443 is used by default. If the command is executed without the VRF parameter, the switch uses the 'default' VRF.

An Aruba Central server location can only be a fully qualified domain name (FQDN) or a valid IP address. If the command is entered without the VRF parameter, the switch uses the default VRF.

Examples

Example of configuring with the aruba-central.com location and VRF red:

```
switch(config-aruba-central)# location-override-alternative aruba-central.com vrf red
switch(config-aruba-central)#
```

Example of a configuration with location only:

```
switch(config-aruba-central)# location-override-alternative aruba-central.com
switch(config-aruba-central)#
```

Example of removing the override configuration:

```
switch(config-aruba-central)# no location-override-alternative
switch(config-aruba-central)# location-override-alternative 10.0.0.1 vrf red
switch(config-aruba-central)# location-override-alternative 10.0.0.1:443 vrf red
switch(config-aruba-central)# location-override-alternative
2001:0db8:85a3:0000:0000:8a2e:0370:7334 vrf red
switch(config-aruba-central)# location-override-alternative
[2001:0db8:85a3:0000:0000:8a2e:0370:7334]:443 vrf red
```

Command History

Release	Modification
10.13.1000	Command updated to reflect OTP scenario.
10.12.1000	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config-aruba-central	Administrators or local user group members with execution rights for this command.

show aruba-central

```
show aruba-central
```

Description

Shows information about Aruba Central connection and the status of the Activate server connection.

Examples

Example of a switch that has the Aruba Central connection:

```
switch# show aruba-central
Central admin state           : enabled
Central location              : 10.0.0.1
VRF for connection            : mgmt
Shared Token                  : N/A
Central connection status     : connected
Central source                : activate
Central source connection status : connected
Central source last connected on : Wed Jun 28 23:07:25 UTC 2023
Main location                 : 10.0.0.1
Main VRF                      : mgmt
Alternative location          : N/A
Alternative VRF               : N/A
Activate Server URL           : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A
Source IP                    : N/A
Source IP Overridden          : False
Central support mode          : disabled
```

Example of a switch when the main CLI location is used:

```
switch# show aruba-central
Central admin state           : enabled
Central location              : 10.0.0.1
VRF for connection            : mgmt
Shared secret                 : N/A
Central connection status     : connected
Central source                : cli
Central source connection status : connected
Central source last connected on : Wed Jun 28 23:07:25 UTC 2023
Main location                 : 10.0.0.1
Main VRF                      : mgmt
Alternative location          : 20.0.0.1
Alternative VRF               : default
Activate server URL           : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A
Source IP                    : N/A
Source IP Overridden          : False
Central support mode          : disabled
```

Example of a switch when the alternative CLI location is used:

```
switch# show aruba-central
Central admin state           : enabled
Central location              : 20.0.0.1
VRF for connection            : default
Shared secret                 : N/A
Central connection status     : connected
Central source                : cli
Central source connection status : connected
Central source last connected on : Wed Jun 28 23:07:25 UTC 2023
Main location                 : 10.0.0.1
Main VRF                      : mgmt
Alternative location          : 20.0.0.1
Alternative VRF               : default
Activate server URL           : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A
```

```

Source IP : N/A
Source IP Overridden : False
Central support mode : disabledswitch# show aruba-central
Central admin state : enabled
Central location : 20.0.0.1
VRF for connection : default
Shared secret : N/A
Central connection status : connected
Central source : cli
Central source connection status : connected
Central source last connected on : Wed Jun 28 23:07:25 UTC 2023
Main location : 10.0.0.1
Main VRF : mgmt
Alternative location : 20.0.0.1
Alternative VRF : default
Activate server URL : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A
Source IP : N/A
Source IP Overridden : False
Central support mode : disabled

```

Example of a switch when the location is obtained from DHCP options:

```

switch# show aruba-central
Central admin state : enabled
Central location : central-western-us.arubanetworks.com
VRF for connection : RED
Shared secret : N/A
Central connection status : connected
Central source : DHCP
Central source connection status : connected
Central source last connected on : Fri Jun 30 20:22:33 UTC 2023
Main location : central-western-us.arubanetworks.com
Main VRF : mgmt
Alternative location : N/A
Alternative VRF : N/A
Activate server URL : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A
Source IP : 100.0.0.1
Source IP Overridden : False
Central support mode : disabled

```

Example of a switch when Aruba Central is disabled:

```

switch# show aruba-central
Central admin state : disabled
Central location : N/A
VRF for connection : N/A
Shared secret : N/A
Central connection status : N/A
Central source : none
Central source connection status : N/A
Central source last connected on : N/A
Main location : N/A
Main VRF : N/A
Alternative location : N/A
Alternative VRF : N/A
Activate server URL : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A

```

```

Source IP                               : N/A
Source IP Overridden                    : False
Central support mode                    : disabled
Central admin state                     : disabled
Central location                        : N/A
VRF for connection                      : N/A
Shared secret                           : N/A
Central connection status                : N/A
Central source                          : none
Central source connection status         : N/A
Central source last connected on        : N/A
Main location                           : N/A
Main VRF                                : N/A
Alternative location                     : N/A
Alternative VRF                          : N/A
Activate server URL                     : devices-v2.arubanetworks.com
System time synchronized from Activate  : N/A
Source IP                               : N/A
Source IP Overridden                    : False
Central support mode                    : disabled

```

Example of a switch when Aruba Central is not reachable:

```

switch# show aruba-central
Central admin state                    : enabled
Central location                       : N/A
VRF for connection                     : N/A
Shared secret                           : N/A
Central connection status               : not-reachable
Central source                         : activate
Central source connection status        : connected
Central source last connected on        : Fri Jun 30 20:22:33 UTC 2023
Main location                           : N/A
Main VRF                                : N/A
Alternative location                     : N/A
Alternative VRF                          : N/A
Activate server URL                     : devices-v2.arubanetworks.com
System time synchronized from Activate  : N/A
Source IP                               : N/A
Source IP Overridden                    : False
Central support mode                    : disabled

```

Command History

Release	Modification
10.12.1000	Enhanced to support more scenarios
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config current-context

show running-config current-context

Description

Shows the running configuration for the current-context. If user is in the context of Aruba-Central (**config-aruba-central**), then Aruba Central running configuration is displayed.

Examples

Shows the running configuration of Aruba Central:

```
switch(config-aruba-central)# show running-config current-context
aruba-central
    disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

The Domain Name System (DNS) is the Internet protocol for mapping domain names or hostnames into IPv4 and/or IPv6 addresses. DNS allows users to enter more readily memorable and intuitive hostnames, rather than IP addresses, to identify devices connected to a network. It also allows a host to keep the same hostname even if it changes its IP address.

Hostname resolution can be either static or dynamic.

- In static resolution, a local table is defined on the switch that associates hostnames with their IP addresses. Static tables can be used to speed up the resolution of frequently queried hosts.
- Dynamic resolution requires that the switch query a DNS server located elsewhere on the network. Dynamic name resolution takes more time than static name resolution, but requires far less configuration and management.

Configuration

VxLAN IPv4 and IPv6 underlay support the DNS client. To ensure configuration, make sure the following are set up:

VxLAN overlay

- Must be configured to establish the VxLAN tunnel.

VxLAN tunnel

- Ensure the tunnel is established between the VTEPS via either static or EVPN using the following command: **show interface VxLAN VTEPS**.



For more information, refer to the [AOS-CX VxLAN EVPN Guide](#).

DNS client

The DNS client resolves hostnames to IP addresses for protocols that are running on the switch. When the DNS client receives a request to resolve a hostname, it can do so in one of two ways:

- Forward the request to a DNS name server for resolution.
- Reply to the request without using a DNS name server, by resolving the name using a statically defined table of hostnames and their associated IP addresses.

Configuring the DNS client

Procedure

1. Configure one or more DNS name servers with the command `ip dns server`.
2. To resolve DNS requests by appending a domain name to the requests, either configure a single domain name with the command `ip dns domain-name`, or configure a list of up to six domain names with the command `ip dns domain-list`.
3. To use static name resolution for certain hosts, associate an IP address to a host with the command `ip dns host`.
4. Review your DNS configuration settings with the command `show ip dns`.

Examples

This example creates the following configuration:

- Defines the domain **switch.com** to append to all requests.
- Defines a DNS server with IPv4 address of **1.1.1.1**.
- Defines a static DNS host named **myhost1** with an IPv4 address of **3.3.3.3**.
- DNS client traffic is sent on the default VRF (named **default**).

```
switch(config)# ip dns domain-name switch.com
switch(config)# ip dns server-address 1.1.1.1
switch(config)# ip dns host myhost1 3.3.3.3
switch(config)# exit
switch# show ip dns
```

```
VRF Name : vrf_mgmt
```

Host Name	Address

```
VRF Name : vrf_default
Domain Name : switch.com
DNS Domain list :
Name Server(s) : 1.1.1.1
```

Host Name	Address

myhost1	

This example creates the following configuration:

- Defines three domains to append to DNS requests **domain1.com**, **domain2.com**, **domain3.com** with traffic forwarding on VRF **mainvrf**.
- Defines a DNS server with an IPv6 address of **c::13**.
- Defines a DNS host named **myhost** with an IPv4 address of **3.3.3.3**.

```
switch(config)# ip dns domain-list domain1.com vrf mainvrf
switch(config)# ip dns domain-list domain2.com vrf mainvrf
switch(config)# ip dns domain-list domain3.com vrf mainvrf
switch(config)# ip dns server-address c::13
switch(config)# ip dns host myhost 3.3.3.3 vrf mainvrf
switch(config)# quit
switch# show ip dns mainvrf
```

```
VRF Name : mainvrf
Domain Name :
```

```
DNS Domain list : domain1.com, domain2.com, domain3.com
Name Server(s) : c::13
```

Host Name	Address
-----	-----
myhost	3.3.3.3

DNS client commands

ip dns domain-list

```
ip dns domain-list <DOMAIN-NAME> [vrf <VRF-NAME>]
no ip dns domain-list <DOMAIN-NAME> [vrf <VRF-NAME>]
```

Description

Configures one or more domain names that are appended to the DNS request. The DNS client appends each name in succession until the DNS server replies. Domains can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF.

The **no** form of this command removes a domain from the list.

Parameter	Description
<code>list <DOMAIN-NAME></code>	Specifies a domain name. Up to six domains can be added to the list. Length: 1 to 256 characters.
<code>vrf <VRF-NAME></code>	Specifies a VRF name. Default: default.

Examples

This example defines a list with two entries: **domain1.com** and **domain2.com**.

```
switch(config)# ip dns domain-list domain1.com
switch(config)# ip dns domain-list domain2.com
```

This example defines a list with two entries, **domain2.com** and **domain5.com**, with requests being sent on **mainvrf**.

```
switch(config)# ip dns domain-list domain2.com vrf mainvrf
switch(config)# ip dns domain-list domain5.com vrf mainvrf
```

This example removes the entry **domain1.com**.

```
switch(config)# no ip dns domain-list domain1.com
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip dns domain-name

```
ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
no ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
```

Description

Configures a domain name that is appended to the DNS request. The domain can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF. If a domain list is defined with the command `ip dns domain-list`, the domain name defined with this command is ignored.

The **no** form of this command removes the domain name.

Parameter	Description
<DOMAIN-NAME>	Specifies the domain name to append to DNS requests. Length: 1 to 256 characters.
vrf <VRF-NAME>	Specifies a VRF name. Default: default.

Examples

Setting the default domain name to `domain.com`:

```
switch(config)# ip dns domain-name domain.com
```

Removing the default domain name `domain.com`:

```
switch(config)# no ip dns domain-name domain.com
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip dns host

```
ip dns host <HOST-NAME> <IP-ADDR> [ vrf <VRF-NAME> ]
no ip dns host <HOST-NAME> <IP-ADDR> [ vrf <VRF-NAME> ]
```


Description

Associates a static IP address with a hostname. The DNS client returns this IP address instead of querying a DNS server for an IP address for the hostname. Up to six hosts can be defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a static IP address associated with a hostname.

Parameter	Description
host <HOST-NAME>	Specifies the name of a host. Length: 1 to 256 characters.
<IP-ADDR>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.
vrf <VRF-NAME>	Specifies a VRF name. Default: default.

Examples

This example defines an IPv4 address of **3.3.3.3** for **host1**.

```
switch(config)# ip dns host host1 3.3.3.3
```

This example defines an IPv6 address of **b::5** for **host 1**.

```
switch(config)# ip dns host host1 b::5
```

This example defines removes the entry for **host 1** with address **b::5**.

```
switch(config)# no ip dns host host1 b::5
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip dns server address

```
ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]  
no ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Configures the DNS name servers that the DNS client queries to resolve DNS queries. Up to six name servers can be defined. The DNS client queries the servers in the order that they are defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a name server from the list.

Parameter	Description
<code><IP-ADDR></code>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.
<code>vrf <VRF-NAME></code>	Specifies a VRF name. Default: default.

Examples

This example defines a name server at **1.1.1.1**.

```
switch(config)# ip dns server-address 1.1.1.1
```

This example defines a name server at **a::1**.

```
switch(config)# ip dns server-address a::1
```

This example removes a name server at **a::1**.

```
switch(config)# no ip dns server-address a::1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show ip dns

```
show ip dns [vrf <VRF-NAME>] [vsx-peer]
```

Description

Shows all DNS client configuration settings or the settings for a specific VRF.

Parameter	Description
vrf <VRF-NAME>	Specifies the VRF for which to show information. If no VRF is defined, the default VRF is used.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

These examples define DNS settings and then show how they are displayed with the `show ip dns` command.

```
switch(config)# ip dns domain-name domain.com
switch(config)# ip dns domain-list domain5.com
switch(config)# ip dns domain-list domain8.com
switch(config)# ip dns server-address 4.4.4.4
switch(config)# ip dns server-address 6.6.6.6
switch(config)# ip dns host host3 5.5.5.5
switch(config)# ip dns host host2 2.2.2.2
switch(config)# ip dns host host3 c::12
switch(config)# ip dns domain-name reddomain.com vrf red
switch(config)# ip dns domain-list reddomain5.com vrf red
switch(config)# ip dns domain-list reddomain8.com vrf red
switch(config)# ip dns server-address 4.4.4.5 vrf red
switch(config)# ip dns server-address 6.6.6.7 vrf red
switch(config)# ip dns host host3 5.5.5.6 vrf red
switch(config)# ip dns host host2 2.2.2.3 vrf red
switch(config)# ip dns host host3 c::13 vrf red
switch# show ip dns
VRF Name : default
```

```
Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6
```

```
Host Name      Address
-----
host2          2.2.2.2
host3          5.5.5.5
host3          c::12
```

```
VRF Name : red
```

```
Domain Name : reddomain.com
DNS Domain list : reddomain5.com, reddomain8.com
Name Server(s) : 4.4.4.5, 6.6.6.7
```

```
Host Name      Address
-----
host2          2.2.2.3
host3          5.5.5.6
host3          c::13
```

```
switch(config)# ip dns domain-name domain.com vrf red
switch(config)# ip dns domain-list domain5.com vrf red
switch(config)# ip dns domain-list domain8.com vrf red
```

```

switch(config)# ip dns server-address 4.4.4.4 vrf red
switch(config)# ip dns server-address 6.6.6.6 vrf red
switch(config)# ip dns host host3 5.5.5.5 vrf red
switch(config)# no ip dns host host2 2.2.2.2 vrf red
switch(config)# ip dns host host3 c::12 vrf red

switch# show ip dns vrf red
VRF Name : red

Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6

Host Name      Address
-----
host3          5.5.5.5
host3          c::12

```

DNS client arbitration on the MGMT interface on a MGMT VRF can be updated via three different methods.

1. Using the **domain-name <name>** or **nameservers <servers>** commands in the command-line interface.
2. Using the **ip dns domain-name <DOMAIN-NAME> vrf MGMT** or **ip dns server-address <SERVER> vrf MGMT** commands in the command-line interface.
3. Using the **ip dhcp** command in the command-line interface (dynamic entries).



AOS-CX gives the following priority levels to these three update methods.

- Priority 1 - standalone CLI configuration
- Priority 2 - static ip dns configuration
- Priority 3 - Dynamic config

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Device discovery and configuration

The switch provides support for LLDP, CDP, and local MAC match by using device profiles to enable automatic discovery and configuration of other devices on the network.

Based on the type of devices connected to the interface, device profiles enable predefined configurations that can be applied to the interface. Connected devices are identified using corresponding protocol packets. When the protocol information on the interface ages, the profile or role is revoked from the interface. Only devices connected directly to the switch are detected and processed to apply a device profile. When a device of a configured type is connected to an interface, the switch automatically applies the corresponding [Device profiles](#).

Local MAC match enables dynamic assignment of client attributes, such as QoS and VLANs by using a locally configured authentication repository. Local MAC match involves creating MAC groups that are used to classify connected devices based on MAC address, MAC address mask, and MAC OUI. Local MAC match feature is useful when you do not want to afford RADIUS infrastructure or when you want to use local authentication as a backup method in case the RADIUS server is unreachable.

The following parameters can be configured for each role:

- `associate`: Used to associate captive-portal-profile or policy with the role.
- `auth-mode`: Used to configure authentication mode for the role.



There is no need to configure `auth-mode` for a plain device profile.

- `mtu`: Used to configure MTU for the role.
- `poe-priority`: Used to configure the PoE priority for the role.
- `trust-mode`: Used to configure trust mode for the role.
- `vlan`: Used to configure VLAN mode for the role.
- `stp-admin-edge-port`: Used to configure STP administrative edge port for the role.

For information on role configurations, see the *Security Guide*.

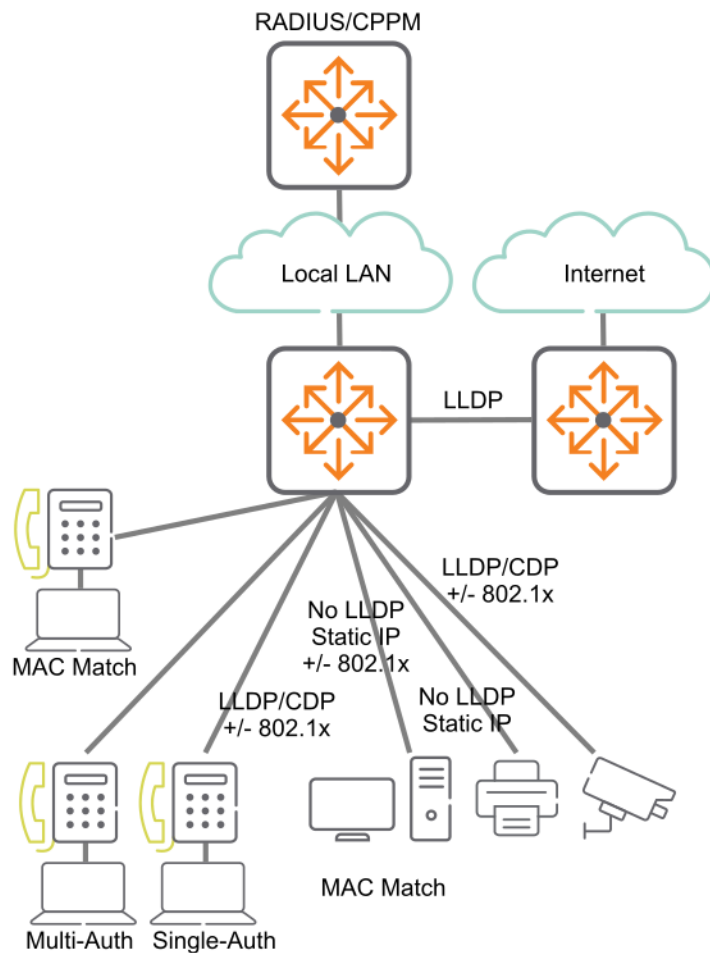
The following commands are not supported in local MAC match feature:

- `aaa authentication port-access mac-auth cached-reauth`
 - `aaa authentication port-access mac-auth cached-reauth-period`
 - `aaa authentication port-access mac-auth quiet-period`
 - `aaa authentication port-access mac-auth reauth-period`
-

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Example configuration of device deployment

The switch provides simplified deployment of devices, such as access points, IP phones, security cameras, and printers, through the use of a locally configured repository that provides authentication and dynamic port assignment, such as QoS, PoE, and tagged VLANs.



Device profiles



Device profiles rely on role configurations. For information on role configurations, see the *Security Guide*.

Device profiles are used to dynamically assign port attributes based on the type of devices connected, without having to create a RADIUS infrastructure. You can map device profiles to device groups. A device group contains various match criteria, which can be obtained from multiple sources, such as LLDP, CDP, and local MAC match. Device profiles contain port attributes to be assigned to the port when a connected device matches a device group.

Device profiles are supported on different scenarios. It can be applied on interfaces that are configured with security (802.1X or MAC authentication), or applied based on L2 port (LLDP, CDP), or applied on standalone ports with the block-until-profile-applied command enabled. All the methods are mutually exclusive of each other. The block-until-profile-applied mode must be configured only when there is a standalone port where no security has been configured and when you want the port to be offline until at least one client is onboarded based on the match and ignore criteria that you configure. Local MAC match is supported when you configure block-until-profile-applied command or device profile with security.



Up to 16 device profiles can be configured on the 6400.

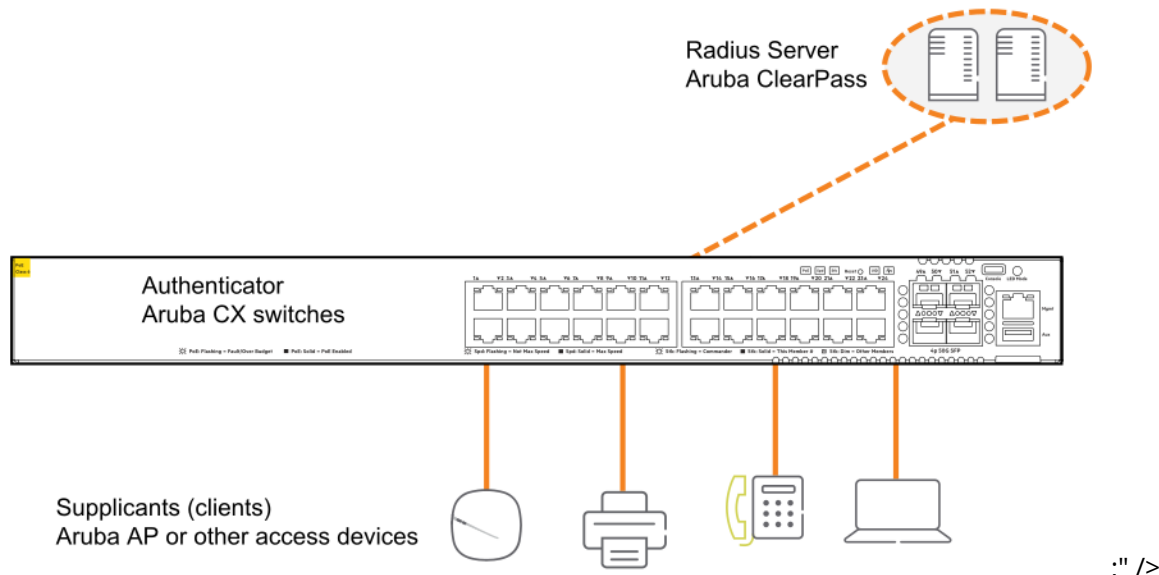
See the *Security Guide* for the following commands:

- The `port-access onboarding-method precedence` command—If you are configuring both security and device profile on the port, and you want to configure the order in which the methods will be executed.
- The `port-access fallback-role` command—If you want to configure a role that must be applied to devices when no other role exists or can be derived for that device.

If you configure a match criteria that matches across multiple device profiles, then the priority considered is LLDP, CDP, and then local MAC match. That is, LLDP precedes over CDP, which in turn precedes over local MAC match.

The following figure displays a simple configuration of device profile and AAA authentication with RADIUS server and Aruba ClearPass Policy Manager. Local MAC match feature is useful when you do not want to afford RADIUS infrastructure or when you want to use local authentication as a backup method in case the RADIUS server is unreachable.

Figure 1 Example of device profile setup along with RADIUS infrastructure



Configuring a device profile for LLDP

Procedure

1. Create an LLDP group with the command `port-access lldp-group`.
2. Define rules for adding devices to an LLDP group with the command `match`.
3. Define rules for ignoring devices so that they are not added to an LLDP group with the command `ignore`.
4. Create a device profile with the command `port-access device-profile`.
5. Add the LLDP group with the command `associate lldp-group`.
6. Add a role to a device profile with the command `associate role`. Make sure that the role is already created. For information on how to create a role, see port access role information in the

7. Enable the device profile with the command `enable`.

Configuring a device profile for CDP

1. Create a CDP group with the command `port-access cdp-group`.
2. Define rules for adding devices to a CDP group with the command `match`.
3. Define rules for ignoring devices so that they are not added to a CDP group with the command `ignore`.
4. Create a device profile with the command `port-access device-profile`.
5. Add a CDP group to a device profile with the command `associate cdp-group`.
6. Add a role to a device profile with the command `associate role`. Make sure that the role is already created. For information on how to create a role, see port access role information in the *Security Guide*.
7. Enable a device profile with the command `enable`.

Configuring a device profile for local MAC match

Procedure

1. Create a MAC group with the `mac-group` command.
2. Define rules for adding devices to a MAC group with the `match (for MAC groups)` command.
3. Define rules for ignoring devices so that they are not added to a MAC group with the `ignore (for MAC groups)` command.
4. Create a device profile with the `port-access device-profile` command.
5. Associate a MAC group with a device profile with the `associate mac-group` command.
6. Add a role to a device profile with the `associate role` command. Make sure that the role is already created. For information on how to create a role, see port access role information in the *Security Guide*.
7. Enable a device profile with the `enable` command.

Device Profile Usage Considerations

The following behaviors and limitations should be taken into consideration when configuring a device profile on a Link Aggregation Group (LAG).

Device Profiles on a LAG

- When using Device profile to onboard a device with multiple interfaces (such as a dual port AP) over a LAG, set the authentication mode in the role applied to the device to **devicemode**. The **multi-domain** authentication mode is not supported with a device profile on a LAG.
- It is not recommended to connect different devices to the same LAG when using a device profile on the LAG.
- When updating a LAG membership, all the device profile clients on the LAG are logged off automatically from the switch.
- Device profile is not supported on a multi-chassis LAG.

- If you issue the command **aaa authentication port-access allow-cdp-proxy-logoff** on the LAG, a proxy-logoff TLV received on any member interface of the LAG will log off all clients onboarded on the LAG.

Security Considerations

- Each individual interface of a LAG must successfully match a Device profile for the corresponding interface to be unblocked on the data plane.
- An onboarded client on the LAG is authorized to send and receive packets on any of the member interface of the LAG that is unblocked on the data plane. This is true even for a different client connected to the same LAG.
- When using LLDP packets to authenticate a device using a device profile on a LAG, issue the command **aaa authentication port-access allow-lldp-auth mac chassis-mac** to set the authentication MAC address as the chassis MAC address
- LLDP BPDUs are blocked by default. Issue the command **aaa authentication port-access allow-lldp-bpdu** to allow the LLDP BPDUs.
- If a Device profile is used to onboard multiple devices connected to the same LAG, the client limit on the LAG must be configured to match the number of clients onboarding the LAG.
- When using LLDP packets to authenticate a Dual Port AP using a device profile on a LAG, issue the command **aaa authentication port-access allow-lldp-auth mac source-mac** to set the authentication MAC address as the port/interface MAC address.

Supported Role Attributes

The following port-access role attributes are supported for clients that onboard via device profile on a LAG.

- Authentication mode
- Client inactivity timeout
- Session timeout
- Access and Trunk VLANs
- Port-Access policy
- MTU
- PoE Priority
- PoE Allocate-By
- Private-VLAN
- STP admin edge port
- Trust mode

Device profile commands

aaa authentication port-access allow-cdp-auth

```
aaa authentication port-access allow-cdp-auth
no aaa authentication port-access allow-cdp-auth
```

Description

Use this command to allow or block authentication on the CDP (Cisco Discovery Protocol) BPDU (Bridge Protocol Data Unit) . This is allowed by default. The **no** form of this command prevents authentication

on CDP packets received on the port. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts.

Examples

On the 6400 Switch Series, interface identification differs.

Allowing authentication on a CDP CPDU on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-cdp-auth
```

Allowing authentication on a CDP CPDU on a LAG port:

```
switch(config)# interface lag 1
switch(config-lag-if)# aaa authentication port-access allow-cdp-auth
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-if config-lag-if	Administrators or local user group members with execution rights for this command.

aaa authentication port-access allow-cdp-bpdu

```
aaa authentication port-access allow-cdp-bpdu
no aaa authentication port-access allow-cdp-bpdu
```

Description

Allows all packets related to the CDP (Cisco Discovery Protocol) BPDU (Bridge Protocol Data Unit) on a secure port or LAG. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts. The **no** form of this command blocks the CDP BPDU on a secure port. On a nonsecure port, the command has no effect.

Examples

On the 6400 Switch Series, interface identification differs.

Allowing a CDP BPDU on secure port **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-cdp-bpdu
switch(config-if)# do show running-config
Current configuration:
```

```

!
!Version AOS-CX 10.0X.0000
led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
    enable
aaa authentication port-access dot1x authenticator
    enable
interface 1/1/1
    no shutdown
    vlan access 1
    aaa authentication port-access allow-cdp-bpdu
    aaa authentication port-access mac-auth
        enable
    aaa authentication port-access dot1x authenticator
        enable

switch(config-if)# do show port-access device-profile interface all
Port 1/1/1, Neighbor-Mac 00:0c:29:9e:d1:20
  Profile Name      : access_switches
  LLDP Group       :
  CDP Group        : aruba-ap_cdp
  Role             : test_ap_role
  Status           : In Progress
  Failure Reason    :

```

Blocking LLDP packet on secure port **1/1/1**:

```

switch(config)# interface 1/1/1
switch(config-if)# no aaa authentication port-access allow-cdp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version AOS-CX 10.0X.0000
led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
    enable
interface 1/1/1
    no shutdown
    vlan access 1
    aaa authentication port-access mac-auth
        enable

```

Allowing a CDP BPDU on LAG **1**:

```

switch(config)# interface lag 1
switch(config-lag-if)# aaa authentication port-access allow-cdp-bpdu

```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300	config-if	Administrators or local user group members with execution rights for this command.
6400	config-lag-if	

aaa authentication port-access allow-cdp-proxy-logoff

```
aaa authentication port-access allow-cdp-proxy-logoff
no aaa authentication port-access allow-cdp-proxy-logoff
```

Description

Allows a client to be logged off from the system via a special TLV in the CDP packet. By default, proxy logoff via CDP packet support is disabled. When **allow-cdp-proxy-logoff** is enabled, TLV received from CDP packets corresponding to logoff processing will be read and logoff is issued to the clients. This only works on client authentication enabled ports and **aaa authentication port-access allow-cdp-bpdu** must be enabled to process. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts.

Examples

On the 6400 Switch Series, interface identification differs.

Allowing a client to be logged off from the system via a special TLV in the CDP packet:

```
switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-cdp-proxy-logoff
switch(config-if)# show running-config interface 1/1/1
interface 1/1/1
    no shutdown

    vlan access 1
    aaa authentication port-access allow-cdp-bpdu
    aaa authentication port-access allow-cdp-proxy-logoff
    aaa authentication port-access allow client-limit 2
    aaa authentication port-access dot1x authenticator
        enable
    aaa authentication port-access mac-auth
        enable
    exit
```

The aaa authentication port-access allow-cdp-proxy-logoff command can also be issued from a LAG port context

```
switch(config)# interface lag 1
switch(config-lag if)# aaa authentication port-access allow-cdp-proxy-logoff
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.09.1000	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config-if	Administrators or local user group members with execution rights for this command.
6400	config-lag-if	

aaa authentication port-access allow-lldp-bpdu

```
aaa authentication port-access allow-lldp-bpdu
no aaa authentication port-access allow-lldp-bpdu
```

Description

Allows all packets related to the LLDP BPDU (Bridge Protocol Data Unit) on a secure port. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts.

The **no** form of this command blocks the LLDP BPDU on a secure port. On a nonsecure port, the command has no effect.

Examples

On the 6400 Switch Series, interface identification differs.

Allowing an LLDP BPDU on secure port **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-lldp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version AOS-CX 10.0X.0000
led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
enable
interface 1/1/1
no shutdown

vlan access 1
aaa authentication port-access allow-lldp-bpdu
aaa authentication port-access mac-auth
enable

switch(config-if)# do show port-access device-profile interface all
Port 1/1/1, Neighbor-Mac 00:0c:29:9e:d1:20
  Profile Name      : access_switches
  LLDP Group        : 2920-grp
  CDP Group         :
  Role              : local_2920_role
```

```
Status          : Profile Applied
Failure Reason   :
```

Blocking LLDP BPDU on secure port **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no aaa authentication port-access allow-lldp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version AOS-CX 10.0X.0000led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
enable
interface 1/1/1
no shutdown

vlan access 1
aaa authentication port-access mac-auth
enable
```

Allowing an LLDP BPDU on a LAG port:

```
switch(config)# interface lag 1
switch(config-lag-if)#aaa authentication port-access allow-lldp-bpdu
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-if config-lag-if	Administrators or local user group members with execution rights for this command.

associate cdp-group

```
associate cdp-group <GROUP-NAME>
no associate cdp-group <GROUP-NAME>
```

Description

Associates a CDP (Cisco Discovery Protocol) group with a device profile. A maximum of two CDP groups can be associated with a device profile.

The **no** form of this command removes a CDP group from a device profile.

Parameter	Description
<i><GROUP-NAME></i>	Specifies the name of the CDP group to associate with this device profile. Range: 1 to 32 alphanumeric characters.

Examples

Associating the CDP group **my-cdp-group** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate cdp-group my-cdp-group
```

Removing the CDP group **my-cdp-group** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate cdp-group my-cdp-group
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-device-profile	Administrators or local user group members with execution rights for this command.

associate lldp-group

```
associate lldp-group <GROUP-NAME>
no associate lldp-group <GROUP-NAME>
```

Description

Associates an LLDP group with a device profile. A maximum of two LLDP groups can be associated with a device profile

The **no** form of this command removes an LLDP group from a device profile.

Parameter	Description
<i><GROUP-NAME></i>	Specifies the name of the LLDP group to associate with the device profile. Range: 1 to 32 alphanumeric characters.

Examples

Associating the LLDP group **my-lldp-group** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate lldp-group my-lldp-group
```

Removing the LLDP group **my-ldp-group** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate lldp-group my-ldp-group
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-device-profile	Administrators or local user group members with execution rights for this command.

associate mac-group

```
associate mac-group <GROUP-NAME>
no associate mac-group <GROUP-NAME>
```

Description

Associates a MAC group with a device profile. A maximum of two MAC groups can be associated with a device profile.

The **no** form of this command removes a MAC group from a device profile.

Parameter	Description
<GROUP-NAME>	Specifies the name of the MAC group to associate with this device profile. Range: 1 to 32 alphanumeric characters.

Examples

Associating the MAC group **mac01-group** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate mac-group mac01-group
```

Removing the MAC group **mac01-group** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate mac-group mac01-group
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-device-profile	Administrators or local user group members with execution rights for this command.

associate role

```
associate role <ROLE-NAME>
no associate role <ROLE-NAME>
```

Description

Associates a role with a device profile. Only one role can be associated with a device profile. For information on how to configure a role, see the port access role information in the *Security Guide*.

The **no** form of this command removes a role from a device profile.

Parameter	Description
<ROLE-NAME>	Specifies the name of the role to associate with the device profile. Range: 1 to 64 alphanumeric characters.

Examples

Associating the role **my-role** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate role my-role
```

Removing the role **my-role** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate role my-role
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-device-profile	Administrators or local user group members with execution rights for this command.

disable

```
disable
no disable
```

Description

Disables a device profile.

The **no** form of this command enables a device profile.

Examples

Disabling a device profile:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# disable
```

Enabling a device profile named profile01:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-device-profile	Administrators or local user group members with execution rights for this command.

enable

enable
no enable

Description

Enables a device profile.

The **no** form of this command disables a device profile.

Examples

Enabling a device profile:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# enable
```

Disabling a device profile named profile01:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-device-profile	Administrators or local user group members with execution rights for this command.

ignore (for CDP groups)

```
ignore [seq <SEQ-NUM>] {platform <PLATFORM> | sw-version <SWVERSION> |
    voice-vlan-query <VLAN-ID>}
no ignore [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |
    voice-vlan-query <VLAN-ID>}
```

Description

Defines a rule to ignore devices for a CDP (Cisco Discovery Protocol) group. Up to 64 match/ignore rules can be defined for a group.

The **no** form of this command removes a rule for ignoring devices from a CDP group.

Parameter	Description
seq <SEQ-ID>	Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.
platform <PLATFORM>	Specifies the hardware or model details of the neighbor. Range: 1 to 128 alphanumeric characters.
sw-version <SWVERSION>	Specifies the software version of the neighbor. Range: 1 to 128 alphanumeric characters.
voice-vlan-query <VLAN-ID>	Specifies the VLAN query value of the neighbor. Range: 1 to 65535.

Examples

Adding a rule to the CDP group **grp01** that ignores a device that transmits **PLATFORM01** in the platform TLV:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# ignore platform PLATFORM01
```

Adding a rule to the CDP group **grp01** that ignores a device that transmits **SWVERSION** in software version TLV:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# ignore sw-version SWVERSION
```

Removing the rule that matches the sequence number **25** from the CDP group named **grp01**.

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# no ignore seq 25
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-cdp-group	Administrators or local user group members with execution rights for this command.

ignore (for LLDP groups)

```
ignore [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
    vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
no ignore [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
    vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
```

Description

Defines a rule to ignore devices for an LLDP group. Up to 64 match/ignore rules can be defined for a group.

The **no** form of this command removes a rule for ignoring devices from an LLDP group.

Parameter	Description
seq <SEQ-ID>	Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.
sys-desc <SYS-DESC>	Specifies the LLDP system description type-length-value (TLV). Range: 1 to 256 alphanumeric characters.
sysname <SYS-NAME>	Specifies the LLDP system name TLV. Range: 1 to 64 alphanumeric characters.
vendor-oui <VENDOR-OUI>	Specifies the LLDP system vendor OUI TLV. Range: 1 to 6 alphanumeric characters.
type <KEY>	Specifies the vendor OUI subtype key. Optional.
value <VALUE>	Specifies the vendor OUI subtype value. Range: 1 to 256 alphanumeric characters.

Examples

Adding a rule to the LLDP group **grp01** that ignores a device that transmits **PLATFORM01** in the system description TLV:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# ignore sys-desc PLATFORM01
```

Removing the rule that matches the sequence number **25** from the LLDP group named **grp01**.

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# no match seq 25
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-lldp-group	Administrators or local user group members with execution rights for this command.

ignore (for MAC groups)

```
[seq <SEQ-ID>] ignore {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
no [seq <SEQ-ID>] ignore {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
```

Description

Defines a rule to ignore devices for a MAC group based on the criteria of MAC address, MAC address mask, or MAC Organizational Unique Identifier (OUI). Up to 64 ignore rules can be defined for a group. The **no** form of this command removes a rule for ignoring devices from a MAC group.

Parameter	Description
seq <SEQ-ID>	Specifies the entry sequence ID of the rule to create or modify a MAC group. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence. Range: 1 to 4294967295 .
mac <MAC-ADDR>	Specifies the MAC address of the device to ignore.
mac-mask <MAC-MASK>	Specifies the MAC address mask to ignore devices in that range. Supported MAC address masks: /32 and /40 .
mac-oui <MAC-OUI>	Specifies the MAC OUI to ignore devices in that range. Supports MAC OUI address of maximum length of 24 bits.

Usage

To achieve the required configuration of matches for devices, it is recommended to first ignore the devices that you do not want to add. Then match the criteria for the rest of the devices that you want to add to the MAC group.

For example, if you want to ignore a specific device but add all the other devices that belong to a MAC OUI, then you must first configure the ignore criteria with a lower sequence number. And then configure match criteria with a higher sequence number.

Examples

Adding a rule to the MAC group **grp01** to ignore a device based on MAC address, but match all other devices belonging to a MAC OUI:

```
switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac 1a:2b:3c:4d:5e:6f
switch(config-mac-group)# match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
!
!
ssh server vrf mgmtdefault
!
!
!
!
vlan 1
interface mgmt
    no shutdown
    ip dhcp
mac-group grp01
    seq 10 ignore mac 1a:2b:3c:4d:5e:6f
    seq 20 match mac-oui 1a:2b:3c
...
```

Adding a rule to the MAC group **grp01** to ignore devices based on MAC address mask, but match all other devices belonging to a MAC OUI:

```
switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac-mask 1a:2b:3c:4d/32
switch(config-mac-group)# match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
!
!
ssh server vrf mgmtdefault
!
!
!
```

```

!
!
vlan 1
interface mgmt
    no shutdown
    ip dhcp
mac-group grp01
    seq 10 ignore mac-mask 1a:2b:3c:4d/32
    seq 20 match mac-oui 1a:2b:3c
...

```

Adding a rule to the MAC group **grp01** that ignores a device based on complete MAC address:

```

switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac 1a:2b:3c:4d:5e:6f

```

Adding a rule to the MAC group **grp02** that ignores devices based on MAC mask:

```

switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac-mask 1a:2b:3c:4d:5e/40
switch(config-mac-group)# ignore mac-mask 18:e3:ab:73/32

```

Adding a rule to the MAC group **grp03** that ignores devices based on MAC OUI:

```

switch(config)# mac-group grp03
switch(config-mac-group)# ignore mac-oui 81:cd:93

```

Adding a rule to the MAC group **grp01** that ignores devices with a sequence number and based on MAC address:

```

switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 ignore mac b2:c3:44:12:78:11
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt
    no shutdown
    ip dhcp
mac-group grp01
    seq 10 ignore mac b2:c3:44:12:78:11
...

```

Removing the rule from the MAC group **grp01** based on sequence number:

```

switch(config)# mac-group grp01
switch(config-mac-group)# no ignore seq 10
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt
    no shutdown
    ip dhcp
mac-group grp01
...

```

Adding a rule to the MAC group **grp01** that ignores devices with MAC entry sequence number and based on MAC OUI:

```

switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 ignore mac b2:c3:44:12:78:11
switch(config-mac-group)# seq 20 ignore mac-oui 1a:2b:3c
switch(config-mac-group)# seq 30 ignore mac-mask 71:14:89:f3/32
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt
    no shutdown
    ip dhcp
mac-group grp01
    seq 10 ignore mac b2:c3:44:12:78:11
    seq 20 ignore mac-oui 1a:2b:3c
    seq 30 ignore mac-mask 71:14:89:f3/32
...

```

Removing the rule from the MAC group **grp01** based on sequence number and MAC OUI:

```

switch(config)# mac-group grp01
switch(config-mac-group)# no seq 20 ignore mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!

```



```

vlan 1
interface mgmt
    no shutdown
    ip dhcp
mac-group grp01
    seq 10 ignore mac b2:c3:44:12:78:11
    seq 30 ignore mac-mask 71:14:89:f3/32

...

```

Removing the rule that matches the sequence number **25** from the MAC group named **grp01**.

```

switch(config)# mac-group grp01
switch(config-mac-group)# no ignore seq 25

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-mac-group	Administrators or local user group members with execution rights for this command.

mac-group

```

mac-group <MAC-GROUP-NAME>
no mac-group <MAC-GROUP-NAME>

```

Description

Creates a MAC group or modifies an existing MAC group. A MAC group is used to classify connected devices based on the MAC address details, such as mask or OUI.

A maximum of 32 MAC groups can be configured on the switch. A maximum of 2 MAC groups can be associated with a device profile. Each group accepts 64 match or ignore commands.

The **no** form of this command removes a MAC group.

Parameter	Description
<MAC-GROUP-NAME>	Specifies the name of the MAC group to create or modify. The maximum number of characters supported is 32.

Examples

Creating a MAC group named **grp01**:

```
switch(config)# mac-group grp01
switch(config-mac-group)# exit
```

Removing a MAC group named **grp01**:

```
switch(config)# no mac-group grp01
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

match (for CDP groups)

```
match [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |
    voice-vlan-query <VLAN-ID>}
no match [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |
    voice-vlan-query <VLAN-ID>}
```

Description

Defines a rule to match devices for a CDP group. A maximum of 32 CDP groups can be configured on the switch. Up to 64 match or ignore rules can be defined for each group.

The **no** form of this command removes a rule for adding devices to a CDP group.

Parameter	Description
seq <SEQ-ID>	Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.
platform <PLATFORM>	Specifies the hardware or model details of the neighbor. Range: 1 to 128 alphanumeric characters.
sw-version <SWVERSION>	Specifies the software version of the neighbor. Range: 1 to 128 alphanumeric characters.
voice-vlan-query <VLAN-ID>	Specifies the VLAN query value of the neighbor. Range: 1 to 65535.

Examples

Adding rules to match a Cisco device with a specific software version on VLAN **512** to the CDP group **grp01**:

```

switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# match platform CISCO
switch(config-cdp-group)# match sw-version 11.2(12)P
switch(config-cdp-group)# match voice-vlan-query 512
switch(config-cdp-group)# match seq 50 platform cisco sw-version 11.2(12)P voice-
vlan-query 512
switch(config-cdp-group)# exit
switch(config)# do show running-config

```

Current configuration:

```

!
!Version AOS-CX Virtual.10.0X.000
!export-password: default
led locator on
!
!
vlan 1
port-access cdp-group grp01
    seq 10 match platform CISCO
    seq 20 match sw-version 11.2(12)P
    seq 30 match voice-vlan-query 512
    seq 50 match platform cisco sw-version 11.2(12)P voice-vlan-query 512

```

Removing a rule that matches the sequence number **25** from the CDP group named **grp01**:

```

switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# no match seq 25

```

Adding a rule that matches the value of vendor-OUI **000b86** to the CDP group named **grp01**:

```

switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# match vendor-oui 000b86

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-cdp-group	Administrators or local user group members with execution rights for this command.

match (for LLDP groups)

```

match [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
    vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
no match [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
    vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}

```

Description

Defines a rule to match devices for an LLDP group. Up to 64 match/ignore rules can be defined for a group.

The **no** form of this command removes a rule.

Parameter	Description
<code>seq <SEQ-ID></code>	Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.
<code>sys-desc <SYS-DESC></code>	Specifies the LLDP system description type-length-value (TLV). Range: 1 to 256 alphanumeric characters.
<code>sysname <SYS-NAME></code>	Specifies the LLDP system name TLV. Range: 1 to 64 alphanumeric characters.
<code>vendor-oui <VENDOR-OUI></code>	Specifies the LLDP system vendor OUI TLV. Range: 1 to 6 alphanumeric characters.
<code>type <KEY></code>	Specifies the vendor OUI subtype key.
<code>value <VALUE></code>	Specifies the vendor OUI subtype value. Range: 1 to 256 alphanumeric characters.

Examples

Adding rules that match the LLDP system description **ArubaSwitch** and system name **Aruba** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match sys-desc ArubaSwitch
switch(config-lldp-group)# match sysname Aruba
switch(config)# do show running-config
```

```
Current configuration:
!
!Version AOS-CX Virtual.10.0X.000
!export-password: default
led locator on
!
!
vlan 1
port-access lldp-group grp01
    seq 10 match sys-desc ArubaSwitch
    seq 20 match sysname Aruba
```

Removing a rule that matches the sequence number **25** from an LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# no match seq 25
```

Adding a rule that matches the value of vendor-OUI **000b86** with type of **1** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match vendor-oui 000b86 type 1
```

Adding a rule that matches the value of vendor-OUI **000c34** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match vendor-oui 000c34
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-lldp-group	Administrators or local user group members with execution rights for this command.

match (for MAC groups)

```
[seq <SEQ-ID>] match {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
no [seq <SEQ-ID>] match {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
```

Description

Defines a rule to match devices for a MAC group based on the criteria of MAC address, MAC address mask, or MAC Organizational Unique Identifier (OUI). Up to 64 match rules can be defined for a group.

You must not configure the following special MAC addresses:

- Null MAC—For example, 00:00:00:00:00:00 or 00:00:00/32
- Multicast MAC
- Broadcast MAC—For example, ff:ff:ff:ff:ff:ff
- System MAC

Although the switch accepts these addresses, it will not process these addresses for the local MAC match feature.

The **no** form of this command removes a rule for adding devices to a MAC group.

The number of clients that can onboard based on the match criteria is configured in the **aaa authentication port-access client-limit** command. For information about this command, see the *Security Guide* for your switch.

Parameter	Description
seq <SEQ-ID>	Specifies the entry sequence ID of the rule to create or modify a MAC group. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command

Parameter	Description
	entered, the rule with the lowest ID takes precedence. Range: 1 to 4294967295 .
mac <MAC-ADDR>	Specifies the MAC address of the device.
mac-mask <MAC-MASK>	Specifies the MAC address mask to add devices in that range. Supported MAC address masks: /32 and /40 .
mac-oui <MAC-OUI>	Specifies the MAC OUI to add devices in that range. Supports MAC OUI address of maximum length of 24 bits.

Examples

Adding a device to the MAC group **grp01** based on complete MAC address:

```
switch(config)# mac-group grp01
switch(config-mac-group)# match mac 1a:2b:3c:4d:5e:6f
switch(config-mac-group)# exit
```

Adding devices to the MAC group **grp02** based on MAC mask:

```
switch(config)# mac-group grp01
switch(config-mac-group)# match mac-mask 1a:2b:3c:4d:5e/40
switch(config-mac-group)# match mac-mask 18:e3:ab:73/32
switch(config-mac-group)# exit
```

Adding devices to the MAC group **grp03** based on MAC OUI:

```
switch(config)# mac-group grp03
switch(config-mac-group)# match mac-oui 81:cd:93
switch(config-mac-group)# exit
```

Adding devices to the MAC group **grp01** with MAC entry sequence number and based on MAC address:

```
switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 match mac b2:c3:44:12:78:11
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt
    no shutdown
    ip dhcp
mac-group grp01
    seq 10 match mac b2:c3:44:12:78:11
...
```

Removing devices from the MAC group **grp01** based on sequence number:

```
switch(config)# mac-group grp01
switch(config-mac-group)# no match seq 10
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt
    no shutdown
    ip dhcp
mac-group grp01
...
```

Adding devices to the MAC group **grp01** with MAC entry sequence number and based on MAC address, MAC address mask, and MAC OUI:

```
switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 match mac b2:c3:44:12:78:11
switch(config-mac-group)# seq 20 match mac-oui 1a:2b:3c
switch(config-mac-group)# seq 30 match mac-mask 71:14:89:f3/32
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt
    no shutdown
    ip dhcp
mac-group grp01
    seq 10 match mac b2:c3:44:12:78:11
    seq 20 match mac-oui 1a:2b:3c
    seq 30 match mac-mask 71:14:89:f3/32
...
```

Removing devices from the MAC group **grp01** based on MAC OUI:

```
switch(config)# mac-group grp01
switch(config-mac-group)# no seq 20 match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
```

```

!
!
vlan 1
interface mgmt
    no shutdown
    ip dhcp
mac-group grp01
    seq 10 match mac b2:c3:44:12:78:11
    seq 30 match mac-mask 71:14:89:f3/32

...

```

Adding devices to the MAC group **grp03** with MAC entry sequence number and based on MAC address mask:

```

switch(config)# mac-group grp03
switch(config-mac-group)# seq 10 match mac-mask 10:14:a3:b7:55/40
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt
    no shutdown
    ip dhcp
mac-group grp03
    seq 10 match mac-mask 10:14:a3:b7:55/40

...

```

Removing devices from the MAC group **grp03** based on MAC address mask:

```

switch(config)# mac-group grp03
switch(config-mac-group)# no seq 10 match mac-mask 10:14:a3:b7:55/40
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt
    no shutdown
    ip dhcp
mac-group grp03

...

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-mac-group	Administrators or local user group members with execution rights for this command.

port-access cdp-group

```
port-access cdp-group <CDP-GROUP-NAME>
no port-access cdp-group <CDP-GROUP-NAME>
```

Description

Creates a CDP (Cisco Discovery Protocol) group or modifies an existing CDP group. A CDP Group is used to classify connected devices based on the CDP packet details advertised by the device. A maximum of 32 CDP groups can be configured on the switch. Each group accepts 64 match/ignore commands.

The **no** form of this command removes a CDP group.

Parameter	Description
<CDP-GROUP-NAME>	Specifies the name of the CDP group to create or modify. The maximum number of characters supported is 32. Required.

Examples

Creating a CDP group named grp01:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# match platform CISCO
switch(config-cdp-group)# match sw-version 11.2(12)P
switch(config-cdp-group)# match voice-vlan-query 512
switch(config-cdp-group)# seq 50 match platform cisco sw-version 11.2(12)P voice-
vlan-query 512
switch(config-cdp-group)# exit
switch(config)# do show running-config

Current configuration:
!
!Version AOS-CX Virtual.10.0X.000
!export-password: default
led locator on
!
!
vlan 1
port-access cdp-group grp01
    seq 10 match platform CISCO
    seq 20 match sw-version 11.2(12)P
    seq 30 match voice-vlan-query 512
    seq 50 match platform cisco sw-version 11.2(12)P voice-vlan-query 512
```

Removing a CDP group named grp01:

```
switch(config)# no port-access cdp-group grp01
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

port-access device-profile

```
port-access device-profile <DEVICE-PROFILE-NAME>  
no port-access device-profile <DEVICE-PROFILE-NAME>
```

Description

Creates a new device profile and switches to the **config-device-profile** context. A maximum of 32 device profiles can be created. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts.

The **no** form of this command removes a device profile.

Parameter	Description
<DEVICE-PROFILE-NAME>	Specifies the name of a device profile. Range: 1 to 32 alphanumeric characters.

Examples

Creating a device profile named **profile01**:

```
switch(config)# port-access device-profile profile01  
switch(config-device-profile)#
```

Removing a device profile named **profile01**:

```
switch(config)# no port-access device-profile profile01
```

Creating a device profile named **profile02** on a LAG port:

```
switch(config)#interface lag 1  
switch(config-lag-if)# port-access device-profile profile01  
switch(config-device-profile)#
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config config-if config-lag-if	Administrators or local user group members with execution rights for this command.

port-access device-profile mode block-until-profile-applied



You must configure this mode in device profile only on standalone ports where there is no security configured and when you not want the port to be offline until one client is onboarded.

```
port-access device-profile mode block-until-profile-applied
no port-access device-profile mode block-until-profile-applied
```

Description

Configures the switch to block the port until a profile match occurs for a device. This configuration is required when no security feature is enabled on the port.

You must enable this mode or security on the port for local MAC match feature to operate. You must not enable both features on the same port at the same time.



You must not combine any other AAA configurations with the block-until-profile-applied mode.

This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts. The **no** form of this command removes a rule for adding devices to a MAC group.

Example

On the 6400 Switch Series, interface identification differs.

Configuring block-until-profile applied mode on port 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access device-profile
switch(config-if-deviceprofile)# mode block-until-profile-applied
switch(config-if-deviceprofile)# end
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-if config-if-deviceprofile config-lag-if	Administrators or local user group members with execution rights for this command.

port-access lldp-group

```
port-access lldp-group <LLDP-GROUP-NAME>  
no port-access lldp-group <LLDP-GROUP-NAME>
```

Description

Creates an LLDP group or modifies an existing LLDP group. An LLDP group is used to classify connected devices based on the LLDP type-length-values (TLVs) advertised by the device. A maximum of 32 LLDP groups can be configured on the switch. Each group accepts 64 match/ignore commands.

The **no** form of this command removes an LLDP group.

Parameter	Description
<LLDP-GROUP-NAME>	Specifies the name of the LLDP group to create or modify. The maximum number of characters supported is 32. Required.

Examples

Creating an LLDP group named grp01:

```
switch(config)# port-access lldp-group grp01  
switch(config-lldp-group)#
```

Removing an LLDP group named grp01:

```
switch(config)# no port-access lldp-group grp01
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

show port-access device-profile

```
show port-access device-profile [[interface {all | <INTERFACE-ID>}  
[client-status <MAC-ADDR>]] | name <DEVICE-PROFILE-NAME>]
```

Description

Shows the client status for a specific MAC address or profile name.

Parameter	Description
<code>interface {all <INTERFACE-ID>}</code>	Select all for all interfaces or specify the name of an interface in the format: member/slot/port .
<code>client-status <MAC-ADDR></code>	Specifies a MAC address (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F.
<code>name <DEVICE-PROFILE-NAME></code>	Specifies the name of the device profile.

Examples

On the 6400 Switch Series, interface identification differs.

Showing the applied state of the device profiles:

```
switch# show port-access device-profile

Profile Name      : accesspoints
LLDP Groups       : 2920-grp
CDP Groups        :
MAC Groups        : 2920-mac-grp1,2920-iot-grp2
Role              : local_role_1
State             : Enabled

Profile Name      : access_switches
LLDP Groups       : 2920-grp
CDP Groups        :
MAC Groups        :
Role              : local_2920_role
State             : Enabled

Profile Name      : iot_devices
LLDP Groups       :
CDP Groups        :
MAC Groups        : iot_camera-grp1,iot_sensors-grp1
Role              : local_2920_role
State             : Enabled

Profile Name      : lobbyaps
LLDP Groups       :
CDP Groups        : lobby_ap_cdp_grp
MAC Groups        :
Role              : test_ap_role
State             : Disabled
```

Showing the applied state of the device profile on interface 1/1/3:

```
switch# show port-access device-profile interface 1/1/3 client-status
00:0c:29:9e:d1:20

Port 1/1/3, Neighbor-Mac 00:0c:29:9e:d1:20
Profile Name      : lobbyaps
LLDP Group        :
CDP Group         : aruba-ap_cdp
```

```
MAC Group      :  
Role           : test_ap_role  
Status        : Failed  
Failure Reason : Failed to apply MAC based VLAN
```

Showing the applied state of a specific device profile:

```
switch# show port-access device-profile name lldp-group  
  
Profile Name      : lldp-group  
LLDP Groups       :  
CDP Groups        :  
MAC Groups        : pc-behind-phone, lldp  
Role              : auth_role  
State             : Enabled
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

LLDP

The IEEE 802.1AB Link Layer Discovery Protocol (LLDP) provides a standards-based method for network devices to discover each other and exchange information about their capabilities. An LLDP device advertises itself to adjacent (neighbor) devices by transmitting LLDP data packets on all interfaces on which outbound LLDP is enabled, and reading LLDP advertisements from neighbor devices on ports on which inbound LLDP is enabled. Inbound packets from neighbor devices are stored in a special LLDP MIB (management information base). This information can then be queried by other devices through SNMP.

LLDP information is used by network management tools to create accurate physical network topologies by determining which devices are neighbors and through which interfaces they connect. LLDP operates at layer 2 and requires an LLDP agent to be active on each interface that sends and receives LLDP advertisements. LLDP advertisements can contain a variable number of TLV (type, length, value) information elements. Each TLV describes a single attribute of a device such as: system capabilities, management IP address, device ID, port ID.

Packet boundaries

When multiple LLDP devices are directly connected, an outbound LLDP packet travels only to the next LLDP device. An LLDP-capable device does not forward LLDP packets to any other devices, regardless of whether they are LLDP-enabled.

An intervening hub or repeater forwards the LLDP packets it receives in the same manner as any other multicast packets it receives. Therefore, two LLDP switches joined by a hub or repeater handle LLDP traffic in the same way that they would if directly connected.

Any intervening 802.1D device or Layer-3 device that is either LLDP-unaware or has disabled LLDP operation drops the packet.

LLDP-MED

LLDP-MED (ANSI/TIA-1057/D6) extends the LLDP (IEEE 802.1AB) industry standard to support advanced features on the network edge for Voice Over IP (VoIP) endpoint devices with specialized capabilities and LLDP-MED standards-based functionality. LLDP-MED in the switches uses the standard LLDP commands described earlier in this section, with some extensions, and also introduces new commands unique to LLDP-MED operation. The show commands described elsewhere in this section are applicable to both LLDP and LLDP-MED operation. LLDP-MED enables:

- Configure Voice VLAN and advertise it to connected MED endpoint devices.
- Power over Ethernet (PoE) status and troubleshooting support via SNMP.

LLDP agent

When you enable LLDP on the switch, it is automatically enabled on all data plane interfaces. You can customize this behavior by manually enabling/disabling support on each interface.

Supported standards

The LLDP agent supports the following standards: IEEE 802.1AB-2005, Station, and Media Access Control Connectivity Discovery.

Supported interfaces

LLDP is supported on interfaces mapped to a physical port, and the Out-Of-Band Management (OOBM) port. It is not supported on logical interfaces, such as loopback, tunnels, and SVIs.

Operating modes

When LLDP is enabled, the switch periodically transmits an LLDP advertisement (packet) out each active port enabled for outbound LLDP transmissions and receives LLDP advertisements on each active port enabled to receive LLDP traffic.

The LLDP agent can operate in one of the following modes:

- Transmit and receive (TxRx): This is the default setting on all ports. It enables a given port to both transmit and receive LLDP packets and to store the data from received (inbound) LLDP packets in the switch's MIB.
- Transmit only (Tx): Enables a port to transmit LLDP packets that can be read by LLDP neighbors. However, the port drops inbound LLDP packets from LLDP neighbors without reading them. This prevents the switch from learning about LLDP neighbors on that port.
- Receive only (Rx): Enables a port to receive and read LLDP packets from LLDP neighbors and to store the packet data in the switch's MIB. However, the port does not transmit outbound LLDP packets. This prevents LLDP neighbors from learning about the switch through that port.
- Disabled: Disables LLDP packet transmissions and reception on a port. In this state, the switch does not use the port for either learning about LLDP neighbors or informing LLDP neighbors of its presence.

An LLDP agent operating in TxRx mode or Tx mode sends LLDP frames to its directly connected devices both periodically and when the local configuration changes.

Sending LLDP frames

Each time the LLDP operating mode of an LLDP agent changes, its LLDP protocol state machine reinitializes. A configurable reinitialization delay prevents frequent initializations caused by frequent changes to the operating mode. If you configure the reinitialization delay, an LLDP agent must wait the specified amount of time to initialize LLDP after the LLDP operating mode changes.

Receiving LLDP frames

An LLDP agent operating in TxRx mode or Rx mode confirms the validity of TLVs carried in every received LLDP frame. If the TLVs are valid, the LLDP agent saves the information and starts an aging timer. The initial value of the aging timer is equal to the TTL value in the Time To Live TLV carried in the LLDP frame. When the LLDP agent receives a new LLDP frame, the aging timer restarts. When the aging timer decreases to zero, all saved information ages out.

TLV support

By default, the agent sends and receives the following mandatory TLVs on each interface:

- Port ID
- Chassis ID
- TTL

By default, the following ANSI/TIA-1057 TLVs for LLDP Media Endpoint Discovery (MED) are enabled on an agent. Sending them depends on the configuration and reception of any MED TLVs:

- MAC/PHY status. Includes the bit rate and auto negotiation status of the link.
- Power Via MDI: Includes Power Over Ethernet related information for supported interfaces.
- Port description
- System name
- System description
- Management address
- System capabilities
- Port VLAN ID

By default, the agent sends and receives the following ANSI/TIA-1057 TLVs for LLDP Media Endpoint Discovery (MED):

- Capabilities: Indicates MED TLV capability.
- Power Via MDI: Includes Power Over Ethernet related information.
- Network Policy: Includes the VLAN configuration for voice application.
- Location: Location identification information.
- Extended Power Via MDI: Power Over Ethernet related information

TLV advertisements

The LLDP agent transmits the following:

- Chassis-ID: Base MAC address of the switch.
- Port-ID: Port number of the physical port.

- Time-to-Live (TTL): Length of time an LLDP neighbor retains advertised data before discarding it.
- System capabilities: Identifies the primary switch capabilities (bridge, router). Identifies the primary switch functions that are enabled, such as routing.
- System description: Includes switch model name and running software version, and ROM version.
- System name: Name assigned to the switch.
- Management address: Default address selection method unless an optional address is configured.
- Port description: Physical port identifier.
- Port VLAN ID: On an L2 port, contains access or native VLAN ID. On an L3 port, contains a value of 0. Trunk allowed VLANs information are not advertised as part of the Port VLAN ID TLV. (Not supported on the OOBM interface)
- Port VLAN Name: Access or native VLAN name. Enabled by default.
- Port MFS: Maximum Frame Size. Enabled by default.

LLDP MED support

LLDP-MED interoperates with directly connected IP telephony (endpoint) clients and provides the following features:

- Advertisement of the voice VLAN configured on the interface which is used by connected IP telephony (endpoint) clients.
- Advertisement of the configured location on the switch that can be used by the connected endpoint.
- Support for the fast-start capability



LLDP-MED is intended for use with VoIP endpoints and is not designed to support links between network infrastructure devices (such as switch-to-switch or switch-to-router links).

LLDP EEE

Energy Efficient Ethernet (EEE) is a mechanism defined by IEEE 802.3az, which is an extension of 802.3 IEEE standards. EEE defines support for the device to operate in the Low Power Idle (LPI) mode during low link utilization. This allows both sides of a link to disable or turn off respective transmit/receive circuitry to save power. EEE uses LLDP for exchanging optimal set of parameters for both devices.

Guidelines

- An LLDP must not contain more than one EEE TLV.
- LLDP may or may not be enabled on the interfaces supporting EEE. If LLDP is not enabled, it might not be in the optimal operational mode.
- EEE TLV should not be exchanged until it negotiates from L1 and it detects peer EEE capabilities.
- EEE TLV is disabled by default and enabled only when EEE is active.
- EEE and EEE TLV exchange can be enabled or disabled at the interface level.

Configuring the LLDP agent

Procedure

1. By default, the LLDP agent is enabled on all active interfaces. If LLDP was disabled, enable it with the command `lldp`.

2. By default, the LLDP agent transmits and receive on all interfaces. To customize LLDP behavior on a specific interface, use the commands `lldp transmit` and `lldp receive`.
3. By default, the LLDP agent sets the management address in all TLVs in the following order:
 - a. LLDP management IP address.
 - b. Loopback interface IP.
 - c. ROP (L3 ports) or SVI (L2 ports).
 - d. OOBM (Management interface IP).
 - e. Base MAC.

On the OOBM port, the following order is used:

- a. LLDP management IP address,
- b. IP address of the management interface (OOBM port).
- c. IP address of the loopback interface.
- d. Base MAC address of the switch.

To specify a different address, use the commands `lldp management-ipv4-address` and `lldp management-ipv6-address`

4. By default, all supported TLVs are sent and received. To customize the list, use the command `lldp select-tlv`.
5. By default, support for the LLDP-MED TLV is enabled. To customize settings, use the commands `lldp med` and `lldp med-location`.
6. If required, adjust LLDP timer, holdtime, reinitialization delay, and transmit delay from their default values with the commands `lldp timer`, `lldp holdtime`, `lldp reinit`, and `lldp txdelay`.

Example

On the 6400 Switch Series, interface identification differs.

This example creates the following configuration:

- Enables LLDP support.
- Disables LLDP transmission on interface **1/1/1**.

```
switch(config)# lldp
switch(config)# interface 1/1/1
switch(config-copp)# no lldp transmit
```

LLDP commands

clear lldp neighbors

```
clear lldp neighbors
```

Description

Clears all LLDP neighbor details.

Examples

Clearing all LLDP neighbor details:

```
switch# clear lldp neighbors
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

clear lldp statistics

```
clear lldp statistics
```

Description

Clears all LLDP neighbor statistics.

Examples

Clearing all LLDP neighbor statistics:

```
switch# clear lldp statistics
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

lldp

```
lldp
no lldp
```

Description

Enables LLDP support globally on all active interfaces. By default, LLDP is enabled.

The **no** form of this command disables LLDP support globally on all active interfaces. It does not remove any LLDP configuration settings.

Examples

Enabling LLDP:

```
switch(config)# lldp
```

Disabling LLDP:

```
switch(config)# no lldp
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp dot3

```
lldp dot3 {poe | macphy}  
no lldp dot3 {poe | macphy}
```

Description

Sets the 802.3 TLVs to be advertised. By default, advertisement of both POE and MAC/PHY TLVs is enabled. Not supported on the OOBM interface.

The **no** form of this command disables advertisement of 802.3 TLVs.

Parameter	Description
poe	Specifies advertisement of power over Ethernet data link classification.
macphy	Specifies advertisement of media access control and physical layer information.

Examples

Enabling advertisement of the POE TLV:

```
switch(config-if)# lldp dot3 poe
```

Disabling advertisement of the POE TLV:

```
switch(config-if)# no lldp dot3 poe
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

lldp dot3 eee

```
lldp dot3 eee
no lldp dot3 eee
```

Description

Sets the 802.3 TLVs for Energy-Efficient Ethernet (EEE) to be advertised. By default, advertisement of EEE TLVs is enabled. Not supported on the OOBM interface.

The **no** form of this command disables advertisement of 802.3 TLVs.

Parameter	Description
eee	Specifies advertisement of 802.3 TLVs for EEE.

Examples

Enabling advertisement of the EEE TLVs:

```
switch(config-if) # lldp dot3 eee
```

Disabling advertisement of the EEE TLVs:

```
switch(config-if) # no lldp dot3 eee
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

lldp dot3 mfs

```
lldp dot3 mfs
no lldp dot3 mfs
```

Description

Enables the 802.3 TLV list in LLDP to advertise for maximum frame size (MFS). Enabled by default. The **no** form of this command disables the advertisement of maximum frame size TLVs.

Examples

Enabling advertisement of maximum frame size TLVs:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dot3 mfs
```

Disabling advertisement of maximum frame size TLVs:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp dot3 mfs
```

Command History

Release	Modification
10.11	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp holdtime-multiplier

```
lldp holdtime-multiplier <multiplier>
no lldp holdtime-multiplier
```

Description

Sets the holdtime TTL multiplier value that is used to calculate the LLDP Time-to-Live value. Time-to-Live defines the length of time that neighbors consider LLDP information sent by this agent as valid. When Time-to-Live expires, the information is deleted by the neighbor. Time-to-live is calculated by multiplying holdtime by the value of **lldp timer**.

The **no** form of this command sets the holdtime TTL multiplier to its default value of 4.

Parameter	Description
<multiplier>	Specifies the TTL multiplier in the range of 2 to 10. Default: 4.

Formula

TTL = Holdtime-multiplier x lldp timer

where:

TTL = Time-to-Live

Holdtime-multiplier = Multiplying holdtime value

Ildp timer = Message transmission interval

Examples

Setting the holdtime to 8 times of the value of Ildp timer:

```
switch(config)# lldp holdtime-multiplier 8
```

Setting the holdtime to the default value of 4 times of the value of Ildp timer:

```
switch(config)# no lldp holdtime-multiplier
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp management-address vlan

```
lldp management-address vlan <VLAN-ID>  
no lldp management-address vlan <VLAN-ID>
```

Description

Sets the VLAN whose IPv4 or IPv6 address is advertised as the LLDP management authority.

The **no** form of this command removes the VLAN whose IPv4 or IPv6 address is advertised as the LLDP management authority.

The following is the precedence for the management IP address TLV in the LLDP packet (in order):

- LLDP management-IPv4-address and management-ipv6-address, if configured.
- LLDP management VLAN's Ipv4 and IPv6 address, if configured.
- Loopback IP address from the smallest configured loopback interface identifier.
- Route-only-port IP address (Layer-3 interface) or IP address of the SVI (Layer-2 interface).
- OOBM IP address.
- Base MAC address of the switch.

Parameter	Description
<VLAN-ID>	Specifies the VLAN ID.

Examples

Setting the management authority for VLAN 10:

```
switch(config)# lldp management-address vlan 10
```

Removing the management authority for VLAN 10:

```
switch(config)# no lldp management-address vlan 10
```

Command History

Release	Modification
10.12	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp management-ipv4-address

```
lldp management-ipv4-address <IPv4-ADDR>  
no lldp management-ipv4-address
```

Description

Defines the IPv4 management address of the switch which is sent in the management address TLV. One IPv4 and one IPv6 management address can be configured.

If you do not define an LLDP management address, then LLDP uses one of the following (in order):

- IP address of the port
- IP address of the management interface
- Base MAC address of the switch

The **no** form of this command removes the IPv4 management address of the switch.

Parameter	Description
<IPv4-ADDR>	Specifies the management address of the switch as an IPv4 format (x.x.x.x), where x is a decimal value from 0 to 255.

Examples

Setting the management address to **10.10.10.2**:

```
switch(config)# lldp management-ipv4-address 10.10.10.2
```

Removing the management address:


```
switch(config)# no lldp management-ipv4-address
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp management-ipv6-address

```
lldp management-ipv6-address <IPV6-ADDR>  
no lldp management-ipv6-address
```

Description

Defines the IPv6 management address of the switch. The management address is encapsulated in the management address TLV.

If you do not define an LLDP management address, then LLDP uses one of the following (in order):

- IP address of the port
- IP address of the management interface
- Base MAC address of the switch

The **no** form of this command removes the IPv6 management address of the switch.

Parameter	Description
<IPV6-ADDR>	Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

Examples

Setting the management address to 2001:db8:85a3::8a2e:370:7334:

```
switch(config)# lldp management-ipv6-address 2001:0db8:85a3::8a2e:0370:7334
```

Removing the management address:

```
switch(config)# no lldp management-ipv6-address
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp med

```
lldp med [poe [priority-override] | capability | network-policy]
no med [poe [priority-override] | capability | network-policy]
```

Description

Configures support for the LLDP-MED TLV. LLDP-MED (media endpoint devices) is an extension to LLDP developed by TIA to support interoperability between VoIP endpoint devices and other networking end-devices. The switch only sends the LLDP MED TLV after receiving a MED TLV from and connected endpoint device.

Not supported on the OOBM interface.

The **no** form of this command disables support for the LLDP MED TLV.

Parameter	Description
poe [priority-override]	Specifies advertisement of power over Ethernet data link classification. The priority-override option overrides user-configured port priority for Power over Ethernet. When both lldp dot3 poe and lldp med poe are enabled, the lldp dot3 poe3 setting takes precedence. Default: enabled.
capability	Specifies advertisement of supported LLDP MED TLVs. The capability TLV is always sent with other MED TLVs, therefore it cannot be disabled when other MED TLVs are enabled. Default: enabled.
network-policy	Network policy discovery lets endpoints and network devices advertise their VLAN IDs, and IEEE 802.1p (PCP and DSCP) values for voice applications. This TLV is only sent when a voice VLAN policy is present. Default: enabled.

Examples

Enabling advertisement of the network policy TLV:

```
switch(config-if)# lldp med network-policy
```

Disabling advertisement of the network policy TLV:

```
switch(config-if)# no lldp med network-policy
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config-if</code>	Administrators or local user group members with execution rights for this command.

lldp med location

```
lldp med location {civic-addr elin-addr }
no med location {civic-addr elin-addr }
```

Description

Configures support for the LLDP-MED TLV. Supports only civic address and emergency location information number (ELIN). Coordinate-based location is not supported.

The **no** form of this command disables support for the LLDP MED TLV.

Parameter	Description
<code>civic-addr</code>	Configures the LLDP MED civic location TLV.
<code>elin-addr</code>	Configures support for the LLDP MED emergency location TLV. This feature is intended for use in ECS applications to support class 3 LLDP-MED VoIP telephones connected to a switch in an MLTS infrastructure. An ELIN is a valid NANP format telephone number assigned to MLTS operators in North America by the appropriate authority. The ELIN is used to route emergency (E911) calls to a PSAP. (Range: 1-15 numeric characters)

The **lldp med location civic-addr** command requires a minimum of one type/value pair, but typically includes multiple type/value pairs as needed to configure a complete set of data describing a given location.

CA-TYPE: This is the first entry in a type/value pair and is a number defining the type of data contained in the second entry in the type/value pair (CA-VALUE.) Some examples of CA-TYPE specifiers include: 3=city 6=street (name) 25=building name (Range: 0 - 255)

CA-VALUE: This is the second entry in a type/value pair and is an alphanumeric string containing the location information corresponding to the immediately preceding CA-TYPE entry. Strings are delimited by either blank spaces, single quotes (' ... '), or double quotes ("... "). Each string should represent a specific data type in a set of unique type/value pairs comprising the description of a location, and each string must be preceded by a CA-TYPE number identifying the type of data in the string.

The following LLDP-MED TLV values are supported. For details on these value types, refer to [RFC 4776](#)

- 1: national subdivisions (state, canton, region, province, prefecture)
- 2: county, parish, gun (JP), district (IN)
- 3: city, township, shi (JP)
- 4: city division, borough, city district, ward, chou (JP)

- 5: neighborhood, block
- 6: group of streets below the neighborhood level
- 16: leading street direction N
- 17: trailing street suffix SW
- 18: street suffix or type
- 19: house number
- 20: house number suffix
- 21: landmark or vanity
- 22: location
- 23: name
- 24: postal/zip code
- 25: building (structure)
- 26: unit (apartment, suite)
- 27: floor
- 28: room
- 29: type of place
- 30: postal community name
- 31: post office box
- 32: additional code 13203000003
- 33: seat (desk, cubicle workstation)
- 34: primary road name 35 road section
- 36: branch road name
- 37: sub-branch road name
- 38: street name pre-modifier
- 39: street name post-modifier

Examples

Enabling support for the LLDP MED emergency location TLV:

```
switch(config-if)# lldp med location elin-addr 408-555-1212
```

Disabling support for the LLDP MED emergency location TLV:

```
switch(config-if)# no lldp med location elin-addr 408-555-1212
```

Enabling support for the LLDP MED civic address TLV:

```
switch(config-if)# lldp med location civic-addr US 1 19 123 6 Fake 18 Street
```

Disabling support for the LLDP MED civic address TLV:

```
switch(config-if)# no lldp med location civic-addr US 1 19 123 6 Fake 18 Street
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

lldp receive

lldp receive
no lldp receive

Description

Enables reception of LLDP information on an interface. By default, LLDP reception is enabled on all active interfaces, including the OOBM interface.

The **no** form of this command disables reception of LLDP information on an interface.

Examples

On the 6400 Switch Series, interface identification differs.

Enabling LLDP reception on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp receive
```

Disabling LLDP reception on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp receive
```

Enabling LLDP reception on the OOBM interface:

```
switch(config)# interface mgmt
switch(config-if)# lldp receive
```

Disabling LLDP reception on the OOBM interface:

```
switch(config)# interface mgmt
switch(config-if)# no lldp receive
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

lldp reinit

```
lldp reinit <TIME>
no lldp reinit
```

Description

Sets the amount of time (in seconds) to wait before performing LLDP initialization on an interface. The **no** form of this command sets the reinitialization time to its default value of 2 seconds.

Parameter	Description
<TIME>	Specifies the reinitialization time in seconds. Range: 1 to 10. Default: 2 seconds.

Examples

Setting the reinitialization time to 5 seconds:

```
switch(config)# lldp reinit 5
```

Setting the reinitialization time to the default value of 2 seconds:

```
switch(config)# no lldp reinit
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp select-tlv

```
lldp select-tlv <TLV-NAME>
no lldp select-tlv <TLV-NAME>
```

Description

Selects a TLV that the LLDP agent will send and receive. By default, all supported TLVs are sent and received.

The **no** form of this command stops the LLDP agent from sending and receiving a specific TLV.



LLDP supports Organization Unique Identifiers (OUI) with the following Organization-specific TLVs:

- IEEE 802.1 (DOT1) (oui:0x00, 0x80, 0xc2)
- IEEE 802.3 (DOT3) (oui:0x00, 0x12, 0x0f)
- Aruba, a Hewlett Packard Enterprise Company (oui:0x88, 0x3a, 0x30)

Parameter	Description
<code>select-tlv <TLV-NAME></code>	Specifies the TLV name to send. The following TLV names are supported: ▪ management-address: Selection is based on priority in the following list (for example if first TLV name isn't selected, the next will be, progressing through this list until a selection is made): 1. IPv4 or IPV6 management address. 2. IP address of the lowest configured loopback interface. 3. If layer 3, then the route-only port IP address. If layer 2, the IP address of the SVI. 4. OOBM interface IP address. 5. Base MAC address of the switch. ▪ port-description: Select port-description TLV. ▪ port-vlan-id: Select port-vlan-id TLV. ▪ port-vlan-name: Select port-vlan-name TLV. ▪ system-capabilities: Select system-capabilities TLV. ▪ system-description: Select system-description TLV. ▪ system-name: Select system-name TLV.

Examples

Stopping the LLDP agent from sending the **port-description** TLV:

```
switch(config)# no lldp select-tlv port-description
```

Enabling the LLDP agent to send the **port-description** TLV:

```
switch(config)# lldp select-tlv oui
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp timer

```
lldp timer <TIME>
no lldp timer
```

Description

Sets the interval (in seconds) at which local LLDP information is updated and TLVs are sent to neighboring network devices by the LLDP agent. The minimum setting for this timer must be four times the value of **lldp txdelay**.

For example, this is a valid configuration:

- **lldp timer** = 16
- **lldp txdelay** = 4

And, this is an invalid configuration:

- **lldp timer** = 5
- **lldp txdelay** = 2

When copying a saved configuration to the running configuration, the value for **lldp timer** is applied before the value of **lldp txdelay**. This can result in a configuration error if the saved configuration has a value of **lldp timer** that is not four times the value of **lldp txdelay** in the running configuration.

For example, if the saved configuration has the settings:

- **lldp timer** = 16
- **lldp txdelay** = 4

And the running configuration has the settings:

- **lldp timer** = 30
- **lldp txdelay** = 7

Then you will see an error indicating that certain configuration settings could not be applied, and you will have to manually adjust the value of **lldp txdelay** in the running configuration.

The **no** form of this command sets the update interval to its default value of 30 seconds.

Parameter	Description
<TIME>	Specifies the update interval (in seconds). Range: 5 to 32768. Default: 30.

Examples

Setting the update interval to 7 seconds:


```
switch(config)# lldp timer 7
```

Setting the update interval to the default value of 30 seconds:

```
switch(config)# no lldp timer
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp transmit

```
lldp transsmmit  
no lldp transmit
```

Description

Enables transmission of LLDP information on specific interface. By default, LLDP transmission is enabled on all active interfaces, including the OOBM interface.

The **no** form of this command disables transmission of LLDP information on an interface.

Examples

On the 6400 Switch Series, interface identification differs.

Enabling LLDP transmission on interface **1/1/1**:

```
switch(config)# interface 1/1/1  
switch(config-if)# lldp transsmmit
```

Disabling LLDP transmission on interface **1/1/1**:

```
switch(config)# interface 1/1/1  
switch(config-if)# no lldp transsmmit
```

Enabling LLDP transmission on the OOBM interface:

```
switch(config)# interface mgmt  
switch(config-if)# lldp transsmmit
```

Disabling LLDP transmission on the OOBM interface:

```
switch(config)# interface mgmt
switch(config-if)# no lldp transsmitt
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

lldp txdelay

```
lldp txdelay <TIME>
no lldp txdelay
```

Description

Sets the amount of time (in seconds) to wait before sending LLDP information from any interface. The maximum value for **txdelay** is 25% of the value of **lldp tx timer**.

The **no** form of this command sets the delay time to its default value of 2 seconds.

Parameter	Description
<TIME>	Specifies the delay time in seconds. Range: 0 to 10. Default: 2.

Examples

Setting the delay time to 8 seconds:

```
switch(config)# lldp txdelay 8
```

Setting the delay time to the default value of 2 seconds:

```
switch(config)# no lldp txdelay
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp trap enable

lldp trap enable
no lldp trap enable

Description

Enables sending SNMP traps for LLDP related events from a particular interface. LLDP trap generation is enabled by default on all the interfaces and has to be disabled for interfaces on which traps are not required to be generated.

The **no** form of this command disables the LLDP trap generation.



LLDP trap generation is disabled by default at the global level and must be enabled before any LLDP traps are sent.

Examples

Enabling LLDP trap generation on global level:

```
switch(config)# lldp trap enable
```

Enabling LLDP trap generation on interface level:

```
switch(config-if)# lldp trap enable
```

Disabling LLDP trap generation on global level:

```
switch(config)# no lldp trap enable
```

Disabling LLDP trap generation on interface level:

```
switch(config-if)# no lldp trap enable
```

Displaying LLDP global configuration:

```
switch# show lldp configuration

LLDP Global Configuration
=====
LLDP Enabled                : No
LLDP Transmit Interval      : 30
LLDP Hold Time Multiplier   : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval  : 2
LLDP Trap Enabled           : No
```

TLVs Advertised

=====

Management Address

Port Description

Port VLAN-ID

System Description

System Name

LLDP Port Configuration

=====

PORT	TX-ENABLED	RX-ENABLED	INTF-TRAP-ENABLED
1/1/1	Yes	Yes	Yes
1/1/2	Yes	Yes	Yes
1/1/3	Yes	Yes	Yes
1/1/4	Yes	Yes	Yes
1/1/5	Yes	Yes	Yes
1/1/6	Yes	Yes	Yes
.....			
.....			
mgmt	Yes	Yes	Yes

Displaying LLDP Configuration for the interface:

```
switch# show lldp configuration 1/1/1
```

LLDP Global Configuration

=====

```
LLDP Enabled           : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled      : No
```

LLDP Port Configuration

=====

PORT	TX-ENABLED	RX-ENABLED	INTF-TRAP-ENABLED
1/1/1	Yes	Yes	Yes

Displaying LLDP Configuration for the management interface:

```
switch# show lldp configuration mgmt
```

LLDP Global Configuration

=====

```
LLDP Enabled           : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled      : Yes
```

LLDP Port Configuration

```

=====
PORT          TX-ENABLED          RX-ENABLED          INTF-TRAP-ENABLED
-----
mgmt          Yes              Yes              Yes

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config and config-if	Administrators or local user group members with execution rights for this command.

show lldp configuration

```
show lldp configuration [<INTERFACE-ID>][vsx-peer]
```

Description

Shows LLDP configuration settings for all interfaces or a specific interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: member/slot/port .
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

On the 6400 Switch Series, interface identification differs.

Showing configuration settings for all interfaces:

```

switch# show lldp configuration

LLDP Global Configuration
=====
LLDP Enabled                : No
LLDP Transmit Interval      : 30
LLDP Hold Time Multiplier   : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval  : 2
LLDP Trap Enabled           : No

TLVs Advertised
=====
Management Address

```

```
Port Description
Port VLAN-ID
System Description
System Name
```

```
LLDP Port Configuration
=====
```

PORT	TX-ENABLED	RX-ENABLED	INTF-TRAP-ENABLED
1/1/1	Yes	Yes	Yes
1/1/2	Yes	Yes	Yes
1/1/3	Yes	Yes	Yes
1/1/4	Yes	Yes	Yes
1/1/5	Yes	Yes	Yes
1/1/6	Yes	Yes	Yes
.....			
.....			
mgmt	Yes	Yes	Yes

This example shows configuration settings for interface **1/1/1**.

```
switch# show lldp configuration 1/1/1
```

```
LLDP Global Configuration
=====
```

```
LLDP Enabled           : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled      : No
```

```
LLDP Port Configuration
=====
```

PORT	TX-ENABLED	RX-ENABLED	INTF-TRAP-ENABLED
1/1/1	Yes	Yes	Yes

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp configuration mgmt

```
show lldp configuration mgmt
```

Description

Shows LLDP configuration settings for the OOBM interface.

Example

Showing configuration settings for all interfaces:

```
switch# show lldp configuration mgmt
```

```
LLDP Global Configuration
```

```
=====
```

```
LLDP Enabled           : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled      : Yes
```

```
LLDP Port Configuration
```

```
=====
```

PORT	TX-ENABLED	RX-ENABLED	INTF-TRAP-ENABLED
mgmt	Yes	Yes	Yes

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp local-device

```
show lldp local-device[vsx-peer]
```

Description

Shows global LLDP information advertised by the switch, as well as port-based data. If VLANs are configured on any active interfaces, the VLAN ID is only shown for trunk native or untagged VLAN IDs on access interfaces.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

Showing global LLDP information only (all ports including OOBM port are administratively down):

```
switch# show lldp local-device

Global Data
=====

Chassis-ID           : 1c:98:ec:e3:45:00
System Name          : switch
System Description    : Aruba JL375A 8400X XL.01.01.0001
Management Address    : 192.168.10.1
Capabilities Available : Bridge, Router
Capabilities Enabled   : Bridge, Router
TTL                   : 120
```

Showing all ports except **1/1/11** and OOBM as administratively down:

```
switch# show lldp local-device

Global Data
=====

Chassis-ID           : 1c:98:ec:e3:45:00
System Name          : switch
System Description    : Aruba
Management Address    : 192.168.10.1
Capabilities Available : Bridge, Router
Capabilities Enabled   : Bridge, Router
TTL                   : 120

Port Based Data
=====

Port-ID              : 1/1/11
Port-Desc             : "1/1/11"
Port Mgmt-Address     : 164.254.21.220
Port VLAN ID         : 1

Port-ID              : mgmt
Port-Desc             : "mgmt"
Port Mgmt-Address     : 164.254.21.220
```

In this example, all the ports except **1/1/11** are administratively down, and VLAN ID 100 is configured on this access interface.

```
switch# show lldp local-device

Global Data
=====

Chassis-ID           : 1c:98:ec:e3:45:00
System Name          : switch
System Description    : Aruba
Management Address    : 192.168.10.1
Capabilities Available : Bridge, Router
Capabilities Enabled   : Bridge, Router
TTL                   : 120

Port Based Data
```



```
=====

Port-ID          : 1/1/11
Port-Desc        : "1/1/11"
Port VLAN ID     : 100
Parent Interface : interface 1/1/11
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp neighbor-info

```
show lldp neighbor-info [<INTERFACE-NAME>] [vsx-peer]
```

Description

Displays information about neighboring devices for all interfaces or for a specific interface. The information displayed varies depending on the type of neighbor connected and the type of TLVs sent by the neighbor.

Parameter	Description
<INTERFACE-NAME>	Specifies the interface for which to show information for neighboring devices. Use the format member/slot/port (for example, 1/3/1).
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

On the 6400 Switch Series, interface identification differs.

Showing LLDP information for all interfaces:

```
switch# show lldp neighbor-info

LLDP Neighbor Information
=====

Total Neighbor Entries       : 3
Total Neighbor Entries Deleted : 0
Total Neighbor Entries Dropped : 0
```

Total Neighbor Entries Aged-Out : 0

LOCAL-PORT	CHASSIS-ID	PORT-ID	PORT-DESC	TTL	SYS-NAME
1/1/1	70:72:cf:a4:7d:50	1/1/1	1/1/1	32	switch
1/1/2	48:0f:cf:af:73:80	1/1/2	1/1/2	120	switch
1/1/46	48:0f:cf:af:73:80	1/1/46	1/1/46	120	switch
mgmt	48:0f:cf:af:73:80	mgmt	mgmt	120	switch

Showing information for interface **1/3/1** when it has only one switch connected as a neighbor:

```
switch# show lldp neighbor-info 1/3/1
```

```
Port : 1/1/1
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : HP-3800-24G-PoEP-2XG
Neighbor Chassis-Description : HP J9587A 3800-24G-PoE+-2XG Switch, revision...
Neighbor Chassis-ID : 10:60:4b:39:3e:80
Neighbor Management-Address : 192.168.1.1
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge
Neighbor Port-ID : 1/1/1
Neighbor Port-Desc : 1/1/1
Neighbor Port VLAN ID : 1
Neighbor Port VLAN Name : DEFAULT_VLAN_1
Neighbor Port MFS : 1500
TTL : 120
```

Showing information for interface **1/3/10** when the neighbor sends a DOT3 power TLV:

```
switch# show lldp neighbor-info 1/3/10
```

```
Port : 1/3/10
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : 84:d4:7e:ce:5d:68
Neighbor Chassis-Description : ArubaOS (MODEL: 325), Version Aruba IAP
Neighbor Chassis-ID : 84:d4:7e:ce:5d:68
Neighbor Management-Address : 169.254.41.250
Chassis Capabilities Available : Bridge, WLAN
Chassis Capabilities Enabled : WLAN
Neighbor Port-ID : 84:d4:7e:ce:5d:68
Neighbor Port-Desc : eth0
TTL : 120
Neighbor Port VLAN ID : 1
Neighbor Port VLAN Name : DEFAULT_VLAN_1
Neighbor Port MFS : 1500
Neighbor PoE information : DOT3
Neighbor Power Type : TYPE2 PD
Neighbor Power Priority : Unkown
Neighbor Power Source : Primary
PD Requested Power Value : 25.0 W
```

```
PSE Allocated Power Value: 25.0 W
Neighbor Power Supported      : Yes
Neighbor Power Enabled       : Yes
Neighbor Power Class         : 5
Neighbor Power Paircontrol   : No
PSE Power Pairs              : Signal
```

Showing information for interface **1/1/1** when it has multiple neighbors (displays a maximum of four):

```
switch# show lldp neighbor-info 1/1/1

Port                               : 1/1/1
Neighbor Entries                   : 4
Neighbor Entries Deleted           : 0
Neighbor Entries Dropped           : 0
Neighbor Entries Aged-Out          : 0
Neighbor Chassis-Name              : switch
Neighbor Chassis-Description       : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID                : 1c:98:ec:fe:25:00
Neighbor Management-Address        : 10.1.1.2
Chassis Capabilities Available     : Bridge, Router
Chassis Capabilities Enabled       : Bridge, Router
Neighbor Port-ID                   : 1/1/1
Neighbor Port-Desc                  : 1/1/1
Neighbor Port VLAN ID              : 1
Neighbor Port VLAN Name             : DEFAULT_VLAN_1
Neighbor Port MFS                   : 1500
TTL                                : 120
Neighbor Chassis-Name              : switch
Neighbor Chassis-Description       : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID                : 1c:98:ec:fe:25:01
Neighbor Management-Address        : 10.1.1.3
Chassis Capabilities Available     : Bridge, Router
Chassis Capabilities Enabled       : Bridge, Router
Neighbor Port-ID                   : 1/1/1
Neighbor Port-Desc                  : 1/1/1
Neighbor Port VLAN ID              : 1
Neighbor Port VLAN Name             : DEFAULT_VLAN_1
Neighbor Port MFS                   : 1500
TTL                                : 120
Neighbor Chassis-Name              : switch
Neighbor Chassis-Description       : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID                : 1c:98:ec:fe:25:02
Neighbor Management-Address        : 10.1.1.4
Chassis Capabilities Available     : Bridge, Router
Chassis Capabilities Enabled       : Bridge, Router
Neighbor Port-ID                   : 1/1/1
Neighbor Port-Desc                  : 1/1/1
Neighbor Port VLAN ID              : 50
Neighbor Port VLAN Name             : VLAN_50
Neighbor Port MFS                   : 1500
TTL                                : 120
Neighbor Chassis-Name              : switch
Neighbor Chassis-Description       : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID                : 1c:98:ec:fe:25:03
Neighbor Management-Address        : 10.1.1.5
Chassis Capabilities Available     : Bridge, Router
Chassis Capabilities Enabled       : Bridge, Router
Neighbor Port-ID                   : 1/1/1
```

```

Neighbor Port-Desc      : 1/1/1
Neighbor Port VLAN ID   : 100
Neighbor Port VLAN Name : VLAN_100
Neighbor Port MFS       : 1500
TTL                     : 120

```

Showing neighbor information for interface 1/3/2 when it has EEE enabled and successfully auto-negotiated:

```

switch# show lldp neighbor-info 1/3/2

Port                               : 1/3/2
Neighbor Entries                   : 1
Neighbor Entries Deleted           : 1
Neighbor Entries Dropped           : 0
Neighbor Entries Aged-Out          : 1
Neighbor Chassis-Name              : BLDG01-F1-6300
Neighbor Chassis-Description       : Aruba JL668A FL.10.07.0001BN
Neighbor Chassis-ID                : 88:3a:30:92:a5:c0
Neighbor Management-Address        : 10.6.9.15
Chassis Capabilities Available     : Bridge, Router
Chassis Capabilities Enabled       : Bridge, Router
Neighbor Port-ID                   : 1/1/1
Neighbor Port-Desc                 : 1/1/1
Neighbor Port VLAN ID              : 1
Neighbor Port VLAN Name            : DEFAULT_VLAN_1
Neighbor Port MFS                   : 1500
TTL                                : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported        : true
Neighbor Auto-Neg Enabled          : true
Neighbor Auto-Neg Advertised       : 1000 BASE_TFD, 100 BASE_T4, 10 BASE_TFD
Neighbor MAU type                   : 1000 BASE_TFD

Neighbor EEE information           : DOT3
Neighbor TX Wake time              : 17 us
Neighbor RX Wake time              : 17 us
Neighbor Fallback time             : 17 us
Neighbor TX Echo time              : 17 us
Neighbor RX Echo time              : 17 us

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp neighbor-info detail

```
show lldp neighbor-info detail [vsx-peer]
```

Description

Shows detailed LLDP neighbor information for all LLDP neighbor connected interfaces.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

On the 6400 Switch Series, interface identification differs.

Showing detailed LLDP information for all interfaces:

```
switch# show lldp neighbor-info detail
```

```
LLDP Neighbor Information
=====
```

```
Total Neighbor Entries      : 6
Total Neighbor Entries Deleted : 2
Total Neighbor Entries Dropped : 0
Total Neighbor Entries Aged-Out : 2
```

```
-----
Port                          : 1/1/1
Neighbor Entries              : 1
Neighbor Entries Deleted      : 0
Neighbor Entries Dropped      : 0
Neighbor Entries Aged-Out     : 0
Neighbor Chassis-Name         : 6300
Neighbor Chassis-Description  : Aruba ...
Neighbor Chassis-ID           : 38:11:17:1a:d5:00
Neighbor Management-Address   : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled  : Bridge, Router
Neighbor Port-ID              : 1/1/4
Neighbor Port-Desc            : 1/1/4
Neighbor Port VLAN ID         : 1
Neighbor Port VLAN Name       : DEFAULT_VLAN_1
Neighbor Port MFS              : 1500
TTL                           : 120
```

```
Neighbor Mac-Phy details
Neighbor Auto-neg Supported   : true
Neighbor Auto-Neg Enabled    : true
Neighbor Auto-Neg Advertised  : 1000 BASE_TFD, 100 BASE_T4, 10 BASE_TFD
Neighbor MAU type             : 1000 BASE_TFD
```

```
-----
Port                          : 1/1/2
Neighbor Entries              : 1
Neighbor Entries Deleted      : 0
```

```

Neighbor Entries Dropped      : 0
Neighbor Entries Aged-Out    : 0
Neighbor Chassis-Name        : 6300
Neighbor Chassis-Description  : Aruba ...
Neighbor Chassis-ID          : 38:11:17:1a:d5:00
Neighbor Management-Address   : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled  : Bridge, Router
Neighbor Port-ID              : 1/1/5
Neighbor Port-Desc            : 1/1/5
Neighbor Port VLAN ID         : 1
Neighbor Port VLAN Name       : DEFAULT_VLAN_1
Neighbor Port MFS              : 1500
TTL                            : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : true
Neighbor Auto-Neg Enabled      : true
Neighbor Auto-Neg Advertised   : 1000 BASE_TFD, 100 BASE_T4, 10 BASE_TFD
Neighbor MAU type               : 1000 BASE_TFD

```

```

-----

Port                          : 1/1/3
Neighbor Entries              : 1
Neighbor Entries Deleted      : 0
Neighbor Entries Dropped      : 0
Neighbor Entries Aged-Out     : 0
Neighbor Chassis-Name         : 6300
Neighbor Chassis-Description   : Aruba ...
Neighbor Chassis-ID           : 38:11:17:1a:d5:00
Neighbor Management-Address    : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/6
Neighbor Port-Desc             : 1/1/6
Neighbor Port VLAN ID         : 1
Neighbor Port VLAN Name       : DEFAULT_VLAN_1
Neighbor Port MFS              : 1500
TTL                            : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : true
Neighbor Auto-Neg Enabled      : true
Neighbor Auto-Neg Advertised   : 1000 BASE_TFD, 100 BASE_T4, 10 BASE_TFD
Neighbor MAU type               : 1000 BASE_TFD

```

```

-----

Port                          : 1/1/46
Neighbor Entries              : 1
Neighbor Entries Deleted      : 0
Neighbor Entries Dropped      : 0
Neighbor Entries Aged-Out     : 0
Neighbor Chassis-Name         : 6300
Neighbor Chassis-Description   : Aruba ...
Neighbor Chassis-ID           : 38:11:17:1a:d5:00
Neighbor Management-Address    : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/19
Neighbor Port-Desc             : 1/1/19

```

```

Neighbor Port VLAN ID      : 1
Neighbor Port VLAN Name    : DEFAULT_VLAN_1
Neighbor Port MFS          : 1500
TTL                        : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported : true
Neighbor Auto-Neg Enabled   : true
Neighbor Auto-Neg Advertised : 1000 BASE_TFD, 100 BASE_T4, 10 BASE_TFD
Neighbor MAU type           : 1000 BASE_TFD

```

```

-----
Port                        : 1/1/47
Neighbor Entries           : 1
Neighbor Entries Deleted   : 0
Neighbor Entries Dropped   : 0
Neighbor Entries Aged-Out  : 0
Neighbor Chassis-Name      : 6300
Neighbor Chassis-Description : Aruba ...
Neighbor Chassis-ID        : 38:11:17:1a:d5:00
Neighbor Management-Address : 38:11:17:1a:d5:00
Chassis Cap

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp neighbor-info mgmt

show lldp neighbor-info mgmt

Description

Displays information about neighboring devices connected to the OOBM interface.

Examples

Showing LLDP information for the OOBM interface:

```

switch# show lldp neighbor-info mgmt

Port                        : mgmt
Neighbor Entries           : 1
Neighbor Entries Deleted   : 0
Neighbor Entries Dropped   : 0
Neighbor Entries Aged-Out  : 0
Neighbor Chassis-Name      : HP-3800-24G-PoEP-2XG

```

```
Neighbor Chassis-Description : HP J9587A 3800-24G-PoE+-2XG Switch, revision...
Neighbor Chassis-ID          : 10:60:4b:39:3e:80
Neighbor Management-Address  : 192.168.1.1
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled  : Bridge
Neighbor Port-ID             : mgmt
Neighbor Port-Desc           : mgmt
TTL                           : 120
```

Showing LLDP information for the OOBM interface when there are four neighbors:

```
switch# show lldp neighbor-info mgmt

Port : mgmt
Neighbor Entries : 4
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : switch
Neighbor Chassis-Description : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID : 1c:98:ec:fe:25:00
Neighbor Management-Address : 10.1.1.2
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/1
Neighbor Port-Desc : 1/1/1
TTL : 120

Neighbor Chassis-Name : switch
Neighbor Chassis-Description : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID : 1c:98:ec:fe:25:01
Neighbor Management-Address : 10.1.1.3
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/1
Neighbor Port-Desc : 1/1/1
TTL : 120

Neighbor Chassis-Name : switch
Neighbor Chassis-Description : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID : 1c:98:ec:fe:25:02
Neighbor Management-Address : 10.1.1.4
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/1
Neighbor Port-Desc : 1/1/1
TTL : 120

Neighbor Chassis-Name : switch
Neighbor Chassis-Description : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID : 1c:98:ec:fe:25:03
Neighbor Management-Address : 10.1.1.5
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/1
Neighbor Port-Desc : 1/1/1
TTL : 120
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp statistics

```
show lldp statistics [<INTERFACE-ID>] [vsx-peer]
```

Description

Shows global LLDP statistics or statistics for a specific interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: member/slot/port .
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

On the 6400 Switch Series, interface identification differs.

Showing global statistics for all interfaces:

```
switch# show lldp statistics
LLDP Global Statistics
=====

Total Packets Transmitted      : 19
Total Packets Received        : 19
Total Packets Received And Discarded : 0
Total TLVs Unrecognized       : 0

LLDP Port Statistics
=====
```

PORT-ID	TX-PACKETS	RX-PACKETS	RX-DISCARDED	TLVS-UNKNOWN
1/1/1	7	7	0	0
1/1/2	7	7	0	0
1/1/3	0	0	0	0
1/1/4	0	0	0	0
1/1/5	0	0	0	0
...				
mgmt	5	5	0	0

...

Showing statistics for interface **1/1/1**:

```
switch# show lldp statistics 1/1/1

LLDP Statistics
=====

Port Name                : 1/1/1
Packets Transmitted      : 159
Packets Received         : 163
Packets Received And Discarded : 0
Packets Received And Unrecognized : 0
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp statistics mgmt

show lldp statistics mgmt

Description

Shows LLDP statistics for the OOBM interface.

Example

Showing LLDP statistics for the OOBM interface:

```
switch# show lldp statistics mgmt

LLDP Statistics
=====

Port Name                : mgmt
Packets Transmitted      : 20
Packets Received         : 23
Packets Received And Discarded : 0
Packets Received And Unrecognized : 0
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp tlv

```
show lldp tlv[vsx-peer]
```

Description

Shows the LLDP TLVs that are configured for send and receive.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

```
switch# show lldp tlv

TLVs Advertised
=====

Management Address
Port Description
Port VLAN-ID
System Capabilities
System Description
System Name
VLAN Name
MFS
OUI
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with

execution rights for this command. Operators can execute this command from the operator context (>) only.

Cisco Discovery Protocol (CDP)

Cisco Discovery Protocol (CDP) is a proprietary layer 2 protocol supported by most Cisco devices. It is used to exchange information, such as software version, device capabilities, and voice VLAN information, between directly connected devices, such as a VoIP phone and a switch.

CDP support

By default, CDP is enabled on each active switch port. This is a read-only capability, which means the switch can receive and store information about adjacent CDP devices, but does not generate CDP packets (except when communicating with Cisco IP phones.)

The switch supports CDPv2 only and does not support SNMP MIB traps.

When a CDP-enabled port receives a CDP packet from another CDP device, it enters data for that device into the CDP Neighbors table, along with the port number on which the data was received. It does not forward the packet. The switch also periodically purges the table of any entries that have expired. (The holdtime for any data entry in the switch CDP Neighbors table is configured in the device transmitting the CDP packet and cannot be controlled in the switch receiving the packet.) A switch reviews the list of CDP neighbor entries every three seconds and purges any expired entries.

Support for legacy Cisco IP phones

Autoconfiguration of legacy Cisco IP phones for tagged voice VLAN support requires CDPv2.

On initial boot-up, and sometimes periodically, a Cisco phone queries the switch and advertises information about itself using CDPv2. When the switch receives the VoIP VLAN Query TLV (type 0x0f) from the phone, the switch immediately responds with the voice VLAN ID in a reply packet using the VoIP VLAN Reply TLV (type 0x0e). This enables the Cisco phone to boot properly and send traffic on the advertised voice VLAN ID.

The switch CDP packet includes these TLVs:

- CDP Version: 2
- CDP TTL: 180 seconds
- Checksum
- Capabilities (type 0x04): 0x0008 (is a switch)
- Native VLAN: The PVID of the port
- VoIP VLAN Reply (type 0xe): voice VLAN ID (same as advertised by LLDP-MED)
- Trust Bitmap (type 0x12): 0x00
- Untrusted port CoS (type 0x13): 0x00

CDP commands

cdp

cdp

Description

Configures CDP support globally on all active interfaces or on a specific interface. By default, CDP is enabled on all active interfaces.

When CDP is enabled, the switch adds entries to its CDP Neighbors table for any CDP packets it receives from neighboring CDP devices.

When CDP is disabled, the CDP Neighbors table is cleared and the switch drops all inbound CDP packets without entering the data in the CDP Neighbors table.

The **no** form of this command disables CDP support globally on all active interfaces or on a specific interface.

Examples

Enabling CDP globally:

```
switch(config)# cdp
```

Disabling CDP globally:

```
switch(config)# no cdp
```

Enabling CDP on interface **1/1/1**:

```
switch(config)# interface 1/1/1s  
switch(config-if)# cdp
```

Disabling CDP on interface **1/1/1**:

```
switch(config)# interface 1/1/1  
switch(config-if)# no cdp
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config config-if	Administrators or local user group members with execution rights for this command.

clear cdp counters

```
clear cdp counters
```

Description

Clears CDP counters.

Examples

Clearing CDP counters:

```
switch(config) clear cdp counters
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

clear cdp neighbor-info

```
clear cdp neighbor-info
```

Description

Clears CDP neighbor information.

Examples

Clearing CDP neighbor information:

```
switch(config) clear neighbor-info
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show cdp

```
show cdp
```

Description

Shows CDP information for all interfaces.

Examples

On the 6400 Switch Series, interface identification differs.

Showing CDP information:

```
switch(config)# show cdp
CDP Global Information
=====
CDP           : Enabled
CDP Mode      : Rx only
CDP Hold Time : 180 seconds

Port          CDP
-----
1/1/1         Enabled
1/1/2         Enabled
1/1/3         Enabled
1/1/4         Enabled
1/1/5         Enabled
1/1/6         Enabled
1/1/7         Enabled
1/1/8         Enabled
1/1/9         Enabled
1/1/10        Enabled
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show cdp neighbor-info

show cdp neighbor-info <INTERFACE-ID>

Description

Shows CDP information for all neighbors or for CDP information on a specific interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: member/slot/port .

Examples

On the 6400 Switch Series, interface identification differs.

Showing all CDP neighbor information:

```
switch(config)# show cdp neighbor-info
Port          Device ID          Platform          Capability
-----
```

Showing CDP information for interface **1/1/1**:

```
switch(config)# show cdp neighbor-info 1/1/1
Local Port : 1/1/1
MAC        : 3c:a8:2a:7b:6b:2b
Device ID   : SEPd4adbd2a30d6
Address     : 2.71.0.230
Platform    : Cisco IP Phone 3905
Duplex      : full
Capability   : host
Voice VLAN Support : Yes
Neighbor Port-ID : Port 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show cdp traffic

show cdp neighbor-info

Description

Shows CDP statistics for each interface.

Examples

On the 6400 Switch Series, interface identification differs.

Showing CDP traffic statistics:

```
switch(config)# show cdp traffic
CDP Statistics
=====
Port          Transmitted Frames    Received Frames    Discarded Frames
-----
1/1/1          0                      4                  0
1/1/2          0                      0                  0
1/1/3          0                      2                  0
1/1/4          0                      0                  0
1/1/5          0                      0                  0
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

Zero Touch Provisioning (ZTP) enables the auto-configuration of factory default switches without a network administrator onsite.

When a switch is booted from its factory default configuration, ZTP autoprovisions the switch by automatically downloading and installing a firmware file, a configuration file, or both. With ZTP, even a nontechnical user (for example: a store manager in a retail chain or a teacher in a school) can deploy devices at a site.

ZTP support

The switch supports standards-based Zero Touch Provisioning (ZTP) operations as follows:

- The switch must be running the factory default configuration.
- The switch can connect to the DHCP server from the OOBM management port.

The switch can connect to the DHCP server from either the OOBM management port, or a data port on the default VLAN.

- ZTP operations are supported over IPv4 connections only. IPv6 connections are not supported for ZTP operations.
- You must configure the DHCP server to provide a standards-based ZTP server solution. Options and features that are specific to Network Management Solution (NMS) tools, such as AirWave, are not supported.
 - Aruba Central on-premise can manage AOS-CX switches on supported models through DHCP ZTP using two approaches:
 - On the DHCP server, configure DHCP option-60 as "ArubaInstantAP" 90 and provide the value in option-43 in the format `<group-details>, <aruba-central-on-prem-ip-or-fqdn>, <shared-token>`.
 - On the DHCP server, configure DHCP option-60 as HPE vendor VCI and provide the value in option-43 in the tag-length-value (TLV) format. Next, enter the Aruba Central on-premise fully qualified domain name (FQDN) or IPv4 address as the value for sub-option code 146 using the format: `<group-details>, <aruba-central-on-prem-ip-or-fqdn>, <shared-token>`.
 - Supported DHCP options are:

DHCP option	Description
43	Vendor Specific Information
43 suboption 144	Name of the configuration file
43 suboption 145	Name of the firmware image file
43 suboption 146	FQDN or IPv4 address of the Aruba Central on-premise server
43 suboption 148	FQDN or IPv4 address of the HTTP Proxy

DHCP option	Description
60	Vendor Class Identifier (VCI)
66	IPv4 address of the TFTP server (Specifying a host name instead of an IP address is not supported.)
67	Name of the configuration file (Option 43 suboption 144 takes precedence over this option.)

- The configuration file is a text file or JSON file that becomes the startup and running configuration on the switch after the ZTP operation is complete. The configuration can be in CLI or in JSON format.
- When the switch is started using the factory default configuration, the ZTP operation is started automatically and is active until any running configuration of the switch is modified. There is no CLI command required to start the operation.

The switch supports the following standards:

- [RFC 2131](#), *Dynamic Host Configuration Protocol*.
- [RFC 2132](#), *DHCP Options and BOOTP Vendor Extensions*. Support is limited to the options listed in the table "Supported DHCP options for ZTP on AOS-CX."

Hewlett Packard Enterprise recommends that you implement ZTP in a secure and private environment. Any public access can compromise the security of the switch, as follows:

- ZTP is enabled only in the factory default configuration of the switch, DHCP snooping is not enabled. The Rogue DHCP server must be manually managed.
- The DHCP offer is in plain data without encryption.

Setting up ZTP on a trusted network

The following procedure is an overview of setting up a Zero Touch Provisioning (ZTP) environment to provision newly installed switches automatically. The procedure is intended for network administrators who are familiar with automatically provisioning switches in a network, and does not provide detailed information about configuring or managing switches.

Procedure

1. For each switch model to be provisioned using ZTP, do the following:
 - a. Obtain the switch firmware image file.
 - b. Prepare the switch configuration file. The configuration file becomes the running configuration and the startup configuration on the switch.
2. Set up a TFTP server and record its IP address. The address is required when you set up the DHCP server. The switch must be able to reach the TFTP server and DHCP server, either on the same subnet, or on a remote subnet via DHCP relay.

Switches support provisioning through a network connected to a data port or through a network connected to the management port.
3. Publish the configuration files and image files to the TFTP server. You need to know the locations of the files and the IP address of the TFTP server when you set up the vendor class options on the DHCP server.

4. On the DHCP server, set up vendor classes for each switch model you plan to provision. To do this you need the following information:
 - The IP address of the TFTP server. Using a host name is not supported.
 - The path to the switch configuration and firmware image files on the TFTP server.
 - The vendor class identifier (VCI) for each switch model.

You can obtain the VCI by entering the `show dhcp client vendor-class-identifier` command from a switch CLI command prompt in the manager context. The VCI is the text string in the response that starts with `Aruba`.

For example:

```
switch# show dhcp client vendor-class-identifier
Vendor Class Identifier:  Aruba xxxxx xxxx
```

Where x indicates the switch model number.

5. At the installation site, provide the switch installer with a Cat6 network cable connected to the network that includes the DHCP and TFTP servers, and information about the switch port to use. The switch installer plugs the cable into the data port you specify.

The ZTP operation begins when power is applied to the switch after the network cable is installed.

6. Assuming the downloaded configuration includes a way to access the CLI of the switch, you can enter the following command to show the options offered by the DHCP server and the status of the ZTP operation:

```
show ztp information
```

ZTP process during switch boot

1. The switch boots up with the factory default configuration.

If the ZTP operation detects that the switch configuration is different from the factory default configuration, the ZTP operation ends. The switch must be configured at the installation site.

2. The switch sends out a DHCP discovery from the management port.

On switches that support ZTP operations on data ports, the switch also sends out a DHCP discovery from all data ports in the default VLAN.

The switch waits to receive DHCP options indefinitely or until the running configuration is modified. If a DHCP IP address is received but no DHCP options are received, the switch waits an additional minute before ending the ZTP operation.

On switches that support ZTP operations on data ports, DHCP options received on the management port have priority over DHCP options received on data ports:

- If DHCP options are received on the management port before being received on a data port, the switch processes those options immediately.
- If DHCP options are received on a data port, the switch waits an additional 30 seconds for options to be received on the management port. If no DHCP options are received on the management port during those 30 seconds, the switch processes the DHCP options it received on the data port.

After the ZTP operation ends, there is no automatic retry. You can either attempt to boot the switch with the factory default configuration again, configure the switch at the installation site, or use the ZTP force-provision CLI to trigger the ZTP process, ignoring the present running configuration of the switch.

- Once force-provision is enabled, new DHCP requests are sent from the switch. Disabling force-provision does not stop the DHCP already in progress, but only changes the switch configuration status of force-provision.
 - If ZTP fails while force-provision is enabled, there is no automatic retry. To retry, `ztp force-provision` should be disabled and re-enabled to clear the current ZTP state and send a new DHCP request. When `ztp force-provision` is already enabled on the switch, re-enabling it results in no operation.
 - If the DHCP server is configured to provide both ZTP image and configuration options and there is a non-default startup configuration present on the switch, clearing the non-default startup configuration before triggering `ztp force-provision` is recommended. If an image is downloaded via ZTP, the switch reboots once the image download is complete and ZTP force-provision configuration is lost, causing ZTP to enter into a failed state. ZTP force-provision will need to be enabled again to continue the process.
3. The DHCP server responds with an offer containing the following:
 - The IPv4 address of the TFTP server
 - One or both of the following:
 - The name of the firmware image file
 - The name of the configuration file
 - Aruba Central Location (optional) including the shared token value for the on-premise server
 - HTTP Proxy Location (optional)
 4. If a firmware image file is offered, the ZTP operation downloads the image file from the TFTP server to the switch. If the current switch image and downloaded firmware image version do not match, then the switch boots with the downloaded image:
 - If the image upgrade fails, the switch retains its original firmware image and the ZTP operation ends with a failed status.
 - If the image upgrade succeeds, the ZTP operation is started again after the switch reboots. Because the downloaded image file matches the image file installed on the switch, the ZTP operation continues, and checks if a configuration file is offered.
 5. If a configuration file is offered, the ZTP operation downloads the configuration file copies the file to the running-config and then to the startup-config of the switch:
 - If the startup configuration update fails, the switch retains its factory-default running configuration and the ZTP operation ends with a failed status.
 - If the copy operation fails, the ZTP operation ends with a failed status.
 - If the copy operation succeeds, the ZTP operation ends successfully.

ZTP VSF switchover support

ZTP status is not synced in the VSF stack. When the VSF stack is formed, configuration changes are applied on the conductor switch, which is then synced to standby switch. When the switchover is performed on the VSF stack, the standby becomes the new conductor switch.

As part of the switchover process, the ZTP daemon starts on the new conductor. The status of the ZTP is failed because there are configuration changes present.

ZTP commands

show ztp information

```
show ztp information
```

Description

Shows information about Zero Touch Provisioning (ZTP) operations performed on the switch.

Usage

When a switch configured to use ZTP is booted from a factory default configuration, the switch contacts a DHCP server, which offers options for obtaining files used to provision the switch:

- The IP address of the TFTP server
- The name of the image file
- The name of the configuration file
- The Aruba Central FQDN or IPv4 address
- The HTTP proxy FQDN or IPv4 address

The **show ztp information** command shows the options offered by the DHCP server and the status of the ZTP operation.

The status of the ZTP operation is one of the following:

Success

The ZTP operation succeeded.

One of the following is true:

- Both the running configuration and the startup configuration were updated.
- The IP address of the TFTP server was received, but the offer did not include a configuration file or a firmware image file.
- Any combination of vendor encapsulated DHCP options are received as configured, along with the firmware image and switch configuration file.
- Only vendor encapsulated DHCP options are configured and are received accordingly.

Failed - Custom startup configuration detected

The switch was booted from a configuration that is not the factory default configuration. For example, the administrator password has been set.

Failed - Timed out while waiting to receive ZTP options

Either the switch received the DHCP IPv4 address but no ZTP options were received within 1 minute or ZTP force-provision is triggered and no ZTP options are received within 3 minutes.

Failed - Detected change in running configuration

The running configuration was modified by a user while the ZTP operation was in progress.

Failed - TFTP server unreachable

The TFTP server is not reachable at the specified IP address.

Failed - TFTP server information unavailable

The image file name or config file name is provided without the TFTP server location to fetch the files from and ZTP enters failed state.

Failed - Invalid configuration file received

Either the file transfer of the configuration file failed, or the configuration file is invalid (an error occurred while attempting to apply the configuration).

Failed - Invalid image file received

Either the file transfer of the firmware image file failed, or the firmware image file is invalid (an error occurred while verifying the image).



In the case of reconnection, connect with a main or alternative location to the COP instance as a user. The current connection is shown in the **Central location** field.

Scenario 1: If the location the device is currently connected on is updated, the system reconnects in order to connect with the new location.

Scenario 2: If the location in which the device is not currently connected on is updated, the DUT does not go through the reconnection process.

Examples

Showing switch image download in progress after receiving ZTP options:

```
switch# show ztp information
TFTP Server           : 10.0.0.2
Image File            : TL_10_02_0001.swi
Configuration File     : config_file
ZTP Status            : In-progress - Image download and verification
Aruba Central Location : secure.arubanetworks.com
Alternative Aruba Central Location: NA
Aruba Central Shared Token : aruba123
Force-Provision       : Disabled
HTTP Proxy Location    : http.proxy.arubanetworks.com
```

Showing switch image download failure after receiving ZTP options:

```
switch# show ztp information
TFTP Server           : 10.0.0.2
Image File            : TL_10_02_0001.swi
Configuration File     : config_file
ZTP Status            : Failed - Unable to download image
Aruba Central Location : secure.arubanetworks.com
Alternative Aruba Central Location: NA
Aruba Central Shared Token : aruba123
Force-Provision       : Disabled
HTTP Proxy Location    : http.proxy.arubanetworks.com
```

Showing switch configuration download in progress after receiving ZTP options:

```
switch# show ztp information
TFTP Server           : 10.0.0.2
Image File            : TL_10_02_0001.swi
Configuration File     : config_file
ZTP Status            : In-progress - Configuration download
Aruba Central Location : secure.arubanetworks.com
Alternative Aruba Central Location: NA
Aruba Central Shared Token : aruba123
Force-Provision       : Disabled
HTTP Proxy Location    : http.proxy.arubanetworks.com
```

Showing switch configuration download failure after receiving ZTP options:

```
switch# show ztp information
```

```
TFTP Server          : 10.0.0.2
Image File           : TL_10_02_0001.swi
Configuration File    : config_file
ZTP Status           : Failed - Unable to download configuration
Aruba Central Location : secure.arubanetworks.com
Alternative Aruba Central Location : NA
Aruba Central Shared Token : aruba123
Force-Provision       : Disabled
HTTP Proxy Location   : http.proxy.arubanetworks.com
```

Showing switch failure to update start-up configuration after downloading configuration received from ZTP options:

```
switch# show ztp information
TFTP Server          : 10.0.0.2
Image File           : TL_10_02_0001.swi
Configuration File    : config_file
ZTP Status           : Failed - Could not copy to start-up configuration
Aruba Central Location : secure.arubanetworks.com
Alternative Aruba Central Location: NA
Aruba Central Shared Token : aruba123
Force-Provision       : Disabled
HTTP Proxy Location   : http.proxy.arubanetworks.com
```

In the following example, the ZTP operation succeeded, and both an image file and a configuration file were provided.

```
VSF-10-Mbr# show ztp information
TFTP Server          : 10.1.84.160
Image File           : FL_10_06_0001CK.swi
Configuration File    : 102720-new-setup-config-updated.txt
Status               : Success
Aruba Central Location : NA
Alternative Aruba Central Location : NA
Aruba Central Shared Token : aruba123
Force-Provision       : Disabled
HTTP Proxy Location   : NA
VSF-10-Mbr#
```

In the following example, the ZTP option succeeded. A configuration file was not provided, but an image file was provided.

```
VSF-10-Mbr# show ztp information
TFTP Server          : 10.1.84.160
Image File           : TL_10_02_0001.swi
Configuration File    : NA
Status               : Success
Aruba Central Location : NA
Alternative Aruba Central Location: NA
Aruba Central Shared Token : aruba123
Force-Provision       : Disabled
HTTP Proxy Location   : NA
VSF-10-Mbr#
```

In the following example, the ZTP operation failed because the TFTP server was unreachable.


```
VSF-10-Mbr# show ztp information
TFTP Server          : 10.1.84.160
Image File           : TL_10_02_0001.swi
Configuration File    : 102720-new-setup-config-updated.txt
Status               : Failed - TFTP server unreachable
Aruba Central Location : NA
Alternative Aruba Central Location: NA
Aruba Central Shared Token : NA
Force-Provision       : Disabled
HTTP Proxy Location   : NA
VSF-10-Mbr#
```

In the following example, the ZTP operation was stopped because the switch did not receive any options from the DHCP server for ZTP within 1 minute of receiving the IP address from the server.

```
VSF-10-Mbr## show ztp information
TFTP Server          : NA
Image File           : NA
Configuration File    : NA
Status               : Failed - Timed out while waiting to receive ZTP
options
Aruba Central Location : NA
Alternative Aruba Central Location: NA
Aruba Central Shared Token : NA
Force-Provision       : Disabled
HTTP Proxy Location   : NA
VSF-10-Mbr#
```

In the following example, the ZTP operation was stopped because the switch was booted from a configuration that was not the factory default configuration.

```
switch# show ztp information
TFTP Server          : 10.0.0.2
Image File           : TL_10_02_0001.swi
Configuration File    : ztp.cfg
Status               : Failed - Custom startup configuration detected
Aruba Central Location : NA
Alternative Aruba Central Location: NA
Aruba Central Shared Token : NA
Force-Provision       : Disabled
HTTP Proxy Location   : NA
```

In the following example, the switch received the image file and the TFTP-server and config file from the DHCP server for ZTP was successful:

```
switch# show ztp information
TFTP Server          : 10.0.0.2
Image File           : TL_10_02_0001.swi
Configuration File    : ztp.cfg
ZTP Status           : Success
Aruba Central Location : NA
Alternative Aruba Central Location: NA
Aruba Central Shared Token : NA
Force-Provision       : Disabled
HTTP Proxy Location   : NA
```

In the following example, the switch received the image file and the TFTP-server and config file from the DHCP server entered the failed state as the TFTP server was not reachable:

```
switch# show ztp information
TFTP Server           : 10.0.0.2
Image File            : TL_10_02_0001.swi
Configuration File    : ztp.cfg
ZTP Status            : Failed - TFTP server unreachable
Aruba Central Location : NA
Alternative Aruba Central Location: NA
Aruba Central Shared Token : NA
Force-Provision       : Disabled
HTTP Proxy Location   : NA
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

ztp force provision

```
ztp force-provision
no ztp force-provision
```

Description

Starts on-demand ZTP.

Usage

DHCP options received are processed independent of the current state of configuration on the switch. Previous ZTP TFTP Server, Image File, Configuration File, Aruba Central Location, and HTTP Proxy location options are cleared and the switch sends a DHCP request.

Examples

In the following example, force-provision is enabled.

```
switch# configure terminal
switch(config)# ztp force-provision
```

In the following example, force-provision status is checked while enabled.

```
switch# show ztp information
TFTP Server           : 10.0.0.2
```

```
Image File           : TL_10_02_0001.swi
Configuration File    : ztp.cfg
Status               : Success
Aruba Central Location : NA
Aruba Central Shared Token : NA
Force-Provision       : Enabled
HTTP Proxy Location   : NA
```

In the following example, force-provision is disabled.

```
switch# configure terminal
switch(config)# no ztp force-provision
```

In the following example, force-provision status is checked while disabled.

```
switch# show ztp information
TFTP Server          : 10.0.0.2
Image File           : TL_10_02_0001.swi
Configuration File    : ztp.cfg
Status               : Success
Aruba Central Location : NA
Aruba Central Shared Token : NA
Force-Provision       : Disabled
HTTP Proxy Location   : NA
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

The HPE Aruba Networking CX vSphere agent allows administrators to trace how Virtual Machines are connected to the physical switches. The ability to visualize how a Virtual Machine connects to a CX port helps to resolve connectivity issues. This functionality helps operators in several troubleshooting scenarios, such as diagnosing a connectivity issue, or ensuring adequate distribution of VMs across the physical infrastructure.

You can install the vSphere agent on one or many of the switches in the fabric. You can perform the following tasks, from any switch where the agent is installed:

Run the `show VMs` command to list all the VMs managed by the vCenter Appliance integrated with the agent. This functionality also helps to view VMs connections to the ports on other switches on the fabric.

Search for a specific VM by name. To find the connection between the VM and a switch port, attach the switch to at least one host managed by the integrated vCenter.

For more information about cloud capabilities of vSphere agent, refer to *New Central Online Help*.



The **show neighbors** command examines the LLDP data that is advertised from any switch to a distributed virtual switch and provides connection details for third-party switches.

Installation of vSphere agent

To install a container, perform the following steps:

1. Download the vSphere agent from ASP and select the arm vs. x86_64 architecture for the switch platform. Refer to the list of supported switches mentioned earlier to select the corresponding architecture.
2. Host the agent on a local network that can be accessed using the HTTP protocol. Usage of the https protocol is a limitation of the container framework. The directory where the images of the agents are placed will be set in step 3.d. The HTTP server must be reachable on the mgmt vrf - or otherwise configured vrf. Any simple HTTP server helps to accomplish this step.

Configure the CLI using the following commands:

1. To enter the config mode, run the container vsphere command.
2. To set the management vrf as the communication path for the container, run the vrf mgmt command. The vrf is mgmt by default, or you can specify any other non-default vrf configured by the user. mgmt is the only supported vrf currently.
3. To limit memory usage of the container, run the restrict memory command. Set 1GB for vSphere instance.
4. To set the installation path to the path accessible by the HTTP protocol noted in Step 2, run the image location command.

5. To pass attributes to be configured for the container from CX CLI, run the env command. Use the following format to pass the attributes to establish the configuration instance:
 - a. VSPHERE_VCENTER_N, where N takes value in the range 1–4.
 - b. The encrypted plaintext command with arguments in the form of vSphere credentials, for example: user@<domain>:password@<fqdn of vcenter>

The following example displays a valid running config:

```
container vsphere
image-location http://X.X.X.X:8000//vsphere-agent.img vrf mgmt restrict
cpu 20
restrict memory 512 vrf mgmt
env VSPHERE_VCENTER_1 encrypted plaintext
USER@vsphere.local:PASSWORD@X.X.X.X
```

Show commands

There are four top-level show commands.

- vms - shows virtual machine connection information.
- connection-status - shows information about configured vSphere vCenters.
- neighbors - shows learned neighbor information (Note: This information is learned from LLDP on vCenter distributed vswitches, not the switch).
- help - shows agent version, description, and configurable environment variables.

show vms

The following show command describes the status of the connection to vSphere, and show the VMs that are attached to switches:

At least one command parameter must be used if the parameters are available.

For example, with the show vms command, we force use of the brief option so the user is aware of all available options.

If not specified, all commands default to brief.

```
igor-sw-02# container vsphere exec show vms brief Hypervisor VM Name Physical
Switch Connections Interface

laser-02-esxi.lab.plexxi.com vm4 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/3 vm2 laser-
sw-02 (b8:d4:e7:da:40:00) 1/1/4
laser-sw-02 (b8:d4:e7:da:40:00) 1/1/3 vm5 laser-sw-02 (b8:d4:e7:da:40:00) 1/1/4
laser-sw-02 (b8:d4:e7:da:40:00) 1/1/3 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/3
laser-01-esxi.lab.plexxi.com vm1 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/1 vm3 laser-
sw-01 (b8:d4:e7:d9:af:00) 1/1/2
laser-sw-02 (b8:d4:e7:da:40:00) 1/1/1
igor-sw-02# container vsphere exec show vms vm-name vm4 Hypervisor VM Name
Physical Switch Connections Interface

laser-02-esxi.lab.plexxi.com vm4 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/3 igor-sw-02#
container vsphere exec show vms vm-name vm4 detailed VM Name : vm4
Hypervisor : laser-02-esxi.lab.plexxi.com Vendor : vmware
Hostname : vm4
OS : CentOS 4/5 or later (64-bit) VM status : on
Virtual NIC : Network adapter 2 IP Address : 172.20.12.204
```

```
MAC Address : 00:50:56:89:3c:bf Virtual Switch : dvs4
Port Group : dp4 Vlan : 0
Physical Switch Connections Interface : 1/1/3
Chassis ID : b8:d4:e7:d9:af:00 Switch Name : laser-sw-01
Switch Description : Aruba JL635A GL10.12.0001G Management Address :
172.20.11.253
```

Container exec commands are top-level, and can be run in any CLI context.

```
igor-sw-02# container vsphere exec show connection-status brief Hypervisor Last
Sync Connection State
```

```
laser-vcsa.lab.plexxi.com 2023-03-17 23:30:03 connected igor-sw-02# config
igor-sw-02(config)# container vsphere exec show connection-status brief Hypervisor
Last Sync Connection State
```

```
laser-vcsa.lab.plexxi.com 2023-03-17 23:30:07 connected igor-sw-02(config)#
interface 1/1/1
igor-sw-02(config-if)# container vsphere exec show connection-status brief
Hypervisor Last Sync Connection State
```

```
laser-vcsa.lab.plexxi.com 2023-03-17 23:30:15 connected
```

show vms

This command displays the connectivity status for the vSphere agent.

The connection timestamp is updated whenever a successful connection is made with the vSphere agent. The sync timestamp is updated whenever a full sync, event poll, or event process occurs.

```
igor-sw-02(config-container-vsphere)# container vsphere exec show connection-
status brief Hypervisor Last Sync Connection State
```

```
laser-vcsa.lab.plexxi.com 2023-03-18 02:36:06 connected
igor-sw-02(config-container-vsphere)# container vsphere exec show connection-
status detailed Hypervisor : laser-vcsa.lab.plexxi.com
Connection State : connected Last Sync : 2023-03-18 02:35:08
Last Connection : 2023-03-18 02:35:09
igor-sw-02(config-container-vsphere)# container vsphere exec show connection-
status detailed Hypervisor : laser-vcsa.lab.plexxi.com
Connection State : disconnected Last Sync : 2023-03-18 02:04:05
Last Connection : 2023-03-18 02:09:30
Connection Failure Reason : Could Not Resolve Hostname
```

show neighbors

show neighbors command displays the virtual network adapter neighbor information.

Neighbors are pulled from distributed virtual switches of the vSphere and cached locally. Neighbors are used to map VMs to physical switch ports.

If you do not see a neighbor on the port, you will not see any VM connectivity on that port either. detailed and brief options are available. Filtering can be done on switch-name and/or interface.

```
igor-sw-02(config-container-vsphere)# container vsphere exec show neighbors brief
Chassis ID Switch Name Interface
```

```
b8:d4:e7:da:40:00 laser-sw-02 1/1/1 1/1/2
b8:d4:e7:d9:af:00 laser-sw-01 1/1/1 1/1/2
```

```
igor-sw-02(config-container-vsphere)# container vsphere exec show neighbors
detailed Chassis ID : b8:d4:e7:da:40:00
Switch Name : laser-sw-02
Switch Description : Aruba JL635A GL.10.10.1000 Management Address : 192.168.200.1
Interface : 1/1/1 Advertisement Type : lldp TTL : 102
Interface : 1/1/2 Advertisement Type : lldp TTL : 102
Chassis ID : b8:d4:e7:d9:af:00 Switch Name : laser-sw-01
Switch Description : Aruba JL635A GL.10.10.1000 Management Address : 192.168.200.0
Interface : 1/1/1 Advertisement Type : lldp TTL : 102
Interface : 1/1/2 Advertisement Type : lldp TTL : 102
```

Supported switches

The following switch platforms are validated for data center. The agent itself has no context about the switch that it is installed on, except that the chipset architecture must be either x86_64 or arm:

- 8100 (x86_64)
- 8320 (x86_64)
- 8325 (x86_64)
- 8360 (arm)
- 9300 (x86_64)
- 10000 (x86_64)



The HPE Aruba Networking CX vSphere agent is a containerized agent developed for data center applications. The plugin consumes memory and CPU that would otherwise be allocated to the switch. Thus, the targeted platforms have higher system resources. Those resources are capped to prevent its interference with performance of the switch.



The solution was tested and qualified on the platforms listed here. Other platforms may work at low scale, but they are not recommended outside of the labor test environments due to the resource requirements for vSphere.

Troubleshooting

Overview

In data center networking, the most advanced step in troubleshooting often involves directly accessing the closest physical device related to the issue at hand. This trouble shooting mechanism allows the diagnosis of problems with connected endpoints. While administrators can use management tools to avoid accessing the physical device, there are scenarios where the tools are not utilized. The lack of utilization is either due to their absence, do not have of familiarity, or some of the administrators prefer device interaction. To address this, it is important to embrace these variations and aim for better integration between device-based approaches and management tools, rather than viewing them as conflicting options.

The complexity of troubleshooting the devices is multiplied by the dynamically varying endpoints within the data center. To troubleshoot, admins might extensively research endpoint metadata before beginning the process. Customers use different methods to store inventory data about their endpoints. This method can result in searching for incorrect or outdated information, causing confusion.

Additionally, admins may struggle to identify the correct network device to start with or make mistakes in the process.

Understanding the reasons behind entering the "Troubleshooting of last resort" scenario is essential for developing more effective solutions. Often, this scenario arises either due to reported connectivity problems or when admins attempt to establish connectivity for a new device or service. Interestingly, the first step in this scenario is to resort to the last option is asking a series of questions about MAC addresses, IP addresses, switches, ports, and more. Users frequently do not have this information and must look it up from spreadsheets, paper records, or outdated inventory systems.

Troubleshooting Tools

Admins can use MAC tables, ARP tables, and ping and trace tools to troubleshoot the issues. You can run the show mac-address-table command to troubleshoot:

```
cedar-sw-01# show mac-address-table MAC age-time : 300 seconds Number of
MAC addresses : 15
```

MAC Address	VLAN	Type	Port
f4:03:43:d3:be:b8	1	dynamic	1/1/4
f4:03:43:c0:e1:38	1	dynamic	lag256
f4:03:43:c0:e1:30	1	dynamic	lag2
00:50:56:5e:7a:69	1	dynamic	lag1
00:50:56:5b:a0:40	1	dynamic	lag1
04:90:81:00:08:a2	10	dynamic	lag256

The MAC address table helps you to check if you have seen the MAC address of the endpoint in the last 5 minutes. If the user does not report a problem or share the MAC address, find out the MAC address used by the endpoint. However, if the device has many virtual network parts such as switches, it can be unclear which MAC it is using.

You can also run the show arp command to troubleshoot the issue.

```
show arp
cedar-sw-01# show arp all-vrfs
```

IPv4 Address	MAC	Port	Physical Port	State	VRF
10.10.3.23	00:50:56:ac:3d:c6	vlan30		lag1	reachable default
1.1.1.12	04:90:81:00:08:a2	vlan99		lag256	reachable default
1.1.1.1	04:90:81:00:08:a2	1/1/48		1/1/48	reachable default

For the following MAC address 04:90:81:00:08:a2, the output shows that there was an ARP request for this MAC address. The device with IPV4 address 1.1.1.1 is the VSX keepalive interface on the peer switch.

Endpoint metadata

Solving the issue involves getting the metadata right down to the switch. When users use specific commands, we can add extra details to the responses using metadata. For instance, if we know that the IP address 1.1.1.1 corresponds to a user VM named "Simon," we must include that information in the output.

```
show arp (decoration)
cedar-sw-01# show arp all-vrfs
```


IPv4 Address	MAC	Port	Physical Port	State	VRF
10.10.3.23	00:50:56:ac:3d:c6	vlan30		lag1	reachable default
1.1.1.12	04:90:81:00:08:a2	vlan99		lag256	reachable default
1.1.1.1 (simons-vm)	04:90:81:00:08:a2	1/1/48			1/1/48 reachable default

The previous example is generic, to show the value of extra meta-data on standard switch screens, but in the vSphere context we have to research further to get more useful results.

VMware Use Cases

If you focus only on physical devices, you must figure out the specific switch to which your endpoint is connected. Other switches in the network might not have complete data about the endpoint, so they might not provide any information. In this scenario, more advanced data and perspectives become valuable for our administrators. By using VMware vSphere data as a foundation for your metadata, you gain a broader range of information beyond individual switches. You acquire a comprehensive set of data about topology of the fabric and endpoints. Operators use this information without requiring them to know the precise location of the endpoints.

Following queries that users might perform to gather more information about the endpoint:

Where can I find this endpoint (whether it is a virtual machine, VMkernel, or host)?

You might have a differing set of data on the endpoint, depending on their situation, and they want to know where it resides based on the limited information they have.

Create this query with a flexible set of data:

- VMs name, DNS name (can be different from VM name), VMs IP (more than one IP is possible), VMs NIC name (VMs version, not ESXi), VMs MAC
 - VMkernel looks like a VM in some ways, just a virtual interface used for some important VMware control functions such as vMotion
- ESXi name, DNS name (often the same as ESXi name), IP, vmnic, vmnic MAC

What does the user want in response to their query?

- Switch, LAG , or Interface endpoint last seen
- When it was last seen
- Previous location (can be useful for detecting loops), will move if vMotion or depending on vmware load balancing

What additional data is available for this endpoint?

There is additional metadata for vSphere on the endpoint that can be useful for the customer. For a complete integration and for full visualization and automation, this data is used and is applied to Switch CLI.

For VMs

- The basic inventory data on this VM
 - VMs name, DNS name (often different from VM name), VMs IP (they can have more than one), VMs NIC name (VMs version, not ESXi), VMs MAC
 - The VMs association with the Virtual network in ESXi/vsphere
 - Portgroup (VLAN, VLAN type), vSwitch, ESXi Host, ESXi Host NIC (name, IP, and MAC)

- For VMkernel
 - Basic inventory data on this VMkernel
 - Name, IP, MAC
 - The VMKs association with the Virtual network in ESXi or vsphere
 - Portgroup (VLAN, VLAN type), vSwitch, ESXi Host, ESXi Host NIC (name, IP, and MAC)
- For ESXi
 - The basic inventory data on this
 - ESXi name, DNS name, ESXi IP, ESXi Nics, and MACs
 - For each ESXi NIC, the association with vSwitch

What devices are connected to this port?

You can run `show mac-table-address interface x/t/z` command, to display active MACs.

The advantage of using VM metadata is that you can find out the location of the VM without active MACs. You can find out the portgroup or vSwitch location through LLDP from the host, and from the VM association you can find out the location of the switch port.

To this end, the `show VMs` interface `x/t/z` could respond with a mix of active and inactive VMs.

Switch system and hardware commands

bluetooth disable

```
bluetooth disable
no bluetooth disable
```

Description

Disables the Bluetooth feature on the switch. The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE). Bluetooth is enabled by default.

The **no** form of this command enables the Bluetooth feature on the switch.

Example

Disabling Bluetooth on the switch. <XXXX> is the switch platform and <NNNNNNNNNN> is the device identifier.

```
switch(config)# bluetooth disable
switch# show bluetooth
Enabled           : No
Device name       : <XXXX>--<NNNNNNNNNN>

switch(config)# show running-config
...
bluetooth disabled
...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

bluetooth enable

```
bluetooth enable
no bluetooth enable
```

Description

This command enables the Bluetooth feature on the switch. The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE).

Default: Bluetooth is enabled by default.

The **no** form of this command disables the Bluetooth feature on the switch.

Usage

The default configuration of the Bluetooth feature is `enabled`. The output of the `show running-config` command includes Bluetooth information only if the Bluetooth feature is disabled.

The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE).

The Bluetooth feature requires the USB feature to be enabled. If the USB feature has been disabled, you must enable the USB feature before you can enable the Bluetooth feature.

Examples

```
switch(config)# bluetooth enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

clear events

```
clear events
```

Description

Clears up event logs. Using the `show events` command will only display the logs generated after the `clear events` command.

Examples

Clearing all generated event logs:

```
switch# show events
-----
show event logs
-----
2018-10-14:06:57:53.534384|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 27
2018-10-14:06:58:30.805504|lldpd|103|LOG_INFO|MSTR|1|Configured LLDP tx-timer to
36
2018-10-14:07:01:01.577564|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
```

```

utilization poll interval is changed to 49

switch# clear events

switch# show events
-----
show event logs
-----
2018-10-14:07:03:05.637544|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 34

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

clear ip errors

```
clear ip errors
```

Description

Clears all IP error statistics.

Example

Clearing and showing ip errors:

```

switch# clear ip errors
switch# show ip errors
-----
Drop reason                Packets
-----
Malformed packets          0
IP address errors          0
...

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

console baud-rate

```
console baud-rate <SPEED>
no console baud-rate <SPEED>
```

Description

Sets the console serial port speed.

The no form of this command resets the console port speed to its default of 115200 bps.

Parameter	Description
<SPEED>	Selects the console port speed in bps, either 9600 or 115200.

Usage

The speed change occurs immediately for the active console session. The console will be inaccessible until the client terminal settings are updated to match the console port speed that you set. After the command is executed you will be prompted to log in again.

Examples

Setting the console port speed to 9600 bps:

```
switch(config)# console baud-rate 9600
This command will configure the baud rate immediately for the active serial
console session. After the command is executed the user will be prompted to
re-login. The serial console will be inaccessible until the terminal client
settings are updated to match the baud rate of the switch.
Continue (y/n)? y
```

Resetting the console port to its default speed 115200 bps:

```
switch(config)# no console baud-rate
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

domain-name

domain-name <NAME>
no domain-name [<NAME>]

Description

Specifies the domain name of the switch.

The **no** form of this command sets the domain name to the default, which is no domain name.

Parameter	Description
<NAME>	Specifies the domain name to be assigned to the switch. The first character of the name must be a letter or a number. Length: 1 to 32 characters.

Examples

Setting and showing the domain name:

```
switch# show domain-name

switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#
```

Setting the domain name to the default value:

```
switch(config)# no domain-name
switch(config)# show domain-name

switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

hostname

hostname <HOSTNAME>
no hostname [<HOSTNAME>]

Description

Sets the host name of the switch.

The **no** form of this command sets the host name to the default value, which is `switch`.

Parameter	Description
<code><HOSTNAME></code>	Specifies the host name. The first character of the host name must be a letter or a number. Length: 1 to 32 characters. Default: <code>switch</code>

Examples

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
myswitch
```

Setting the host name to the default value:

```
myswitch(config)# no hostname
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

module admin-state

```
module <SLOT-ID> admin-state {diagnostic | down | up}
no module <SLOT-ID> [admin-state [diagnostic | down | up]]
```

Description

Sets the administrative state of the specified line module.

The **no** form of the command configures administrative state to the default **up**.

Parameter	Description
<code><SLOT-ID></code>	Specifies the member and slot of the module. For example, to

Parameter	Description
	specify the module in member 1, slot 3, enter the following: 1/3
diagnostic	Selects the diagnostic administrative state. Network traffic does not pass through the module.
down	Selects the down administrative state. Network traffic does not pass through the module.
up	Selects the up administrative state. The line module is fully operational. The up state is the default administrative state.

Example

Setting the administrative state of the module in slot **1/3** to **down**:

```
switch(config)# module 1/3 admin-state down
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6400	config	Administrators or local user group members with execution rights for this command.

module product-number

```
module <SLOT-ID> product-number [<PRODUCT-NUM>]
no module <SLOT-ID> [product-number [<PRODUCT-NUM>]]
```

Description

Changes the configuration of the switch to indicate that the specified member and slot number contains, or will contain, a line module.

The **no** form of this command removes the line module and its interfaces from the configuration. If there is a line module installed in the slot, the line module is powered off and then powered on.

Parameter	Description
<SLOT-ID>	Specifies the member and slot in the form m/s , where m is the member number, and s is the slot number.
<PRODUCT-NUM>	Specifies the product number of the line module. For example: JL363A

Parameter	Description
	If there is a line module installed in the slot when you execute this command, <i><PRODUCT-NUM></i> is optional. The switch reads the product number information from the module that is installed in the slot. If there is no line module installed in the slot when you execute this command, <i><PRODUCT-NUM></i> is required.

Usage

The default configuration associated with a line module slot is:

- There is no module product number or interface configuration information associated with the slot. The slot is available for the installation with any supported line module.
- The Admin State is Up (which is the default value for Admin State).

To add a line module to the configuration, you must use the **module** command either before or after you install the physical module.

If you execute the **module** command after you install a line module in an empty slot, you can omit the *<PRODUCT-NUM>* variable. The switch reads the product information from the installed module.

If the module is not installed in the slot when you execute the module command, you must specify a value for the *<PRODUCT-NUM>* variable:

- The switch validates the product number of the module against the slot number you specify to ensure that the right type of module is configured for the specified slot.

For example, the switch returns an error if you specify the product number of a line module for a slot reserved for management modules.

- You can configure the line module interfaces before the line module is installed.

When you install the physical line module in a preconfigured slot, the following actions occur:

- If a product number was specified in the command and it matches the product number of the installed module, the switch initializes the module.
- If a product number was specified in the command and the product number of the module does not match what was specified, the module device initialization fails.

The **no** form of the command removes the line module and its interfaces from the configuration and restores the line module slot to the default configuration.

If there is a line module installed in the slot when you execute the **no** form of the command, the command also powers off and then powers on the module. Traffic passing through the line module is stopped. Management sessions connected through the line module are also affected.

If the slot associated with the line module is in the default configuration, you can remove the module from the chassis without disrupting the operation of the switch.

Examples

Configuring slot 1/1 for future installation of a line module:

```
switch(config)# module 1/1 product-number j1363a
```

Configuring a line module that is already installed in slot 1/1:

```
switch(config)# module 1/1 product-number
```

Attempting to configure slot 1/1 for the future installation of a line module without specifying the product number (returned error shown):

```
switch(config)# module 1/1 product-number
Line module '1/4' is not physically available. Please provide the product
number to preconfigure the line module.
```

Removing a module from the configuration:

```
switch(config)# no module 1/1
This command will power cycle the specified line module and restore its default
configuration. Any traffic passing through the line module will be interrupted.
Management sessions connected through the line module will be affected. It
might take a few minutes to complete this operation.

Do you want to continue (y/n)? y
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6400	config	Administrators or local user group members with execution rights for this command.

mtrace

```
mtrace <IPv4-SRC-ADDR> <IPv4-GROUP-ADDR> [lhr <IPv4-LHR-ADDR>] [ttl <HOPS>]
[vrf <VRF-NAME>]
```

Description

Traces the specified IPv4 source and group addresses.

Parameter	Description
<i>IPv4-SRC-ADDR</i>	Specifies the source IPv4 address to trace.
<i>IPv4-GROUP-ADDR</i>	Specifies the group IPv4 address to trace.
<i>lhr <IPv4-LHR-ADDR></i>	Specifies the last hop router address from which to start the trace.
<i>ttl <HOPS></i>	Specifies the Time-To-Live duration in hops. Range: 1 to 255 hops. Default: 8 hops.

Parameter	Description
<code>vrf <VRF-NAME></code>	Specifies the name of the VRF. If a name is not specified the default VRF will be used.

Examples

Tracing with source, group, and LHR addresses and TTL:

```
(switch)# mtrace 20.0.0.1 239.1.1.1 lhr 10.1.1.1 ttl 10

Type escape sequence to abort.
Mtrace from 10.0.0.1 for Source 20.0.0.1 via Group 239.1.1.1
From destination(?) to source (?)...
Querying ful reverse path...
0 10.0.0.1
-1 30.0.0.1 PIM 0 ms
-2 40.0.0.1 PIM 2 ms
-3 50.0.0.1 PIM 100 ms
-4 60.0.0.1 PIM 156 ms
-5 20.0.0.1 PIM 123 ms
```

Tracing with source and group addresses:

```
(switch)# mtrace 200.0.0.1 239.1.1.1

Type escape sequence to abort.
Mtrace from self for Source 200.0.0.1 via Group 239.1.1.1
From destination(?) to source (?)...
Querying ful reverse path...
0 10.0.0.1
-1 30.0.0.1 PIM 0 ms
-2 40.0.0.1 PIM 2 ms
-3 50.0.0.1 PIM 100 ms
-4 60.0.0.1 PIM 156 ms
-5 200.0.0.1 PIM 123 ms
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

power consumption-average-period

`power consumption-average-period <PERIOD-IN-SECONDS>`

Description

Configures a time period for average power consumption in seconds.

Parameter	Description
<code><PERIOD-IN-SECONDS></code>	Specifies the period in seconds for average power consumed. Range: 60-3600. Default: 600

Example

Configuring a time period of 60 seconds for average power consumption:

```
switch(config)# power consumption-average-period 60
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

show bluetooth

```
show bluetooth
```

Description

Shows general status information about the Bluetooth wireless management feature on the switch.

Usage

This command shows status information about the following:

- The USB Bluetooth adapter
- Clients connected using Bluetooth
- The switch Bluetooth feature.

The output of the **show running-config** command includes Bluetooth information only if the Bluetooth feature is disabled.

The device name given to the switch includes the switch serial number to uniquely identify the switch while pairing with a mobile device.

The management IP address is a private network address created for managing the switch through a Bluetooth connection.

Examples

Example output when Bluetooth is enabled but no Bluetooth adapter is connected. <XXXX> is the switch platform and <NNNNNNNNNN> is the device identifier.

```
switch# show bluetooth
Enabled          : Yes
Device name      : <XXXX>-<NNNNNNNNNN>
Adapter State    : Absent
```

Example output when Bluetooth is enabled and there is a Bluetooth adapter connected:

```
switch# show bluetooth
Enabled          : Yes
Device name      : <XXXX>-
Adapter State    : Ready
Adapter IP address : 192.168.99.1
Adapter MAC address : 480fcf-af153a

Connected Clients
-----
Name           MAC Address      IP Address      Connected Since
-----
Mark's iPhone  089734-b12000    192.168.99.10   2018-07-09 08:47:22 PDT
```

Example output when Bluetooth is disabled:

```
switch# show bluetooth
Enabled          : No
Device name      : <XXXX>-<NNNNNNNNNN>
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show boot-history

```
show boot-history [all|{vsf member <1-10>}]
```

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.

For switches that support line modules (such as 6400 switch series) including the **all** parameter displays information for the active management module and all available line modules.



To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
all	Optional. Shows boot information for the active management module. For switches that support line modules, including this parameter displays information for and all available line modules.
vsf member <1-10>	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
Index	The position of the boot in the history file. Range: 0 to 3.
Boot ID	A unique ID for the boot . A system-generated 128-bit string.
Current Boot, up for <time>	For the current boot, the show boot-history command shows the number of seconds the module has been running on the current software.
<Timestamp>: boot reason	For previous boot operations, the show boot-history command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values: ▪ <DAEMON-NAME> crash: The daemon identified by <DAEMON-NAME> caused the module to boot. ▪ Kernel crash: The operating system software associated with the module caused the module to boot. ▪ Uncontrolled reboot: The reason for the reboot is not known. ▪ Reboot requested through database: The reboot occurred because of a request made through the CLI or other API. For details, see show boot-history

Table 1: *Description of reboots handled through the database*

Boot History String	Description
Reboot requested by user	A user requested a switch reboot through the CLI or web UI.
Reset button pressed	The switch detected a short-press of the reset button
Backplane fault	A backplane fault occurred.
Configuration change	A configuration change resulted in a reboot.
Configuration version migration	A configuration version migration occurred which required a reboot.
Console error	The console failed to start.
Fabric fault	A fabric fault occurred.
All line modules faulted	A zero line card condition occurred.
Redundancy switchover requested	A user requested a redundancy switchover.
Redundant Management communication timeout	The standby management module has taken over from an unresponsive active management module.
Redundant Management election timeout	A failure to elect a standby management module in the allotted time.
Critical service fault (error)	A daemon critical to switch operation has stopped functioning. An extra error string may be present to describe the error in detail.
VSF autojoin renumber	Reset triggered by VSF autojoin.
VSF member renumbered	A user requested a renumber of a VSF member.
VSF switchover requested	A user requested a VSF switchover.
VSX software update	Reset triggered by a VSX software update.
Chassis critical temperature	Chassis operating temperature exceeded.
Chassis low critical temperature	Chassis temperature below the minimum operating threshold.
Chassis insufficient fans	Insufficient fans to cool the chassis.
Chassis unsupported PSUs/fans	Unsupported or misconfigured PSUs or system fans.
Management module critical temperature	Management module operating temperature exceeded.
ISSU SMM update	Standby management module reboot triggered by an In-Service Software Upgrade (ISSU).
ISSU switchover	Redundancy switchover triggered by an In-Service Software Upgrade.

Boot History String	Description
ISSU aborted	Standby management module reset triggered by failure during an In-Service Software Upgrade.
Rollback timer expired	Reset triggered by the ISSU rollback timer expiring.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=====

Index : 2
Boot ID : c34a2c2499004a02bbeeff4992elfdbd
Current Boot, up for 1 days 13 hrs 13 mins 27 secs

Index : 1
Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2

Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811ddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1

Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 : Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 : Switch boot count is 3
25 Aug 23 10:29:02 : Primary overtemperature fault detected with 0x2 in PSU 1/1
```

(For 6400 Switch series) Showing the boot history of the active management module and all line modules:

```
switch#
Management module
=====

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
```

```

07 Aug 18 13:00:46 : Reboot requested through database

Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...

Management module
=====

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database

Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...

```

In the event of a reset triggered by a power supply unit (PSU), or a PSU input fault, the output of this command also displays information about why the PSU initiated a reboot. The following example displays the boot history of a switch with a reboot initiated by a PSU.

```

switch# show boot-history

Management module
=====
Index : 2
Boot ID : a61ad00d10864c748bc7893a5d4af2e4
15 Dec 23 19:02:02 : Power on reset with 0x1, Version: FL.10.13.1000AF

15 Dec 23 19:02:02 : Switch boot count is 0
15 Dec 23 19:02:17 : PSU 1/1: Fault detected

Index : 1
Boot ID : 30d831bbfdfa425baf50a629ee01b185
15 Dec 23 19:01:58 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:01:58 : Switch boot count is 0

```

The following example displays the boot history for the VSF member 2.

```

switch# show boot-history vsf member 2

Member-2
=====

```

```
Index : 0
Boot ID : df99026c194a44f1944a3e7685fb4d90
Current Boot, up for 3 hrs 31 mins 39 secs

Index : 3
Boot ID : 7bf4104903fe4ad1ba4bce40e8099c76
10 Aug 17 10:02:24 : Reboot requested through database
10 Aug 17 10:02:13 : Switch boot count is 2
```

Command History

Release	Modification
10.13.1000	The output of this command is enhanced to display additional information about the reason for the reboot, if available.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show capacities

`show capacities <FEATURE> [vsx-peer]`

Description

Shows system capacities and their values for all features or a specific feature.

Parameter	Description
<FEATURE>	Specifies a feature. For example, aaa or vrp .
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

Capacities are expressed in user-understandable terms. Thus they may not map to a specific hardware or software resource or component. They are not intended to define a feature exhaustively.

Examples

Showing all available capacities for BGP:

```
switch# show capacities bgp
```

```

System Capacities: Filter BGP
Capacities Name                                     Value
-----
-
Maximum number of AS numbers in as-path attribute
32
...

```

Showing all available capacities for mirroring:

```

switch# show capacities mirroring

System Capacities: Filter Mirroring
Capacities Name                                     Value
-----
-
Maximum number of Mirror Sessions configurable in a system
4
Maximum number of enabled Mirror Sessions in a system
4

```

Showing all available capacities for MSTP:

```

switch# show capacities mstp

System Capacities: Filter MSTP
Capacities Name                                     Value
-----
-
Maximum number of mstp instances configurable in a system
64

```

Showing all available capacities for VLAN count:

```

switch# show capacities vlan-count

System Capacities: Filter VLAN Count
Capacities Name                                     Value
-----
-
Maximum number of VLANs supported in the system
4094

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show capacities-status

show capacities-status <FEATURE> [vsx-peer]

Description

Shows system capacities status and their values for all features or a specific feature.

Parameter	Description
<FEATURE>	Specifies the feature, for example aaa or vrrp for which to display capacities, values, and status. Required.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Showing the system capacities status for all features:

```
switch# show capacities-status

System Capacities Status
Capacities Status Name                                     Value Maximum
-----
Number of active gateway mac addresses in a system         0           16
Number of aspath-lists configured                          0           64
Number of community-lists configured                       0           64
...
```

Showing the system capacities status for BGP:

```
switch# show capacities-status bgp

System Capacities Status: Filter BGP
Capacities Status Name                                     Value Maximum
-----
Number of aspath-lists configured                          0           64
Number of community-lists configured                       0           64
Number of neighbors configured across all VRFs             0           50
Number of peer groups configured across all VRFs           0           25
Number of prefix-lists configured                          0           64
Number of route-maps configured                            0           64
Number of routes in BGP RIB                                0      256000
Number of route reflector clients configured across all VRFs 0           16
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show console

show console

Description

Shows the serial console port current speed.

Examples

Showing the console port current speed:

```
switch# show console
Baud Rate: 9600
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show core-dump

show core-dump [all | <SLOT-ID>]

Description

Shows core dump information about the specified module. When no parameters are specified, shows only the core dumps generated in the current boot of the management module. When the **all** parameter is specified, shows all available core dumps.

Parameter	Description
all	Shows all available core dumps.
<SLOT-ID>	Shows the core dumps for the management module or line module in <SLOT-ID>. <SLOT-ID> specifies a physical location on the switch. Use the format member/slot/port (for example, 1/3/1) for line

Parameter	Description
	modules. Use the format member/slot for management modules.
	You must specify the slot ID for either the active management module, or the line module.

Usage

When no parameters are specified, the `show core-dump` command shows only the core dumps generated in the current boot of the management module. You can use this command to determine when any crashes are occurring in the current boot.

If no core dumps have occurred, the following message is displayed: **No core dumps are present**

To show core dump information for the standby management module, you must use the **standby** command to switch to the standby management module and then execute the **show core-dump** command.

In the output, the meaning of the information is the following:

Daemon Name

Identifies name of the daemon for which there is dump information.

Instance ID

Identifies the specific instance of the daemon shown in the **Daemon Name** column.

Present

Indicates the status of the core dump:

Yes

The core dump has completed and available for copying.

In Progress

Core dump generation is in progress. Do not attempt to copy this core dump.

Timestamp

Indicates the time the daemon crash occurred. The time is the local time using the time zone configured on the switch.

Build ID

Identifies additional information about the software image associated with the daemon.

Examples

Showing core dump information for the current boot of the active management module only:

```
switch# show core-dump
=====
Daemon Name      | Instance ID | Present   | Timestamp           | Build ID
=====
hpe-fand         | 1399        | Yes       | 2017-08-04 19:05:34 | 1246d2a
hpe-sysmond      | 957         | Yes       | 2017-08-04 19:05:29 | 1246d2a
=====
Total number of core dumps : 2
=====

=====
Daemon Name      | Instance ID | Present   | Timestamp           | Build ID
=====
hpe-fand         | 1399        | Yes       | 2017-08-04 19:05:34 | 1246d2a
hpe-sysmond      | 957         | Yes       | 2017-08-04 19:05:29 | 1246d2a
=====
Total number of core dumps : 2
=====
```

Showing all core dumps:

```

switch# show core-dump all
=====
Management Module core-dumps
=====
Daemon Name      | Instance ID | Present      | Timestamp                | Build ID
=====
hpe-sysmond      513          Yes          2017-07-31 13:58:05      e70f101
hpe-tempd        1048         Yes          2017-08-13 13:31:53      e70f101
hpe-tempd        1052         Yes          2017-08-13 13:41:44      e70f101

Line Module core-dumps
=====
Line Module : 1/1
=====
dune_agent_0     18958        Yes          2017-08-12 11:50:17      e70f101
dune_agent_0     18842        Yes          2017-08-12 11:50:09      e70f101
=====
Total number of core dumps : 5
=====

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show domain-name

show domain-name [vsx-peer]

Description

Shows the current domain name.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

If there is no domain name configured, the CLI displays a blank line.

Example

Setting and showing the domain name:


```
switch# show domain-name

switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show environment fan

```
show environment fan [vsf | vsx-peer]
```

Description

Shows the status information for all fans and fan trays (if present) in the system.

Parameter	Description
vsf	Shows output from the VSF member-id on switches that support VSF.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

For fan trays, **Status** is one of the following values:

- ready:The fan tray is operating normally.
- fault:The fan tray is in a fault event. The status of the fan tray does not indicate the status of fans.
- empty:The fan tray is not installed in the system.

For fans:

Speed :Indicates the relative speed of the fan based on the nominal speed range of the fan. Values are:

Slow:The fan is running at less than 25% of its maximum speed.
Normal:The fan is running at 25-49% of its maximum speed.
Medium:The fan is running at 50-74% of its maximum speed.
Fast:The fan is running at 75-99% of its maximum speed.
Max:The fan is running at 100% of its maximum speed.

N/A:The fan is not installed.
 Direction: The direction of airflow through the fan. Values are:
 front-to-back:Air flows from the front of the system to the back of the system.
 N/A:The fan is not installed.
 Status: Fan status. Values are:
uninitialized:The fan has not completed initialization.
ok: The fan is operating normally.
fault: The fan is in a fault state.
empty: The fan is not installed.

Examples

Showing output for systems with fan trays for 6300 switch series:

```
switch# show environment fanFan tray information
-----
Name Description Status Serial Number
Fans
-----
1/1 JL669A Aruba X751 FB Fan Tray ready
CN97KN9131 2
1/2 JL669A Aru
-----
1/1/1      N/A      N/A      slow    front-to-back  ok      5371
1/1/2      N/A      N/A      slow    front-to-back  ok      5320
1/1/3      N/A      N/A      slow    front-to-back  ok      5328
1/1/4      N/A      N/A      slow    front-to-back  ok      5256
1/2/1      N/A      N/A      slow    front-to-back  ok      5371
1/2/2      N/A      N/A      slow    front-to-back  ok      5349
1/2/3      N/A      N/A      slow    front-to-back  ok      5292
1/2/4      N/A      N/A      slow    front-to-back  ok      5349
1/3/1      N/A      N/A      slow    front-to-back  ok      5313
1/3/2      N/A      N/A      slow    front-to-back  ok      5371
1/3/3      N/A      N/A      slow    front-to-back  ok      5379
1/3/4      N/A      N/A      slow    front-to-back  ok      5379
1/4/1      N/A      N/A      slow    front-to-back  ok      5313
1/4/2      N/A      N/A      slow    front-to-back  ok      5299
1/4/3      N/A      N/A      slow    front-to-back  ok      5285
1/4/4      N/A      N/A      slow    front-to-back  ok      5371
```

Showing output for a system without a fan tray:

```
switch# show environment fan
Fan information
-----
Fan    Serial Number  Speed    Direction    Status    RPM
-----
1      SGXXXXXXXXXX    slow     front-to-back  ok        6000
2      SGXXXXXXXXXX    normal   front-to-back  ok        8000
3      SGXXXXXXXXXX    medium   front-to-back  ok        11000
4      SGXXXXXXXXXX    fast     front-to-back  ok        14000
5      SGXXXXXXXXXX    max      front-to-back  fault     16500
6      N/A             N/A      N/A           empty
...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show environment led

```
show environment led <MEMBER-ID> [vsx-peer]
```

Description

Shows state and status information for all the configurable LEDs in the system.

Parameter	Description
<MEMBER-ID>	Shows output from the specified VSF member ID on switches that support VSF.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

Showing state and status for LED:

```
switch# show environment led
Mbr/Name      State      Status
-----
1/locator     off        ok
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show environment power-consumption

```
show environment power-consumption <DETAIL>
show environment power-consumption member <MEMBER-ID>
```

Description

Displays power consumption information.

Parameter	Description
<DETAIL>	Displays detailed power consumption information.
<MEMBER-ID>	For VSF supported platforms only. Displays the power consumption information for the specified VSF member. Range: 1-10.

Usage

Power consumed values are updated every minute. The total power consumed is the total power used in a chassis. The power consumed average is calculated from the total power consumed as a running average over a period of time. The average period has a default of 10 minutes. The period can be configured using **power-consumption-average-period**.

For VSF supported platforms, this command displays the power consumed for all modules of the member in a given argument.

The following information is provided for a summary of power consumption:

- line module: power used by line module
- management module: power used by management module
- fabric module: power used by fabric module
- chassis module: power used by chassis module
- fan module: power used by fan module
- power total: total power consumption
- average power: average for total power consumption that is calculated over a given period
- average period: time to calculate power average

Example

Showing the power consumption for a switch with a single line card:

```
switch# show environment power-consumption

Power Consumption Averaging Period: 60 seconds

Name      Description      Instantaneous Power (W)      Average Power (W)
-----
1         8360-32YAC Switch      300.00                       311.50
```

Showing the power consumption for a VSF stack:

```
switch# show environment power-consumption
```

Power Consumption Averaging Period: 60 seconds

Name	Description	Instantaneous Power (W)	Average Power (W)
1	6200M 24G 24G 4SPF+ SW	300.00	311.50
2	6200M 24G 24G 4SPF+ SW	280.00	275.50

Showing the power consumption for a switch with multiple line cards, in brief:

```
switch# show environment power-consumption
```

Power Consumption Averaging Period: 60 seconds

Name	Description	Instantaneous Power (W)	Average Power (W)
1	6410 v2 Chassis	1300.00	1311.50

Showing the power consumption for a switch with multiple line cards, in detail:

```
switch# show environment power-consumption detail
```

Power Consumption Averaging Period: 60 seconds

Name	Module Type	Instantaneous Power (W)	Average Power (W)
1	chassis total	1000.00	1002.50
1/1	fabric	40.00	
1/1	line	120.00	
1/2	line	350.00	
1/1	management	20.00	
	other	470.00	

Showing the power consumption for 4 member stack, in detail:

```
switch# show environment power-consumption detail
```

Power Consumption Averaging Period: 60 seconds

Name	Module Type	Instantaneous Power (W)	Average Power (W)
1	chassis total	300.00	302.50
2	chassis total	320.00	319.50
3	chassis total	280.00	282.50
4	chassis total	310.00	311.50

Total power consumption 1210.00

Showing the power consumption for VSF member 2:

```
switch# show environment power-consumption member 2
```

Power Consumption Averaging Period: 60 seconds

Name	Module Type	Instantaneous Power (W)	Average Power (W)
2	6200M 24G 4SFP+ SW	280.00	275.00

Showing the power consumption for VSF invalid member:

```
switch# show environment power-consumption member 5
Member 5 is not present
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show environment power-consumption

Not supported on the 6300 Switch Series.

```
show environment power-consumption [vsx-peer]
```

Description

Shows the power being consumed by each management module, line card, and fabric card subsystem, and shows power consumption for the entire chassis.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

This command is only applicable to systems that support power consumption readings.

The power consumption values are updated once every minute.

The output of this command includes the following information:

Parameter	Description
Name	Shows the member number and slot number of the management module, line module, or fabric card module.
Type	Shows the type of module installed at the location specified by Name.
Description	Shows the product name and brief description of the module.
Usage	Shows the instantaneous power consumption of the module.

Parameter	Description
	Power consumption is shown in Watts.
Module Total Power Usage	Shows the total power consumption of all the modules listed. Power consumption is shown in Watts.
Chassis Total Power Usage	Shows the total instantaneous power consumed by the entire chassis, including modules and components that do not support individual power reporting. Power consumption is shown in Watts.
Chassis Total Power Available	Shows the total amount of power, in Watts, that can be supplied to the chassis.
Chassis Total Power Allocated	Shows total power, in Watts, that is allocated to powering the chassis and its installed modules.
Chassis Total Power Unallocated	Shows the total amount of power, in Watts, that has not been allocated to powering the chassis or its installed modules. This power can be used for additional hardware you install in the chassis.

Example

Showing the power consumption for an Aruba 6400 switch:

```
switch> show environment power-consumption
```

Name	Type	Description	Power Usage
1/1	management-module	R0X31A 6400 Management Module	18 W
1/2	management-module		0 W
1/3	line-card-module		0 W
1/4	line-card-module	R0X39A 6400 48p 1GbE CL4 PoE 4SFP56 Mod	54 W
1/5	line-card-module		0 W
1/6	line-card-module	R0X39A 6400 48p 1GbE CL4 PoE 4SFP56 Mod	56 W
1/7	line-card-module	R0X39A 6400 48p 1GbE CL4 PoE 4SFP56 Mod	51 W
1/1	fabric-card-module	R0X24A 6405 Chassis	71 W
Module Total Power Usage			250 W
Chassis Total Power Usage			294 W
Chassis Total Power Available			1800 W

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6400	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show environment power-supply

```
show environment power-supply [vsf | vsx-peer]
```

Description

Shows status information about all power supplies in the switch.

Parameter	Description
vsf	Shows output from the VSF member-id on switches that support VSF.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

The following information is provided for each power supply:

Parameter	Description
Mbr/PSU	Shows the member and slot number of the power supply.
Product Number	Shows the product number of the power supply.
Serial Number	Shows the serial number of the power supply, which uniquely identifies the power supply.
PSU Status	The status of the power supply. Values are: ▪ OK:Power supply is operating normally. ▪ OK*: Power supply is operating normally, but it is the only power supply in the chassis. One power supply is not sufficient to supply full power to the switch. When this value is shown, the output of the command also shows a message at the end of the displayed data. ▪ Absent: No power supply is installed in the specified slot. ▪ Input fault: The power supply has a fault condition on its input. ▪ Output fault: The power supply has a fault condition on its output. ▪ Warning: The power supply is not operating normally. ▪ Wattage Maximum: Shows the maximum amount of wattage that the power supply can provide.

Example

Showing the output when only one power supply is installed in an Aruba 6400 switch chassis:

```
switch# show environment power-supply
      Product  Serial      PSU      Wattage
Mbr/PSU Number  Number      Status      Maximum
-----
1/1    R0X36A   CN91KMM2H3   OK         3000
1/2    N/A      N/A          Absent      0
1/3    N/A      N/A          Absent      0
1/4    N/A      N/A          Absent      0
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show environment rear-display-module

```
show environment rear-display-module [vsx-peer]
```

Description

Shows information about the display module on the back of the switch.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

Showing the rear display module information on the back of the switch:

```
switch> show environment rear-display-module

Rear display module is ready
Description: 8400 Rear Display Mod
Full Description: 8400 Rear Display Module
Serial number: SG00000000
Part number: 5300_0272
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show environment temperature

```
show environment temperature [detail] [vsf | vsx-peer]
```

Description

Shows the temperature information from sensors in the switch that affect fan control.

Parameter	Description
detail	Shows detailed information from each temperature sensor.
vsf	Shows output from the VSF member-id on switches that support VSF
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

Temperatures are shown in Celsius.

Valid values for status are the following: \

Parameter	Description
normal	Sensor is within nominal temperature range.
max	Highest temperature from this sensor.
low_critical	Lowest threshold temperature for this sensor.
critical	Highest threshold temperature for this sensor.
fault	Fault event for this sensor.
emergency	Over temperature event for Over temperature event for this sensor.

Examples

Showing current temperature information for a 6300 switch:

```
switch# show environment temperature
```

```
Temperature information
```

Mbr/Slot-Sensor	Module Type	Current temperature	Status	

1/1-PHY-01-04	line-card-module	45.00 C	normal	
1/1-PHY-05-08	line-card-module	45.00 C	normal	
1/1-PHY-09-12	line-card-module	46.00 C	normal	
1/1-PHY-13-16	line-card-module	47.00 C	normal	
1/1-PHY-17-20	line-card-module	47.00 C	normal	
1/1-PHY-21-24	line-card-module	50.00 C	normal	
1/1-PHY-25-28	line-card-module	45.00 C	normal	
1/1-PHY-29-32	line-card-module	47.00 C	normal	
1/1-PHY-33-36	line-card-module	48.00 C	normal	
1/1-PHY-37-40	line-card-module	47.00 C	normal	
1/1-PHY-41-44	line-card-module	48.00 C	normal	
1/1-PHY-45-48	line-card-module	49.00 C	normal	
1/1-Switch-ASIC-Internal	line-card-module	56.25 C	normal	
1/1-CPU-Zone-0	management-module	50.00 C	normal	
1/1-CPU-Zone-1	management-module	50.00 C	normal	
1/1-CPU-Zone-2	management-module	50.00 C	normal	
1/1-CPU-Zone-3	management-module	51.00 C	normal	
1/1-CPU-Zone-4	management-module	51.00 C	normal	
1/1-CPU-diode	management-module	53.12 C	normal	
1/1-DDR	management-module	45.25 C	normal	
1/1-Inlet-Air	management-module	24.88 C	normal	
1/1-MB-IBC	management-module	45.62 C	normal	
1/1-Switch-ASIC-diode	management-module	58.06 C	normal	

Showing detailed temperature information for a 6300 switch:

```
switch# show environment temperature detail
```

```
Detailed temperature information
```

```
-----
Mbr/Slot-Sensor      : 1/1-PHY-01-04
Module Type          : line-card-module
Module Description    : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Status               : normal
Fan-state            : normal
Current temperature   : 45.00 C
Minimum temperature   : 41.00 C
Maximum temperature   : 50.00 C
```

```
Mbr/Slot-Sensor      : 1/1-PHY-05-08
Module Type          : line-card-module
Module Description    : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Status               : normal
Fan-state            : normal
Current temperature   : 45.00 C
Minimum temperature   : 41.00 C
Maximum temperature   : 50.00 C
```

```
...
```

```
Detailed temperature information
```

```
-----
Mbr/Slot-Sensor      : 1/1-PHY-01-04
Module Type          : line-card-module
Module Description    : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Status               : normal
```

```

Fan-state           : normal
Current temperature : 45.00 C
Minimum temperature : 41.00 C
Maximum temperature : 50.00 C

Mbr/Slot-Sensor     : 1/1-PHY-05-08
Module Type          : line-card-module
Module Description   : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Status               : normal
Fan-state            : normal
Current temperature  : 45.00 C
Minimum temperature  : 41.00 C
Maximum temperature  : 50.00 C

...

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show events

```

show events [ -e <EVENT-ID> |
    -s {emergency | alert | critical | error | warning | notice | info | debug} |
    -r |
    -a |
    -n <COUNT> |
    -i <MEMBER-SLOT> |
    -m {active | standby} |
    -c {lldp | ospf | ...} |
    -d {lldpd | bgpd | fand | ...}]

```

Description

Shows event logs generated by the switch modules since the last reboot.

Parameter	Description
-e <EVENT-ID>	Shows the event logs for the specified event ID. Event ID range: 101 through 99999.
-s {emergency alert critical error warning notice info debug}	Shows the event logs for the specified severity. Select the severity from the following list: emergency: Displays event logs with severity emergency only.

Parameter	Description
	▪ alert: Displays event logs with severity alert and above. ▪ critical: Displays event logs with severity critical and above. ▪ error: Displays event logs with severity error and above. ▪ warning: Displays event logs with severity warning and above. ▪ notice: Displays event logs with severity notice and above. ▪ info: Displays event logs with severity info and above. ▪ debug: Displays event logs with all severities.
<code>-r</code>	Shows the most recent event logs first.
<code>-a</code>	Shows all event logs, including those events from previous boots.
<code>-n <COUNT></code>	Displays the specified number of event logs.
<code>-i <MEMBER-SLOT></code>	On a 6400: Shows the event logs for the specified slot ID.
<code>-i <MEMBER-SLOT></code>	On a 6300: Shows the event logs for the specified VSF member ID.
<code>-m {active standby}</code>	On a 6400: Shows the event logs for the specified management card role. Selecting active displays the event log for the AMM management card role and standby displays event logs for the SMM management card role.
<code>-m {active standby}</code>	On a 6300: Shows the event logs for the specified role. Selecting active displays the event log for the VSF conductor role and standby displays event logs for the VSF standby role.
<code>-c {lldp ospf ...}</code>	Shows the event logs for the specified event category. Enter <code>show event -c</code> for a full listing of supported categories with descriptions.
<code>-d {lldpd bgpd fand ...}</code>	Shows the event logs for the specified process. Enter <code>show event -d</code> for a full listing of supported daemons with descriptions.

Examples

Showing event logs:

```
switch# show events
-----
show event logs
-----
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to
up for bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1
in Hardware
```

Showing the most recent event logs first:

```
switch# show events -r
-----
show event logs
-----
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1
in Hardware
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to
up for bridge_normal interface
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
```

Showing all event logs:

```
switch# show events -a
-----
show event logs
-----
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to
up for bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1
in Hardware
```

Showing event logs related to LACP:

```
switch# show events -c lacp
-----
show event logs
-----
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
```

Showing event logs as per the specified management card role for a 6400 switch:

```
switch# show events -m active
-----
show event logs
-----
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to
up for bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1
in Hardware
```

Showing event logs as per the specified member/slot ID:

```
switch# show events -i 1/1
-----
show event logs
-----
2017-08-17:22:32:25.743991|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource
utilization poll interval is changed to 313
2017-08-17:22:33:01.692860|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource
utilization poll interval is changed to 23
```

```
2017-08-17:22:33:06.181436|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource
utilization poll interval is changed to 512
2017-08-17:22:33:06.181436|systemd-coredump|1201|LOG_CRIT|LC|1/1|hpe-sysmond
crashed due to signal:11
```

Showing event logs as per the specified process:

```
switch# show events -d lacpd
-----
show event logs
-----
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
```

Displaying the specified number of event logs:

```
switch# show events -n 5
-----
show event logs
-----
2018-03-21:06:12:15.500603|arpmgrd|6101|LOG_INFO|AMM|-|ARPMGRD daemon has started
2018-03-21:06:12:17.734405|lldpd|109|LOG_INFO|AMM|-|Configured LLDP tx-delay to 2
2018-03-21:06:12:17.740517|lacpd|1307|LOG_INFO|AMM|-|LACP system ID set to
70:72:cf:d4:34:42
2018-03-21:06:12:17.743491|vrfmgrd|5401|LOG_INFO|AMM|-|Created a vrf entity
42cc3df7-1113-412f-b5cb-e8227b8c22f2
2018-03-21:06:12:17.904008|vrfmgrd|5401|LOG_INFO|AMM|-|Created a vrf entity
4409133e-2071-4ab8-adfe-f9662c06b889
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

show fabric

Not supported on the 6300 Switch Series.

```
show fabric [<SLOT-ID>] [vsx-peer]
```

Description

Shows information about the installed fabrics.

Parameter	Description
<SLOT-ID>	Specifies the member and slot of the fabric to show. For example, to show the module in member 1, slot 2, enter the following: 1/2
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Showing all fabrics on Aruba 6400 switches that have two fabrics:

```
switch# show fabric

Fabric Modules
=====
```

	Product		Serial	
Name	Number	Description	Number	Status
1/1	R0X25A	6410 Chassis	SG9ZKM9999	Ready
1/2	R0X25A	6410 Chassis	SG9ZKM9999	Ready

Showing all fabrics on Aruba 6400 switches that have one fabric:

```
switch# show fabric

Fabric Modules
=====
```

	Product		Serial	
Name	Number	Description	Number	Status
1/1	R0X24A	6405 Chassis	SG9ZKM9076	Ready

Showing a single fabric module on Aruba 6400 switches:

```
switch# show fabric 1/1

Fabric module 1/1 is ready
Admin state: Up
Description: 6405 Chassis
Full Description: 6405 Chassis
Serial number: SG00000000
Product number: R0X24A
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6400	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show hostname

show hostname [vsx-peer]

Description

Shows the current host name.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show images

show images [vsx-peer]

Description

Shows information about the software in the primary and secondary images.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

Showing the primary and secondary images on a 6300 switch:

```
switch(config)# show images
-----
AOS-CX Primary Image
-----
Version : FL.xx.xx.xxxx
Size    : 722 MB
Date    : 2019-10-22 17:00:46 PDT
SHA-256 : 4c84e49c0961fc56b5c7eab064750a333f1050212b7ce2fab587d13469d24cfa
-----
Primary Image
-----
Version : FL.xx.xx.xxxx
Size    : 722 MB
Date    : 2019-10-22 17:00:46 PDT
SHA-256 : 4c84e49c0961fc56b5c7eab064750a333f1050212b7ce2fab587d13469d24cfa
-----
AOS-CX Secondary Image
-----
Version : FL.xx.xx.xxxx
Size    : 722 MB
Date    : 2019-10-22 17:00:46 PDT
SHA-256 : 4c84e49c0961fc56b5c7eab064750a333f1050212b7ce2fab587d13469d24cfa
-----
Default Image : secondary
-----
Management Module 1/1 (Active)
-----
Active Image      : secondary
Service OS Version : FL.01.05.0001-internal
BIOS Version      : FL.01.0001
```

Showing the primary and secondary images on a 6400 switch:

```
switch(config)# show images
-----
AOS-CX Primary Image
-----
Version : FL.xx.xx.xxxxQ-2710-gd4ac39f30c9
Size    : 766 MB
Date    : 2019-10-30 17:22:01 PDT
SHA-256 : e560ca9141f425d19024d122573c5ff730df2a9a726488212263b45ea00382cf
-----
AOS-CX Secondary Image
-----
```

```
Version : FL.xx.xx.xxxx
Size    : 722 MB
Date    : 2019-10-21 19:36:26 PDT
SHA-256 : 657e28adc1b512217ce780e3523c37c94db3d3420231deac1ab9aaa8324dc6b9
```

```
Default Image : secondary
```

```
-----
Management Module 1/1 (Active)
-----
```

```
Active Image      : secondary
Service OS Version : FL.01.05.0001-internal
BIOS Version      : FL.01.0001
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ip errors

```
show ip errors [vsx-peer]
```

Description

Shows IP error statistics for packets received by the switch since the switch was last booted.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

IP error info about received packets is collected from each active line card on the switch and is preserved during failover events. Error counts are cleared when the switch is rebooted.

Drop reasons are the following:

■ Malformed packet

The packet does not conform to TCP/IP protocol standards such as packet length or internet header length. A large number of malformed packets can indicate that there are hardware malfunctions such as loose cables, network card malfunctions, or that a DOS (denial of service) attack is occurring.

■ IP address error

The packet has an error in the destination or source IP address. Examples of IP address errors include the following:

- The source IP address and destination IP address are the same.
- There is no destination IP address.
- The source IP address is a multicast IP address.
- The forwarding header of an IPv6 address is empty.
- There is no source IP address for an IPv6 packet.

■ Invalid TTLS

The TTL (time to live) value of the packet reached zero. The packet was discarded because it traversed the maximum number of hops permitted by the TTL value.

TTLS are used to prevent packets from being circulated on the network endlessly.

Example

Showing ip error statistics for packets received by the switch:

```
switch# show ip errors
-----
Drop reason          Packets
-----
Malformed packets    1
IP address errors    10
...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show module

show module [<SLOT-ID>] [vsx-peer]

Description

Shows information about installed line modules and management modules.

Parameter	Description
<SLOT-ID>	Specifies the member and slot numbers in format member/slot. For example, to show the module in member 1, slot 3, enter 1/3.

Parameter	Description
<code>vsx-peer</code>	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

Identifies and shows status information about the line modules and management modules that are installed in the switch.

If you use the `<slot-id>` parameter to specify a slot that does not have a line module installed, a message similar to the following example is displayed:

```
Module 1/4 is not physically present.
```

To show the configuration information—if any—associated with that line module slot, use the **show running-configuration** command.

Status is one of the following values:

`Active`

This module is the active management module.

`Standby`

This module is the standby management module.

`Deinitializing`

The module is being deinitialized.

`Diagnostic`

The module is in a state used for troubleshooting.

`Down`

The module is physically present but is powered down.

`Empty`

The module hardware is not installed in the chassis.

`Failed`

The module has experienced an error and failed.

`Failover`

This module is a fabric module or a line module, and it is in the process of connecting to the new active management module during a management module failover event.

`Initializing`

The module is being initialized.

`Present`

The module hardware is installed in the chassis.

`Ready`

The module is available for use.

`Updating`

A firmware update is being applied to the module.

Examples

Showing all installed modules on a 6300 switch:

```
switch(config)# show module
```

```
Management Modules
```

```
=====
```

	Product		Serial	
Name	Number	Description	Number	Status
----	-----	-----	-----	-----

```
1/1  JL659A  6300M 48SR5 CL6 PoE 4SFP56 Swch          ID9ZKHN090 Active (local)
```

Line Modules
=====

Name	Number	Description	Serial Number	Status
1/1	JL659A	6300M 48SR5 CL6 PoE 4SFP56 Swch	ID9ZKHN090	Ready

Management Modules
=====

Name	Number	Description	Serial Number	Status
1/1	JL659A	6300M 48SR5 CL6 PoE 4SFP56 Swch	ID9ZKHN090	Active (local)

Line Modules
=====

Name	Number	Description	Serial Number	Status
1/1	JL659A	6300M 48SR5 CL6 PoE 4SFP56 Swch	ID9ZKHN090	Ready

Showing a line module on a 6400 switch:

```
switch# show module 1/3

Line module 1/3 is ready
Admin state: Up
Description: 6400 24p 10GT 4SFP56 Mod
Full Description: 6400 24-port 10GBASE-T and 4-port SFP56 Module
Serial number: SG9ZKMS045
Product number: R0X42A
Power priority: 128
```

Showing a slot that does not contain a line module:

```
switch(config)# show module 1/3
Module 1/3 is not physically present
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager	Operators or Administrators or local user group members with

Platforms	Command context	Authority
	(#)	execution rights for this command. Operators can execute this command from the operator context (>) only.

show running-config

show running-config [<FEATURE>] [all] [vsx-peer]

Description

Shows the current nondefault configuration running on the switch. No user information is displayed.

Parameter	Description
<FEATURE>	Specifies the name of a feature. For a list of feature names, enter the <code>show running-config</code> command, followed by a space, followed by a question mark (?). When the <code>json</code> parameter is used, the <code>vsx-peer</code> parameter is not applicable.
<i>all</i>	Shows all default values for the current running configuration.
<i>vsx-peer</i>	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Showing the current running configuration:

```
switch> show running-config
Current configuration:
!
!Version AOS-CX 10.0X.XXXX
!
lldp enable
linecard-module LC1 part-number JL363A
vrf green
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
vlan 1
    no shutdown
vlan 20
    no shutdown
vlan 30
    no shutdown
interface 1/1/1
```

```

no shutdown
no routing
vlan access 30
interface 1/1/32
no shutdown
no routing
vlan access 20
interface bridge_normal-1
no shutdown
interface bridge_normal-2
no shutdown
interface vlan20
no shutdown
vrf attach green
ip address 20.0.0.44/24
ip ospf 1 area 0.0.0.0
ip pim-sparse enable

interface vlan30
no shutdown
vrf attach green
ip address 30.0.0.44/24
ip ospf 1 area 0.0.0.0
ip pim-sparse enable

ip pim-sparse hello-interval 100

```

Showing the current running configuration in json format:

```

switch> show running-config json
Running-configuration in JSON:
{
  "Monitoring_Policy_Script": {
    "system_resource_monitor_mm1.1.0": {
      "Monitoring_Policy_Instance": {
        "system_resource_monitor_mm1.1.0/system_resource_monitor_
mm1.1.0.default": {
          "name": "system_resource_monitor_mm1.1.0.default",
          "origin": "system",
          "parameters_values": {
            "long_term_high_threshold": "70",
            "long_term_normal_threshold": "60",
            "long_term_time_period": "480",
            "medium_term_high_threshold": "80",
            "medium_term_normal_threshold": "60",
            "medium_term_time_period": "120",
            "short_term_high_threshold": "90",
            "short_term_normal_threshold": "80",
            "short_term_time_period": "5"
          }
        }
      }
    },
    ...
    ...
    ...
    ...

```

Show the current running configuration without default values:


```

!
snmp-server vrf mgmt
snmp-server agent-port 161
snmp-server community public
!
!
!
!
!
vlan 1
switch(config)#

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config current-context

show running-config current-context

Description

Shows the current non-default configuration running on the switch in the current command context.

Usage

You can enter this command from the following configuration contexts:

- Any child of the global configuration (`config`) context. If the child context has instances—such as interfaces—you can enter the command in the context of a specific instance. Support for this command is provided for one level below the `config` context. For example, entering this command for a child of a child of the `config` context not supported. If you enter the command on a child of the `config` context, the current configuration of that context and the children of that context are displayed.
- The global configuration (`config`) context. If you enter this command in the global configuration (`config`) context, it shows the running configuration of the entire switch. Use the `show running-configuration` command instead.

Examples

On the 6400 Switch Series, interface identification differs.

Showing the running configuration for the current interface:

```
switch(config-if)# show running-config current-context
```

```

interface 1/1/1
vsx-sync qos vlans
    no shutdown
    description Example interface
vlan access 1
    exit

```

Showing the current running configuration for the management interface:

```

switch(config-if-mgmt)# show running-config current-context
interface mgmt
    no shutdown
    ip static 10.0.0.1/24
    default-gateway 10.0.0.8
    nameserver 10.0.0.1

```

Showing the running configuration for the external storage share named `nasfiles`:

```

switch(config-external-storage-nasfiles)# show running-config current-context
external-storage nasfiles
    address 192.168.0.1
    vrf default
    username nasuser
    password ciphertext
    AQBapalKj+XMsZumHEwIc9OR6YcOw5Z6Bh9rV+9ZtKDKzvbaBAAAAB1CTrM=
    type scp
    directory /home/nas
    enable
switch(config-external-storage-nasfiles)#

```

Showing the running configuration for a context that does not have instances:

```

switch(config-vsx)# show run current-context
vsx
    inter-switch-link 1/1/1
    role secondary
    vsx-sync sflow time

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code> or a child of <code>config</code> . See Usage.	Administrators or local user group members with execution rights for this command.

show startup-config

```
show startup-config [json]
```

Description

Shows the contents of the startup configuration.



Switches in the *factory-default* configuration do not have a startup configuration to display.

Parameter	Description
<code>json</code>	Display output in JSON format.

Examples

Showing the startup-configuration in non-JSON format for a 6300 switch:

```
switch(config)# show startup-config
Startup configuration:
!
!Version AOS-CX FL.xx.xx.xxxx
!export-password: default
hostname BLDG01-F1
user admin group administrators password ciphertext

AQBapWl8I2ZunZ43NE/8KlbQ7zYC4gTT6uSFYi6n6wyY9PdBYgAAACONCR/3+AcNvzRBch0DoG7W9z84Lp
JA+6C9SKfNwCqi5/
nUPk/ZOvN91/EQXvPNkHtBtQWYyZqfkebbEH78VVRHfWZjApv4II9qmQfxpA79wEvzshdzZmuAKrm
user ateam group administrators password ciphertext

AQBapcPqMXoF+H10NkrqAedXLvlSRwf4wUEL22hXGD6ZBhicYgAAAGsbh70DKglu+Ze1wxgmDXjkGO3bse
YiR3LKQg66vrfrqR/
M3oLl1iPdZDnq9XMMvCL+7jBbYhYes8+uDxuSTh8kdkd/qj3lo5FUuC5fENGcjU0YI1l7qtU+YEnsJ
!
!
!
!
radius-server host 10.10.10.15
!
radius dyn-authorization enable
ssh server vrf default
ssh server vrf mgmt
!
!
!
!
!
router ospf 1
    router-id 1.63.63.1
    area 0.0.0.0
vlan 1
vlan 66
    name vlan66
vlan 67
    name vlan67
vlan 999
    name vlan999
vlan 4000
spanning-tree
interface mgmt
    no shutdown
```

```
ip static 10.6.9.15/24
default-gateway 10.6.9.1
```

Showing the startup-configuration in JSON format:

```
switch# show startup-config json
Startup configuration:
{
  "AAA_Server_Group": {
    "local": {
      "group_name": "local"
    },
    "none": {
      "group_name": "none"
    }
  },
  ...
}
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show system error-counter-monitor

```
show system error-counter-monitor {basic <PORT-NUM> | extended} [vsx-peer]
```

Description

Shows error counter statistics.

Parameter	Description
basic <PORT-NUM>	Specifies a physical port on the switch. Use the format member/slot/port (for example, 1/3/1).
extended	Shows statistics for all interfaces.

Examples

Showing error counter statistics for interface 1/1/1:

```
switch# show system error-counter-monitor basic 1/1/1
```

```
Interface error counter statistics for 1/1/1
```

Error Counter	Value
EtherStatsOversizePkts	983
EtherStatsUndersizePkts	1024
EtherStatsJabbers	10
Dot3StatsAlignmentErrors	462
Dot3StatsFCSErrors	321
Dot3StatsLateCollisions	2024
EtherStatsFragments	121
Dot3StatsExcessiveCollisions	1025
IfInBroadcastPkts	2001

Showing error counter statistics for all interfaces:

```
switch# show system error-counter-monitor extended
```

```
Interface error counter statistics for 1/1/1
```

Error Counter	Value
EtherStatsOversizePkts	983
EtherStatsUndersizePkts	1024
EtherStatsJabbers	10
Dot3StatsAlignmentErrors	462
Dot3StatsFCSErrors	321
Dot3StatsLateCollisions	2024
EtherStatsFragments	121
Dot3StatsExcessiveCollisions	1025
IfInBroadcastPkts	2001

```
...  
...
```

```
Interface error counter statistics for 1/8/32
```

Error Counter	Value
EtherStatsOversizePkts	0
...	

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

show system

```
show system [serviceos password-prompt] [vsx-peer]
```

Description

Shows general status information about the system.

Parameter	Description
<code>serviceos password-prompt</code>	Shows the Service OS password prompt status.
<code>vsx-peer</code>	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

CPU utilization represents the average utilization across all the CPU cores.

System Description, System Contact, and System Location can be set with the `snmp-server` command.

When `vsx-peer` is specified, the `Up Time` value is not shown because it is not synchronized between VSX peers.

Examples

Showing system information:

```
switch# show system
Hostname                : switch
System Description      : switch description
System Contact          : contact
System Location         : location

Vendor                  : Aruba
Product Name            : XXXXXX ...
Chassis Serial Nbr      : XXXXXXXXXXXX
Base MAC Address        : xxxxxx-xxxxxx
AOS-CX Version          : XX.99.99.9999

Time Zone               : UTC

Up Time                 : 1 week, 5 hours, 28 minutes
CPU Util (%)            : 5
CPU Util (% avg 1 min) : 11
CPU Util (% avg 5 min) : 10
Memory Usage (%)        : 35
```

Showing the Service OS password prompt status:

```
switch# show system serviceos password-prompt
password-prompt: disabled
```

Showing system information for a VSX primary and secondary (peer) switch:

```
switch# show system
Hostname                : vsx-primary
System Description      : switch description
System Contact          : contact
System Location         : location
```

```

Vendor           : Aruba
Product Name     : XXXXXX ...
Chassis Serial Nbr : XXXXXXXXXXXX
Base MAC Address  : xxxxxx-xxxxxx

Hostname         : vsx-primary
System Description : switch description
System Contact    : contact
System Location   : location

Vendor           : Aruba
Product Name     : XXXXXX ...
Chassis Serial Nbr : XXXXXXXXXXXX
Base MAC Address  : xxxxxx-xxxxxx
AOS-CX Version    : XX.99.99.9999

Time Zone        : UTC

Up Time          : 1 week, 2 hours, 15 minutes
CPU Util (%)     : 15
CPU Util (% avg 1 min) : 12
CPU Util (% avg 5 min) : 8
Memory Usage (%) : 37

switch# show system vsx-peer
Hostname         : vsx-secondary
System Description : switch description
System Contact    : contact
System Location   : location

Vendor           : Aruba
Product Name     : XXXXXX ...
Chassis Serial Nbr : XXXXXXXXXXXX
Base MAC Address  : xxxxxx-xxxxxx
AOS-CX Version    : XX.99.99.9999

Time Zone        : UTC

CPU Util (%)     : 7
CPU Util (% avg 1 min) : 13
CPU Util (% avg 5 min) : 9
Memory Usage (%) : 32

```

Command History

Release	Modification
10.12	Added CPU Util (% avg 1 min) and CPU Util (% avg 5 min).
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager	Operators or Administrators or local user group members with

Platforms	Command context	Authority
	(#)	execution rights for this command. Operators can execute this command from the operator context (>) only.

show system resource-utilization

Applies to the 6300 Switch Series:

```
show system resource-utilization [all | daemon <DAEMON-NAME>] |
standby | member <MEMBER-NUM>] [vsx-peer]
```

Applies to the 6400 Switch Series:

```
show system resource-utilization [all | daemon <DAEMON-NAME>] |
standby | module <SLOT-ID>] [vsx-peer]
```

Description

Shows the system resource utilization data.

Parameter	Description
all	Shows the resource utilization data for the entire switch.
daemon <DAEMON-NAME>	Shows only the resource utilization data for the process identified by <DAEMON-NAME>.
standby	Shows only the resource utilization data for the standby management module.
member <MEMBER-NUM>	(Applies to the 6300 Switch Series.) Shows only the resource utilization data for the VSF member identified by <MEMBER-NUM>.
module <SLOT-ID>	(Applies to the 6400 Switch Series.) Shows only the resource utilization data for the line module identified by <SLOT-ID>.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

For a list of daemons that log events, enter `show events -d ?` from a switch prompt in the manager (#) context.

Examples

Showing system resource utilization data:

```
switch# show system resource-utilization
System Resources:
Processes                : 144
CPU usage(%)             : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%)          : 22
```

```

Open FD's                : 1358
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b),
Health = normal

Data written to various partitions since boot
Nos      : 5 MB
Log      : 1 MB
Coredump : 23 MB
Security : 2 MB
Selftest : 405 KB
Swap     : 14 MB

Storage partition usage(%)
Nos      : 5
Log      : 60
Coredump : 23
Security : 2
Selftest : 1
Swap     : 0

Process                CPU Usage(%)    Memory Usage(%)    Open FD's
-----
hpe-sysmond            1                2                11
hpe-mgmdd              0                1                 5
...

```

Attempting to show resource utilization data when system resource utilization polling is disabled:

```

switch# show system resource-utilization
System resource utilization data poll is currently disabled

```

Showing the resource utilization data for a particular process:

```

switch# show system resource-utilization daemon hpe-sysmond
Process                CPU Usage(%)    Memory Usage(%)    Open FD's
-----
hpe-sysmond            1                2                11

```

(Applies to the 6300 Switch Series.) Showing resource utilization data for all VSF members:

```

aaa

```

(Applies to the 6300 Switch Series.) Showing resource utilization data for a particular VSF member:

```

switch# show system resource-utilization member 2
-----
Resource utilization data for vsf member 2
-----

System Resources:
Processes                : 244

```

```

CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 1854
Storage 1: Endurance utilization = 0-10% (mmc-type-a), 0-10% (mmc-type-b),
Health = normal

Data written to various partitions since boot
Nos : 15 MB
Log : 1 MB
Coredump : 23 MB
Security : 2 MB
Selftest : 0 KB
Swap : 0 MB

Storage partition usage(%)
Nos : 5
Log : 60
Coredump : 23
Security : 2
Selftest : 1
Swap : 0

Process CPU Usage(%) Memory Usage(%) Open FD's
-----
(sd-pam) 0 0 7
agetty 0 0 4
ata_sff 0 0 0
...

```

```

switch# show system resource-utilization all
-----
Resource utilization data for vsf member 1
-----

System Resources:
Processes : 244
CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 1854
Storage 1: Endurance utilization = 0-10% (mmc-type-a), 0-10% (mmc-type-b),
Health = normal

Data written to various partitions since boot
Nos : 15 MB
Log : 1 MB
Coredump : 23 MB
Security : 2 MB
Selftest : 405 KB
Swap : 14 MB

Storage partition usage(%)
Nos : 5
Log : 60
Coredump : 23
Security : 2
Selftest : 1

```

```

Swap      :    0

Process           CPU Usage(%)    Memory Usage(%)    Open FD's
-----
(sd-pam)          0                0                  7
aaa
utilspamcfg       0                1                 10
-----

Resource utilization data for vsf member 2
-----

System Resources:
Processes          : 244
CPU usage(%)       : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%)    : 11
Open FD's          : 1854
Storage 1: Endurance utilization = 0-10% (mmc-type-a), 0-10% (mmc-type-b),
Health = normal

Data written to various partitions since boot
Nos      : 15 MB
Log      : 1 MB
Coredump : 23 MB
Security : 2 MB
Selftest : 0 KB
Swap     : 0 MB

Storage partition usage(%)
Nos      : 5
Log      : 60
Coredump : 23
Security : 2
Selftest : 1
Swap     : 0

Process           CPU Usage(%)    Memory Usage(%)    Open FD's
-----
(sd-pam)          0                0                  7
agetty            0                0                  4
ata_sff           0                0                  0

```

(Applies to the 6400 Switch Series.) Showing resource utilization data for the standby management module:

```

switch# show system resource-utilization standby
System Resources:
Processes          : 244
CPU usage(%)       : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%)    : 11
Open FD's          : 1854
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b)
Health = normal

Data written to various partitions since boot
Nos      : 15 MB

```

```

Log      : 1 MB
Coredump : 23 MB
Security : 2 MB
Selftest : 405 KB
Swap     : 14 MB

```

Storage partition usage(%)

```

Nos      : 5
Log      : 60
Coredump : 23
Security : 2
Selftest : 1
Swap     : 0

```

Process	CPU Usage(%)	Memory Usage(%)	Open FD's
hpe-sysmond	1	2	11
hpe-mgmdd	0	1	5
...			

(Applies to the 6400 Switch Series.) Showing resource utilization data for a line module:

```

switch# show system resource-utilization module 1/5
-----
System Resource utilization for line card module: 1/5
-----
CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 754

```

(Applies to the 6400 Switch Series.) Showing resource utilization data for all modules:

```

switch# show system resource-utilization all
-----
Resource utilization data for Management Module
-----

System Resources:
Processes : 244
CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 1854
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b),
Health = normal

Data written to various partitions since boot
Nos : 15 MB
Log : 1 MB
Coredump : 23 MB
Security : 2 MB
Selftest : 405 KB

```

Swap : 14 MB

Storage partition usage(%)

Nos : 5
Log : 60
Coredump : 23
Security : 2
Selftest : 1
Swap : 0

Process	CPU Usage(%)	Memory Usage(%)	Open FD's
(sd-pam)	0	0	7
aaa			
utilspamcfg	0	1	10

Resource utilization data for Standby Management Module

System Resources:

Processes : 244
CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 1854
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b),
Health = normal

Data written to various partitions since boot

Nos : 15 MB
Log : 1 MB
Coredump : 23 MB
Security : 2 MB
Selftest : 405 KB
Swap : 0 KB

Storage partition usage(%)

Nos : 5
Log : 60
Coredump : 23
Security : 2
Selftest : 1
Swap : 0

System Resource utilization for line card module: 1/7

CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 854

System Resource utilization for line card module: 1/8

CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15

```
Memory usage (%)      :    11
Open FD's             :   980
```

Command History

Release	Modification
10.12	The output of this command includes CPU usage(% average over 1 minute) and CPU usage(% average over 5 minute) .
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show tech

```
show tech [basic | <FEATURE>] [local-file]
```

Description

Shows detailed information about switch features by automatically running the `show` commands associated with the feature. If no parameters are specified, the `show tech` command shows information about all switch features. Technical support personnel use the output from this command for troubleshooting.

Parameter	Description
<code>basic</code>	Specifies showing a basic set of information.
<code><FEATURE></code>	Specifies the name of a feature. For a list of feature names, enter the <code>show tech</code> command, followed by a space, followed by a question mark (?).
<code>local-file</code>	Shows the output of the <code>show tech</code> command to a local text file.

Usage

To terminate the output of the `show tech` command, enter **Ctrl+C**.

If the command was not terminated with **Ctrl+C**, at the end of the output, the `show tech` command shows the following:

- The time consumed to execute the command.
- The list of failed `show` commands, if any.

To get a copy of the local text file content created with the `show tech` command that is used with the `local-file` parameter, use the `copy show-tech local-file` command.

Example

Showing the basic set of system information:

```
switch# show tech basic
=====
Show Tech executed on Wed Sep  6 16:50:37 2017
=====
[Begin] Feature basic
=====

*****
Command : show core-dump all
*****
no core dumps are present

...

[End] Feature basic
=====

1 show tech command failed
=====
Failed command:
  1. show boot-history
=====
Show tech took 3.000000 seconds for execution
```

Directing the output of the **show tech basic** command to the local text file:

```
switch# show tech basic local-file
Show Tech output stored in local-file. Please use 'copy show-tech local-file'
to copy-out this file.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show usb

show usb [vsx-peer]

Description

Shows the USB port configuration and mount settings.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

If USB has not been enabled:

```
switch> show usb
Enabled: No
Mounted: No
```

If USB has been enabled, but no device has been mounted:

```
switch> show usb
Enabled: Yes
Mounted: No
```

If USB has been enabled and a device mounted:

```
switch> show usb
Enabled: Yes
Mounted: Yes
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show usb file-system

```
show usb file-system [<PATH>]
```

Description

Shows directory listings for a mounted USB device. When entered without the <PATH> parameter the top level directory tree is shown.

Parameter	Description
<PATH>	Specifies the file path to show. A leading "/" in the path is optional.

Usage

Adding a leading "/" as the first character of the <PATH> parameter is optional.

Attempting to enter '.' as any part of the <PATH> will generate an invalid path argument error. Only fully-qualified path names are supported.

Examples

Showing the top level directory tree:

```
switch# show usb file-system
/mnt/usb:
'System Volume Information'  dir1'

/mnt/usb/System Volume Information':
IndexerVolumeGuid  WPSettings.dat

/mnt/usb/dir1:
dir2  test1

/mnt/usb/dir1/dir2:
test2
```

Showing available path options from the top level:

```
switch# show usb file-system /
total 64
drwxrwxrwx 2 32768 Jan 22 16:27 'System Volume Information'
drwxrwxrwx 3 32768 Mar  5 15:26 dir1
```

Showing the contents of a specific folder:

```
switch# show usb file-system /dir1
total 32
drwxrwxrwx 2 32768 Mar  5 15:26 dir2
-rwxrwxrwx 1      0 Feb  5 18:08 test1

switch# show usb file-system dir1/dir2
total 0
-rwxrwxrwx 1 0 Feb  6 05:35 test2
```

Attempting to enter an invalid character in the path:

```
switch# show usb file-system dir1/../../
Invalid path argument
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show version

show version [vsx-peer]

Description

Shows version information about the network operating system software, service operating system software, and BIOS.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

Showing version information for a 6300 switch:

```
6300# show version
-----
ArubaOS-CX
(c) Copyright 2017-2022 Hewlett Packard Enterprise Development LP
-----
Version       : FL.10.10.0001BJ
Build Date    : 2022-05-25 10:22:06 UTC
Build ID      : ArubaOS-CX:FL.10.10.0001BJ:16d4d3ca52e9:202205908
Build SHA     : 16d4d349695b50298f34b21a8c67637ae0
Hot Patches   : hpe-routing_FL_10_10_0001BJ.patch
Active Image  : primary

Service OS Version : FL.01.11.0001-internal
BIOS Version      : FL.01.0004
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

system resource-utilization poll-interval

system resource-utilization poll-interval <SECONDS>

Description

Configures the polling interval for system resource information collection and recording such as CPU and memory usage.

Parameter	Description
<SECONDS>	Specifies the poll interval in seconds. Range: 10-3600. Default: 10.

Example

Configuring the system resource utilization poll interval:

```
switch(config)# system resource-utilization poll-interval 20
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

top cpu

top cpu

Description

Shows CPU utilization information.

Example

Showing top CPU information:

```
switch# top cpu
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
```

```

Tasks: 76 total, 2 running, 74 sleeping, 0 stopped, 0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total, 2487508 free, 897040 used, 661948 buff/cache
KiB Swap: 0 total, 0 free, 0 used, 2859196 avail Mem

  PID USER      PRI  NI   VIRT   RES   SHR S  %CPU  %MEM     TIME+ COMMAND
...

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

top memory

top memory

Description

Shows memory utilization information.

Example

Showing top memory:

```

switch> top memory
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks: 76 total, 2 running, 74 sleeping, 0 stopped, 0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total, 2487508 free, 897040 used, 661948 buff/cache
KiB Swap: 0 total, 0 free, 0 used, 2859196 avail Mem

  PID USER      PRI  NI   VIRT   RES   SHR S  %CPU  %MEM     TIME+ COMMAND
...

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

usb

usb
no usb

Description

Enables the USB ports on the switch. This setting is persistent across switch reboots and management module failovers. Both active and standby management modules are affected by this setting.

The **no** form of this command disables the USB ports.

Example

Enabling USB ports:

```
switch(config)# usb
```

Disabling USB ports when a USB drive is mounted:

```
switch(config)# no usb
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

usb mount | unmount

usb {mount | unmount}

Description

Enables or disables the inserted USB drive.

Parameter	Description
mount	Enables the inserted USB drive.
unmount	Disables the inserted USB drive in preparation for removal.

Usage

If USB has been enabled in the configuration, the USB port on the active management module is available for mounting a USB drive. The USB port on the standby management module is not available. An inserted USB drive must be mounted each time the switch boots or fails over to a different management module.

A USB drive must be unmounted before removal.

The supported USB file systems are FAT16 and FAT32.

Examples

Mounting a USB drive in the USB port:

```
switch# usb mount
```

Unmounting a USB drive:

```
switch# usb unmount
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	https://www.arubanetworks.com/support-services/
AOS-CX Switch Software Documentation Portal	https://www.arubanetworks.com/techdocs/AOS-CX/help_portal/Content/home.htm
HPE Aruba Networking Support Portal	https://networkingsupport.hpe.com/home
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe-aruba-networking-aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS-CX Software Technical Update channel on	Videos on new features introduced in this release: https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP-Q_UL3CskS

YouTube.	
HPE Aruba Networking Hardware Documentation and Translations Portal	https://www.arubanetworks.com/techdocs/hardware/DocumentationPortal/Content/home.htm
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End-of-Life information	https://www.arubanetworks.com/support-services/end-of-life/

Accessing Updates

You can access updates from the Aruba Support Portal or the HPE My Networking Website.

Aruba Support Portal

<https://networkingsupport.hpe.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>

Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://networkingsupport.hpe.com/notifications/subscriptions> (requires an active HPE Aruba Networking support account to manage subscriptions). Security notices are viewable without a networking support account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

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